Vom Fachbereich Produktionstechnik der
UNIVERSITÄT BREMEN

zur Erlangung des Grades
Doktor der Ingenieurwissenschaften (Dr.-Ing.)
genehmigte Dissertation

Dynamic Electrochemical Impedance Spectroscopy for Energy Storage Applications: from Electrode Kinetics to Cell Degradation

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Tag der mündlichen Prüfung: 3 September 2026

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Date	Federico Scarpioni

Scientific contributions

Peer-reviewed journals

- *Nicolò Pianta, Federico Scarpioni, Richard Chukwu, Fabio La Mantia, Riccardo Ruffo*
"Evaluation of kinetic parameters of non-faradic processes in carbon-based electrodes using multisine dynamic electrochemical impedance spectroscopy"
Electrochemical Acta (2023)

- *Federico Scarpioni, Shahid Khalid, Richard Chukwu, Nicolò Pianta, Fabio La Mantia, Riccardo Ruffo*
"Electrochemical Impedance Spectroscopy for Electrode Process Evaluation: Lithium Titanium Phosphate in Concentrated Aqueous Electrolyte"
ChemElectroChem (2023)

- *Noël Hallemans, David Howey, Alberto Battistel, Nessa Fereshteh Saniee, Federico Scarpioni, Benny Wouters, Fabio La Mantia, Annick Hubin, Widanalage Dhammika Widanage, John Lataire*
"Electrochemical impedance spectroscopy beyond linearity and stationarity—A critical review"
Electrochemical Acta (2023)

- *Lorenz F. Olbrich, Nicoloò Pianta, Ben Jagger, Yiming Xu, Manav Kakkanat, Federico Scarpioni, Christopher Allen, Fabio La Mantia, Riccardo Ruffo, and Mauro Pasta*
"Electrochemical impedance spectroscopy beyond linearity and stationarity—A critical review"
ACS Electrochemistry (2026)

- *Federico Scarpioni, Cécile Pot d'or, Doriano Brogioli, Fabio La Mantia*
"Critical Review on Dynamic Electrochemical Impedance Spectroscopy"
Current Opinion in Electrochemistry (2026)

Scientific congresses

- Oral contribution: "Towards an on-line methodology for lithium-ion batteries state of health estimation using dynamic impedance and Machine Learning: from a novel experimental set-up to feature extraction"
FEMS Euromat 2023, Frankfurt (Germany)

- Oral contribution: "Dynamic Electrochemical Impedance Spectroscopy for characterization of the state of health of graphite electrodes"
244^{th} ECS Meeting 2023, Gothenburg (Sweden)

- Pitch presentation: "Dynamic Electrochemical Impedance Spectroscopy for the Investigation of Materials for Energy Storage in Three-Electrode Cells" - **Best Pitch Award supported by SCI**
37^{th} Topical Meeting of the International Society of Electrochemistry 2024, Stresa (Italy)

- Poster presentation: "Dynamic Electrochemical Impedance Spectroscopy for the Investigation of Materials for Energy Storage in Three-Electrode Cells"
  Electrochemistry 2024 - GDCh, Braunschweig (Germany)
- Oral contribution: "Broad-band multi-sine non-stationary impedance spectroscopy for the monitoring of battery state"
  76^{th} Annual Meeting of the International Society of Electrochemistry 2025, Mainz (Germany)

Abstract

Battery degradation is governed by electrochemical processes evolving across multiple time scales and operating conditions. Electrochemical impedance spectroscopy (EIS) is widely used to study these processes because it provides frequency-resolved information on interfacial kinetics, transport phenomena, and charge storage mechanisms. However, classical EIS relies on stationary conditions and long acquisition times, which restrict its use during realistic battery operation and limit its applicability in long-term aging studies.

This work addresses these limitations by developing a framework for the estimation of instantaneous impedance under non-stationary conditions, referred to as dynamic electrochemical impedance spectroscopy (DEIS). The method is grounded in a system-theoretical interpretation of impedance as the linearized response of a time-varying electrochemical system, and relies on broadband multi-frequency perturbations and Fourier-based analysis to track impedance continuously during operation.

A dedicated experimental and software framework was developed to enable the design, execution, and interpretation of dynamic impedance experiments. The methodology introduces automated experiment management and online impedance estimation, enabling long-duration measurements and substantially reducing data storage requirements. The frequency bandwidth achieved for battery measurements spans up to seven decades, extending the range typically explored in dynamic impedance studies.

The applicability of DEIS is demonstrated across different electrochemical systems, chemistries, and cell configurations, including full cells, half-cells, and three-electrode setups revealing the time evolution of interfacial properties. For batteries, these measurements highlight clear relationships between impedance features, state of charge, and degradation mechanisms, supporting the use of DEIS as a diagnostic tool for tracking state of health during aging.

Zusammenfassung

Die Alterung von Batterien wird durch elektrochemische Prozesse bestimmt, die sich über mehrere Zeitskalen und Betriebsbedingungen hinweg entwickeln. Die elektrochemische Impedanzspektroskopie (EIS) wird häufig zur Untersuchung dieser Prozesse eingesetzt, da sie frequenzaufgelöste Informationen über Grenzflächenkinetik, Transportphänomene und Ladungsspeichermechanismen liefert. Klassische EIS setzt jedoch stationäre Bedingungen und lange Messzeiten voraus, was ihren Einsatz während des realistischen Batteriebetriebs einschränkt und ihre Anwendbarkeit in Langzeit-Alterungsstudien begrenzt.

Diese Arbeit adressiert diese Einschränkungen durch die Entwicklung eines Rahmens zur Bestimmung der momentanen Impedanz unter nichtstationären Bedingungen, bezeichnet als Dynamic Electrochemical Impedance Spectroscopy (DEIS). Die Methode basiert auf einer systemtheoretischen Interpretation der Impedanz als linearisierten Antwort eines zeitvarianten elektrochemischen Systems und nutzt breitbandige Mehrfrequenz-Anregungen sowie Fourier-basierte Auswerteverfahren, um die Impedanz kontinuierlich während des Betriebs zu verfolgen.

Zur Umsetzung wurde ein spezielles experimentelles und softwarebasiertes Framework entwickelt, das die Planung, Durchführung und Interpretation dynamischer Impedanzexperimente ermöglicht. Die Methodik umfasst ein automatisiertes Versuchsmanagement sowie eine Online-Schätzung der Impedanz, wodurch Langzeitmessungen realisierbar werden und der Speicherbedarf erheblich reduziert wird. Der für Batteriemessungen erreichte Frequenzbereich erstreckt sich über bis zu sieben Dekaden und erweitert damit den in dynamischen Impedanzstudien üblicherweise untersuchten Bereich.

Die Anwendbarkeit von DEIS wird an verschiedenen elektrochemischen Systemen, Chemien und Zellkonfigurationen demonstriert, einschließlich Vollzellen, Halbzellen und Drei-Elektroden-Aufbauten, wobei die zeitliche Entwicklung der Grenzflächeneigenschaften sichtbar gemacht wird. Für Batteriesysteme zeigen diese Messungen klare Zusammenhänge zwischen Impedanzmerkmalen, Ladezustand und Degradationsmechanismen auf und unterstützen damit den Einsatz von DEIS als diagnostisches Werkzeug zur Verfolgung des Gesundheitszustands während der Alterung.

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Chapter 1

Introduction

The transition to electric mobility and renewable energy storage demands batteries with high energy density, fast charging capability, long operational lifetime, and stringent safety requirements. Lithium-ion batteries have dominated the market for portable electronics and are now the technology of choice for electric vehicles, electrical grid stabilization and renewable energy storage. Yet, significant challenges remain in understanding and predicting their degradation mechanisms during operation. Battery aging is a complex interplay of processes occurring at multiple time scales from fast charge-transfer reactions at electrode-electrolyte interfaces (microseconds to milliseconds) to slow solid-state diffusion and mechanical degradation (seconds to hours). The understanding and prediction of a battery, a dynamical system, can be performed from characterization in the time or frequency domain; the latter, allowing a much more effective way of separating processes at multiple time scales with one simple experiment, is the subject of this work.

In electrochemistry, the frequency domain identification technique is referred to as electrochemical impedance spectroscopy (EIS), extensively used in battery research. By measuring the frequency-dependent response of a battery to small alternating current or voltage perturbations, EIS separates process responses based on their characteristic time constants [1]. A single impedance spectrum contains information about charge-transfer resistance, double-layer capacitance, ion diffusion, and adsorption phenomena across a frequency range typically spanning from mHz to kHz. This multi-scale view makes EIS invaluable for mechanistic studies and performance diagnosis.

However, conventional EIS measurements require the battery to be in thermodynamic equilibrium, where the system state (the set of its properties) does not change during the measurement time. For a full spectrum covering six or seven decades of frequency in the band mHz to kHz, this requirement translates to measurement times of 30 minutes to several hours. During battery cycling, interrupting the charge or discharge to perform EIS introduces significant overhead time and alters the electrochemical and thermal state of the cell. As a result, operando impedance measurements have been limited to either narrow frequency bands in the middle to high range (above 1 Hz), which sacrifices information about transport, or sparse equilibrium measurements at fixed states of charge, which miss the continuous evolution across voltage and cycles. This gap becomes critical when studying phenomena that evolve continuously during cycling: solid-electrolyte interphase growth, lithium plating, particle cracking, or state-dependent kinetics. Furthermore, measuring impedance during oper-

ation provides *in situ* information under realistic electrochemical conditions (varying state of charge, current load, thermal state) that cannot be replicated in equilibrium measurements.

Electrochemical impedance spectroscopy, however, is not merely an experimental technique but a system identification method with strict underlying assumptions. In its classical formulation, impedance is defined only for linear and time-invariant systems, implying that the electrochemical state must remain effectively constant over the duration of the measurement. Batteries operated under realistic conditions violate this assumption by construction: their state of charge, interfacial composition, temperature, and reaction rates evolve continuously during charge and discharge. As a consequence, extending impedance spectroscopy to operando battery measurements is not simply a matter of faster data acquisition, but requires rethinking how impedance is defined, estimated, and validated for non-stationary systems. This challenge motivates a unified treatment of electrochemistry, frequency-domain system theory, digital signal processing, and experimental instrumentation, which together form the conceptual foundation of this thesis.

1.1 Scope

This thesis addresses this gap by developing a complete methodology for broadband electrochemical impedance spectroscopy in non-stationary conditions. By combining commercial instrumentation with custom automation software and online signal processing, the setup enabled continuous impedance acquisition across seven decades of frequency (10 mHz to 100 kHz) throughout multi-week battery cycling experiments. The methodology removes the stationary-state requirement through multi-frequency perturbation techniques and time-resolved impedance estimation, while solving the practical challenges of data volume (reducing storage requirements by four orders of magnitude through online processing) and multi-channel operation. The open-source software stack makes this capability accessible to other research groups using standard laboratory equipment.

The development of this methodology required integrating knowledge from multiple disciplines. Electrochemistry, describing the physical processes at electrode-electrolyte interfaces, was necessary to design appropriate perturbation signals, interpret the resulting impedance spectra, and design reference electrodes. Battery technology knowledge informed the choice of active materials, cell geometries, and cycling protocols relevant to practical applications, as well as device performance evaluation. The hardware implementation demanded familiarity with potentiostat operation and electronics to configure it with a waveform generator and signal acquisition device. Digital signal processing theory was necessary for the design of multi-sine excitations, the implementation of Fourier transform-based impedance estimation, and strategies for managing spectral leakage and noise. System identification theory provided concepts for understanding the classification of systems by stationarity and linearity, and the concept of transfer function representations. Finally, the automation and online processing required advanced programming in Python, including multi-threaded operation, efficient numerical computation with NumPy, and custom hardware interfaces, while the fitting of instantaneous impedance spectra demanded expertise in mathematical optimization. This thesis therefore

sits at the intersection of experimental electrochemistry, electrical engineering, and computational methods, reflecting the inherently multidisciplinary nature of modern battery research.

The primary focus of this work was the development and validation of the measurement methodology itself. The thesis demonstrates the technique's applicability across multiple electrochemical energy storage systems: commercial batteries, laboratory cells with reference electrodes, and Faradaic (intercalation) and non-Faradaic (adsorption) processes in single electrode materials. While physical models are applied to well-known systems (chapters 10-11), the work does not attempt comprehensive mechanistic modeling or systematic correlation of impedance features to specific battery degradation modes (chapter 9). Such applications require the foundation that this work provides: a robust, accessible platform for generating the large datasets needed for data-driven electrochemical and battery studies.

This thesis is organized as follows. Part I provides the theoretical framework: chapter 2 reviews fundamental electrochemistry concepts, chapter 3 discusses battery technology and characterization methods, chapter 4 introduces system identification theory for frequency-domain characterization, chapter 5 and chapter 6 survey the scientific literature state-of-the-art in non-stationary impedance spectroscopy.

Part II describes the methodologies developed in this work: chapter 7 details the automated data acquisition system, multi-sine signal design, and chapter 8 the method for offline and online impedance estimation.

Part III reports results and discussion: chapter 9 validates the methodology on commercial coin cells throughout their cycling lifetime, chapters 10-11 demonstrate applications to single-electrode studies (lithium titanium phosphate and electrical double-layer capacitors), and chapter 12 presents exploratory measurements on lithium and sodium metal anodes that reveal methodological challenges for future development. Chapters 13 and 14 conclude with a summary of achievements and outlook for future work.

Part I

Theoretical framework

Chapter 2

Electrochemistry

Electrochemistry provides the fundamental framework for understanding the operation and aging of rechargeable batteries. All essential processes in a battery (ion intercalation, charge transfer, solid-electrolyte interphase formation, and lithium plating) are governed by electrochemical thermodynamics and kinetics of the redox reactions occurring at the interfaces between electrodes and electrolyte. The voltage, efficiency, power capability, and long-term stability of a cell arise from these reactions and transport mechanisms taking place at these interfaces.

To characterize these processes in electrochemistry, the electrochemical impedance spectroscopy (EIS) technique is used for its capability of separating processes occurring at different time scales in one single experiment while being non-destructive to the system and operating *in situ*, meaning that the system is kept in its operational state (in contrast to other spectroscopy analyses that require specific sample preparation).

EIS does not probe isolated physicochemical phenomena, but rather the linearized response of a coupled system involving charge transfer, mass transport, and interfacial storage. A meaningful interpretation of impedance spectra therefore requires identifying which electrochemical processes can contribute to the measured response, over which time and length scales, and under which thermodynamic and kinetic assumptions.

For this reason, the present chapter introduces only those electrochemical concepts that directly govern the small-signal response of an electrochemical interface, with particular emphasis on interfacial kinetics, capacitive storage, and transport phenomena that manifest as distinct features in the impedance spectrum. First, the thermodynamics and kinetics of interfacial redox reactions are discussed, referring to the basic equations describing electrode potential and charge-transfer resistance. Then, the structure of the electrical double-layer is described, including the adsorption capacitance. The chapter follows with a brief description of electrochemical methods based on direct or alternating current control; the emphasis is on electrochemical impedance spectroscopy.

2.1 Thermodynamics

Electrochemical systems convert chemical energy into electrical work through redox reactions occurring at electrode-electrolyte interfaces. At equilibrium, when no net current flows in the cell, the driving force for electron transfer is quantified by the electrode potential, which originates from differences in the electrochemical potential of ionic and electronic

species in the two phases.

2.1.1 Electrochemical potential

For a species i with charge number z_i, the electrochemical potential is defined as

$$\tilde{\mu}_i = \mu_i + z_i F \phi, \tag{2.1}$$

where μ_i is the chemical potential, F is Faraday's constant, and ϕ is the electric potential. At thermodynamic equilibrium, the electrochemical potentials of all species involved in a half-reaction must be equal on both sides of the interface. For a generic half-reaction

$$\sum_j \nu_j X_j + n e^- \rightleftharpoons \sum_k \nu_k X_k,$$

the associated change in Gibbs free energy is related to the equilibrium electrode potential E through

$$\Delta G = -nFE, \tag{2.2}$$

which provides the thermodynamic link between chemical composition and measurable cell voltage.

2.1.2 The Nernst equation

When the activities of the reacting species deviate from standard equilibrium conditions, the electrode potential shifts according to the Nernst equation:

$$E = E^\circ + \frac{RT}{nF} \ln \left(\frac{\prod_k a_k^{\nu_k}}{\prod_j a_j^{\nu_j}} \right), \tag{2.3}$$

where E° is the standard redox potential and a_i is the activity of species i. Equation 2.3 shows that the electrode potential is determined not only by the intrinsic thermodynamics of the redox couple but also by the composition and temperature of the system. The activity of the species can be approximated with their concentrations for dilute solutions, where the ion-ion interactions are negligible.

2.1.3 Measurement of electrode potentials

Only potential differences can be measured experimentally. Therefore, determining the potential of an electrode requires a second electrode with a known and stable potential value. Historically, the standard hydrogen electrode (SHE) was used to define the zero of the electrochemical reduction potential scale, but practical measurements typically rely on secondary references such as Ag/AgCl or calomel electrodes, which offer adequate stability with simpler construction and maintenance. To keep the potential stable, the reference electrode must remain unaltered under small current drawn by the electrochemical instrumentation.

In systems where a classical reference electrode cannot be implemented for geometrical constraints, a pseudo-reference can be realized by immersing a metal wire directly into the electrolyte. Although convenient, the potential of a pseudo-reference depends on the local ionic composition and often drifts during operation. Its use is acceptable only when the resulting uncertainty is quantified and compatible with the intended measurements.

A third option is the use of a quasi-reference electrode that is similar to a pseudo-reference in the sense that the electrode is immersed directly into the electrolyte solution without ionic separation, but it is constituted by a metal covered by its halogen forming a solid state redox couple system for better potential stability.

Accurate knowledge of the electrode potential is essential not only for thermodynamic characterization but also for interpreting electrochemical impedance spectra. Because impedance measurements rely on small perturbations applied around a defined operating point, an uncertainty in the underlying equilibrium potential directly affects the meaning and reproducibility of the response.

2.2 Kinetics

While thermodynamics defines the equilibrium potential of an electrode, practical electrochemical systems operate away from equilibrium. When a current flows, the interfacial concentrations of reacting species deviate from their equilibrium values, leading to a measurable shift in potential known as the overpotential. Electrode kinetics provides a quantitative description of how charge-transfer, mass transport, and interfacial phenomena contribute to this deviation.

2.2.1 Overpotential and activation kinetics

For an ideal planar electrode, the overpotential η is defined as the difference between the actual electrode potential and its equilibrium value,

$$\eta = E - E_{\mathrm{eq}}. \tag{2.4}$$

A positive overpotential accelerates oxidation, whereas a negative overpotential accelerates reduction. For small departures from equilibrium, the current is limited by the intrinsic kinetics of electron transfer between the electrode and the redox-active species. At sufficiently large overpotentials, mass transport may become rate-limiting.

2.2.2 The Butler-Volmer equation

The activation-controlled charge-transfer is described by the Butler-Volmer equation,

$$i = i_0 \left[\exp\left(\frac{\alpha_a F \eta}{RT} \right) - \exp\left(-\frac{\alpha_c F \eta}{RT} \right) \right], \tag{2.5}$$

where i_0 is the exchange current density and α_a and α_c are the anodic and cathodic transfer coefficients. The exchange current density quantifies the intrinsic reactivity of the interface and depends on the concentrations of reacting species at the surface. Equation 2.5 connects the near-equilibrium regime, where the response is linear, with the high-overpotential regime, where one exponential term dominates. This equation highlights the intrinsic non-linearity of electrochemical systems, especially far from equilibrium; an important point for the following discussion of non-stationary impedance spectroscopy.

2.2.3 Mass transport

Charge-transfer kinetics alone cannot describe the behavior of an electrochemical interface under operating conditions. When a net current flows, the interfacial concentrations of reacting species deviate from their bulk values, and mass transport determines how quickly these deviations are restored. The flux of an ionic species i in solution is governed by the Nernst–Planck equation,

$$\mathbf{J}_i = -D_i \nabla c_i - z_i u_i F c_i \nabla \phi + c_i \mathbf{v}, \tag{2.6}$$

where D_i is the diffusion coefficient, u_i the ionic mobility, z_i the charge number, c_i the concentration, ϕ the electric potential, and $\mathbf{v}$ the solvent velocity. The three terms represent diffusion, migration in an electric field, and convection, respectively. In a typical compact battery, convection is absent, so $\mathbf{v} = \mathbf{0}$.

The time evolution of concentration follows from the continuity equation,

$$\frac{\partial c_i}{\partial t} = -\nabla \cdot \mathbf{J}_i, \tag{2.7}$$

which, in the absence of migration, reduces to Fick's second law:

$$\frac{\partial c_i}{\partial t} = D_i \nabla^2 c_i. \tag{2.8}$$

Diffusion describes the spontaneous relaxation of concentration gradients, and the associated characteristic time scale is proportional to the square of the diffusion length. This slow relaxation becomes relevant whenever the rate of the electrochemical reaction is high enough to deplete or enrich the surface concentration relative to the bulk.

2.2.4 Mass-transfer limitations and diffusion-controlled kinetics

The Butler–Volmer equation assumes that the concentrations of reacting species at the electrode surface remain equal to their bulk values. When the reaction rate is high compared to the rate at which diffusion can replenish or remove reactive species, concentration gradients develop near the interface. This leads to a maximum sustainable current, known as the limiting current density,

$$i_{\mathrm{lim}} = n F k_m c_{\infty}, \tag{2.9}$$

where k_m is the mass-transfer coefficient and c_{∞} is the bulk concentration of the reactant. As the surface concentration approaches zero (for reduction) or saturation (for oxidation), the current no longer increases with overpotential.

Figure 2.1 illustrates the nonlinear current–overpotential relationship described by the Butler–Volmer equation, highlighting both the diffusion-limited regime and the near-equilibrium region where linearization is valid. The latter regime forms the basis of impedance spectroscopy, as small perturbations around the operating point lead to a proportional relationship between current and overpotential.

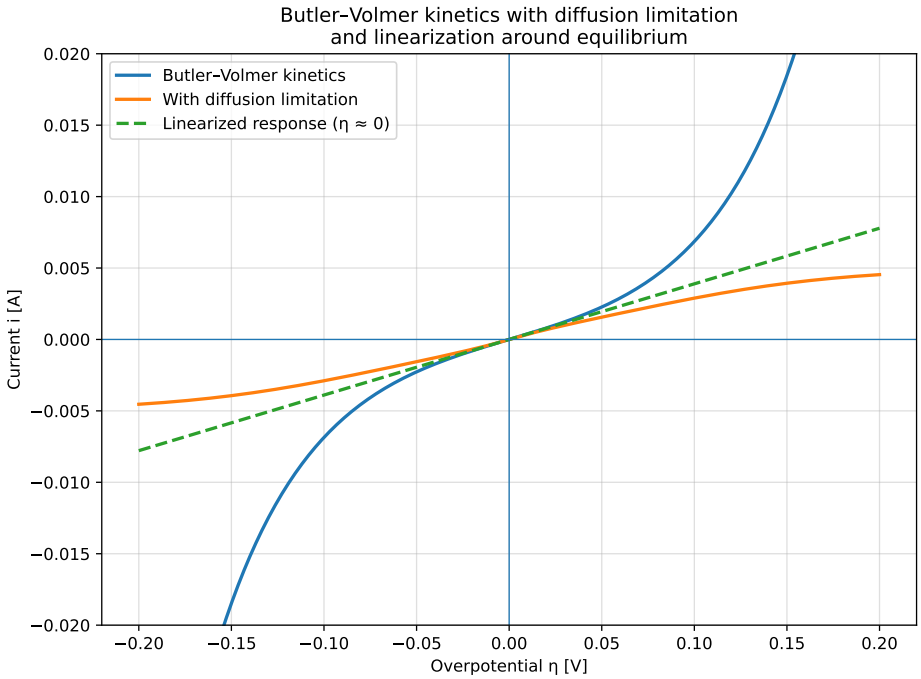


Figure 2.1: Current–overpotential relationship described by the Butler–Volmer equation. The nonlinear response transitions to diffusion-limited behavior at large overpotentials, while the linearized region around $\eta = 0$ defines the charge-transfer resistance used in small-signal impedance analysis.

2.3 Double-layer structure and interfacial capacitance

At the interface between an electronic conductor and an ionic solution, a separation of charge develops. Electrons in the electrode and ions in the electrolyte arrange themselves such that the electrochemical potentials of all species are equal across the interface. The resulting region, known as the electrical double layer, stores charge electrostatically and therefore contributes a capacitive component to the interfacial impedance.

The double layer is commonly described as consisting of a compact region adjacent to the electrode surface and a diffuse region that extends into the electrolyte. In the compact layer, often referred to as the Helmholtz layer, specifically adsorbed ions and oriented solvent molecules form a thin dielectric layer of thickness on the order of molecular dimensions (tens of Å). Its capacitance per unit area can be approximated by that of a parallel-plate capacitor,

$$C_{\mathrm{H}} \approx \frac{\varepsilon_{\mathrm{H}} \varepsilon_0}{d_{\mathrm{H}}}, \tag{2.10}$$

where ε_{H} is the effective dielectric constant of the compact layer and d_{H} its thickness.

Beyond the compact layer, ions distribute according to the balance of electrostatic and entropic forces. Assuming a symmetric $z : z$ electrolyte and a planar interface, the ion concentrations follow Boltzmann statistics, which, combined with Poisson's equation, leads to the Gouy–Chapman description of the diffuse region. The surface charge density σ associated

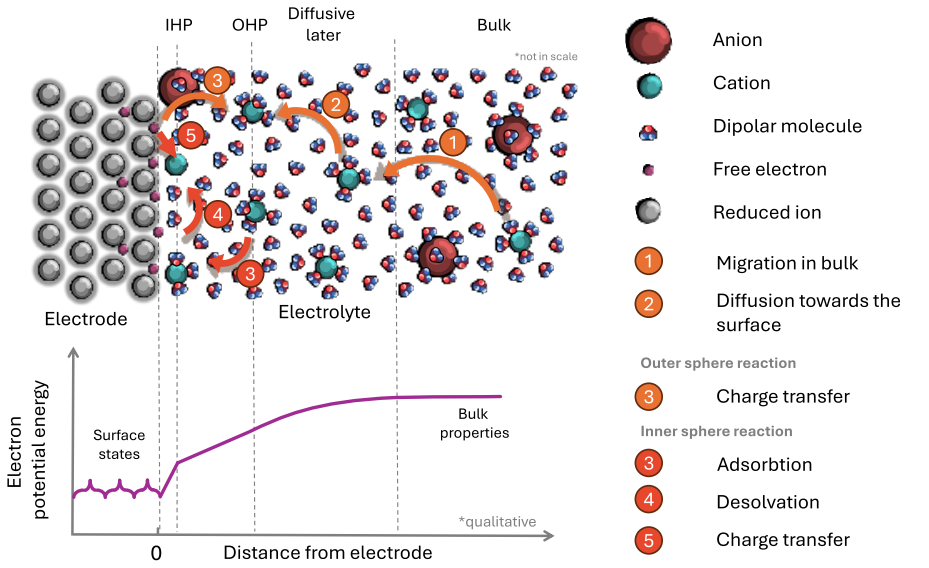


Figure 2.2: Schematic representation of the electrode–electrolyte interface and electrical double-layer structure. The separation of charge across the compact and diffuse layers gives rise to the interfacial capacitance that contributes to the frequency-dependent impedance response.

with the potential drop across the diffuse layer ψ^d (Stern potential) is then

$$\sigma = \sqrt{8\varepsilon\varepsilon_0 RTc_\infty}\,\sinh\left(\frac{zF\psi^d}{2RT}\right), \tag{2.11}$$

where ε is the permittivity of the electrolyte, c_∞ the bulk ion concentration and z the ionic charge number. The corresponding differential capacitance is obtained by differentiation with respect to ψ_0,

$$C_{\mathrm{GC}}(\psi_0) = \frac{d\sigma}{d\psi_0} = \frac{\varepsilon\varepsilon_0}{\lambda_D}\cosh\left(\frac{zF\psi^d}{2RT}\right), \tag{2.12}$$

with λ_D the Debye length,

$$\lambda_D = \sqrt{\frac{\varepsilon\varepsilon_0 RT}{2z^2F^2c_\infty}}. \tag{2.13}$$

The Stern model combines these two contributions. Since the compact and diffuse layers are arranged in series, the total differential double-layer capacitance is

$$\frac{1}{C_{\mathrm{dl}}} = \frac{1}{C_{\mathrm{H}}} + \frac{1}{C_{\mathrm{GC}}(\psi^d)}. \tag{2.14}$$

For highly concentrated electrolytes, as typically found in batteries, the Debye length becomes very small and the diffuse-layer capacitance grows correspondingly large, making the compact layer the dominant contributor to C_{dl} [2]. Conversely, in dilute electrolytes the diffuse layer significantly affects the interfacial response, and the potential dependence of C_{GC} can lead to pronounced nonlinearity.

Figure 2.2 schematically represents the electrode–electrolyte interface, highlighting the compact (Helmholtz) and diffuse layers and their role in charge storage and interfacial polarization.

2.4 Electrochemical methods

In the electrochemical characterization of energy storage materials and devices, a wide classification is between time-domain and frequency-domain methods. This distinction highlights a crucial difference: time-domain methods typically probe macro-kinetic[1] and thermodynamic parameters under large perturbations, while frequency-domain methods resolve electrode processes under small-signal control (perturbative).

[1] Overall, measured reaction rate that accounts for all physical steps (charge transfer, diffusion, ohmic losses) at the scale of the entire electrode

2.4.1 Time-domain methods

Time-domain methods involve applying a constant or time-varying potential or current input to the cell and observe the output's time variation. Historically, cyclic voltammetry (CV) has been fundamental in electroanalytical chemistry, probing the thermodynamic potential of redox species and characterizing chemical reversibility via voltage sweep across a determined voltage window. The sweep scan rate determines the capability of resolve different processes during a measurement.

An alternative approach to voltage-controlled techniques is the potentiostatic method, in which the electrode potential is held constant, hence fixing the overpotential η and allowing the resulting current response to be solely governed by the rate-limiting kinetics. The potentiostatic method is also used to operate electrochemical devices such as electrolyzers and fuel cells. The current output varies depending on the interface electrochemistry so the methods are also called chrono-amperometry (CA). This is ideal for studying reaction mechanisms where the interfacial potential needs to be tightly controlled.

In contrast, in a galvanostatic technique the current flowing through the cell is fixed and, as a consequence, it controls the macro-kinetics of the system. The output voltage is composed of the electrode's thermodynamic potential and the overpotential generated by the imposed current. This is the standard method for simulating battery operation, yielding data for device-relevant metrics such as capacity, energy density, and cycle life (discussed in chapter 3) rather than isolated fundamental kinetic parameters.

2.4.2 Electrochemical impedance spectroscopy

The use of alternating excitation signals in electrochemistry originated from the need to resolve the electrified double-layer. Early works showed that by using high-frequency periodic voltage perturbation, the large interfacial capacitance could be "short-circuited", allowing the isolation of the pure Ohmic resistance of the electrolyte [3]. Crucially, the frequency-dependent measurements of capacitance provided direct experimental evidence for the electrical double-layer, demonstrating that alternating current techniques are unique in their ability to resolve fundamental physical elements based on their distinct *time constants* ($\tau = RC$ for a parallel resistor and capacitor circuit).

The application of different frequency excitations on the same system allows one to resolve the processes and their time constants. This type of experimental technique is called electrochemical impedance spectroscopy (EIS). Its main characteristics are the non-destructive nature because of the small-amplitude sinusoidal perturbation (typically $\pm 5 - 10$ mV) and the wide range of frequencies probed, from mHz to MHz. The result of an

EIS measurement is the frequency-dependent impedance function obtained with the formula

$$Z(\omega) = \frac{\tilde{E}(\omega)}{\tilde{I}(\omega)}, \tag{2.15}$$

where ω is the angular frequency. The tilde notation over $\tilde{E}$ and $\tilde{I}$ denotes the phasor representation proposed by Orazem and Tribollet [1]. By varying the frequency of the sinusoidal perturbation, the impedance response of different interfacial processes is measured; hence the name spectroscopy. The primary advantage of impedance spectroscopy is its ability to separate and quantify phenomena occurring at vastly different time scales without altering the system state.

In the simplest interfacial model, the impedance of an electrode under activation control consists of a charge-transfer resistance R_{ct} in parallel with the double-layer capacitance C_{dl}. At high frequencies, the capacitive branch dominates and the impedance becomes primarily imaginary. At low frequencies, the capacitive reactance becomes large and the impedance approaches R_{ct}. Deviations from ideal capacitive behavior, commonly observed in rough or porous electrodes, are captured using a constant-phase element to represent distributed interfacial properties. The representation of the interface as charge-transfer resistance R_{ct} and double-layer capacitance C_{dl} is obtained analytically from the linearization of the Butler-Volmer equation and the double-layer capacitance of the Stern model.

Before the advent of microcomputers in the 70s, impedance was estimated frequency by frequency manually with the aid of an analog circuit called a lock-in amplifier. The impedance $Z(\omega_k)$ is calculated from the amplitude ratio and phase shift $\Delta\phi$ between input (reference) and output signals using the formula

$$Z(\omega_k) = A_r e^{j\omega\Delta\phi}, \tag{2.16}$$

where A_r is the ratio between input and output signal amplitudes.

Later, digital signal processing was integrated into microcontrollers, creating the digital Frequency Response Analyzer (FRA) capable of automatic frequency sweeping, averaging, automatic current range selection, and application of voltage or current bias (Solartron 1174 in 1978 and Solartron 1250 in 1983). These instruments set the standard for impedance measurements that we still find today in commercial devices.

2.4.3 Linearization and charge-transfer resistance

In the limit of small perturbations, the nonlinear Butler-Volmer expression may be linearized by expanding Eq. 2.5 to first order about $\eta = 0$, yielding:

$$i \approx \left(\frac{\partial i}{\partial \eta}\right)_{\eta=0} = \frac{nFi_0}{RT}\,\eta. \tag{2.17}$$

The inverse of this proportionality defines the charge-transfer resistance,

$$R_{\mathrm{ct}} = \frac{RT}{nFi_0}, \tag{2.18}$$

which represents the kinetic contribution to the interfacial impedance. In the absence of mass-transfer limitations, the small-signal impedance of an activation-controlled interface therefore contains a resistive contribution R_{ct} in parallel with the capacitive behavior of the double layer.

2.4.4 Double-layer capacitance

Under small perturbations, the double layer behaves as an effective capacitor whose value reflects the adsorbed charges and electrode potential. Together with the charge-transfer resistance arising from activation kinetics, it forms a parallel combination that constitutes the fundamental interfacial element in equivalent-circuit models: the RC element.

Deviations from ideal capacitive behavior, often observed in porous or heterogeneous electrodes, require adapting the description of the capacitance impedance with the constant-phase-element (CPE) as

$$Z(\omega) = \frac{1}{Q(j\omega)^{\alpha}}, \tag{2.19}$$

where Q is the relative capacitance when the parameter α is equal to 1 [4]. Although the CPE provides a better mathematical fitting of some impedance data, the correlation with the physical processes governing the system is often not clear [1].

2.4.5 Diffusion impedance

When mass transport contributes to the interfacial response, the impedance acquires a characteristic frequency dependence that reflects the dynamics of concentration relaxation. Starting from the diffusion equation under small-signal perturbation,

$$\frac{\partial c}{\partial t} = D\nabla^2 c,$$

a sinusoidal perturbation of frequency ω leads to concentration profiles that vary as $\exp(-\sqrt{j\omega/D}\,x)$. The associated flux at the surface is therefore proportional to $(j\omega)^{1/2}$, giving rise to the classical diffusion impedance.

In the complex plane, diffusion impedance manifests as characteristic frequency-dependent trajectories whose shape reflects the underlying transport geometry as described in the rest of the section.

Semi-infinite (infinite-length) Warburg impedance

For a planar electrode with a semi-infinite diffusion domain, the small-signal diffusion impedance is

$$Z_{\mathrm{W}}(\omega) = \sigma_{\mathrm{W}} \frac{1-j}{\sqrt{\omega}}, \tag{2.20}$$

where σ_{W} is the Warburg coefficient. The real and imaginary parts are equal, producing a line at 45° in the complex plane. This form arises when the perturbation penetrates only a small fraction of the electrolyte or electrode, such that the concentration field does not reach a physical boundary during the oscillation period.

Finite-length Warburg impedance

When diffusion occurs in a bounded domain—such as within porous electrodes, solid diffusion in active particles, or electrolyte confined between electrodes—the diffusion layer cannot grow indefinitely. The impedance then takes one of two canonical forms depending on the boundary conditions.

For reflective (blocking) boundaries, the finite-length Warburg impedance is

$$Z_{\mathrm{W,\,ref}}(\omega) = \sigma_{\mathrm{W}} \frac{\tanh\left(\sqrt{j\omega\tau_D}\right)}{\sqrt{j\omega}}, \tag{2.21}$$

while for transmissive (open) boundaries,

$$Z_{\mathrm{W,\,open}}(\omega) = \sigma_{\mathrm{W}} \frac{\coth\left(\sqrt{j\omega\tau_D}\right)}{\sqrt{j\omega}}, \tag{2.22}$$

where $\tau_D = L^2/D$ is the characteristic diffusion time for a layer of thickness L.

Diffusion-related impedance elements are essential for interpreting battery electrode behavior, where mass transport in the electrolyte, within porous electrode structures, and through solid particles strongly influences the dynamic response. The distinction between semi-infinite and finite-length diffusion is particularly important when analyzing non-stationary or transient impedance measurements, as the effective diffusion domain may change with state of charge, cycling history, or electrode microstructure.

2.4.6 Visualize impedance

The complex impedance can be represented graphically in two equivalent forms: the Bode representation and the Nyquist representation. In the Bode plot, the magnitude $|Z(\omega)|$ and phase $\phi(\omega)$ are displayed as functions of frequency on logarithmic axes. This format is particularly useful for identifying characteristic time constants and separating processes occurring over different frequency decades.

Alternatively, the Nyquist plot represents the negative imaginary part of the impedance as a function of its real part for each excitation frequency. Each point on the curve corresponds to a specific frequency, although frequency is not explicitly shown on the axes. For electrochemical systems, this representation is often preferred because it allows a direct geometric interpretation of the impedance. Ohmic resistances appear as intercepts on the real axis, while interfacial processes characterized by parallel resistance-capacitance elements produce semicircular arcs whose diameter corresponds to the associated resistance. Diffusion-controlled processes generate straight lines or curved tails at approximately 45° or tending toward the vertical direction at low frequency.

For complex electrochemical systems containing multiple overlapping processes, the Nyquist representation provides an immediate visual assessment of the relative contributions of resistive and capacitive elements and of their evolution with operating conditions. For this reason, Nyquist plots are used extensively throughout this work to analyze time-varying impedance data. A comparison between Bode and Nyquist is provided in Figure 2.3.

When interpreting Nyquist plots geometrically, it is essential that the real and imaginary axes are plotted with identical scaling. Only with equal axis scaling do semicircles, straight diffusion lines, and intercepts retain their correct geometric meaning; any distortion of the aspect ratio can lead to misinterpretation of resistances and time constants.

2.5 Equivalent circuit models

The linearization of the electrochemical equations yields processes that behave as electrical elements connected in series or parallel, depending on

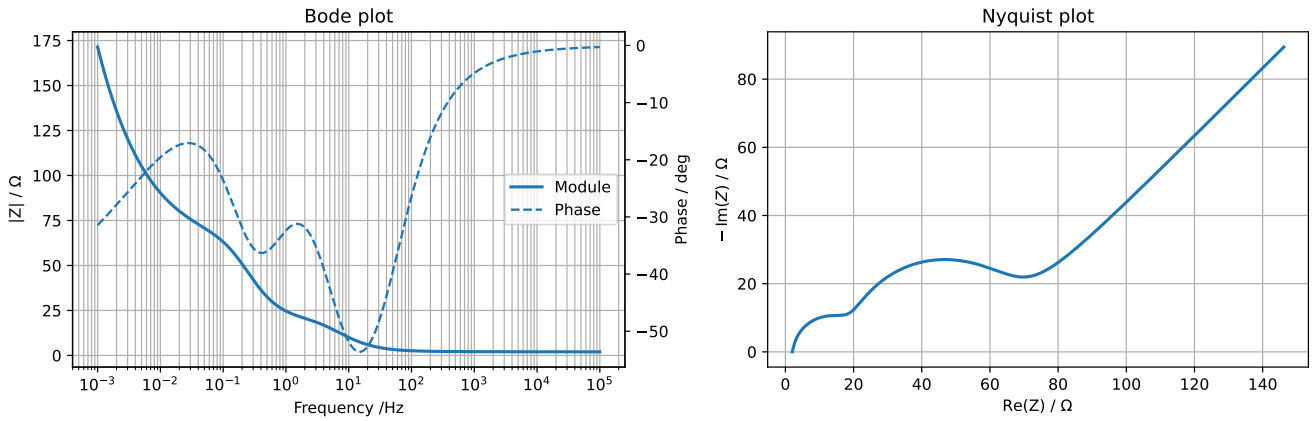


Figure 2.3: Different representation of the same impedance with two types of visualization. The Bode plot shows separately phase and amplitude as function of frequency while the Nyquist plot shows the impedance in the complex plane keeping the frequency implicit. The simulated circuit is made of two RC elements in series followed by a semi-infinite Warburg element.

the system characteristics. To extract the values of the physical parameters, one typically translates the set of equations into a circuit and applies the laws of series and parallel connection of electrical elements to obtain a simplified and linearized model of the interface. This leads to a physically derived equivalent circuit model of the electrochemical interface, an idea introduced by Randles [3, 5].

The correspondence between electrochemical processes and their electrical analogues is summarized in Figure 2.4, where charge transfer, double-layer charging, and diffusion are mapped to resistive, capacitive, and dispersive circuit elements.

2.6 Validity and consistency of electrochemical impedance

Electrochemical Impedance Spectroscopy relies on a set of assumptions that define the conditions under which impedance is a physically meaningful quantity.

First, the system must be *linear* with respect to the applied perturbation. This requires that the amplitude of the AC excitation be sufficiently small so that the response can be accurately described by the first-order linearization of the governing equations. In practice, this condition implies that the measured impedance must be independent of the perturbation amplitude and that higher harmonics in the response are negligible.

Second, the system must be *stationary* during the measurement, meaning that the underlying thermodynamic state (electrode potential, surface composition, temperature, state of charge) does not change significantly over the duration of the impedance experiment. This assumption is often implicit in classical EIS but becomes critical in battery systems, where slow processes such as diffusion, phase transitions, and interphase growth continuously evolve.

Third, the system must be *stable* and *causal*. Stability ensures that bounded perturbations produce bounded responses, while causality requires that the response at any time depends only on present and past inputs.

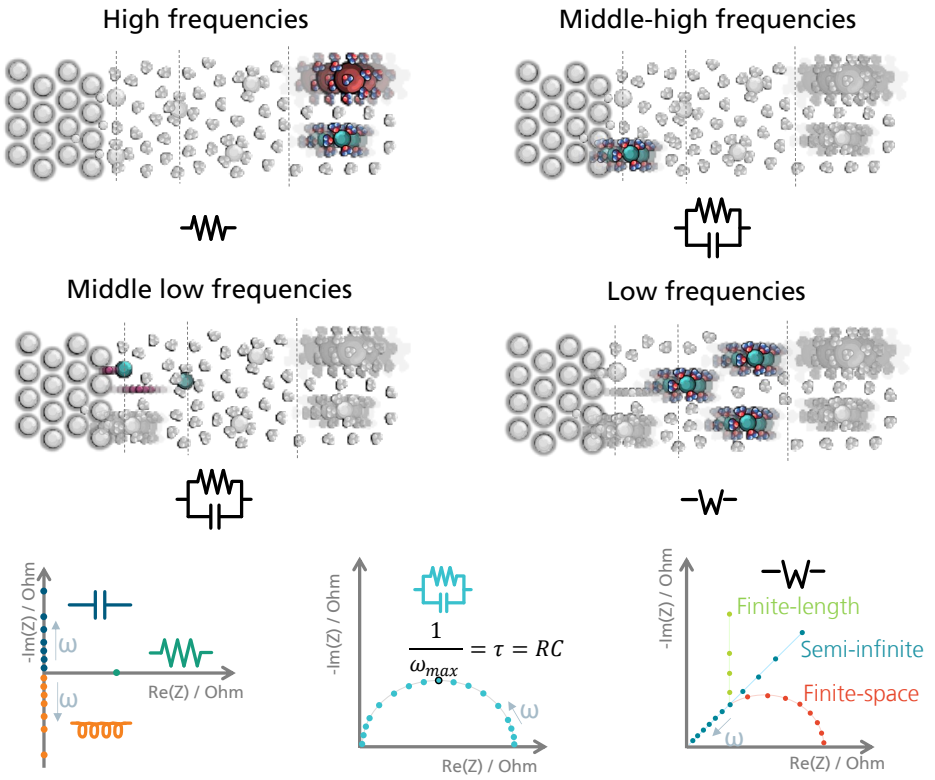


Figure 2.4: Mapping between electrochemical interfacial processes and their equivalent electrical impedance elements. Bulk mass transport (high frequencies), adsorption (middle-high frequencies), charge-transfer kinetics (middle-low frequencies), and diffusive layer transport (low frequencies) contribute distinct features to the impedance spectrum and can be represented using lumped circuit components. The lower part represents the impedance of the electrical or electrochemical elements and the RC circuit.

These properties are automatically satisfied for most passive electrochemical systems under equilibrium or near-equilibrium conditions, but may be violated in the presence of strong feedback, parasitic reactions, or measurement artefacts.

2.6.1 Kramers–Kronig relations

A fundamental consequence of linearity, causality, and stability is that the real and imaginary parts of the impedance are not independent. They are linked by the Kramers–Kronig (KK) relations, which express each component as a Hilbert transform of the other.

For an impedance function $Z(\omega) = Z'(\omega) + jZ''(\omega)$, the KK relations read

$$Z'(\omega) = \frac{2}{\pi}\mathcal{P}\int_0^\infty \frac{\omega' Z''(\omega')}{\omega'^2 - \omega^2}\, d\omega', \tag{2.23}$$

$$Z''(\omega) = -\frac{2\omega}{\pi}\mathcal{P}\int_0^\infty \frac{Z'(\omega')}{\omega'^2 - \omega^2}\, d\omega', \tag{2.24}$$

where $\mathcal{P}$ denotes the Cauchy principal value.

In electrochemistry, the KK relations provide a powerful consistency check: if an experimentally measured impedance violates KK relations beyond experimental uncertainty, at least one of the underlying assumptions of EIS has been violated.

In practice, KK relations cannot be exactly verified by the above equation because they require spectral information from zero to infinite frequency. Fitting experimental data to a measurement model consisting of a series of RC elements produces a close approximation to the formula and it is usually referred to as a KK test [1, 6, 7].

From an experimental standpoint, deviations from KK consistency can arise from several sources: non-stationarity of the system, artifacts introduced by cell design, or the instrumentation electronics.

Chapter 3

Batteries

Electrochemical energy storage systems convert chemical energy into electrical work through redox reactions occurring at electrode interfaces. Among these systems, rechargeable (or secondary) batteries have become essential across portable electronics, stationary storage, and electric mobility. Lithium-ion batteries, in particular, have dominated the field due to their high operating voltage, large specific energy, and excellent cycle efficiency.

Electrochemical impedance spectroscopy serves as an ideal technique to monitor the internal resistance of the battery and to estimate its state of health (SoH) [8–10]. Without the separation of electrodes, impedance measurements on batteries do not probe a single interface or reaction, but rather the collective response of multiple electrodes, electrolyte phases, and transport pathways embedded within a specific cell architecture. As a consequence, the measured impedance depends not only on intrinsic electrochemical properties, but also on the state of charge, operating history, and degradation state of the battery as a system. Hence, in contrast to classic electrochemical characterization, the impedance of a battery is measured at different states and changes are monitored throughout its lifetime rather than considering its absolute value.

This chapter introduces the fundamental principles of rechargeable batteries required to interpret impedance spectra measured at the cell level, including electrode configuration, operating states, and common degradation processes. The focus is not on a comprehensive description of battery chemistry, but on those aspects that directly influence the structure, evolution, and variability of impedance spectra during operation and aging.

3.1 Basic principles of operation

A battery consists of two electrodes separated by an electronically insulating, but ionically conductive, electrolyte. During discharge, oxidation occurs at the negative electrode, releasing electrons to the external circuit and cations to the electrolyte. Concurrently, reduction occurs at the positive electrode, accepting electrons and intercalating the cations arriving through the electrolyte. A positive current flows through the external circuit until the electrodes reach the same potential and no driving force remains. The reaction is spontaneous, and this system is referred to as a *galvanic cell*. During charging, the current reverses direction, producing the reductive redox half-reaction at the positive electrode and oxidation at the negative electrode. Such a system is called an *electrolytic cell*, and the

reaction is not spontaneous but requires external work.

The open-circuit voltage (OCV) of the cell is the difference between the half-reactions' equilibrium potentials at the positive and negative electrodes:

$$U_{\mathrm{OCV}} = E_{+} - E_{-}. \tag{3.1}$$

3.2 Performance metrics

Battery performance is evaluated using energy, power and capacity.

The specific energy (gravimetric)

$$E_{\mathrm{sp}} = \int U_{\mathrm{load}}(t)\, I(t)\, dt, \tag{3.2}$$

and specific power

$$P_{\mathrm{sp}} = \frac{U_{\mathrm{load}}(t) I(t)}{m_{\mathrm{cell}}}, \tag{3.3}$$

The theoretical capacity of an intercalation material accommodating n electrons per formula unit is

$$Q_{\mathrm{th}} = \frac{nF}{M}, \tag{3.4}$$

where M is the molar mass. For a composite electrode of active mass m_{act}, the theoretical specific capacity is $q_{\mathrm{th}} = Q_{\mathrm{th}} m_{\mathrm{act}}$.

The practical capacity is limited by side reactions, rate capability and electrode macroscopic structure (porosity and thickness).

In battery technology the current flowing in the cell is expressed as C-rate, defined as the charge or discharge current normalized by the nominal capacity Q_{nom}:

$$I = C\, Q_{\mathrm{nom}}. \tag{3.5}$$

A 1C current charges or discharges the battery in 1 hour, while C/10 corresponds to 10 hours.

3.3 Storage mechanisms

The type of redox reaction at each of the battery electrodes can be of different nature. The first "pile" of Alessandro Volta used electro-dissolution and deposition of zinc and copper, respectively, while modern rechargeable batteries rely on the far more efficient process of ion intercalation. These mechanisms are not specific to lithium-based systems and can, in principle, involve different alkali or multivalent ions depending on the electrolyte and host structure [11].

3.3.1 Intercalation

In intercalation, guest metal ions reversibly insert into the crystal lattice of a host material without disrupting its structure. This principle applies to inorganic materials where crystal planes are stacked and held together by electrostatic forces. When an element of the structure undergoes oxidation or reduction to balance the overall charge, an ion is either inserted or removed from the interlayer volume. Examples of such material structures include octahedral transition metal oxides, olivines, and spinel structures [12].

In layered oxides such as cobalt or manganese oxides, host ions occupy octahedral sites between transition-metal oxide layers. As examples, the intercalation half-reactions of lithium in CoO_2 and graphite are[13]:

$$\mathrm{LiCoO_2} \rightleftharpoons \mathrm{Li_{1-x}CoO2 + nLi^{+}ne^{-}}, \tag{3.6}$$

and

$$n\mathrm{Li^{+}}ne^{-} + n\mathrm{C_6} \rightleftharpoons n\mathrm{LiC_6}. \tag{3.7}$$

In olivine phosphates like $LiFePO_4$, ions occupy sites within one-dimensional channels running through the crystal. The two-phase behavior (coexistence of intercalated and de-intercalated domains) arises from the crystal structure and electrostatic interactions, producing the characteristic flat voltage plateau.

3.3.2 Alloying

In the alloying mechanism metal ions are stored in the active material through formation of intermetallic phases rather than lattice insertion. Silicon, for instance, forms a series of phases with lithium:

$$\mathrm{Si} + \beta\mathrm{Li^{+}} + \beta e^{-} \rightleftharpoons \mathrm{Li_\beta Si}, \tag{3.8}$$

where β can range up to 4.4, giving a theoretical specific capacity of 4200 mAh g^{-1}, nearly ten times higher than the graphite used in state-of-the-art commercial batteries [14].

However, the dramatic volume expansion accompanying alloying ($\sim$ 300% for silicon) induces mechanical stress, particle fragmentation and rapid electrode degradation. Modern high-capacity anodes employ silicon-graphite composites and carbon coatings with a small percentage of silicon to mitigate damage and maintain electrical contact across cycling.

3.3.3 Conversion

Conversion reactions involve reduction of a metal oxide or fluoride to metallic form, coupled with oxide/fluoride ion migration [15]. These reactions typically exhibit multiple voltage plateaus corresponding to sequential reduction steps and offer very high theoretical capacities. However, conversion materials suffer from large voltage hysteresis, low coulombic efficiency, and rapid capacity fade due to electrolyte decomposition and material restructuring during cycling. Consequently, they remain largely in research phase and are not employed in commercial cells.

3.3.4 Electrodeposition

Ions can be directly converted to their metallic form and avoid any host structure through electrodeposition and dissolution of the metal itself. Lithium metal as negative electrode offers the highest specific capacity (3861 mAh g^{-1}) and lowest potential for lithium-ion batteries, enabling very high energy density cells. Also sodium and other alkali metals provide similar benefits for their respective battery chemistries [16]. However, metallic deposition is prone to dendritic growth, which increases surface area and accelerates electrolyte decomposition and loss of active material, reducing cycle life and raising safety concerns.

3.4 Electrolytes

Electrolytes must provide high ionic conductivity, wide electrochemical stability and compatibility with both electrodes.

Most Li-ion cells use organic carbonate solvents (EC, DEC, EMC) with lithium salts such as $LiPF_6$ [16]. Their voltage stability window extends to $\sim 4.5 - 5.0$ V vs Li but decompose at low potentials, forming the SEI.

The ionic conductivity

$$\sigma = \sum_i z_i^2 F^2 u_i c_i, \tag{3.9}$$

depends on mobility, concentration and viscosity. Solvent mixtures balance stability and conductivity.

Aqueous electrolytes have excellent conductivity but a thermodynamic stability window around 1.23 V. Highly concentrated systems ("water-in-salt") extend this window to ~ 3 V by reducing water activity [17].

3.5 Solid electrolyte interphase

The solid electrolyte interphase (SEI) is a nanometer-scale passivation layer that forms spontaneously on the surface of negative electrodes in non-aqueous alkali-ion batteries when the electrode potential is outside the thermodynamic stability window of the electrolyte. It originates from the reductive decomposition of electrolyte components, including solvents, salts, and additives, and is a defining feature of lithium-ion and sodium-ion battery operation [18].

Although initially regarded as a static and protective film, the SEI is now understood as a chemically and structurally heterogeneous, dynamically evolving interphase whose properties strongly depend on electrode material, electrolyte composition, operating conditions, and cycling history. Its formation and evolution govern several key battery characteristics, including irreversible capacity loss, Coulombic efficiency, rate capability, and long-term degradation.

3.5.1 Formation and composition

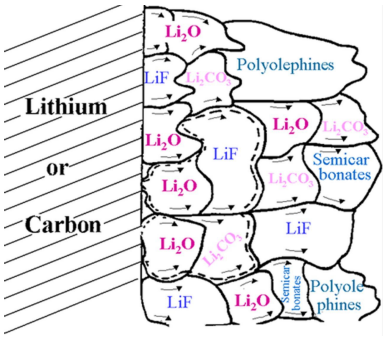


Figure 3.1: Illustration of the composition of the SEI layer on lithium or carbon. Used under a CC BY License [18]

SEI formation occurs predominantly during the first reduction of a negative electrode, when the electrode potential approaches the reduction potential of the electrolyte. At this stage, electrolyte components undergo electrochemical and chemical reactions that generate insoluble products, which precipitate onto the electrode surface and form a passivating layer. Typical inorganic SEI constituents include LiF, Li_2CO_3, Li_2O, and Li_2SO_3, while organic components often consist of polymeric or oligomeric species derived from solvent decomposition [19].

The resulting SEI is generally described as a multilayer structure, with an inorganic-rich inner layer in contact with the electrode and a more organic, polymeric outer layer interfacing with the electrolyte. This structure is not uniform across the electrode surface and varies locally depending on surface chemistry, morphology, and current distribution.

From a functional standpoint, the SEI must satisfy two requirements: it must be electronically insulating to prevent continuous electrolyte reduction, while remaining sufficiently permeable to Li^+ or Na^+ ions to enable

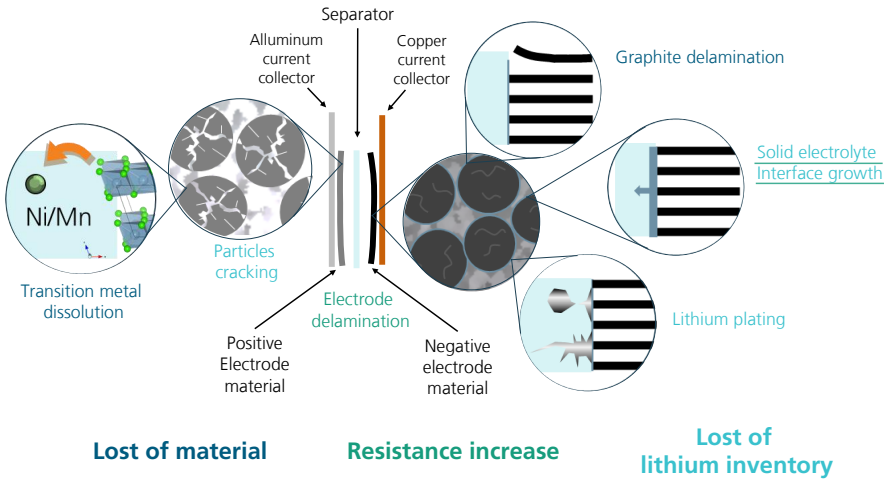


Figure 3.2: Summary of the principal degradation mechanisms in a single-layer lithium-ion battery. The cell, in the middle, is represented as parallel current collectors, active materials and separators. The colors group the various processes into three general categories: loss of lithium inventory, loss of active material, and impedance growth.

reversible charge transfer. When these conditions are met, the SEI stabilizes the electrode–electrolyte interface and enables long-term cycling.

3.5.2 SEI characterization with EIS

The SEI is not a static element with fixed properties. Its resistance, capacitance, and transport characteristics evolve with time, state of charge, current direction, and cycling history. Processes such as SEI cracking and reformation, electrolyte consumption, and accumulation of resistive inorganic phases introduce slow but continuous changes in interfacial kinetics.

These time-dependent changes directly affect the electrochemical impedance response of the system. In classical stationary electrochemical impedance spectroscopy (EIS), SEI-related features are typically represented by series connection of resistive–capacitive elements [20, 21], but XPS measurements reveal inhomogeneity of the layers that makes the modelling of impedance more challenging [22, 23].

3.6 Degradation mechanisms

Battery degradation arises from a combination of chemical, mechanical, and interfacial processes that progressively reduce the accessible lithium inventory, active material, and ionic/electronic transport pathways. The evolution of degradation with cycling or calendar time directly determines the state of health (SOH). Although multiple mechanisms coexist, they are commonly grouped into three major families: loss of shuttling ions (called loss of lithium inventory or LLI in lithium-ion batteries), loss of active material (LAM), and impedance growth [24]. Figure 3.2 schematically summarizes the main degradation mechanisms for a lithium-ion battery and the categories into which they are classified.

3.6.1 Loss of shuttling ions

Loss of shuttling ions is primarily caused by parasitic reactions that consume the ions. In organic electrolyte base batteries a major contributor is the continuous formation and repair of the solid-electrolyte interphase (SEI) at the negative electrode. Although a stable SEI is essential for suppressing further electrolyte reduction, mechanical stress, particle cracking, and volume changes can expose uncovered surface and produce additional SEI growth. Furthermore, temperature affects the composition of SEI resulting in new growth. The amount of metallic ions irreversibly trapped in the SEI increases with cycling and causes a progressive loss in capacity [25].

Metal plating also consumes shuttling ions. When the negative electrode potential reaches values below 0 V vs the standard metal reduction potential, metallic ions can deposit on the surface. Plating is favored by high current density, low temperature or large concentration overpotentials. If the deposited metal remains electronically connected, it can be re-dissolved in the electrolyte in the following cycle; otherwise, it may lose contact with the current collector, forming electrically isolated phases[2] that irreversibly reduce the shuttling ion inventory [26].

[2] Called inactive or "dead" lithium in lithium-ion batteries.

3.6.2 Loss of active material

Mechanical degradation of electrode materials results in reduced electrochemically active surface. In positive electrodes such as layered NMCs, repeated lithium intercalation and de-intercalation induce anisotropic lattice strain, phase transitions, and microcrack formation. Cracking leads to particle fragmentation and loss of contact with the conductive carbon network or current collector, increasing both the resistance and the inaccessible fraction of active material.

Graphite particles also undergo structural degradation under fast cycling due to expansion/contraction associated with staging transitions. Silicon-based anodes suffer even more severe mechanical degradation due to their large volumetric expansion, requiring advanced composite architectures to maintain structural integrity.

3.6.3 Impedance growth

With aging, the ohmic, charge-transfer and diffusion-related impedance components increase [27] due to:

- thickening of SEI,
- buildup of resistive reaction products,
- loss of electrolyte conductivity,
- increase of charge-transfer resistance due to surface restructuring,
- reduced effective diffusion coefficients due to material degradation.

Impedance growth not only reduces power capability but also induces non-uniform current distribution, accelerating further degradation. Electrochemical impedance spectroscopy (EIS) is therefore a powerful diagnostic tool for quantifying these aging phenomena.

3.7 Electrochemical characterization of batteries

Understanding battery performance and degradation requires controlled electrochemical testing. The most widely used methods rely on galvanostatic or potentiostatic excitation and analysis of voltage, current responses with time.

3.7.1 Galvanostatic cycling

A popular method to determine the capacity, coulombic efficiency, and cycle life of a battery is a recursive galvanostatic charge and discharge technique. The cell voltage $U(t)$ produced by the constant current I is the sum of OCV and overpotentials due to kinetic, ohmic and mass-transfer limitations:

$$U(t) = U_{\mathrm{OCV}}(x) + \eta_{\mathrm{kin}} + \eta_{\mathrm{ohm}} + \eta_{\mathrm{mt}}. \tag{3.10}$$

The charge flown during the galvanostatic technique from a time t_0 to t is

$$Q = \int_{t_0}^{t} I \, dt. \tag{3.11}$$

When the time frame $t - t_0$ is a complete half-cycle, one gets the charging Q_{cha} and discharging Q_{dis} charge. The half-cycle's charge is referred as *capacity* in the battery community. The coulombic efficiency (CE) of a cycle is

$$\mathrm{CE} = \frac{|Q_{\mathrm{dis}|}}{|Q_{\mathrm{chg}}|}. \tag{3.12}$$

For a commercial battery, efficiency should exceed 99.98% to achieve a thousand-cycle lifetime with 80% capacity retention.

In galvanostatic cycling, the cell is kept between a lower and higher operation voltage (voltage limits) to avoid degradation of electrodes or electrolyte. At high states of charge, the diffusional overpotentials limit the achievable capacity of a cell in galvanostatic operation by reaching the voltage limit. Therefore, Li-ion cells are typically charged using a constant-current and constant-voltage (CC-CV) protocol: first a constant current up to an upper voltage limit, then a constant voltage stage at V_{max} until the current tapers below a threshold. The CC-CV method enables to fully charge a battery without undesired side reactions.

3.7.2 Rate capability tests

By cycling a cell at progressively increasing C-rates, the dependence of delivered capacity on current reveals transport limitations. The characteristic rate at which the capacity drops significantly corresponds to limitations in solid-state diffusion, electrolyte transport, or internal resistance.

3.7.3 Electrochemical impedance spectroscopy (EIS)

Electrochemical Impedance Spectroscopy is extensively used in battery characterization for its capability to identify and separate processes such as ohmic resistance, charge-transfer kinetics, double-layer capacitance, and diffusion. These kinetics evolve with the aging of the cell and the occurrence of degradation phenomena. The impedance of a cell is measured at different states of charge (SoC) and states of health (SoH) by interrupting the constant-current constant-voltage (CC-CV) cycling protocol at fixed points

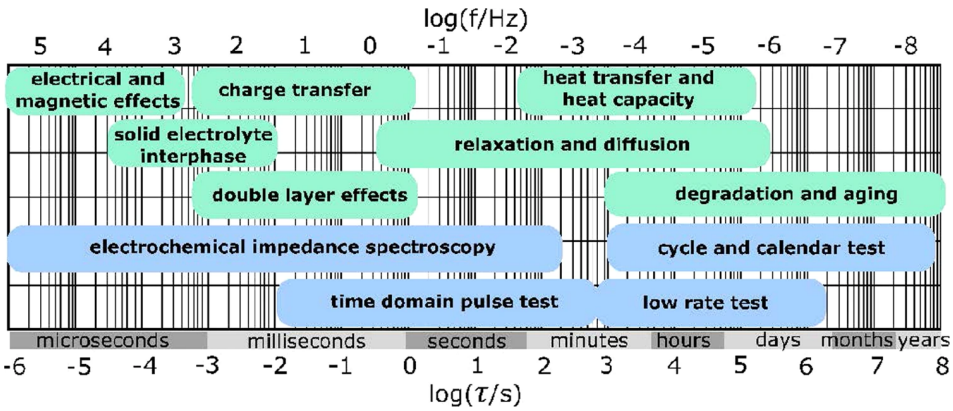


Figure 3.3: Battery processes time constants. Some processes are expresses in the time frame of cycling test, while many requires EIS to probe them, especially the fastest that are not reachable with time-domain system identification. This will be discussed in chapter 4. Used under CC BY Licence ***[Danzer2019GeneralizedDRT]***

and waiting for interface relaxation. A correct impedance measurement using classical sinusoidal perturbation across a decreasing frequency range may take 30 minutes or longer to probe the complete frequency response of a typical lithium-ion battery (10 mHz to 1 MHz), as schematized in Figure 3.3. This time requirement significantly increases the overall experiment duration, limiting its use for long-term explorations. This limitation is the focus of the present work and requires an introduction to the concepts of System Identification and Dynamic System Theory provided in the next chapter to frame the problem and justify the proposed solutions.

Chapter 4

Identification of dynamical systems

Electrochemical systems such as batteries are dynamical systems in the sense that their behavior is governed by differential equations describing the time evolution of internal variables, including ionic concentrations, electrode potentials and temperature, under given operating conditions.

Electrochemical impedance is commonly interpreted as a frequency-dependent material or interfacial property, yet its definition originates from linear system theory. Impedance corresponds to the frequency response of a system obtained by linearizing its input-output behavior around a given operating point, under assumptions of linearity, causality, and time invariance.

These assumptions determine whether an impedance spectrum is physically valid. When they are violated, as in the presence of drift, nonlinearity, or strong operating transients, the measured frequency response function may no longer correspond to a true impedance, but to an approximation whose validity must be carefully assessed.

This chapter introduces the elements of system theory required to formalize the concept of impedance, including linear time-invariant systems, transfer functions, and frequency-domain representations. The aim is not to develop system theory in full generality, but to establish a precise framework within which electrochemical impedance measurements can be defined, and later extended to non-stationary conditions.

4.1 Dynamical systems

A dynamical system is defined as a system whose evolution in time is governed by a set of differential equations. In general form, the dynamics can be expressed as

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t), t)\,, \tag{4.1}$$

where $\mathbf{x}(t)$ is the vector of state variables and $\mathbf{u}(t)$ represents external inputs.

In electrochemical systems, the state variables include quantities such as ionic concentrations, electric potentials, temperature and interfacial coverages. The governing equations typically consist of coupled partial differential equations arising from mass transport, charge conservation and reaction kinetics.

4.2 System identification

System identification is the discipline concerned with building mathematical models of dynamical systems based on measured input–output data. Rather than deriving models exclusively from first principles, system identification seeks to infer system behavior experimentally by applying controlled excitations and analyzing the resulting responses.

The identified model does not necessarily coincide with the full physical description of the system. Instead, it represents an effective input–output relationship that is valid within a specified operating regime and under given assumptions. In this sense, impedance spectroscopy constitutes a system identification experiment performed in the frequency domain.

4.3 Linearization and time-invariant approximation

Electrochemical systems are inherently nonlinear and time-varying. However, system identification commonly relies on linear approximations around a given operating point.

Let $\mathbf{x}_0$ denote a nominal state. For sufficiently small perturbations, the system dynamics can be linearized as

$$\delta\left(\frac{\mathrm{d}\mathbf{x}(t)}{\mathrm{d}t}\right) = \mathbf{A}\,\delta\mathbf{x}(t) + \mathbf{B}\,\delta\mathbf{u}(t), \tag{4.2}$$

where $\mathbf{A}$ and $\mathbf{B}$ are Jacobian matrices evaluated at the operating point.

If the operating point evolves slowly compared to the timescale of the excitation, the system can be treated as locally time-invariant. This assumption underlies the interpretation of impedance as a linear frequency response.

A system is said to be *linear* if it satisfies the superposition principle, meaning that the response to a linear combination of inputs equals the corresponding linear combination of the individual responses. Formally, for any inputs $\mathbf{u}_1(t)$ and $\mathbf{u}_2(t)$ and scalars α, β, linearity implies

$$\mathbf{y}(\alpha\mathbf{u}_1 + \beta\mathbf{u}_2) = \alpha\mathbf{y}(\mathbf{u}_1) + \beta\mathbf{y}(\mathbf{u}_2)\,.$$

A system is *time-invariant* if its input–output relationship does not depend explicitly on absolute time. If an input $\mathbf{u}(t)$ produces an output $\mathbf{y}(t)$, then a time-shifted input $\mathbf{u}(t-t_0)$ produces the correspondingly shifted output $\mathbf{y}(t-t_0)$ for any t_0.

When both properties hold, the system is referred to as a *linear time-invariant* (LTI) system. Under the assumptions of small-signal linearization and sufficiently slow evolution of the operating point, electrochemical systems can be locally approximated as LTI over finite time intervals.

4.4 Time-domain representation

In linear time-invariant form, a dynamical system can be represented in the so called "state-space model" as

$$\frac{\mathrm{d}\mathbf{x}(t)}{\mathrm{d}t} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \tag{4.3}$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t), \tag{4.4}$$

where $\mathbf{y}(t)$ denotes the measured output.

In electrochemistry, time-domain identification methods are commonly employed to estimate transport parameters from transient experiments. A notable example is the galvanostatic intermittent titration technique (GITT), in which diffusion coefficients are estimated from the relaxation of the electrode potential following a current step. However, electrochemical systems exhibit a wide spectrum of characteristic timescales. Charge-transfer reactions, adsorption, and interfacial transport phenomena occur at timescales several orders of magnitude shorter than bulk diffusion. Capturing the full system dynamics in the time domain would therefore require both very high temporal resolution to resolve fast processes and very long experiments to observe slow relaxation, which is often impractical. Moreover, the superposition of multiple coupled processes leads to transient responses that are difficult to model using simple exponential or power law functions. While diffusion-dominated behavior can be interpreted in the time domain under restrictive assumptions, the simultaneous presence of faster interfacial phenomena complicates parameter identification.

Frequency-domain identification offers a better solution for addressing this complexity separating processes according to their characteristic time constant.

4.5 Frequency-domain identification

4.5.1 From differential equations to frequency domain

Linearized electrochemical models are commonly expressed as ordinary or partial differential equations in time. The Laplace transform was introduced as a method to convert linear differential equations into algebraic equations in a complex variable, exploiting the well known mathematical framework of linear algebra to solve them. In doing so, it replaces differentiation with multiplication by the complex frequency variable s.

For a time-dependent function $x(t)$, defined for $t \geq 0$, the Laplace transform is defined as

$$X(s) = \mathcal{L}\{x(t)\} = \int_0^\infty x(t)\, e^{-st}\, \mathrm{d}t, \tag{4.5}$$

where $s = \sigma + j\omega$ is a complex variable.

A key property of the Laplace transform is that differentiation in time corresponds to multiplication in the Laplace domain:

$$\mathcal{L}\left\{\frac{\mathrm{d}x}{\mathrm{d}t}\right\} = sX(s) - x(0). \tag{4.6}$$

Under zero initial conditions, which are typically assumed in impedance analysis, this reduces to a simple algebraic relation.

Furthermore, when the real part σ of the complex variable $s = \sigma + j\omega$ is set to zero, the Laplace transform formally reduces to the Fourier transform, provided that the integral converges. This condition is satisfied for linear time-invariant and stable systems subjected to bounded excitations, as typically assumed in impedance spectroscopy.

Under these assumptions, the system dynamics can be characterized by evaluating the transfer function along the imaginary axis $s = j\omega$. This evaluation corresponds to the steady-state response to sinusoidal perturbations and motivates the term *frequency-domain* characterization.

4.5.2 Fourier Transform

The Fourier transform provides a mathematical framework to represent time-dependent signals as a superposition of sinusoidal components. Given a continuous-time signal $x(t)$, its Fourier transform $X(\omega)$ is defined as

$$X(\omega) = \int_{-\infty}^{+\infty} x(t)\, e^{-j\omega t}\, dt, \tag{4.7}$$

while the inverse Fourier transform reconstructs the original signal:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\omega)\, e^{j\omega t}\, d\omega. \tag{4.8}$$

In the context of system identification, the Fourier transform enables the analysis of linear systems in the frequency domain by converting differential equations in time into algebraic relations in frequency. For a linear time-invariant system, the response to a sinusoidal excitation at angular frequency ω is also sinusoidal at the same frequency, differing only in amplitude and phase. This property allows the system behavior to be fully characterized by its frequency response.

4.5.3 Definition of the transfer function

Consider a linearized dynamical system described by a differential equation of the form

$$\frac{\mathrm{d}y(t)}{\mathrm{d}t} + a\, y(t) = b\, u(t), \tag{4.9}$$

where $u(t)$ is an input perturbation and $y(t)$ is the measured response.

Applying the Laplace transform under zero initial conditions yields

$$sY(s) + aY(s) = bU(s), \tag{4.10}$$

which can be rearranged to give

$$Y(s) = \frac{b}{s+a} U(s). \tag{4.11}$$

The transfer function is defined as the ratio between the Fourier-transformed output and input:

$$G(s) = \frac{Y(s)}{U(s)} \tag{4.12}$$

and it describes how the system responds with the output $Y(s)$ to a certain input $U(s)$. Equation 4.12 resembles the definition of impedance reported in chapter 2, which is, in fact, the same mathematical object for electrochemical systems.

4.5.4 Connection to state-space representations

The same transfer function can be derived from a state-space model defined by equations 4.3 and 4.4 applying the Laplace transform, which yields the expression

$$G(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}, \tag{4.13}$$

where $\mathbf{I}$ is the identity matrix. This expression demonstrates that state-space and transfer-function models are mathematically equivalent descriptions of the same linear system. The choice between time or frequency domain characterization depends on the intended application.

4.6 Experimental estimation of frequency response functions

The transfer function is a mathematical concept deduced from a first-principles model of a system subjected to a small periodic perturbation of frequency $f = \omega/2\pi$. In practical settings, a system at steady state can be perturbed with a small sinusoidal excitation to establish the linear and time-invariant conditions and estimate the *frequency response function* (FRF) at frequencies f_k. In other words, the frequency response function is the practical realization of the transfer function.

While the transfer function is a theoretical object and LTI by definition, an experimental frequency response function is an approximation of it. The validity of the approximation depends on the conditions in which the experiment is performed. The Kramers–Kronig relations as a test for validity of the transfer functions were discussed in section 2.6.1 (p. 18).

4.7 Parameter estimation from impedance spectra

Once a frequency response function has been measured for a given system and validated, the goal of system identification is to infer a mathematical model that represents the underlying transfer function. In this work, the focus is placed on parametric models, which are defined by a finite set of parameters. Such models may be derived from first-principles physical assumptions or formulated as more general mathematical expressions, such as rational functions or polynomials. In both cases, the resulting models are typically nonlinear with respect to their parameters.

4.7.1 Fitting procedure

After selecting a model structure, parameter estimation is performed by fitting the model to the experimentally measured complex-valued impedance data. The fitting procedure consists of solving an optimization problem to determine the parameter vector $\boldsymbol{\theta}$ that minimizes the discrepancy between the model response $Z_{\text{model}}(\omega_k; \boldsymbol{\theta})$ and the experimental data $Z_{\text{exp}}(\omega_k)$, up to a measurement error ϵ_Z. This relationship can be expressed as

$$Z_{\text{exp}}(\omega_k) = Z_{\text{model}}(\omega_k; \boldsymbol{\theta}) + \epsilon_Z. \tag{4.14}$$

The optimization problem is commonly formulated as a least-squares minimization [1, 28]. Since the impedance is a complex quantity, the real and imaginary parts are treated jointly in the definition of the objective function:

$$J(\boldsymbol{\theta}) = \sum_{k=1}^{N} w_k \left\| Z_{\text{exp}}(\omega_k) - Z_{\text{model}}(\omega_k; \boldsymbol{\theta}) \right\|^2, \tag{4.15}$$

where w_k is a weighting factor associated with the measurement uncertainty at angular frequency ω_k. This approach is commonly referred to as complex nonlinear least squares (CNLS).

In practice, the measurement uncertainty is not uniform across the frequency range. At low frequencies, impedance measurements are generally affected by larger noise and drift, often proportional to $1/|Z|$. Appropriate weighting of the residuals is therefore essential to avoid overemphasizing

low-frequency data and to obtain physically meaningful parameter estimates.

The objective function is evaluated iteratively for different parameter values in order to locate its minimum. Most impedance fitting routines rely on derivative-based optimization algorithms, which exploit local gradient information in the parameter space. These methods require the computation of the Jacobian matrix of the residuals with respect to the parameters, either analytically or by numerical differentiation.

4.7.2 Initial parameters and local minima

A major challenge in impedance model fitting arises from the fact that the objective function $J(\boldsymbol{\theta})$ is non-convex in non-linear problems. The associated hypersurface in parameter space may exhibit multiple local minima, strong parameter correlations and flat directions, and narrow valleys and sharp ridges.

As a consequence, gradient-based optimization algorithms converge to a local minimum that depends on the initial parameter guess. There is no guarantee that this minimum corresponds to the global optimum or to a physically meaningful parameter set.

This issue becomes more pronounced for models with a large number of parameters or when fitting broadband impedance spectra spanning several decades of frequency.

The choice of initial parameters therefore plays a critical role in the success of nonlinear optimization. Well-informed initial guesses can significantly improve convergence speed and increase the likelihood of identifying the global minimum.

In principle, global optimization strategies based on random or systematic sampling of the parameter space can be employed, followed by local refinement of the best candidates. While such approaches improve robustness, they are often computationally expensive and may become impractical for complex models or large datasets.

Chapter 5

Digital signal processing

Electrochemical impedance spectra are obtained by processing time-domain voltage and current signals measured under controlled perturbations. As a consequence, impedance is not directly observed, but estimated through signal processing operations that transform finite, noisy, and discretely sampled data into a frequency-domain representation.

The accuracy, resolution, and physical validity of impedance measurements therefore depend not only on the electrochemical system and the applied excitation, but also on the signals processing. While system theory defines impedance as the frequency response of a linear system, digital signal processing determines how this response can be experimentally accessed in practice.

This chapter introduces the foundamental signal processing concepts required to manipulate discrete digital signals, including the discrete Fourier transform, spectral leakage, and noise considerations. These tools form the basis for understanding both classical impedance spectroscopy and the advanced time-resolved methods discussed in the following chapters.

5.1 Sampling and quantization

In the real world, any quantity changes continuously in time. When performing an experiment, an electronic instrumentation measures the quantity's value at a defined time interval using an electronic circuit called analog-to-digital converter (ADC).

In mathematical terms, the digital measurement system samples a continuous signal $x(t)$ at discrete times $t = nT_s$, producing the sampled discrete signal $x[n]$; where T_s is the sampling time. The most important parameter to consider in sampling is the time interval or its reciprocal: the sampling frequency $f_s = 1/T_s$. To analyze the spectral content of a signal, the sampling frequency must be at least twice the highest oscillation frequency. This is known as the Nyquist–Shannon sampling theorem. Sampling below this threshold causes higher-frequency components to "fold back" into lower frequencies—a phenomenon known as aliasing—which corrupts the true spectral content of the signal.

5.2 Discrete Fourier transform

After acquiring a digital signal, the role of digital signal processing is to analyze or act on the frequency components of the signal. As introduced in the previous chapter, the Fourier transform is used for spectral analysis of

signals, which is the process of decomposing a time-domain signal into its constituent frequencies and quantifying the amplitude and phase associated with each.

One converts a signal $x(t)$ or $x[n]$ into $X(\omega)$ or $X[k]$, revealing how its energy or variance is distributed across frequency.

Since digital signals are discrete vectors, the discrete version of the Fourier transform is used, called the discrete Fourier transform (DFT):

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j 2\pi k n / N}. \tag{5.1}$$

The inverse discrete Fourier transform (iDFT) takes the form:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j 2\pi k n / N}. \tag{5.2}$$

The discrete Fourier transform of a signal is calculated in a computer using the fast Fourier transform (FFT) algorithm, the most efficient implementation developed by Cooley and Tukey. The computational complexity is $O(N \log N)$ compared to the $O(N^2)$ of direct summation, although it requires the signal length to be a power of 2, i.e., $N = 2^n$ for some integer n.

5.2.1 Spectral leakage

Spectral leakage is the effect in the discrete Fourier transform of spectral components "leaking" or "spilling" into neighboring frequency bins, producing spurious components where the spectrum baseline should be flat (at noise level). Multiple causes produce spectral leakage. The most common cause is a discrete signal $x[n]$ that does not contain an integer number of periods of the oscillatory components. Other causes of spectral leakage derive from practical hardware limitations or user errors in the experimental design, such as:

- lack of synchronization between ADC and DAC clocks;
- instability of the ADC (data acquisition) or DAC (signal generator) clock;
- ADC jitter and slew-rate;
- low resolution of the ADC compared to the oscillation amplitude, which may produce spurious harmonics;
- aliasing due to improper choice of sampling frequency without adequate filtering.

5.2.2 Noise

Noise denotes random, undesired fluctuations superimposed on the signal originating from physical sources, instrumentation (quantization, clock jitter) or environment (EM interference).

Signal-to-noise ratio (SNR) quantifies the relative power between the signal components and the noise:

$$SNR = \frac{P_{\text{signal}}}{P_{\text{noise}}}$$

and dictates how reliably the signal can be detected or processed.

When increasing the excitation amplitude is not possible, averaging is an alternative tool for improving SNR: by repeating the same measurement or subdividing a long record into M independent segments, uncorrelated noise averages toward zero while the coherent signal adds constructively. This concept is used in electrochemical instrumentation where impedance spectroscopy is implemented. The user can choose the number of sinusoidal wave periods to average. The net SNR gain scales with the square root of the number of independent measurements. In the time domain, this is called ensemble or block averaging; in the frequency domain, it appears as spectral averaging. Averaging can alternatively be obtained by increasing the acquisition time of the periodic signal or the window length. In fact, narrowing the spectral bin bandwidth reduces the noise power in it linearly (i.e., doubling the signal length halves Δf, and thus the power in each bin, while the coherent signal amplitude in the bin does not change).

5.3 Filters

A filter in signal processing is a system that selectively modifies an input signal's amplitude and/or phase as a function of frequency and it is a core subject of signal processing.

By their effect on frequency content, filters are classified as low-pass (passing frequencies below a cutoff f_c), high-pass (passing frequencies above f_c), band-pass (passing a band between two cutoffs), or band-stop/notch (rejecting a specific band).

Filters can be distinguished between analog, a physical circuit in the signal acquisition pipeline, or digital, applied on the digitized signal by a micro-controller or computer.

There are different types of digital implementation of filters, but for the purpose of this work it is only necessary to consider the effect on the frequency content of a signal and performance metrics such as passband ripples (amplitude variation within the passband), stopband attenuation (suppression of unwanted bands), and transition bandwidth (width of the roll-off region).

5.4 Signal synthesis

To perform certain experiments, one might want to perturb the system with non-constant signals such as pulses, ramps, sinusoids, or other arbitrary shapes. This requires designing a digital waveform and generating it in analog form using electronic instrumentation. This process is called digital-to-analog conversion, and the electronic component performing it is the digital-to-analog converter (DAC). Several techniques exist for implementing a digital-to-analog converter on hardware with different complexity and properties; the simplest is *direct digital synthesis* (DDS). This method relies on a 'look-up table' in which the desired waveform is stored as a table of value-time pairs and at constant time intervals the value is read and converted to an analog voltage. In arbitrary waveform generators, the table of points is usually stored in limited buffer memory of a few thousand samples. The generated signal is typically passed through a series of filters to smooth it and avoid producing harmonics or causing spectral leakage.

The synthesis procedure is crucial for the quality of the excitation in system identification.

The length of the arbitrary waveform can be extended by storing the signal in larger memory and sending it directly to the digital-to-analog converter using the direct memory access (DMA) protocol; however, this approach requires dedicated micro-controller units and knowledge of digital signal processor implementation that may be outside the typical background of electrochemists.

Chapter 6

Electrochemical impedance spectroscopy beyond stationarity

Classical electrochemical impedance spectroscopy relies on the assumption that the system under investigation is linear and time invariant (LTI) over the duration of the measurement. Under these conditions, frequency-domain techniques introduced in the previous chapter allow the impedance to be estimated reliably from periodic perturbations. In many electrochemical systems of practical interest, however, this assumption does not hold. Processes such as charging and discharging of batteries, surface modification, aging, or phase transformations introduce a slow but continuous evolution of the system state during the measurement. When the characteristic time scale of this evolution becomes comparable to the duration required to probe low frequencies, the interpretation of impedance spectra obtained with conventional methods becomes ambiguous.

This chapter reviews the methodological developments that extend impedance spectroscopy beyond the stationary regime. Emphasis is placed on multi-frequency excitation, time-frequency analysis, and time-resolved impedance estimation, highlighting the assumptions, limitations, and validity domains of each approach. These methods form the conceptual basis for the experimental implementation described in Chapter 7.

6.1 Reducing acquisition time

The challenge of measuring impedance in non-stationary electrochemical systems has been recognized since the early development of EIS. Over the decades, several conceptual strategies have been proposed to overcome the stationarity constraint. All methods share a common principle: reducing the acquisition time for all frequencies so that the system can be considered "temporarily" time-invariant. One of the earliest and most widely adopted strategies to address non-stationarity is reducing measurement time through broadband excitation containing multiple frequencies simultaneously, rather than a single sinusoidal excitation. The most widely used broadband excitation is the multi-sine, a sum of sinusoidal waves with different frequencies, because it yields the best signal-to-noise ratio [29].

In classical single-sine EIS, the impedance spectrum is obtained by sequentially exciting the system at individual frequencies. The time required

to collect one spectrum is

$$T = n \sum_k \frac{1}{f_k}, \tag{6.1}$$

where f_k are the excited frequencies and n is the number of repetitions used for averaging to compute the impedance. On the other hand, in an experiment with multi-sine the time required is

$$T = \frac{1}{f_{min}}, \tag{6.2}$$

where f_{min} is the minimum excited frequency. Depending on the choice of n and the frequency grid, the stepped method requires 3 to 10 times longer [30]. Figure 6.1 illustrates this difference by comparing the total acquisition time required for stepped single-sine excitation with different numbers of repetitions to a single multi-sine period covering the same frequency range.

The impedance at each angular frequency ω_k is then obtained by isolating the corresponding spectral components of the voltage and current signals through Fourier analysis:

$$Z(\omega_k) = \frac{\mathcal{F}\{v(t)\}(\omega_k)}{\mathcal{F}\{i(t)\}(\omega_k)}. \tag{6.3}$$

The use of multi-sine excitation was explored as early as the 1970s, notably by Kojima and Fujiwara [31], Glover and Smith [32], and Creason and Smith [33], who demonstrated its advantages in terms of measurement speed and signal-to-noise ratio. While initially developed for stationary systems, multi-sine methods naturally lend themselves to applications involving slow system drift.

6.1.1 Short-time Fourier transform impedance

When a non-stationary system drifts slower than the period of the applied multi-sine, the stationarity assumption is valid over the time window equal to the inverse of the period. An "instantaneous" impedance can be defined as the ratio between the short-time Fourier transform (denoted $\mathcal{F}$) of the voltage $v(t)$ and current $i(t)$ signals:

$$Z(\omega, \tau) = \frac{\mathcal{F}[v(t)w(t-\tau)](\omega)}{\mathcal{F}[i(t)w(t-\tau)](\omega)}, \tag{6.4}$$

where $w(t)$ is the window function and τ is the time resolution of the impedance, equivalent to the window length. This formulation is schematically illustrated in Figure 6.2, which shows how a sliding time window is used to extract a local frequency response during a dynamic experiment.

This approach was introduced by Bond, Schwall, and Smith in 1977 [34], whom used a micro-controller-based online Fourier analysis during cyclic voltammetry experiments, and it is here called short-time Fourier transform (STFT)impedance.

With the advancement in computing power in the 1990s, Popkirov and Schindler developed a versatile instrumental setup for dynamic impedance spectroscopy, assembling a home-made potentiostat, an arbitrary waveform generator, and a digital acquisition device that streams the multi-frequency voltage and current signals to a personal computer. This configuration is still used today, and it is how the measurements in this work were carried out.

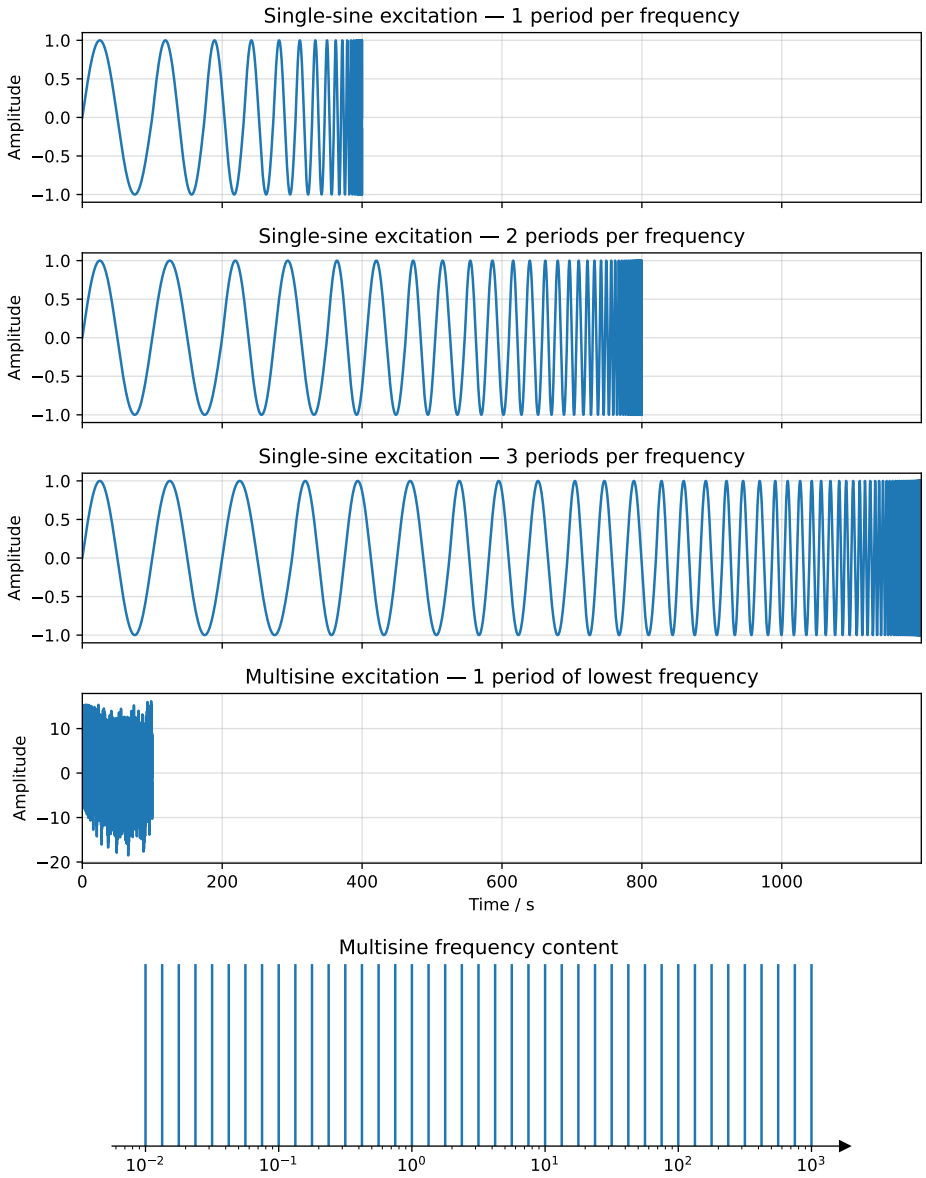


Figure 6.1: Comparison between stepped single-sine and multi-sine excitation strategies for the same frequencies. Top: time-domain representation of sequential single-sine excitations with increasing number of periods per frequency. Bottom: frequency distribution of a multi-sine excitation spanning 10 mHz to 1 kHz with 8 points per decade. A single multi-sine period enables simultaneous excitation of all frequencies, drastically reducing acquisition time compared to stepped methods.

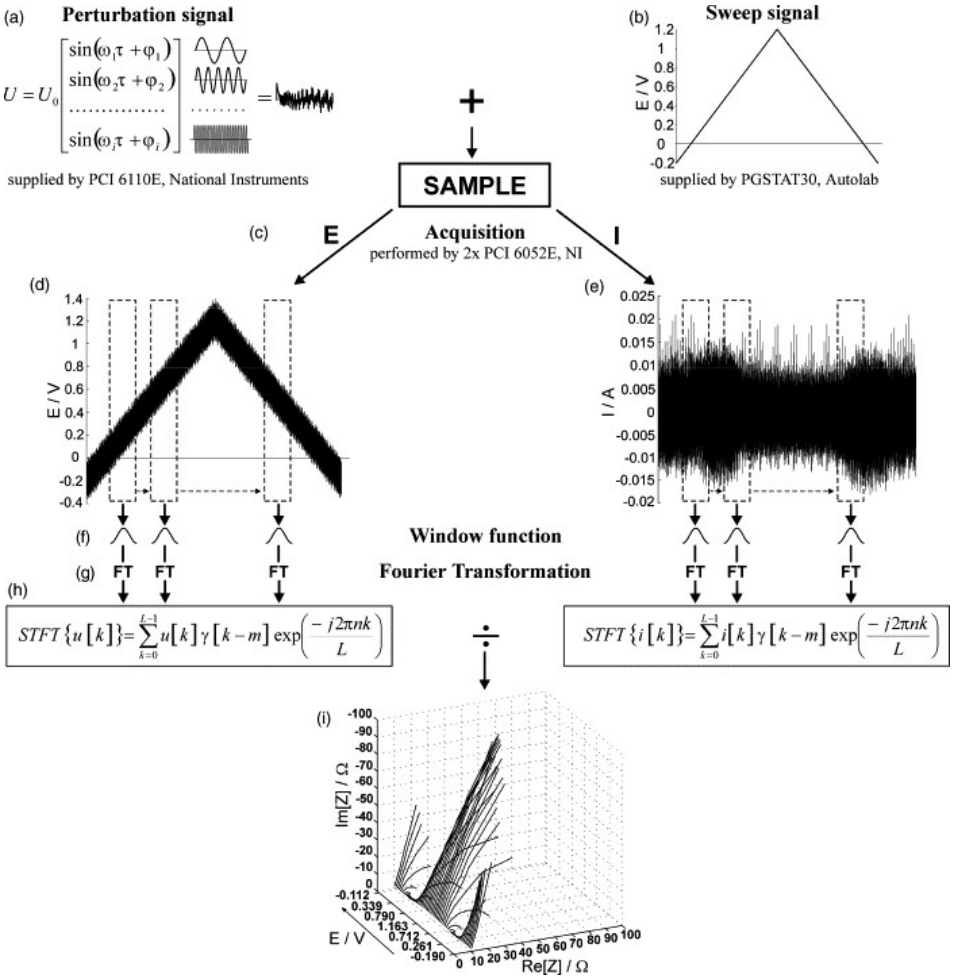


Figure 6.2: Principle of short-time Fourier transform (STFT) impedance estimation during a dynamic electrochemical experiment. The voltage and current signals are multiplied by a sliding window function, allowing the impedance to be estimated locally in time under the assumption of temporary stationarity. Reproduced with permission [35].

Various authors have implemented their own versions of this setup in the past three decades using devices from different manufacturers, without changing the basic formula:

- Hazi and Bond [36];
- Roy, Garland, and Pettit [37];
- Darowicki and Slepski [38];
- Schouckens, Pintelon, and Hubin [39];
- Sacci and Harrington [40];
- Pajkossy and Mészáros [41].

What distinguishes the work of different groups is the studied system, the drift speed, and the frequency components of the multi-sine. The choice of the frequency range and distribution is important for obtaining the full electrochemical response of the cell while avoiding intermodulation between harmonics and fundamental components. Intermodulation refers to the phenomenon where two or more different frequency signals interact in a

non-linear system, producing additional frequencies that are the sum and difference of the original signals. This leads to errors in the estimation of frequency components amplitudes.

The multi-sine prepared by Popkirov and Schindler consisted of three decades (100 kHz to 100 Hz, or 0.76 Hz to 0.076 mHz) with 42 prime-number components [42]. Roy, Garland, Pettit, and colleagues proposed a multi-sine with 133 components of only odd frequencies, random phase, and equal amplitude [37]. The choice of the phase and amplitude of each sinusoidal component of the multi-sine affects the quality of the estimated impedance and it will be discussed in detail in chapter 7. Schouckens, Pintelon, Hubin, and colleagues also proposed a multi-sine using odd-number harmonics with 220 components between 10 mHz and 2 kHz [43]. Sacci and Harrington designed a multi-sine in the frequency band 1 Hz to 1.3 kHz with removed specific harmonics that cause intermodulation, following the work of Popkirov, with amplitude decreasing by a factor of two every two decades.

Table 6.1 highlights that, despite decades of development, STFT-based impedance measurements are typically limited to minimum frequencies on the order of 1 Hz, reflecting the fundamental trade-off between time resolution and low-frequency access. A detailed chronological review of dynamic impedance spectroscopy work is provided in Appendix B.

Method	f_{min}(Hz)	f_{max}(Hz)	Drift type	Drift speed	Publication
STFT	1.8	3.1k	CV	8.5mV/s	[44]
STFT	40	10k	LSV	4mV/s	[45]
STFT	5	10k	CV	10mV/s	[46]
STFT	20	16k	CV	30mV/s	[35]
STFT	0.7	4.5k	CV	0.01mV/s	[47]
STFT	0.03 or 0.170	4.5k	CP	1/2.5 C	[48]
STFT	0.1	1.3k	CV	5mV/s	[49]
STFT	1	100k	CV	5mV/s	[50]
STFT	10	13.2k	CV	50mV/s	[51]
STFT	18.3	17.8k	CV	200mV/s	[41]

Table 6.1: Representative parameters of STFT-based dynamic impedance measurements reported in the literature. The table highlights the frequency range, drift conditions, and experimental protocols used by different authors, illustrating the practical lower-frequency limitations imposed by time-frequency resolution.

The technique of dynamic impedance spectroscopy can be applied to any electrochemical system under different type of drift, such cyclic voltammetry, chronopotentiometry, spontaneous oxidation or even mechanical strain. Examples of system studied with the STFT method are:

- redox couples [**macia2015orp**, 52–54];
- conductive polymers [35, 55];
- electrodeposition [56];
- molecule electro-oxidation [57, 58];
- stainless steel corrosion [59, 60];
- metal alloy corrosion [61];

- polymeric coatings [62];
- hydride battery discharge [48];
- fuel cells [63];
- redox-flow batteries [64].

6.2 The limiting lower frequency

The STFT method is based on the idea that the system is temporarily stationary for the time of the window length, but how to determine the lowest frequency that respects the LTI condition? In the literature, a couple of formulas have been proposed that apply to cyclic voltammetry multi-sine experiments on redox couples[49][45]:

$$\frac{F\nu}{RT} \ll 2\pi f_{min} \tag{6.5}$$

based on the scan rate ν of the voltage input and

$$\frac{\mathrm{d}i}{\mathrm{d}t} \ll 2\pi A_{min} f_{min} \tag{6.6}$$

based on the derivative of the output current i, and A_{min} is the amplitude of the system output at frequency f_{min}. These formulas present a general inequality, and it is not possible to define how much greater the minimum frequency should be compared to the speed of the drift. A few time might be sufficient or maybe one or more orders of magnitude is necessary. Furthermore they don't generalize to other electrochemical systems or other other drift types outside cyclic voltammetry.

A straightforward practical solution is to proceed by tentatively acquiring the impedance at a low desired frequency and test the linearity and stationarity condition via a Kramers-Kronig test.

6.3 Dynamic Multi-Frequency Analysis

Even when the lowest frequency of the multi-sine is fast enough to estimate a correct frequency response function, the STFT method has a limitation due to the reciprocity of time and frequency. In fact, if the window is short, the frequency resolution in the Fourier spectrum is low and it is not possible to identify low frequencies. The limitation imposed by time-frequency reciprocity is illustrated in Figure 6.3, which shows how frequency resolution improves only when multiple periods of a signal are sampled. This problem is worsened by the presence of strong pink noise[3] in this region as well as the zero-frequency skirt produced by the drift.

To increase the frequency resolution, La Mantia, Battistel, and Koster introduced *dynamic multi-frequency analysis* (DMFA) as a method to estimate instantaneous impedance spectra during fast electrochemical experiments while retaining access to low-frequency information [66, 67].

The key idea of DMFA is to record the entire multi-frequency signals of the experiment that spans hundreds of periods and perform a *frequency-selective analysis* of the measured signals. Rather than windowing the signal in time, DMFA isolates individual frequency components in the frequency domain using narrow band-pass filters and reconstructs their time

[3] Pink noise, also known as flicker noise, is a type of electronic noise characterized by a power spectral density that is inversely proportional to frequency, meaning it is stronger at lower frequencies. It commonly occurs in electronic devices due to fluctuations in the properties of materials, impacting the performance of components like transistors and sensors.

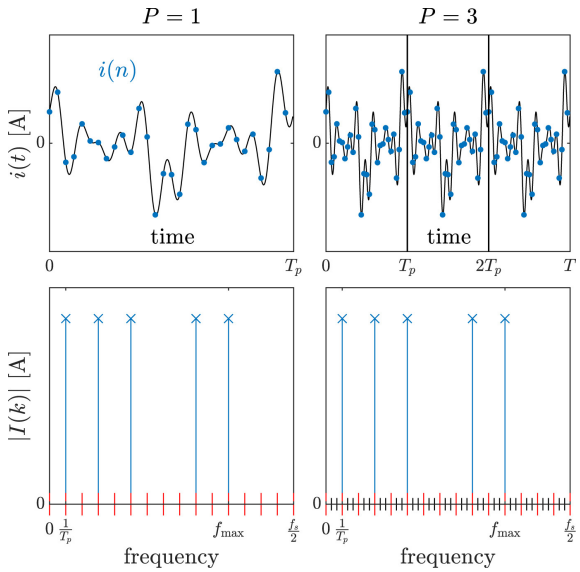


Figure 6.3: Illustration of the time-frequency resolution trade-off. Sampling an increasing number of periods improves frequency resolution (red markers) but requires longer observation times, limiting access to low frequencies in non-stationary systems when using time-windowed Fourier analysis. Used under CC BY License [65]

evolution via inverse Fourier transformation. The time resolution of the instantaneous impedance is no longer determined by the window length, but by the bandwidth of the band-pass filter.

Let $V(\omega)$ and $I(\omega)$ denote the Fourier transforms of the measured voltage and current signals obtained during a dynamic experiment with a multi-sine excitation. For a given excitation frequency ω_k, DMFA estimates the instantaneous impedance as

$$Z(\omega_k, \tau) = \frac{\mathcal{F}^{-1}[V(\omega)\, F(\omega - \omega_k)]\,(\tau)}{\mathcal{F}^{-1}[I(\omega)\, F(\omega - \omega_k)]\,(\tau)}, \tag{6.7}$$

where $F(\omega - \omega_k)$ is a flat-top filter function centered at ω_k, $\mathcal{F}^{-1}$ denotes the inverse Fourier transform, and τ represents time. This operation extracts the contribution of a single excitation frequency and reconstructs its time-dependent amplitude and phase. The individual processing steps of the DMFA algorithm are summarized in Figure 6.4, highlighting the frequency-selective filtering and reconstruction of time-dependent impedance components. In Step 3, it is evident that the peaks corresponding to the frequency components are not sharp but broadened: that is the effect of the system's time variance.

As a filter function, the author proposed a symmetric Fermi-Dirac function [67], defined as

$$g(\omega - \omega_k) = \frac{[1 + \exp(-n)]^2}{[1 + \exp(-n(\omega - \omega_k + bw)/bw)][1 + \exp(n(\omega - \omega_k - bw)/bw)]}, \tag{6.8}$$

where the parameter n controls the steepness of the filter edges and bw is the bandwidth and its minimum value correspond to the maximum time resolution τ. The effect of the filter shape on frequency selectivity and temporal smoothing is illustrated in Figure 6.5.

The choice of the bandwidth bw and attenuation parameter n plays a big role in the quality of the estimated impedance. In fact, they shape the

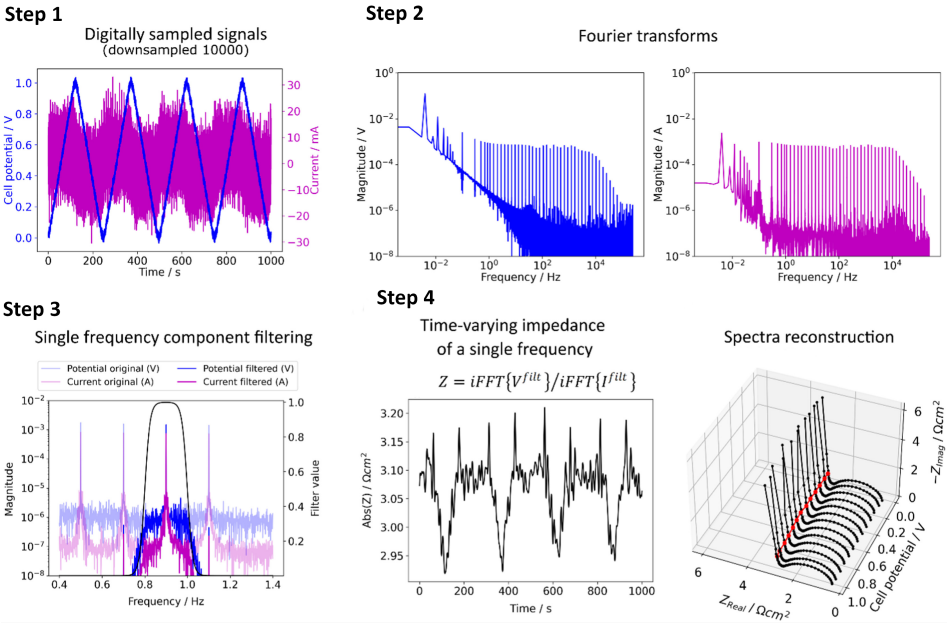


*Figure 6.4: Schematic overview of the dynamic multi-frequency analysis (DMFA) algorithm. The full voltage and current spectra are filtered around individual excitation frequencies and transformed back into the time domain to reconstruct the instantaneous impedance. Spectral broadening reflects deviations from strict time invariance. Adapted with permission [**piantaEvaluationKineticParameters2023a**].*

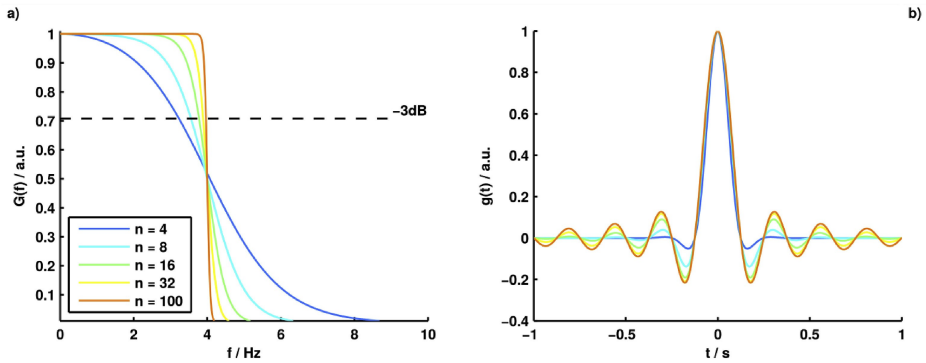


Figure 6.5: Symmetric Fermi-Dirac filter used in DMFA. (a) Frequency-domain representation for different values of the steepness parameter n. (b) Corresponding impulse responses in the time domain. The filter bandwidth and roll-off control the trade-off between time resolution, frequency isolation, and noise amplification. Used under CC BY LIcence [67].

width and lateral oscillatory behavior of the equivalent time-domain window [68]. An optimal combination of bw and n ensures that two consecutive impedances are independent from the information of the other.

When using this type of filter with the DMFA, the resulting impedance dataset suffers from high noise at the extremes because the filter function brings the signal values to zero before the inverse transform is performed. As a consequence, some of the first and last spectra of a dataset are always discarded. Other filters should be investigated to mitigate this problem.

The method was used to study redox-active species [66, 69], dissolution of p-type silicon [70], the intercalation of cations in nickel hexacyanoferrate [71], lithium manganese oxide [72, 73], and lithium titanium phosphate [74], and for the adsorption of ions on porous carbon electrodes. (The latter two are part of the discussion of this thesis). In reference [75], the DMFA is performed on simulated multifrequency signals and a deviation of the estimated impedance from the ideal model one at low frequencies is found.

6.3.1 Relation to STFT and interpretation

From a signal-processing perspective, DMFA can be interpreted as a generalization of the STFT. When the filter bandwidth is increased to include the full excitation spectrum, the DMFA formulation analytically converges to the STFT-based impedance estimation. Conversely, narrow filters allow individual frequency components to be tracked independently over time.

In the time domain, the filtering operation corresponds to a convolution of the original signal with the impulse response of the filter, resulting in a smoothed, frequency-specific representation. This behavior is conceptually related to wavelet transforms, although DMFA is derived directly from Fourier-domain considerations and tailored to multi-sine excitation.

6.4 Best linear time-varying approximation

While STFT and DMFA rely on signal-processing techniques applied to temporarily stationary data segments, an alternative perspective is to explicitly model the system as linear, yet time-varying (LTV). In 2012, Bruegelmans, Latier, and their colleagues proposed the idea of reconstructing the best linear time-varying approximation of the output signal of a non-stationary system exposed to multi-sine perturbation [76] and using it to compute the instantaneous impedance.

The principle is that a linear time-varying system can be expressed as a sum of N_p parallel connected LTI systems G_p multiplied by time-varying gains $b_p(t)$ such that the relation between an input $u(t)$ and output $y(t)$ is given by:

$$y(t) = \sum_{p=0}^{N_p} G_p\{u(t)\} b_p(t). \tag{6.9}$$

In practice, the BLTVA method consists of performing a parametric fitting of the output Fourier spectrum to approximate the linear and time-invariant response. Noise, non-linearities [77, 78], and non-stationarities [79] are estimated from the residues of the fitting.

After the fitting, the instantaneous impedance is estimated as a ratio between input and (modelled) output:

$$Z_{inst}(\omega_k, \tau) = \frac{U(\omega_k)}{Y(\omega_k, \tau)} = \frac{U(\omega_k)}{\sum_{p=0}^{N_p} \theta_{p,k}\, b_p(\tau)}, \tag{6.10}$$

where U and Y are Fourier transforms of u and y, respectively, and $\theta_{p,k}$ are optimized parameters of the output non-parametric fitting.

The group applied the method for the identification of the transfer function of many highly non-stationary electrochemical systems under constant current drift: metal electrodeposition [80] [81], the conversion of zinc to zirconia [82], to study fast charging [83] and SEI formation in Li-ion batteries [84].

6.4.1 Comparison with DMFA

The BLTVA estimation is similar to DMFA in the sense that the voltage and current signals are Fourier transformed in their entirety before the fitting. Furthermore the fitting is performed in the frequency band ($\{\omega_{k-1}, \omega_k, \omega_{k+1}\}$) which is equivalent to a rectangular band-pass filter.

The advantage of the BLTVA is that the resulting impedance is LTI by definition and doesn't require a validation, but its execution is computationally more expensive because it requires an iterative regression procedure.

Table 6.2 emphasizes that DMFA and BLTVA address complementary regions of the non-stationary impedance problem, trading computational complexity for access to lower frequencies under stronger drift conditions. A complete comparison is not possible since the drift and frequency ranges are so different.

An interesting study would be to perform all the methods for impedance estimation on the same simulated data under different conditions to uncover the effect of drift speed on stationarity and how well the experimentally estimated impedance approximates the ideal one.

Method	f_{min}(Hz)	f_{max}(Hz)	Drift type	Drift speed	Publicatio
DMFA	10	125k	CV	200mV/s	[71]
DMFA	2.5	210k	CV	16mV/s	[73]
DMFA	0.1	100k	CV	8mV/s	[**piantaEvaluationKinetic**
DMFA	0.1	10k	CV	0.8mV/s	[74]
BLTVA	0.0056	67	CP	C/2	[83]
BLTVA	0.1	10k	LSV	5mV/s	[84]

Table 6.2: Representative applications of DMFA and BLTVA in non-stationary electrochemical impedance spectroscopy. The table compares achievable frequency ranges, drift conditions, and experimental protocols, illustrating the complementary strengths and limitations of the two approaches.

Part II

Methodologies

Chapter 7

Acquisition of non-stationary impedance

The previous chapters illustrate the state of the art of dynamic impedance spectroscopy with emphasis on the instrumentation, a custom set-up comprising a potentiostat, a waveform generator, and a digital data sampler, and the methods for impedance estimation from multi-frequency voltage and current signals. This chapter focuses on the hardware and software architecture I developed in this work to perform continuous, broadband impedance measurements under controlled drifts.

7.1 A map of the dynamic impedance experiment

This work proposes a complete pipeline for dynamic impedance spectroscopy measurements covering:

- design and generation of multi-sine excitation signals;
- acquisition of multi-frequency signals with automated set-up;
- estimation of instantaneous impedance from multi-frequency voltage and current signals;
- on-line estimation of the high frequency band of the impedance;
- analysis of the impedance data with physical or non-physical models.

To clarify the scope and modular structure of the proposed methodology, Figure 7.1 summarizes the complete experimental and analytical workflow adopted in this work. The schematic can be simplified in three steps:

$$\text{Acquisition of signals} \rightarrow \text{Estimation of } Z \rightarrow \text{Analysis of } Z$$

This chapter describes the methodology of signal acquisition; chapter 8 describes the estimation of impedance from multi-frequency signals and its online implementation within the acquisition software. The analysis of impedance data is adressed separately for each application showed in Part III of this work.

Before the dynamic impedance spectroscopy experiment, I usually perform classic electrochemical measurements to inform the choice of parameters. Classic EIS measurement dictates the frequency interval and makes the multi-sine optimization informed, while direct current methods give information on the macro-kinetics and cycle life.

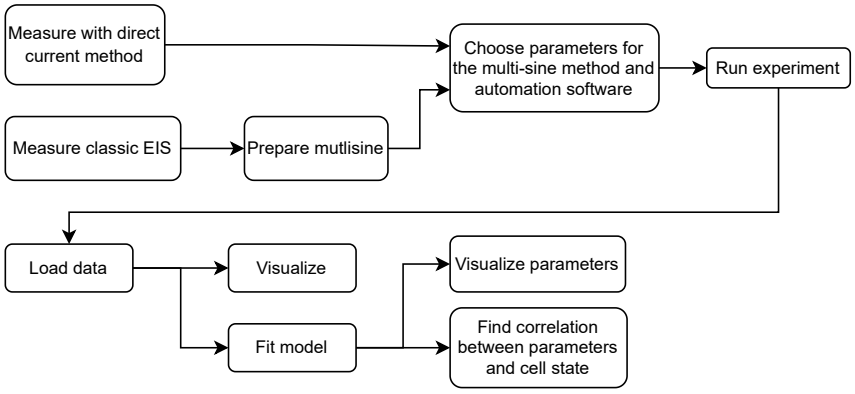


Figure 7.1: Conceptual pipeline of dynamic electrochemical impedance spectroscopy. The workflow spans multi-sine excitation design, synchronized signal acquisition, time-resolved impedance estimation, and subsequent model-based analysis.

7.2 Electrochemical instrumentation

In chapter 6, I stated that the typical experimental setup for dynamic impedance consists of a potentiostat/galvanostat (or potentiostat for short), an arbitrary waveform generator, and a digital acquisition device. While the latter provides high-fidelity signal generation and sampling, the core of the setup is the potentiostat, which applies those signals and controls the cell terminals. The potentiostat provides the user with built-in perturbation waveforms used in cyclic voltammetry, potentiostatic, and galvanostatic experiments. The same hardware is used for classic electrochemical experiments and for multi-frequency ones; it is important to discuss the inner components of a typical instrument to address how the setup was implemented for this work.

7.2.1 Generalities of potentiostats

Typical electrochemical methods consist of the controlled drift of one cell variable among voltage and current while recording the other as function of time, with the use of a potentiostat or galvanostat. The purpose of a potentiostat is to maintain the potential of a working electrode (WE) at a prescribed value relative to a reference electrode (RE). This is achieved using a high-gain feedback loop. The potential difference between WE and RE is measured through a high-impedance[4] voltage follower connected to the RE. This element ensures that only a negligible current flows through the RE, preserving its equilibrium potential. The measured potential is compared to a user-defined setpoint, and the resulting error signal drives a power amplifier connected to the counter electrode (CE). By adjusting the CE potential, the amplifier forces the WE potential toward the setpoint. The output stage must source or sink the full cell current while maintaining stability of the feedback loop.

[4] Usually in the range 1-10 $T\Omega$

This check is performed continuously at a rate dictated by the bandwidth of the operational amplifier. Usually, this value is arbitrary on a scale of integer values. High values mean fast loops, while low values result in slower loops. To reduce noise or fluctuation, a low bandwidth is an ideal choice; for measuring the dynamics of fast systems, it is necessary to use

high values and accept more noise in the sampling.

The current flowing through the WE-CE path is measured by a current-to-voltage converter, typically implemented using a transimpedance amplifier. The converter's output provides a low-noise measurement of the cell current and is digitized for acquisition and further processing. The transimpedance gain determines the current range, and switching networks or programmable gain amplifiers are often used to accommodate multiple current's order of magnitude.

Instead, a galvanostat controls the current flowing between WE and CE. The measured current is compared to a desired setpoint, and the feedback loop adjusts the CE potential to maintain the prescribed value. The WE potential is measured against the RE.

Because the implementation of dynamic impedance spectroscopy relies on injecting external perturbations and accessing raw analogue signals, it is essential to understand the internal structure of a potentiostat and its control loop. Figure 7.2 shows a schematic of the connections for a typical electrochemical device with highlights of the electronics for potentiostatic and galvanostatic modes. It does not include the circuitry for measuring impedance and the frequency response analyzer, but it is not interesting for this work.

7.2.2 Auxiliary analogue inputs and outputs

Some commercial potentiostats incorporate auxiliary input and output connections to enable external excitation or synchronization with additional hardware. Analog input ports are typically connected upstream of the main control loop, allowing externally generated waveforms to be superimposed on the programmed potential or current (see Figure 7.2, top left of the schematic). This functionality is essential for experiments requiring arbitrary perturbations and high-speed signal recording.

Auxiliary analog inputs provide direct access to measured analog signals before digital conversion, bypassing part of the instrument's internal filtering and signal conditioning. Recording these signals enables high-resolution acquisition beyond the default sampling configuration of the potentiostat with a dedicated device.

7.2.3 Digital control and communication

Although the core functionality of the potentiostat is analog circuitry, the control and signal acquisition is digital and implemented with field-programmable gate arrays (FPGAs[5]) or micro-controllers. Each channel of the potentiostat is connected with a communication board that sends signals and receives commands to and from a personal computer via USB or Ethernet. This interface sets on the hardware the measurement parameters chosen by the user and retrieves the sampled data. The instrument is then capable of working independently of the computer, storing data temporarily in a buffer memory.

[5] In FPGAs, logic blocks can be programmed to execute simple digital operations like sums and multiplication. The advantage is the possibility of parallelizing these operations for real-time data processing. They are more expensive than microcontrollers

7.3 Dynamic impedance spectroscopy setup

The instrumental setup used in our laboratory follows the same design introduced before: a potentiostat interfaced with a dedicated waveform generator for excitation signal generation and a dedicated acquisition device

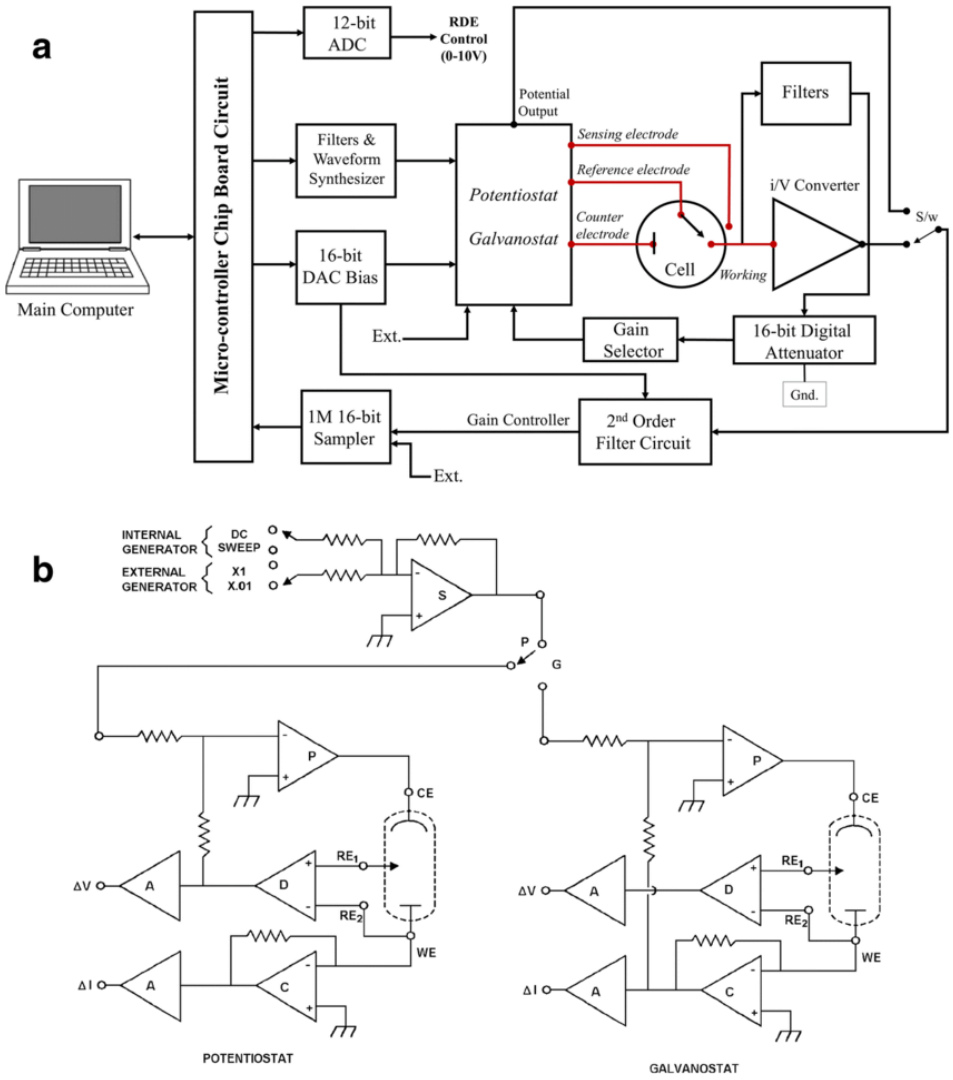


Figure 7.2: Simplified schematic of a potentiostat/galvanostat architecture. The feedback loops governing potentiostatic and galvanostatic control, the transimpedance current measurement, and the analogue summing nodes are highlighted. Understanding these internal blocks is essential for safely injecting external multi-sine excitations and accessing raw voltage and current signals in dynamic impedance experiments. (Source: CHI 700 Instruction manual)

for high-frequency sampling and streaming to a computer. The multi-sine excitation is generated as an analog voltage signal and injected into the potentiostat, where it is superimposed on the main drift. The external signal is applied before the voltage/current converter (see Figure 7.2), so for current-controlled experiments, the current flowing through the circuit is calculated from the external applied voltage and the measurement resistor following Ohm's law. Not all commercial potentiostats have the capability to accept external signals. In this work, I used devices from BioLogic, which provides an (unsupported) library for controlling the instrument through open-source programming languages. This capability was crucial for the automation of dynamic impedance measurements.

As arbitrary waveform generators, we use dedicated dual-channel benchtop devices from Keysight. The notable features are:

- a big memory of 64MSa per channel of the digital to analog converter;
- capability to sum and synchronize the output of the two channels into

one signal;

- jitter-free analog conversion that preserve the original signal reducing spectral leakage.

These devices can be controlled with computers using the *VISA* communication framework and *SCPI* language, both supported in Python with dedicated libraries.

The cell signals were sampled with USB acquisition devices from PicoTech that allow the capture of long continuous signals via streaming capability at high sampling rates (up to 20 MSa/s, though for electrochemistry 1–3 MSa/s is sufficient). The manufacturer provides a software development kit with support for multiple languages for streaming data to a computer and integrating into existing code.

BioLogic potentiostats have an auxiliary 9-pin port for the input and output of analog signals. For the purpose of this project, I was interested in only three:

- Auxiliary In 2 for the injection of the excitation signal;
- voltage monitor to record the analog signal with a dedicated external device;
- current monitor.

The manufacturer supplies adapters for the 9-pin to BNC connectors, but for me it was more practical to build them myself with longer and independent cables. Instructions for the construction of these cables are available in Appendix C.

The modular separation between electrochemical control, excitation generation, and signal acquisition is illustrated schematically in Figure 7.3, while Figure 7.4 shows a photograph of the instrumentation in our laboratory. It is possible to see how the auxiliary cables interface the single potentiostat's channels with the external devices.

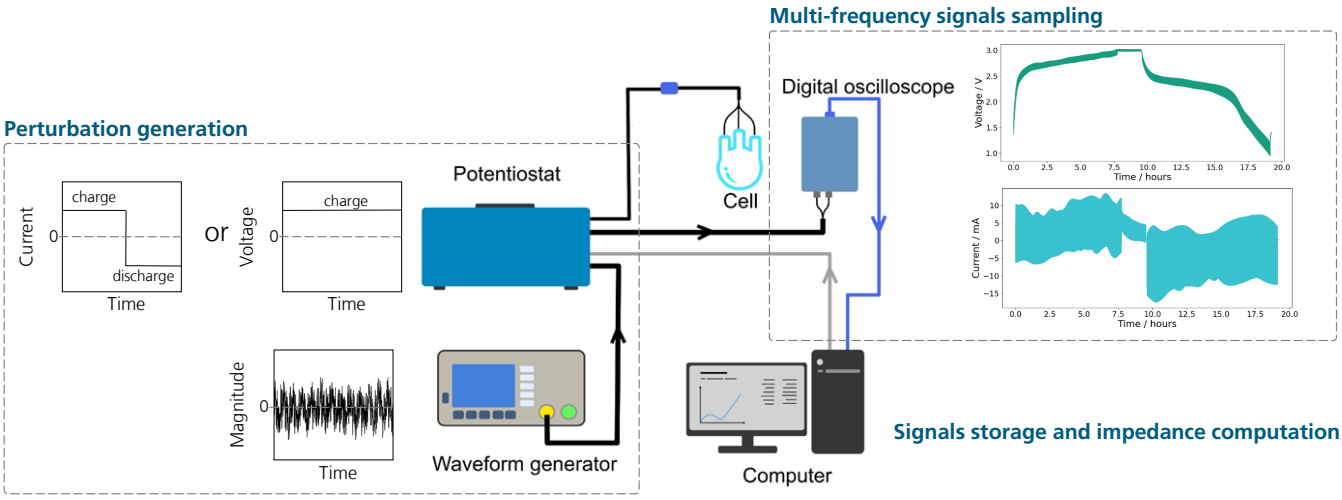


Figure 7.3: Schematic of the experimental setup for dynamic impedance spectroscopy. A potentiostat imposes the electrochemical operating trajectory, while an external arbitrary waveform generator injects the multi-sine excitation and a dedicated acquisition device records voltage and current signals with high temporal resolution. This modular architecture enables full control over excitation, synchronization, and data access.

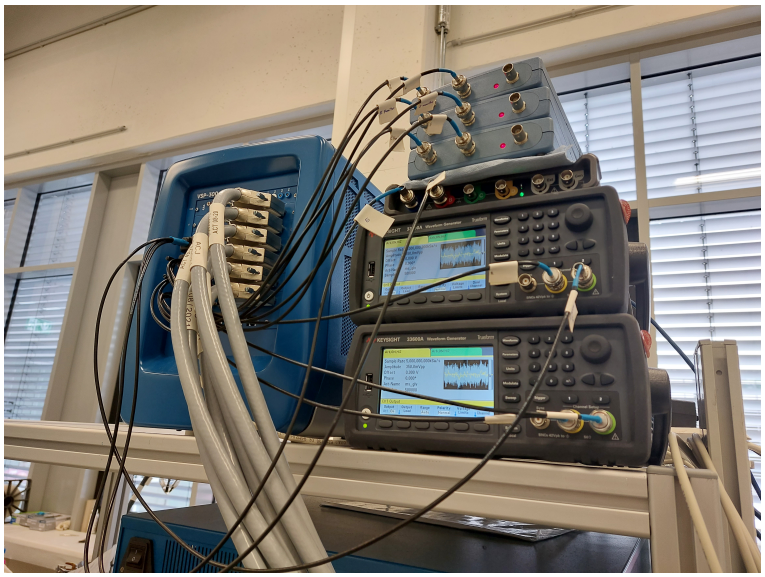


Figure 7.4: Photograph of the laboratory setup used for dynamic impedance measurements. A multi-channel potentiostat is interfaced with external waveform generators and acquisition devices through auxiliary analogue ports, enabling parallel dynamic impedance experiments on multiple electrochemical cells.

7.4 Multi-sine design

A multi-sine is the superposition of sinusoidal components:

$$u(t) = \sum_{k=1}^{K} A_k \sin(2\pi f_k t + \phi_k).$$

It allows simultaneous excitation of a large set of frequencies with a single time-domain waveform, reducing the experiment duration by at least a third in battery studies.

Obtaining high-quality instantaneous impedance in non-stationary conditions, however, requires careful design of the multi-sine: frequency grid, crest-factor minimization, amplitude choice and memory constraints of the AWG.

7.4.1 Frequencies selection

The frequencies f must be chosen such that one period of the whole multi-sine contains an integer number of oscillations to avoid spectral leakage due to non-periodicity. The period is then defined by one oscillation of the sine wave with lowest frequency: $T = 1/f_{min}$. Every other frequency must satisfy the condition $f_k = m \times f_{min}$, where m is an integer. In this work, we generate the desired frequencies from a harmonic series of integers to ensure the above condition and the possibility of reusing the sequence for multi-sine excitations with different f_{min}. With the term harmonic sequence, I refer to $h_k = f_k/f_{min}$, such that it always starts from 1 and contains only integers. The sequence is generated as a logarithmic distribution with a number of points per decade between six and ten. Lower values give sparse spectra, while higher values give denser ones. Denser spectra are more beneficial for data regression because they provide more points to fit in the cost function, but the signal power increases. For this work, I obtained sufficient impedance density with eight points per decade.

7.4.2 Amplitude optimization

When the amplitudes are all the same, the multi-sine is said to be "flat". The question is: how should the amplitudes be decided for the best signal-to-noise ratio? It can be achieved via analytical formulas or experimental optimization. The first case was used in Battistel [85] and Hallemans [65], with opposite results, making the amplitudes drop logarithmically with increasing or decreasing frequency.

In this work, I followed an empirical strategy: the input multi-sine components amplitudes are chosen such to give a constant output's amplitude knowing system's impedance. To clarify the concept, let's take the example of a voltage-controlled experiment in which the multi-sine is generate by the waveform generator and directly added to the drift voltage. The sinusoidal components amplitudes of the multi-sine $V_{ms}(\omega_k)$ are chosen to have desired outputs amplitudes $I_{des}(\omega_k)$ according to the formula

$$V_{ms}(\omega_k) = |Z(\omega_k)| \times I_{des}(\omega_k), \tag{7.1}$$

where $|Z(\omega_k)|$ is the system's impedance module. $I_{des}(\omega_k)$ should be a small value (5–10 mV) to ensure linearity of the response. $Z(\omega_k)$ is a measured impedance spectrum of the system[6], but for non-stationary conditions I use the average impedance $Z_{avg}(\omega_k)$ over a complete DEIS experiment performed beforehand without optimizing the multi-sine.

For current-controlled experiments, there is one more step to consider because the generated waveform is converted to current within the galvanostat. The conversion is made through the measurement resistance R_m[7] following Ohm's law:

$$V_{ms}(\omega_k) = R_m \times I_{ms}(\omega_k). \tag{7.2}$$

In practice, one computes first the desired output amplitudes $I_{ms}(\omega_k)$ with the inverse of formula 7.1:

$$I_{ms}(\omega_k) = \frac{V_{des}(\omega_k)}{|Z(\omega_k)|}, \tag{7.3}$$

and then generates the excitation with the amplitudes $V_{ms}(\omega_k)$ obtained with formula 7.2.

The choice of the multi-sine components' amplitudes is a difficult engineering problem that affects the signal-to-noise ratio of cell voltage and current Fourier spectra, and therefore the quality of the estimated impedance.

7.4.3 Phase optimization

The process of identification should be carried out with small-amplitude excitation around the operating point. Multi-sine waveforms can often have large peak-to-peak total amplitude after optimizing the relative amplitudes, which derive from the superposition of multiple sinusoidal components at the value of their peaks. In most cases, however, it can be reduced by acting on the phase of each sinusoidal component in the total multi-sine. The quantity of interest to reduce is the *crest factor* (CF), defined as

$$CF = \frac{|x_{max}|}{x_{rms}}$$

where x_{rms} is the root-mean-square, and the peak-to-average power ratio (PAPR) equals the square of the crest factor. The crest factor of a

[6] Can be measured with the classic stepped-sine method. When the frequency grid does not match the multi-sine one, I interpolate or extrapolate the impedance module at the desired frequency from a spline fit.

[7] Contained in the i/R converter in series with the working electrode; see Figure 7.2

pure sinusoidal wave is $\sqrt{2}$, which serves as a lower ideal boundary (it is matched only by a chirp waveform; a square wave has a CF of 1, which is also ideal).

In this work, iterative-stochastic optimization was used in all experiments presented later; although not optimal, it gives a sufficiently low CF for the exploratory stage of this work. The method consists of sampling the phases from a randomly generated distribution and computing the crest factor. The process is repeated a certain number of times (e.g., 50 or 100) and the phase that produces the lowest crest factor is kept. A summary of the main methods for crest-factor minimization present in the literature is described in Appendix D.

7.4.4 Multi-sine design library

To streamline the design of multi-sines, I created an open-source Python library named `multisine`[1]. It provides the `Multisine` object that generates multi-sines, optionally optimizes phases for minimal crest factor via iterative randomization, and provides tools for visual spectral analysis.

First, one imports the `Multisine` object and defines the interested parameters as follows:

```
from multisine.multisine import Multisine
import numpy as np

frequencies = [1, 10, 100, 1000] # in Hz
amplitudes = np.ones(frequencies.size) * 0.05  # in V
sampling_frequency = 10000 # in Hz
```

`Multisine` offers a compact way to generate multi-sines, normalize, and analyze their frequency components[8] with a few lines of code:

[8] Useful to spot errors in the parameters that may cause spectral leakage

```
ms = Multisine(sampling_frequency, frequencies, amplitudes,)
ms.best_random_phases(500)
print(ms.crest_factor)
ms.normalize_waveform(1)
ms.plot('voltage')
num_period = 10
ms.fourier_analysis(num_period)
ms.plot_dft((0.001,sampling_frequency//2))
```

7.4.5 AWG memory limitations and splitting strategy

Benchtop waveform generators consist of electronic circuits constructed around a digital-to-analog converter (DAC). An integrated modern DAC chip has a buffer memory in which the waveform values are written, and during generation, such memory is read in a loop (called look-up table generation). Commonly, commercial devices have a buffer memory consisting of 12 to 32 kSa, which is quite small for the excitations used in batteries that might span 6–7 decades (at least at the laboratory scale). A solution used in digital signal processing devices (used, for example, for the playback of audio files in any electronic device) is called direct digital synthesis (DDS), which creates direct communication between a DAC and a microcontroller memory. This option requires proficiency in the programming and setting of micro-controllers, specifically synchronizing the components to the same clock.

[1] Available at https://github.com/federicoscarpioni/multisine

In our laboratory, we use instead a dedicated benchtop Arbitrary Waveform Generator (AWG) with two channels, each with 64 MSa; a huge upgrade from conventional integrated DACs. This amount of memory allows multi-sine waveforms containing at least 6 decades. To study batteries, however, my goal was to have seven or more decades of frequency range in the multi-sine and to be able to switch between two waveforms optimized specifically for chronopotentiometry and chronoamperometry. The solution I developed is to divide a multi-sine waveform into two waveforms and use the ability of the generator to sum the signals of the two channels ("dual channel mode") at the output. The two separate waves have fewer samples, and the two types of multi-sines can easily fit in the memory of the device.

As an example, let's consider a seven-decade multi-sine waveform from 10 mHz to 100 kHz, generated at 1 MHz. As generation frequency, using 10 times the value of the maximum frequency multi-sine component gives good results, although the literature advises using even 100 times to avoid spectral leakage. Keysight devices have a proprietary "TrueForm" technology to reduce jitter that might help limit spectral leakage in the final signal.

Generating one single broadband waveform at 1 MHz produces a file of 100 MSa (obtained as $period \times f_{gen}$). Dividing the frequency band at 100 Hz creates two smaller multi-sines: one from 10 mHz to 100 Hz (not included), generated at 1000 Hz (size 100 kSa), and a second one from exactly 100 Hz to 100 kHz, generated at 1 MHz, which, when choosing a period of 1 s, leads to a waveform of 100 kSa. The two smaller multi-sines take up only 200 kSa of the generator memory to produce the same excitation as the complete multi-sine of 100 MSa. Figure 7.5 reports the original and reduced multi-sines in the time domain.

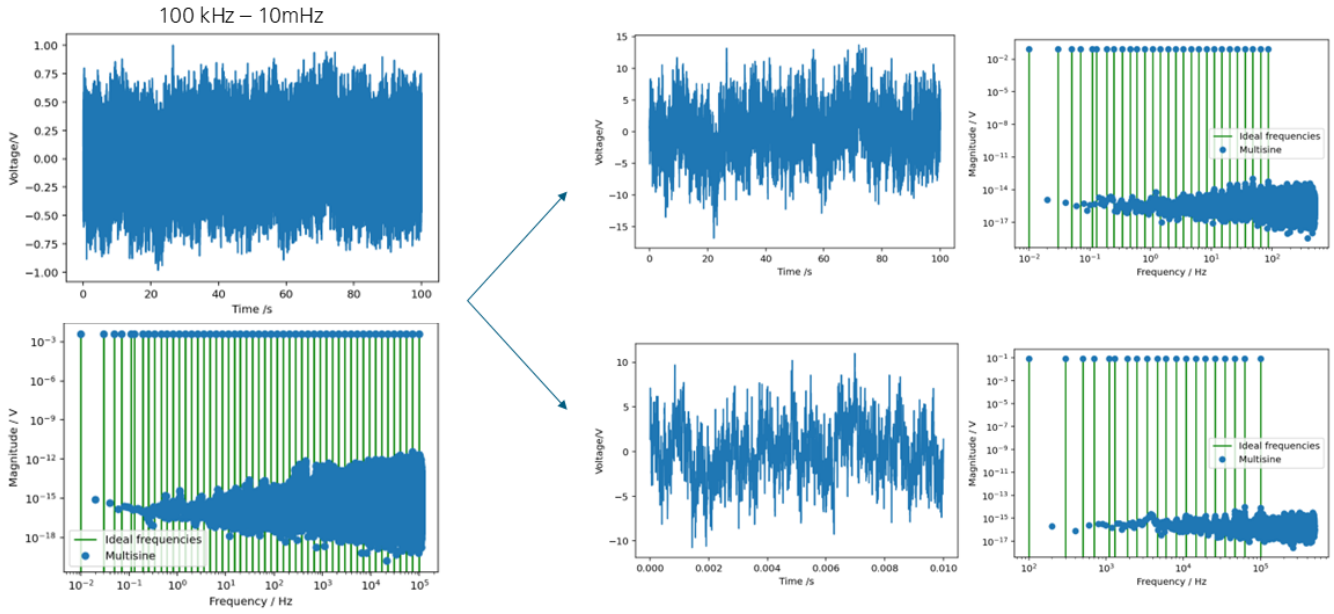


Figure 7.5: Broadband multi-sine generation using frequency-band splitting. A single wideband multi-sine (right) is reproduced by summing two narrower-band multi-sines (left), each fitting within the memory constraints of the waveform generator. This strategy enables excitation spanning multiple decades of frequency without loss of spectral coherence.

In the example, I used two arbitrary parameters: the 100 Hz boundary frequency and the 1 s period of the high-band waveform. In this application,

100 Hz sits in the middle; this choice can be adjusted case by case using the same rationale. Regarding the period of the second waveform, one might consider using 0.01 seconds to have exactly one period of the lowest frequency, but it must be considered that the frequencies of the two waveforms should follow the same rules described before to avoid intermodulation artifacts; being made from sequences of harmonics. Logarithmic sequences optimized for intermodulation tend to be sparser on the low-frequency part and denser at higher frequencies. If both waveforms are generated with frequencies starting from the harmonic of 1, the measured impedance spectra would have a sparser point distribution in the decade from 100 Hz to 1 kHz, where the high-band multi-sine waveform starts. See Figure 7.6 second row.

To get a uniform distribution of points, I generated the high-frequency series starting from 1 Hz and discarded the frequencies until 100 Hz (not included), since they are already present in the low-band waveform. Then, the period of the waveform must be 1 Hz; otherwise, there will be spectral leakage in the final excitation (note that all frequencies of the high-frequency series are integer multiples of 1 Hz). The sum of the two shorter waveforms generated using this method has the exact frequency distribution of a single generated waveform, as shown in the bottom part of Figure 7.6. The effect of different strategies for restarting harmonic sequences when splitting a broadband multi-sine is illustrated in the same Figure 7.6.

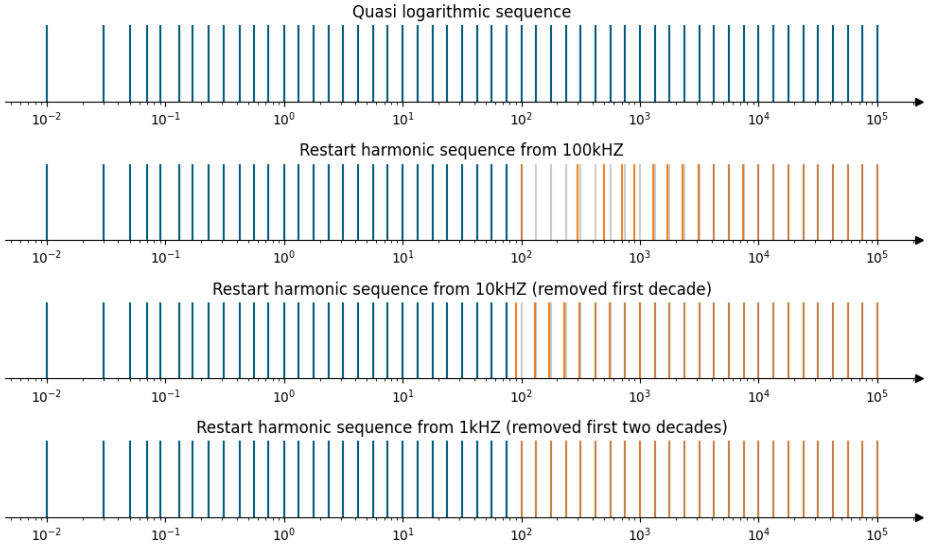


Figure 7.6: Effect of frequency-band splitting on harmonic density in broadband multi-sine design. Different restart strategies for the high-frequency band lead to non-uniform spectral density near the band boundary. Starting the harmonic sequence at 1 Hz and discarding overlapping decades preserves a uniform logarithmic distribution across the full frequency range.

To summarize: for a CC-CV battery experiment, one channel of the generator contains in memory two low-band multi-sine waveforms, one optimized for chronoamperometry and the other for chronopotentiometry, and the other channel contains the two optimized high-band multi-sine waveforms.

7.5 Automation software

7.5.1 Problem statement

Dynamic impedance spectroscopy measurements performed during battery aging experiments extend over time scales of days to weeks and generate continuous, high-bandwidth voltage and current data streams. When broadband excitation is applied throughout cycling, the resulting acquisition rate rapidly exceeds the available volatile memory of standard data acquisition systems. As a consequence, the feasibility of long-duration non-stationary impedance experiments is not limited by electrochemical stability, but by the ability to sustain uninterrupted signal acquisition and storage over arbitrarily long time horizons.

In conventional laboratory workflows, data acquisition is often constrained to fixed-length recordings whose duration is dictated by the available RAM of the acquisition device or host computer. This approach is incompatible with aging studies, where impedance must be tracked continuously across hundreds of electrochemical techniques and repeated charge-discharge cycles. Without an automated strategy for segmenting, storing, and reinitializing acquisition buffers, dynamic impedance measurements would be restricted to short snapshots, preventing systematic observation of slow degradation phenomena.

For this reason, the automation in the present work is conceived primarily as a memory-management and scalability mechanism. The software framework developed here enables the acquisition of multi-frequency signals over arbitrarily long experiments by periodically transferring voltage and current data blocks from volatile memory to persistent storage, while maintaining phase coherence and continuity of the excitation.

To save the signals to disk at the right time, a computer program must follow the execution of the sequence of techniques on the potentiostat with a control loop, execute the saving to disk, and immediately restart the data streaming without losing samples. A *technique* is the building block of an electrochemical experiment. The most basic techniques are chronoamperometry, chronopotentiometry, voltage sweep, and open-circuit voltage. A *sequence* of techniques is a set of those building blocks concatenated to perform a desired experiment. Most of the time, a "loop" technique is inserted in the sequence to repeat it a certain number of times. The instrumnets control loop is schematized in Figure 7.7

7.5.2 Software solution

A control loop must monitor the sequence advancement and loop number returned by querying the potentiostat communication board. BioLogic EC-Lab and EC-Lab Express do not allow automating this process. The output triggers functionality is too limited for fine tuning the acquisition as described here. BioLogic provides a simple software development kit (SDK) to communicate with the potentiostats using various programming languages, which I expanded into a more sophisticated object-oriented Python library that I called `pyeclab`[2].

Object-oriented programming (OOP) is a software design paradigm in which complex systems are represented as collections of interacting *objects*.

[2] Available at https://github.com/federicoscarpioni/pyeclab

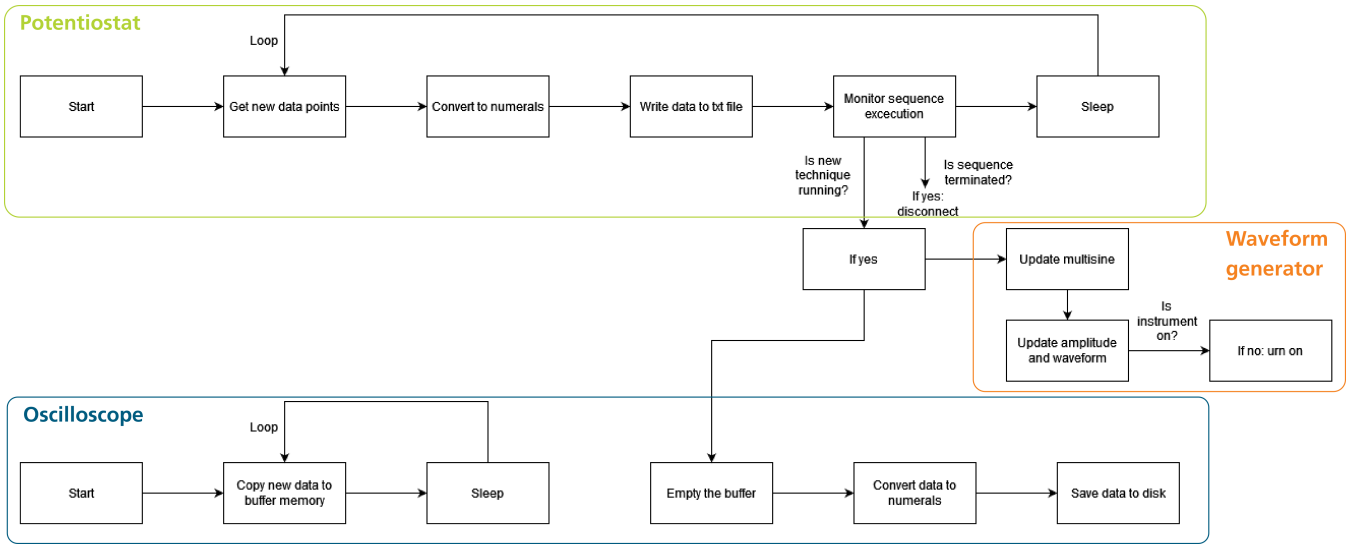


Figure 7.7: Signal flow in the automated dynamic impedance spectroscopy framework. Continuous voltage and current streams are acquired in memory blocks, transferred to persistent storage, and reinitialized without interrupting excitation or acquisition. This strategy enables arbitrarily long experiments without RAM limitations.

Each object encapsulates both data and the operations that can be performed on that data, allowing physical devices or logical processes to be represented as modular software entities. Typically, objects are schematized as in Figure 7.8 to highlight their name, attributes and methods. For simplicity, in the rest of this discussion, only the methods (that incapsulate the behavior) are represented.

In the present implementation, each laboratory instrument (potentiostat, waveform generator, and oscilloscope) is abstracted as an independent object exposing a well-defined *interface* for configuration, status monitoring, and data exchange. This abstraction layer separates the experimental logic from the hardware-specific communication details, enabling the same high-level control code to operate across different hardware configurations with minimal modification.

Encapsulation and modularity are particularly advantageous for long-duration experiments, where robustness and maintainability are essential. By structuring the automation software as a set of interacting objects, individual components can be extended, replaced, or debugged without affecting the entire acquisition framework. This design also enables *hierarchical composition*, where higher-level objects coordinate the behavior of lower-level instrument objects, reflecting the physical structure of the experimental setup.

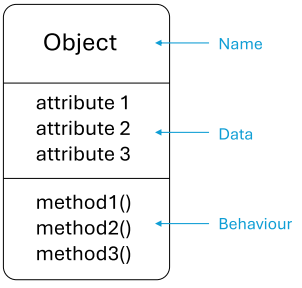


Figure 7.8: Structure of a generic object as used in object oriented programming (OOP).

Through the `Channel` object, the status and data points are recursively retrieved from each channel of the potentiostat to the computer program, which converts, stores, and plots data, and executes a *callback* function when a technique is finished.

The callback is the heart of the automation because it allows triggering special behavior of the program execution. To synchronize actions with the waveform generator[3] and the scope[4], I developed Python libraries for both.

A comprehensive library is then used to coordinate the devices, called

[3] Available at https://github.com/federicoscarpioni/trueformawg
[4] Available at https://github.com/federicoscarpioni/pypicostreaming

`DEIStools`[5]. The `DEISChannel` object provided in it coordinates the workflow of the setup according to Figure 7.9.

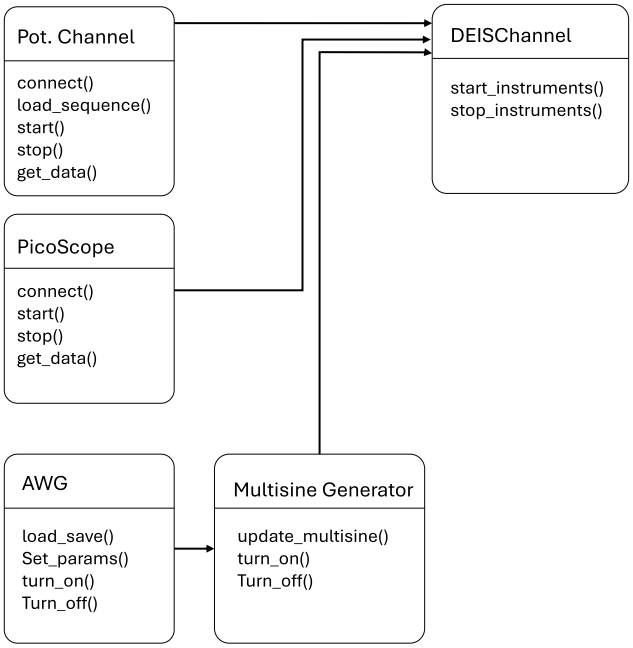


Figure 7.9: Object-oriented software architecture for dynamic impedance acquisition. Each instrument is encapsulated in a dedicated worker object, coordinated by a central control loop inside DEISChannel. This modular design enables synchronization, extensibility, and hardware abstraction across different experimental configurations.

The coordination of the electrochemical sequence, excitation waveform, and data acquisition is implemented through a central software control loop. This loop continuously monitors the execution state of the potentiostat, synchronizes the activation of the multi-sine excitation, and manages the transfer of acquired data from volatile memory to persistent storage. By reacting to technique completion events and loop counters, the control logic ensures uninterrupted acquisition while preserving phase coherence between excitation and response signals. The structure of this control loop, implemented in Python and used throughout the battery aging experiments presented in this thesis, is illustrated in Figure 7.10.

Because the potentiostat, waveform generator, and acquisition device operate simultaneously and exchange data in real time, the automation software must manage multiple processes concurrently. Sequential execution of instrument commands would introduce delays and risk data loss, particularly during high-bandwidth streaming. For this reason, the control software employs concurrent execution through *multithreading*.

In multithreaded programming, multiple execution threads run within the same process and share access to memory and program state. This approach allows independent tasks to proceed in parallel while remaining coordinated through a central control logic.

To make the automation possible, each instrument must independently execute its routine in a dedicated thread, with its actions being triggered from the main control loop. The typical approach is to implement each instrument as a *worker* object, which executes operations inside a *while:*

[5]Available at https://github.com/federicoscarpioni/DEIStools

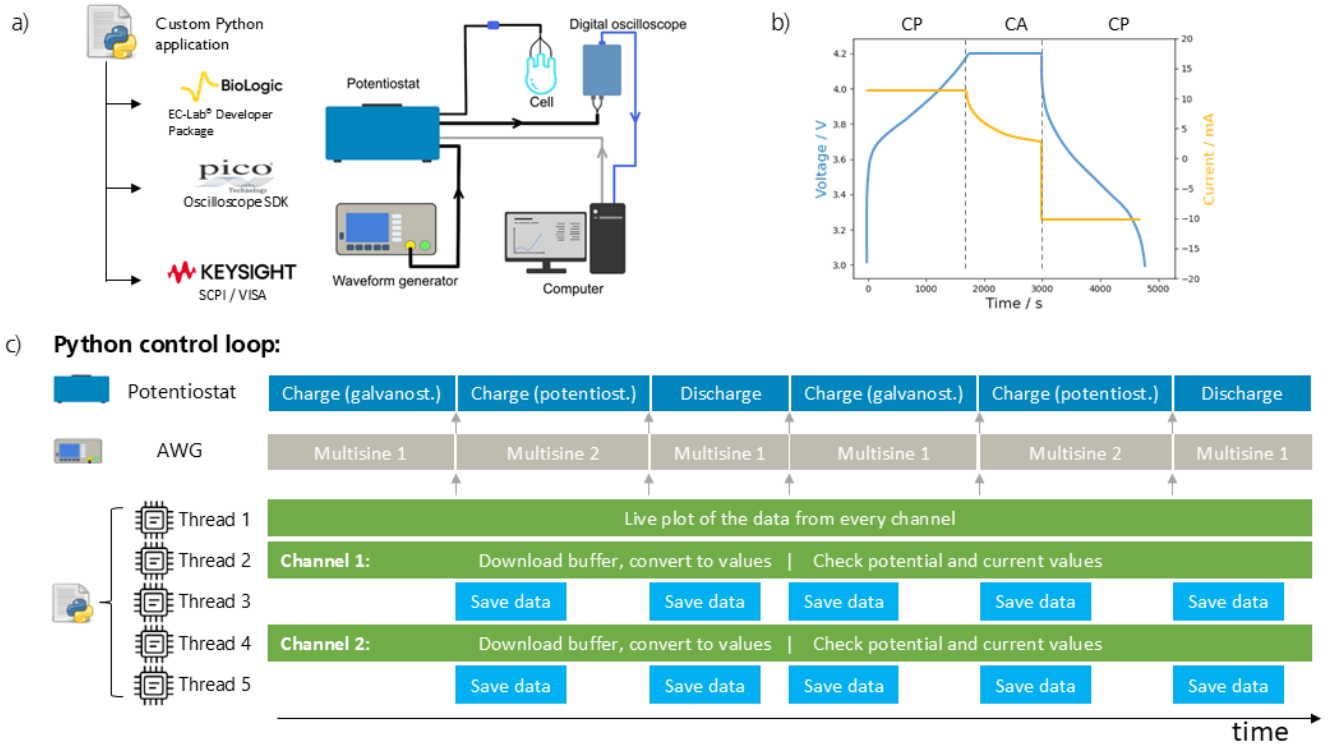


Figure 7.10: Python-based control loop for a CCCV battery experiment with embedded dynamic impedance spectroscopy. a) Scheme of the software components to interact with the single devices. b) Voltage and current time series of a typical CC-CV experiment on a battery as an example for a complex sequence monitoring. c) Visualization of the multithreaded software approach in multichannel mode. The loop monitors technique progression, triggers data transfer and multi-sine activation, and maintains synchronization between electrochemical control and signal acquisition.

True loop[9]. The multi-threading standard library of Python, **`threading`**, is capable of automatically managing all objects without user indication in the code (compared to other libraries like **`asyncio`**).

A worker object has a property with the name **`running`** that can be used by other objects to check their status. This idea is the basis of all the libraries used to synchronize the potentiostat, waveform generator, and acquisition device. This shared-state approach allows the main control loop to coordinate the execution of all instruments without blocking, ensuring real-time synchronization between electrochemical control, excitation, and data acquisition.

The automatic switching of the multi-sine between galvanostatic and potentiostatic electrochemical experiments (see Figure 7.10, the AWG timeline) can still be done by **`DEISChannel`** for the case of combined multi-sine waveforms, by passing the **`MultisineWaveCombined`** object, also provided in the package **`TrueFormAWG`**, during initialization. To allow a safe update between galvanostatic and potentiostatic waveforms in the generator, a 3 s open-circuit period is added to the sequence when the type of technique is changed.

The multithreaded capability is not only used to coordinate the three instruments for dynamic impedance spectroscopy, but also allows using multiple channels of the potentiostat (and corresponding waveform generators and acquisition devices) independently and simultaneously. Figure 7.11 shows four cells connected to the same multi-channel potentiostat and

[9] It always contains a pause during which the main thread is allowed to execute code for another worker object while new data are accumulated in another.

the streaming of data from the acquisition devices to a single computer.

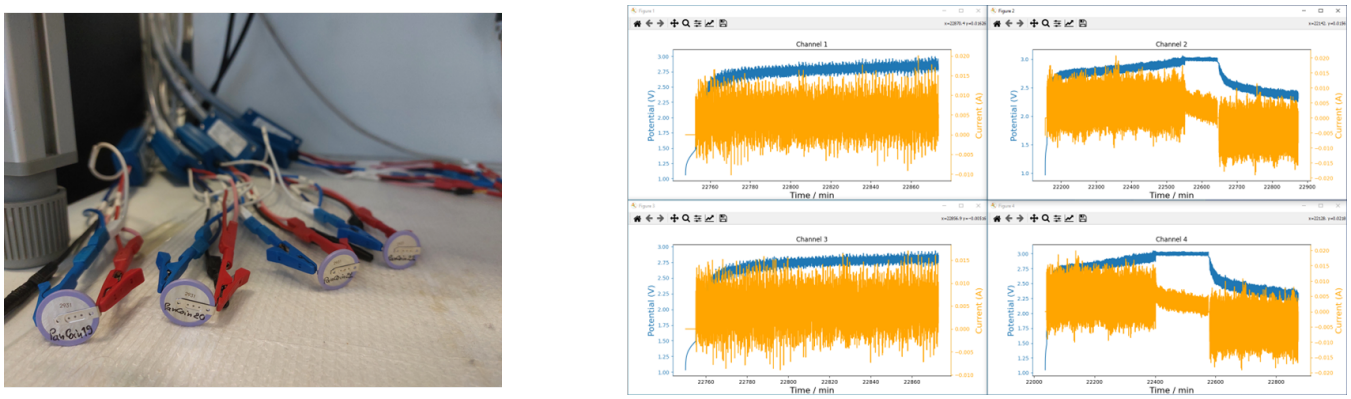


Figure 7.11: Multi-channel execution of dynamic impedance experiments. Left: photo of four identical lithium-ion coin cells connected to the same potentiostat. Each channel is interfaces with independent waveform generator and acquisition device. Right: screenshot of the computer screen with the real-time visualization of the multi-frequency voltage and current signals from the acquisition device.

Given the modularity of the approach presented here, it is possible to use different hardware to perform the experiment in the same manner, provided a similar interface is available.

In Appendix E are reported examples of scripts to load and use the libraries and perform dynamic impedance spectroscopy experiments with broadband multi-sine and online impedance computation.

7.6 Summary

This chapter presents the experimental framework developed for the acquisition of multi-frequency signals for the estimation of instantaneous impedance in non-stationary conditions. The focus was deliberately placed on the hardware integration, excitation design, sampling strategy, and software automation required to enable dynamic electrochemical impedance spectroscopy under realistic battery operating conditions.

A central technical contribution of this work is the development of a strategy to generate long, arbitrary broadband excitation signals using commercial laboratory instrumentation. By exploiting the dual-channel capabilities and large onboard memory of modern arbitrary waveform generators, and by introducing a frequency-subdivision approach, it was possible to synthesize multi-sine excitations spanning up to seven decades of frequency while preserving spectral coherence and avoiding intermodulation artifacts. This capability exceeds what is typically achievable with conventional embedded DAC architectures and is essential for broadband impedance characterization of electrochemical systems.

Equally important is the automation layer developed to coordinate the potentiostat, waveform generator, and high-speed acquisition device over long-duration cycling experiments. A modular, open-source software ecosystem was created to manage instrument synchronization, multi-sine activation, continuous data streaming, buffering, and metadata logging without user intervention. This infrastructure enables uninterrupted acquisition of multi-frequency voltage and current signals over many charge–discharge cycles, a prerequisite for studying impedance evolution during battery aging. The ability to reliably acquire long, synchronized, broad-

band datasets under controlled electrochemical conditions was a limiting factor in dynamic impedance spectroscopy up to this point. This work establishes a practical and extensible platform for time-resolved impedance analysis in batteries and other electrochemical systems.

Chapter 8

Impedance estimation

The previous chapter showed how to design a multi-frequency experiment and digitally acquire the voltage and current signals of the electrochemical cell, while this chapter discusses the methods used in this work to estimate instantaneous impedance from the multi-frequency signals. Chapter 6 covers the three main methods for estimating impedance: STFT, DMFA, and BLTVA. For its simplicity of implementation, I used mainly the DMFA method and the STFT for the online computation of part of the impedance. The novelty of this work is the use of the combination of DMFA and STFT to exploit the advantages of both: high frequency resolution of the DMFA for the low-frequency band of the impedance spectrum and fast computation of the STFT for the high frequency band. The STFT method was implemented online in the experiment dataflow thanks to its modular design and multi-threaded execution.

This chapter describes how impedance is estimated offline as used in the experiments presented later in the results part (chapters 10, 11, 12), and then I will describe the signal workflow to estimate part of the impedance online with the STFT method, as used for the main results chapter on commercial lithium-ion batteries (chapter 9).

8.1 Off-line estimation with DMFA

In this work I use the term *off-line* to emphasize the difference with the *on-line* approach that is discussed later in the chapter. The off-line method is the traditional method used in system identification for which input and output multi-frequency signals are sampled for an integer number of periods and the instantaneous frequency response function is estimated through a Fourier-transform-based analysis.

In practice, the acquired multi-frequency data are stored in single files that contain either a whole electrochemical experiment (as for chapters 10, 11, 12) or the data of one technique of the entire sequence that defined the experiment. The latter is the case for long battery cycling (as in chapter 9), so the estimation of the instantaneous impedance for the whole experiment requires looping through single datasets and executing the DMFA algorithm for each. In both cases, before starting the estimation algorithm the signal baseline was removed to reduce the Gibb's effect. This procedure, referred as *detrending*, is important in system identification to obtain clean Fourier spectra. The linear detrending used here is the simplest of the detrending methods; better solution exist [86, 87], but they are computationally difficult for long signals like the ones in this work.

When executing the DMFA, the important parameters to choose are the filter function, its strength, and its bandwidth. Following the study of Chukwu et al.[68] as presented in Section 6.3, in this work I used the symmetric Fermi-Dirac function as filter with a value n of 8 that produces a flat bandpass and fast roll-off. The bandwidth, instead, varies by experiment since it is chosen as the inverse of the multi-sine period, corresponding to the maximum temporal resolution of the DMFA method.

8.2 On-line processing and data reduction

8.2.1 Motivation

Using the DMFA method requires storing the complete signal of an experiment or a technique. When the sample rate of the acquisition is high or the acquisition time is long, the amount of disk space occupied by the signals becomes significant. The sample rate is determined by the maximum frequencies of the excitation wave (at least twice according to the Nyquist theorem), while the experiment time depends on the system under analysis; characterization of redox couples via cyclic voltammetry might last a few minutes, while characterizing intercalation in layered materials takes hours, and battery aging phenomena appear over weeks.

Typically, batteries and other electrochemical systems respond to excitations from millihertz up to a few thousand hertz. Sampling, for example, the signals of a system at 500 kHz creates 7.2 GB per hour, filling up even the most spacious hard drives in a few days. Employing dynamic impedance spectroscopy with high-frequency excitations and for long durations becomes impractical under these circumstances, especially when interested in performing replicates or measurements of parallel cells with the same computer. Furthermore, integrating such measurements in more modest hardware, like micro-controllers, for real-time estimation of energy storage or generation devices is practically impossible.

8.2.2 Strategy of the on-line processing

This work provides a solution to the problem of data volume by computing the high-frequency band impedance online via the STFT method over a short time window, then filtering and downsampling the windowed signal for offline estimation of the low-band impedances via DMFA.

The online processing of the multi-frequency signals is possible thanks to a software component executed inside the automation software described in the previous chapter. The online processing consists of the object `PicoCalculator`, part of the `DEIStools` module, which adds the possibility of executing operations on the data streamed by the PicoScope device. At its core, it simply adds a condition in the streaming loop that checks whether the buffered data is long enough (i.e., the window of the STFT) to be processed. The baseline of the signal is removed before executing the algorithm, and no bell-shaped function is multiplied since it makes no difference for systems drifting slowly, as for the case in chapter 9.

The combination of on-line and off-line impedance estimation requires a carefully structured data-processing pipeline. High-frequency components must be extracted immediately to reduce storage requirements, while low-frequency components must be preserved for later analysis. This separation

imposes a strict ordering of operations on the streamed data and motivates the hybrid processing architecture illustrated in Figure 8.1.

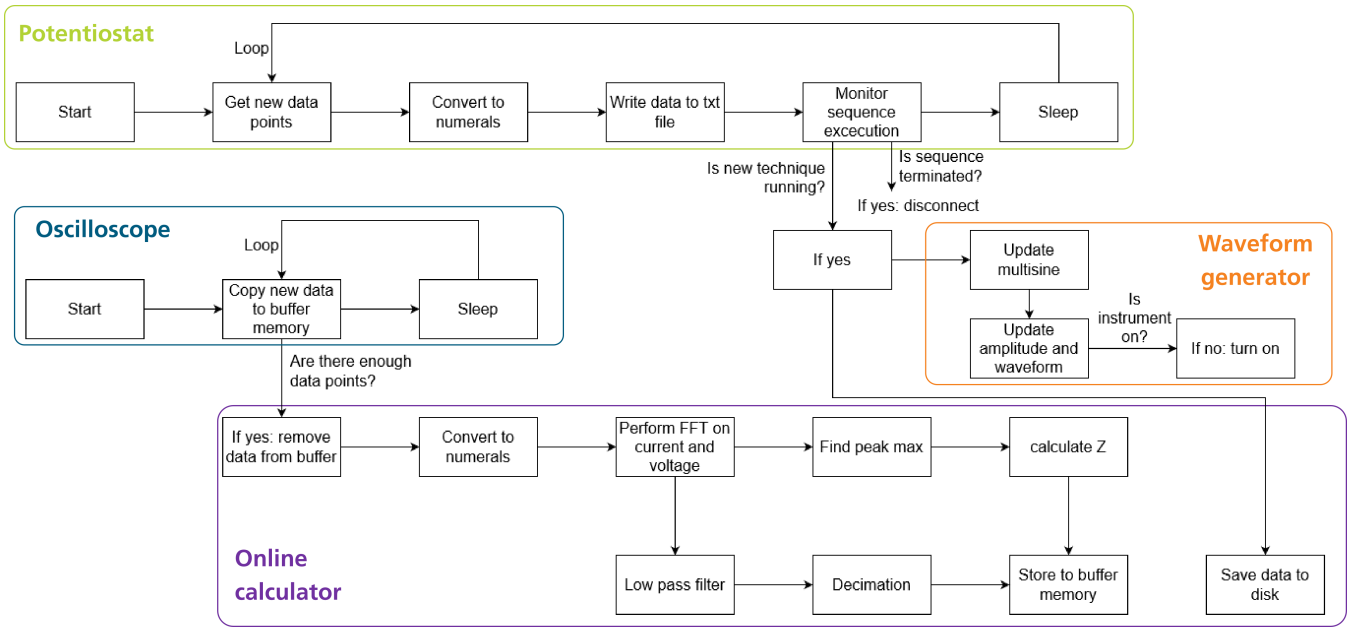


Figure 8.1: Data-flow architecture for dynamic impedance spectroscopy with on-line estimation of the high-frequency impedance band. Voltage and current signals are streamed continuously, processed in short-time windows for STFT-based impedance estimation, filtered to remove high-frequency components, and downsampled for subsequent off-line DMFA analysis of the low-frequency band.

8.2.3 Details of the data processing

When voltage and current data blocks equivalent to the window length are acquired, `PicoCalculator` copies the signals to another object, `BlockCalculator`, that performs the following actions:

- remove the linear baseline of both signals;
- perform the DFT of the signals;
- estimate the impedance of the high-frequency band as the ratio of the two Fourier spectra at the frequencies of the excitation and store it;
- apply a Fermi-Dirac function as a low-pass filter;
- perform inverse DFT of the filtered signals;
- decimate the signals and store them.

The user must choose the low-pass filter critical frequency to remove all the frequency components estimated with the STFT method.

In this work, I characterized commercial lithium-ion batteries in the frequency range 10 mHz–100 kHz with this combined method. The online STFT was configured to compute the impedance of all frequencies above 10 Hz, so the critical frequency of the low-pass filter was set to 8 Hz. For ease of implementation, I performed the filtering by multiplying the spectra with the filter function. The filter was the same symmetric Fermi-Dirac function used in the DMFA and was set with an n value of 25 that produces a gain roll-off fast enough to bring the 10 Hz signal to noise level. Later, the DMFA

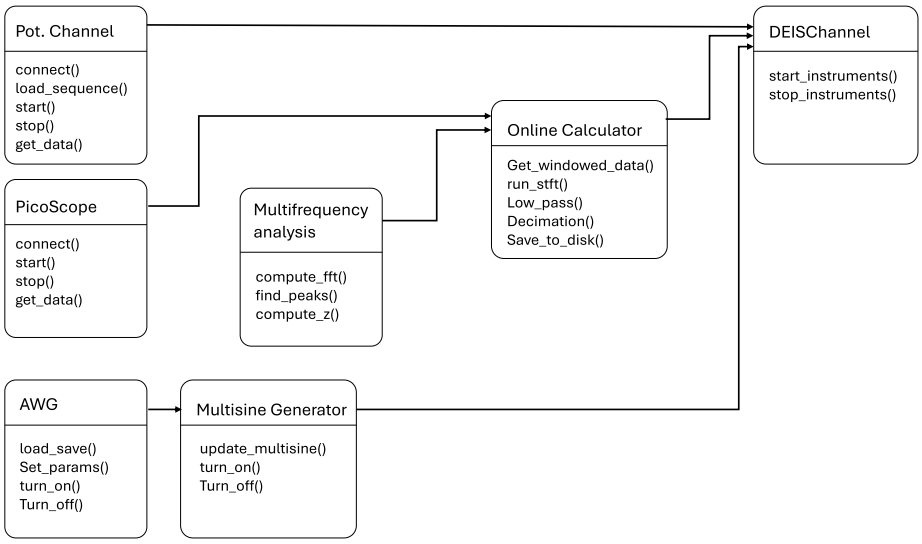


Figure 8.2: Object-oriented software architecture for dynamic impedance acquisition with online computation of the impedance. Each instrument is encapsulated in a dedicated worker object, coordinated by a central control loop inside DEISChannel. This modular design enables synchronization, extensibility, and hardware abstraction across different experimental configurations.

is used to estimate impedance from 10 mHz to 10 Hz (excluded), and the two are merged together to obtain the complete broadband impedance. If the time resolution of the DMFA and the window length are the same, there is no need to process the instantaneous impedances to match the time steps, and the results of the two methods are simply concatenated.

In Appendix E are reported examples of how to write scripts to configure the online computation of impedance and downsampling and integrate it in the automation software.

8.2.4 Software limits

The presence of a broadband excitation superimposed on a drifting operating point complicates the definition of voltage and current limits. Generally, electrochemical methods apply an external perturbation between upper and lower limit of voltage (either in cyclic voltammetry or galvanostatic) or current (in potentiostatic). When a multi-sine is applied on top of the drift, it is important to know the average of the signals to accurately compare it with the chosen limit instead of solely their latest value. In this work I implemented an average signal calculator. This average limit can be added to `DEISChannel` (see Appendix E) for the values of working electrode voltage, cell voltage, or current returned by the potentiostat. When the object `PicoCalculator` is used for the online computation of the impedance, the average of the signal can be estimated directly from the Fourier spectra of the windowed signals.

8.2.5 Validation against DMFA

Before implementing the STFT online in the acquisition dataflow, I simulated the process offline to verify that the output of the STFT and downsampling procedure was not altering the results. As an example dataset, I used the multi-frequency charging voltage and current obtained from a

galvanostatic experiment at C/10 rate with a multi-sine in the range 10 mHz–1 kHz. The cell used for the experiment is a lithium-ion coin cell of the same type used in the study presented in chapter 9. Figure 8.3 compares the original and processed signals in both the time and frequency domains. As a low-pass filter, I applied a Bessel filter of order 7 and cut-off frequency 100 Hz on data blocks of 100 s length.

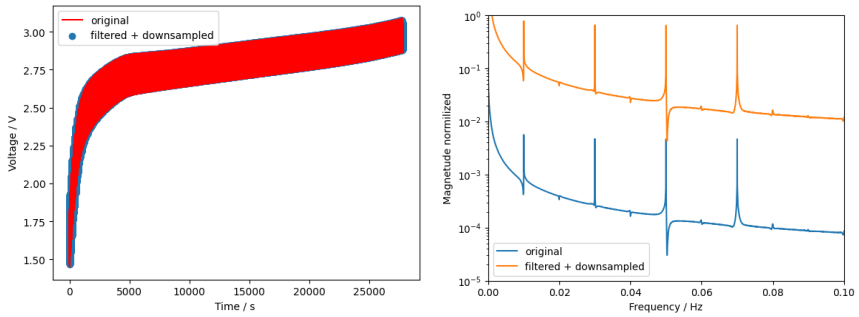


Figure 8.3: Validation of spectral integrity after filtering and downsampling. Left: comparison of the original and downsampled voltage signals in the time domain. Right: corresponding Fourier spectra demonstrating preservation of the excitation frequencies and absence of spectral distortion introduced by the processing chain.

I estimated the impedance for all the frequencies in the multi-sine with the DMFA on the original dataset; then the impedance of high frequencies on consecutive data windows with the STFT and estimated the low-frequency impedance on the downsampled signals with the DMFA. To validate the hybrid impedance estimation approach, the impedance obtained from the combined STFT–DMFA method was compared against a reference impedance estimated using DMFA alone on the full-resolution dataset. The results of this comparison are shown in Figure 8.4, where no difference is observed.

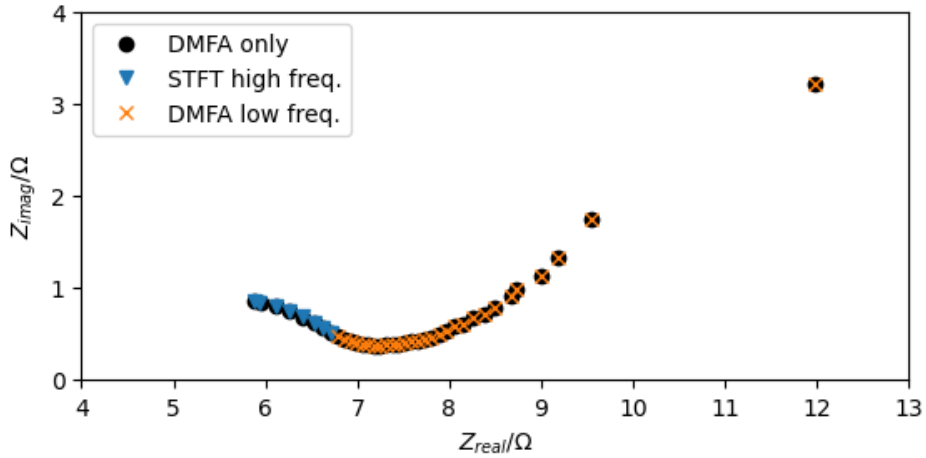


Figure 8.4: Validation of the hybrid impedance estimation method. Impedance spectra obtained using the combined STFT (high-frequency band) and DMFA (low-frequency band) approach are compared with reference spectra estimated using DMFA alone. The agreement across the entire frequency range confirms that the on-line processing and downsampling do not introduce measurable distortion. This instantaneous impedance is the middle spectrum of the time-varying impedance dataset.

8.3 Summary and limitations

This chapter presented a hybrid strategy for estimating non-stationary electrochemical impedance over an extended frequency range. By combining online STFT-based estimation for the high-frequency band with offline DMFA for the low-frequency band, it is possible to reduce data volume by several orders of magnitude while preserving spectral accuracy. This approach enables continuous broadband impedance tracking during long-duration experiments such as battery aging. The use of a hybrid impedance estimation method allows for true broadband dynamic impedance spectroscopy to characterize the complete frequency response of energy storage systems.

The analysis in blocks of data has a drawback in the use of a multiplicative filter because it produces small discontinuities of approximately 6 mV between two consecutive signal blocks. Continuous processing of the voltage and current data, using for example the low-pass Bessel filter, would improve the downsampled signal. The modularity of the software allows for easy optimization of each of the steps when needed in the future.

Part III

Results and discussions

Chapter 9

Identification of battery aging

In this chapter, I present measurements of dynamic impedance on commercial batteries during operation. The aim is to demonstrate the applicability of the mixed method for instantaneous impedance estimation under non-stationary conditions.

I chose to conduct this experiment on commercial coin cells with a theoretical capacity of 45 mAh because of the convenience and low cost of purchasing large quantities. I acquired a set of identical commercial coin cells and tested them using the dynamic impedance method during cycling, alongside conventional electrochemical measurements for comparison. The results from the dynamic impedance experiment are compared with a series of classical stationary impedance spectroscopy measurements. Furthermore, I demonstrate that the presence of a multi-sine perturbation continuously applied during the dynamic impedance spectroscopy test does not affect cell aging.

Figure 9.1: Panasonic ML2020 coin cell format battery used to test long-term cycling with dynamic impedance spectroscopy.

This study comprises the following measurements:

Measurement type	Number of cells	Number of cycles
DEIS during CC-CV	9	30
Traditional CC-CV	12	22–30
EIS during interrupted CC-CV	3	1

9.1 Experimental section

9.1.1 Cell choice

For this research, I acquired a set of identical Panasonic ML2020 coin cell batteries containing manganese dioxide (MnO_2) as the positive electrode and a lithium–aluminum alloy as the negative electrode. The lithium content is approximately 0.02 g. These cells utilize an electrolyte solution composed of 1,2-dimethoxyethane and ethylene glycol dimethyl ether (EGDME) combined with an unspecified lithium salt.

According to manufacturer specifications, these batteries have a nominal capacity of 45 mAh and are designed to operate within a voltage window of 1–3 V, with a recommended charging current of 100 μA.

The cells exhibit relatively slow kinetics compared to NMC/graphite systems due to the viscosity of the electrolyte, and they experience rapid

capacity fade—a characteristic of low-power designs at modest C-rates. They are designed as buffers for low-power electronics; thus, a viscous solvent is used as the electrolyte to reduce self-discharge. When cycling at a C-rate of C/10, the cells degrade rapidly, reaching end-of-life in approximately 22 cycles, with short-circuits often observed. I attribute this to lithium plating in dendritic form.

These limitations are actually favorable for rapidly iterating optimizations of the set-up. The use of commercial cells ensures reproducibility for validation of the method.

9.1.2 Dynamic impedance spectroscopy measurements

Setup description

I conducted the measurements of this chapter using the setup configured for broadband multi-sine waves: two channels of the generator apply two multi-sine excitations covering a total of 7 frequency decades, and the computer processes the streamed data from the oscilloscope in real time as described in Section 8.2. Three of the six channels of the BioLogic SP-300 potentiostat were measured simultaneously thanks to the multi-threading capability of the software `EIStools`.

Multi-sine design

The multi-sine waveform excitation contains 49 frequency components in the range 10 mHz–100 kHz, generated at 250 kSa/s, thus one period is of 100 s. The total waveform is split into two parts as described in Section 7.4.5 to work around the memory limitation of one generator channel. The amplitude of the multi-sine was optimized from a previous experiment as described in Section 7.4.2.

Applied drift

The potentiostat applies a constant current–constant voltage (CC-CV) sequence with an upper voltage limit of 3 V during charge and the same value for the constant voltage step with a lower current limit of 1 mA. The lower voltage limit for discharge was 1 V. The applied current was 4.5 mA, equivalent to a C-rate of C/10. Because of the multi-sine perturbation, the limits are checked during the online computation algorithm as the average value of the data block (as described in Section 8.2.4). The cells are cycled for 30 cycles continuously or until reaching end-of-life.

The voltage and current signals are sampled with a two-channel PicoScope 4000 series at a sampling rate of 500 kSa/s.

After the experiment, the automated software returns the impedance of the high frequencies (10 Hz–100 kHz) computed online using the STFT method and the voltage and current signals from the cells at the reduced sampling rate of 50 Sa/s. The signals contain the lowest 21 frequency components, ranging from 10 mHz to approximately 8 Hz.

To illustrate the output of the online computation and data-reduction pipeline, Figure 9.2 shows the reconstructed voltage and current signals after filtering and downsampling as they are saved in the computer storage after the acquisition step. These signals retain the low-frequency components while discarding high-frequency content already processed online.

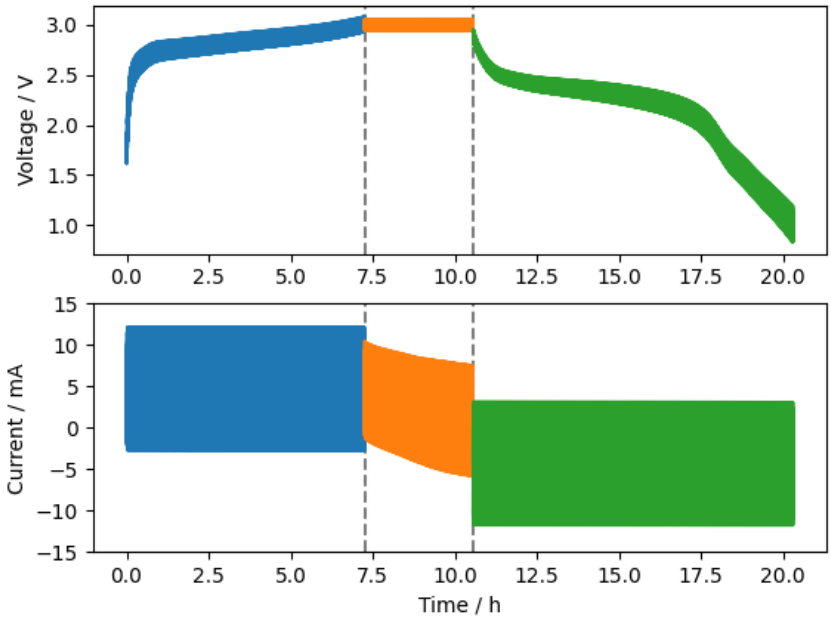


Figure 9.2: Voltage and current signals sampled at 50 Sa/s after online computation and downsampling. The signals contain the low-frequency components (10mHz–8Hz) of the original broadband excitation, which are used for off-line DMFA estimation of the impedance and reconstruction of the electrochemical drift.

The impedance of the lower frequency band and the drift are estimated using the DMFA with a filter bandwidth of 0.01 Hz, which yields a time resolution of 100 s, equal to one period of the multi-sine and the signal block used in the online computation. The total impedance spectrum is the concatenation of the parts computed with STFT and DMFA and reported in Figure 9.3.

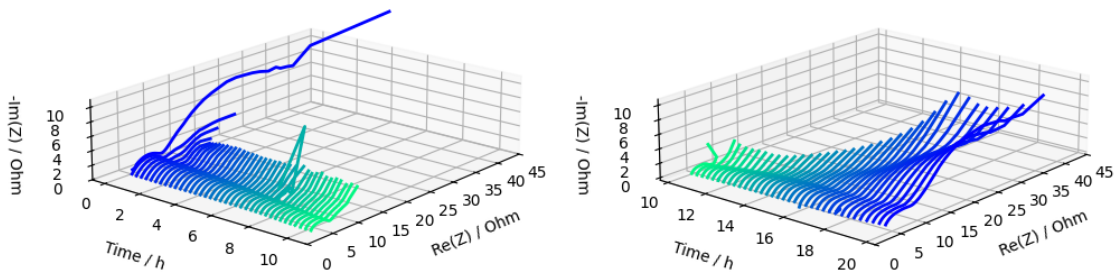


Figure 9.3: Impedance of charge and discharge estimated with DMFA and online STFT combined. One spectrum per 10 cycles is plotted; the first 2 plots are discarded.

The effectiveness of the DMFA in separating the slow electrochemical drift from the superimposed multi-sine excitation is illustrated in Figure 9.4.

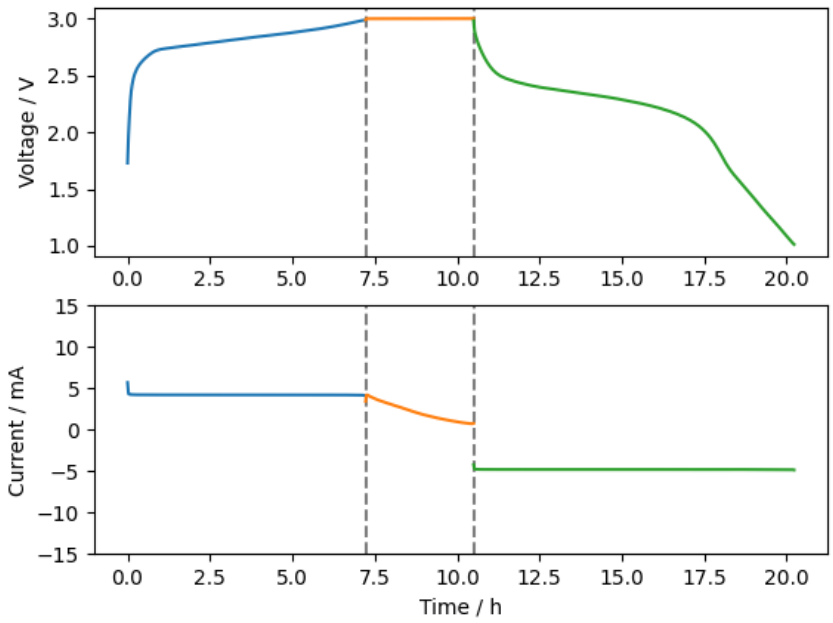


Figure 9.4: Drift reconstructed using the DMFA from the downsampled voltage and current signals. The y-axis scale is kept the same as in Figure 9.2 for comparison.

9.1.3 Impedance analysis

General transfer function model

A battery contains at least two principal interfaces (positive and negative electrodes), whose impedances are connected in series in an equivalent circuit model. This makes it challenging to develop a physical model that effectively describes the system without too many parameters.

Instead of attempting to identify its physical parameters with a physical model (referred to as a white-box model), it is equally interesting to identify the system with a mathematical model (referred to as a black-box model). Thanks to the continuous instantaneous impedance spectra generated with dynamic impedance spectroscopy, the time variation of the mathematical model's parameters remains informative for the purpose of control and prediction.

In this context of system identification, the battery is considered a generic linear and time-variant system, hence the input $u(t)$ and the output $y(t)$ of a system are connected by the linear ordinary differential equation

$$\sum_N a_n \frac{\mathrm{d}^n y(t)}{dt} = \sum_M b_m \frac{\mathrm{d}^m y(t)}{dt}. \tag{9.1}$$

As discussed in Section 4.5.1, the differential equation can be solved by applying the Laplace transform, which yields

$$\sum_N a_n s^n Y(s) = \sum_M b_m s^m U(s), \tag{9.2}$$

and so the transfer function for the system is given by

$$G(s) = \frac{Y(s)}{U(s)} = \frac{\sum_N a_n s^n}{\sum_M b_m s^m}. \tag{9.3}$$

This equation for the transfer function is the same as Equation 4.12 from section 4.5.3 (p. 32).

The Laplace variable s for the case of impedance is simplified to $s = j\omega$, but since the system is electrochemical, the homogeneous differential equations describing diffusion are semi-integer [88] and lead to solutions as functions of $\sqrt{j\omega}$. Therefore, for electrochemical systems the generic transfer function has the form:

$$G(\sqrt{j\omega}) = \frac{\sum_N a_n (\sqrt{j\omega})^n}{\sum_M b_m (\sqrt{j\omega})^m}. \tag{9.4}$$

In this work, I used such a rational polynomial function of order $n = m$ as a measurement model to fit impedance data. The concept is not new [89–91], but it is the first time this model has been used for instantaneous impedance of time-varying systems. Polynomial functions can flexibly fit impedances with any complex shapes and also have the advantage of working as a test for the linear Kramers-Kronig relations [92].

Regression procedure

A fundamental procedure for the regression of rational polynomial function is to add L1 regularization to the objective function, also called LASSO (*least absolute shrinkage and selection operator*). In practice, the objective function to minimize is

$$obj = \sum_n r_n^2 + \lambda ||\theta||_1, \tag{9.5}$$

where r_n^2 is the weighted complex squared residual of the n-th point of the spectra as described in Section 4.7.1, θ is the parameters array, and λ is a penalty factor that controls the regularization strength. Regularization is important because rational function parameters can grow unnecessarily large and be compensated by parameters in the denominator. The choice of penalty factor is arbitrary, and I found that values as small as 1×10^{-12} are sufficient. Removing the regularization terms makes the optimization procedure ineffective (parameters diverge) or causes the solver to fail.

Before fitting, I removed the first three and last two spectra from the dataset, as well as the first three and last frequency points (remaining range: 30 mHz–42 kHz).

In the least squares objective function introduced in Section 4.7.1, each frequency component residue is weighted by a factor equal to the experimental error. The error in a frequency response function increases with the lowering of the frequency. Orazem et al.[1] originally proposed to use the inverse of the impedance modulus as weighting factor, but I found that using the square root of the modulus is more effective in this case. It weights the middle-to-high frequency range more heavily, allowing the model to fit the impedance's features better.

A better strategy would be to estimate the error from error propagation rules through elementary mathematical operations, as Reference [68] did for admittance data estimated via DMFA. Unfortunately, I was unable

to replicate his derivation for my impedance data. Furthermore, error propagation for the FFT impedance method is not available in the literature and must be derived from first principles.

For the optimization, I adopted a classical derivative-based approach (trust region reflective method) as implemented in the Python package `LMFIT` [93] to solve such minimization problems. I first fitted the spectrum at the middle of the dataset using random initial parameters (seed 42) and then used the optimized parameters as the starting points for the first spectrum of the dataset. Subsequently, I proceeded by passing optimized parameters as initial points for the next fitting.

The difficulty in fitting dynamic impedance data is identifying one set of values for polynomial order, regularization penalty factor, and least-squares weighting factor that work effectively across all electrochemical techniques in the experiment, all cycles, and are general enough to apply to multiple cells.

9.1.4 Classical electrochemical test

I performed cycling experiments using the same constant current–constant voltage protocol on a BioLogic battery cycler with 12 cells from new to aged condition to compare degradation with and without multi-sine excitation.

I also prepared an experiment performing classical electrochemical impedance spectroscopy at fixed states of charge. For this purpose, I created an experiment sequence that interrupts the cycling protocol every 10% of the nominal capacity and includes a 1 h rest before impedance measurement in the frequency range 10 mHz–20 kHz (8 points/decade).

9.2 DEIS results

9.2.1 Impedance at different battery states

Continuous impedance acquisition provides a complete view of battery interface resistance at any state of charge and state of health. Figures 9.5 and 9.6 show this information for charge (including the potentiostatic technique) and discharge. I compared the impedance at four states of health corresponding to the beginning, midpoint, just before the 80% threshold of first-cycle capacity, and after it. It can be observed at least two consecutive semicircles and a diffusive tail. At lower states of charge, represented in red in the colored scale, a third semicircle appears, shifting the diffusion to lower frequencies. Another observation is the change in ohmic drop resistance with state of charge. The evolution of the impedance spectrum with battery aging during charge is summarized in Figure 9.5. A corresponding representation for discharge is reported in Figure 9.6.

The cell is provided from the manufacturer at 2.7 V, so in the first cycle of the measurement, the cell voltage quickly reaches the charging limit and then proceeds with the discharge as shown in Figure 9.7. The impedance of the first cycle is reported in Figure 9.8

All behaviors observed here are reproduced in the other 8 cells of the set, including the higher impedance in the first discharge. Detailed results are reported in the Appendix. The reproducibility of the dynamic impedance measurements across nominally identical cells is assessed in Figure 9.9.

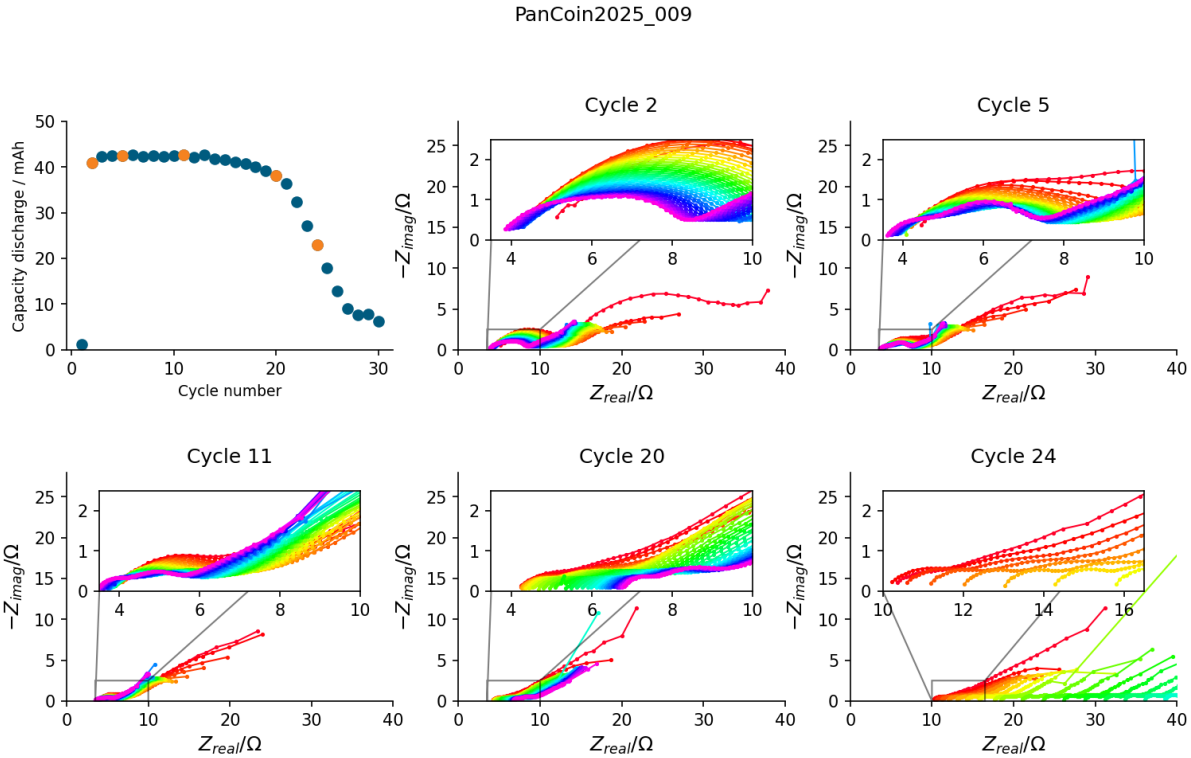


Figure 9.5: Dynamic impedance spectra during charge at selected states of health. Spectra are shown for four aging stages: beginning of life, mid-life, just before the 80% capacity threshold, and after crossing it. Color coding indicates state of charge.

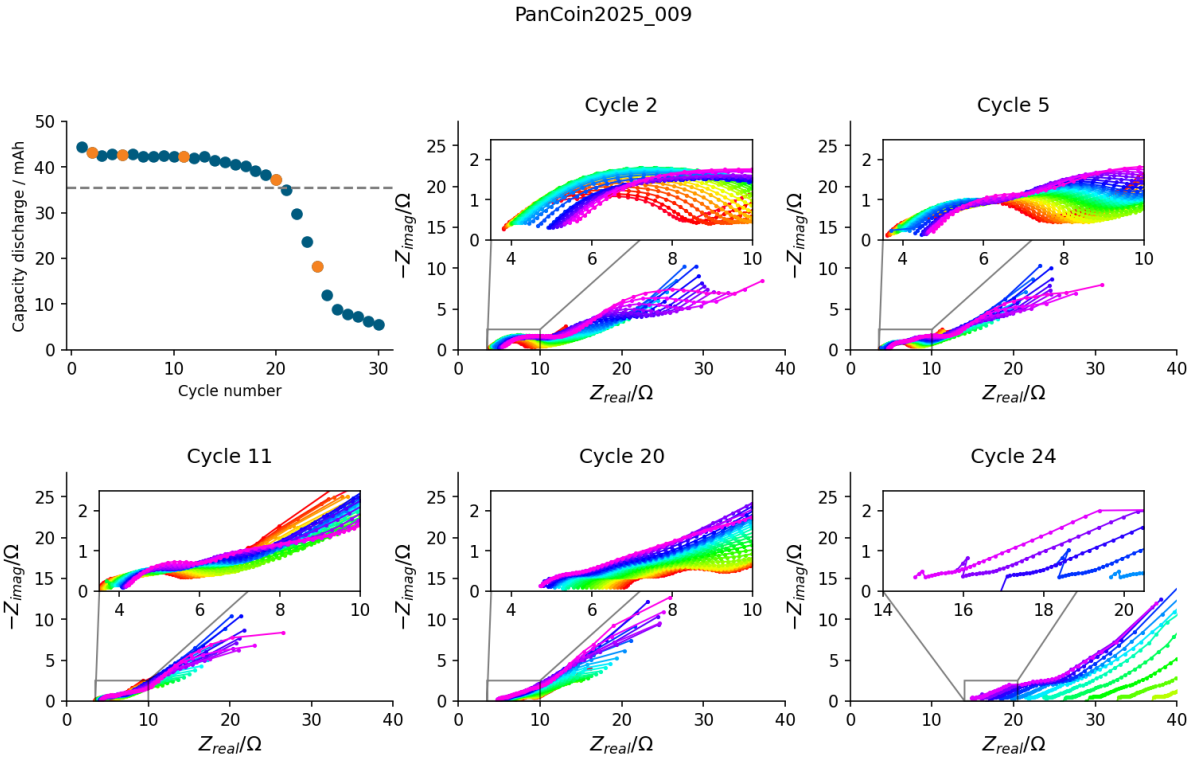


Figure 9.6: Dynamic impedance spectra during discharge at the same aging stages as Figure 9.5. The appearance and shift of multiple semicircles and diffusive features reflect changes in interfacial and transport processes with aging.

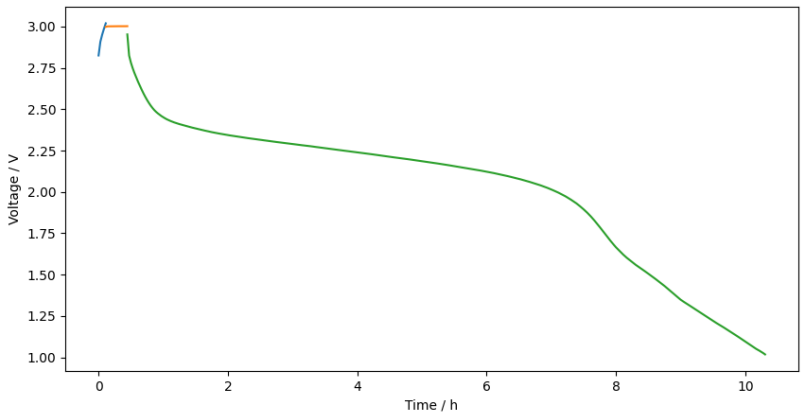


Figure 9.7: Voltage profile of the first charge–discharge cycle. The cell reaches the upper voltage limit early during the first charge, indicating a conditioning process that differs from subsequent cycles.

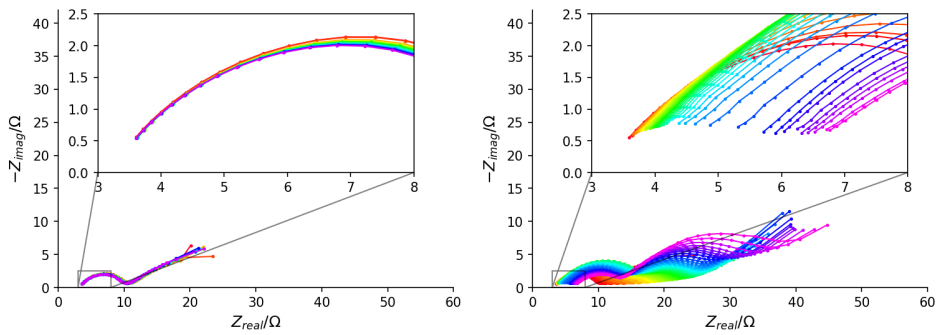


Figure 9.8: Impedance spectra during the first cycle. Left: all charge spectra; right: discharge spectra with one spectrum every ten cycles. Both magnitude and shape differ markedly from later cycles, indicating transient interfacial processes during initial conditioning.

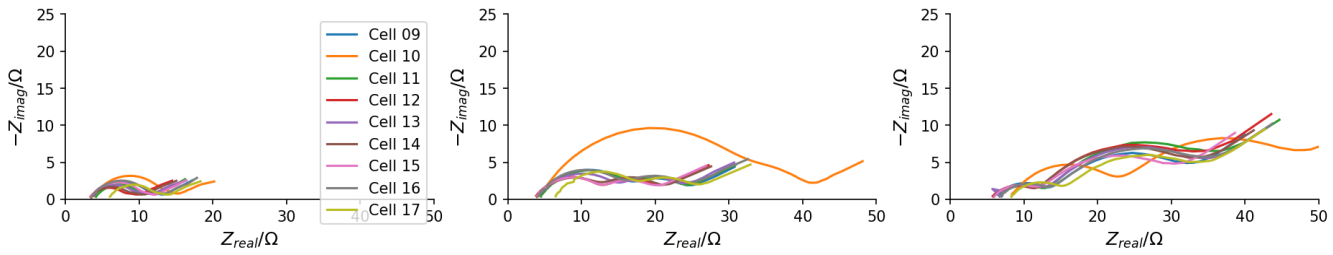


Figure 9.9: Dynamic impedance variability across nine cells at 90%, 50%, and 10% state of charge during the second discharge half-cycle. While overall spectral features are consistent, variations in magnitude and ohmic resistance reflect manufacturing variability.

As with any production process, cells do not respond identically but show similar behavior. One can notice impedance magnitude variations for Cell 10 and variations in the ohmic drop, which is notably higher in Cell 17 than in the others.

9.2.2 Effect of the multi-sine on aging

To assess whether continuous multi-sine excitation influences degradation, capacity retention is compared statistically in Figure 9.10.

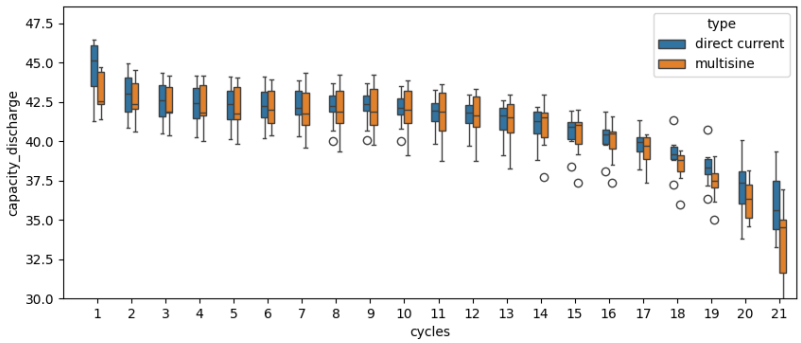


Figure 9.10: Comparison of discharge capacity retention for cells cycled with DC-only excitation and DC with superimposed multi-sine perturbation. Boxplots summarize the distribution across 12 DC-only cells and 9 DEIS cells, showing no statistically significant difference under the tested conditions.

The reduced y-axis scale highlights the marginal differences between the two datasets, therefore I conclude that the multi-sine is not impacting the cycling performance and can be applied continuously for system identification; at least for these small cells at low C-rate.

9.2.3 Effect of the drift on the impedance

Static impedance measured at different states of charge also shows good consistency. Figure 9.11 reports data from three cells at the same states of charge as Figure 9.9. For reference, the variability of stationary impedance measurements obtained during interrupted cycling is shown in Figure 9.11.

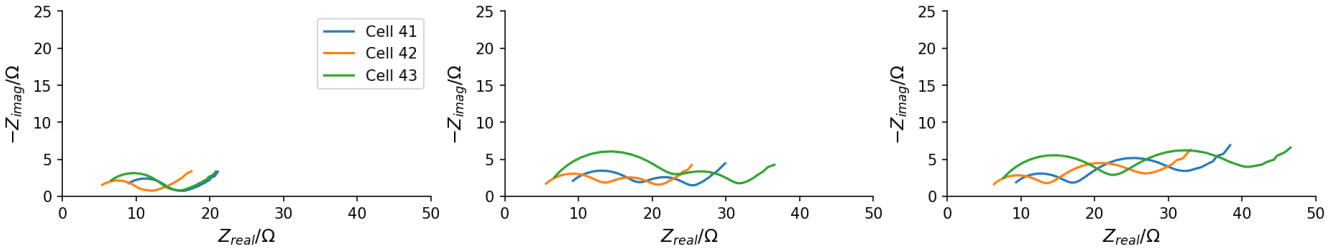


Figure 9.11: Static impedance variability across three cells at selected states of charge. Impedance was measured after a 1 h rest at each state of charge using conventional stepped-sine EIS.

A direct comparison between static and dynamic impedance measurements is presented in Figure 9.12.

This result suggests that at such a low C-rate (C/10), the electrode interfaces are in a similar state to the static case, while significantly reducing measurement time.

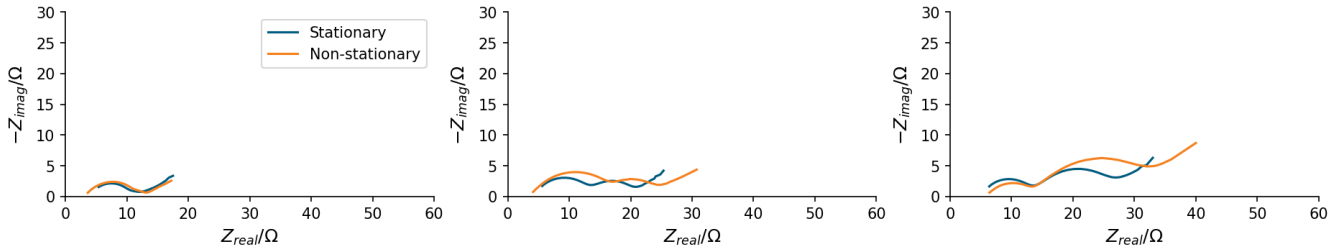


Figure 9.12: Comparison of impedance measured under stationary conditions and during non-stationary operation at the same states of charge. The close agreement indicates that, at C/10, dynamic impedance measurements capture interfacial properties comparable to equilibrium EIS.

Comparing the classic and non-stationary measurements, it is clear how much more convenient the multi-sine perturbation is over the single sine, as reported in Figure 9.13. The latter takes around 27 hours to produce 10 spectra, while the multi-sine-based method outputs 270 spectra in 10 hours.

9.3 Results of fitting

Evaluating the fitting quality of dynamic impedance data involves multiple assessments: qualitative visualization, magnitude and structure of residuals, and consistency of model parameters across cycles.

In this work, I performed the fitting of rational polynomial functions with orders from 4 to 9. From the visual inspection of the fitting results, I arbitrarily selected the order 5 because it can represent all the different shapes of instantaneous impedance, as shown in Figure 9.14 with the minimum number of parameters. The residuals are small for the real part, approximately 1%, but larger for the imaginary part. The systematic structure of the residuals, especially the pronounced deviations at the highest three frequencies, indicates that the model does not fully reproduce all spectral features of the dataset. I believe this issue is both related to the arbitrary choice of the weighting factors in the least-square objective function and the solver itself.

Despite the small residual magnitudes, their oscillatory behavior persists for polynomial orders up to nine (the highest tested). In a well-posed regression problem, residuals are expected to be randomly distributed with magnitude comparable to the measurement variance. The persistence of structured oscillations therefore indicates that the limitation is not simply insufficient model order. The fitting with a smaller order of 4 does not lead to a correct representation of the impedance since the model is not capable of resolving the two semicircles at higher frequency. This issue could not be connected with the model structure, but rather the arbitrary weighting factor used in the regression problem.

Despite the criticalities, it is interesting to look at the fitting result for order 5 to have an idea of the advantage of parameterization. The fitted parameters vary smoothly across state of charge and state of health although some abrupt magnitude changes are observed when the impedance manifests a new feature (or the features disappear) and the model must adapt to it. Representing complex-valued impedance through real-valued model parameters provides a compact description of the spectrum and facilitates subsequent data-driven modeling of cell aging. The fitted coefficients can

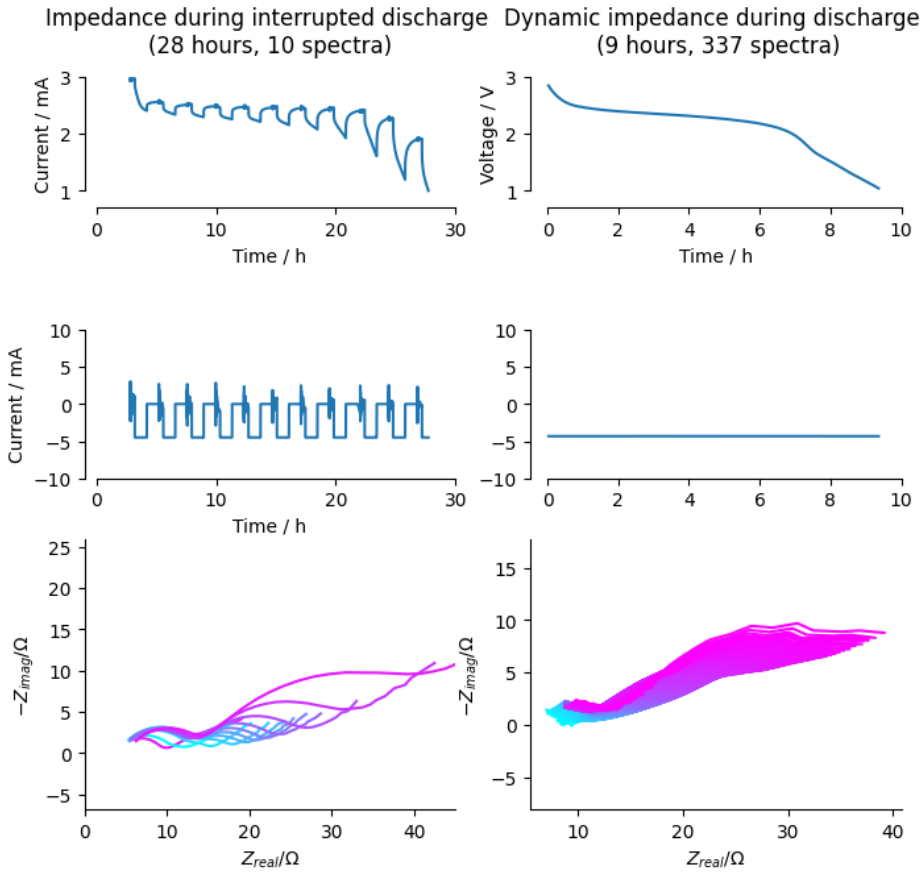


Figure 9.13: Comparison between classic impedance measurement at different equilibrium voltages and a DEIS measurement. The classic experiments takes much longer time and produces fewer spectra. On the right hand side the multi-sine components are filtered out from voltage and current signals and only one over ten of the instantaneous impedances is plotted.

therefore be used as features for machine-learning models to predict state of health.

The evolution of the fitted parameters during cycle 2 across all electrochemical techniques is illustrated in Figure 9.15. Parameters a_1 (orange line) and a_3 (red line) exhibit abrupt transitions when a new spectral feature becomes significant: a semicircle emerges at high state of charge and gradually merges into another high-frequency semicircle. The parameter jump occurs because the regularized least-squares procedure allows the parameters to increase only when the residual contribution in that region becomes sufficiently large to significantly impact the cost function.

As shown in Figures 9.5 and 9.6, after cycle 5 the impedance shape stabilizes, showing two consecutive semicircles followed by a diffusive feature. This stabilization is reflected in the corresponding fitted parameters for cycle 5 (Figure 9.16) and subsequent cycles (e.g., cycles 10, 15, 20, and 22). Parameter magnitudes and trends are largely preserved across cycles and remain mostly continuous within the techniques of the same cycle. These observations suggest that, despite their numerical limitations, the fitted parameters capture systematic changes correlated with state of charge and state of health. The same qualitative behavior is observed in the results of the other cells.

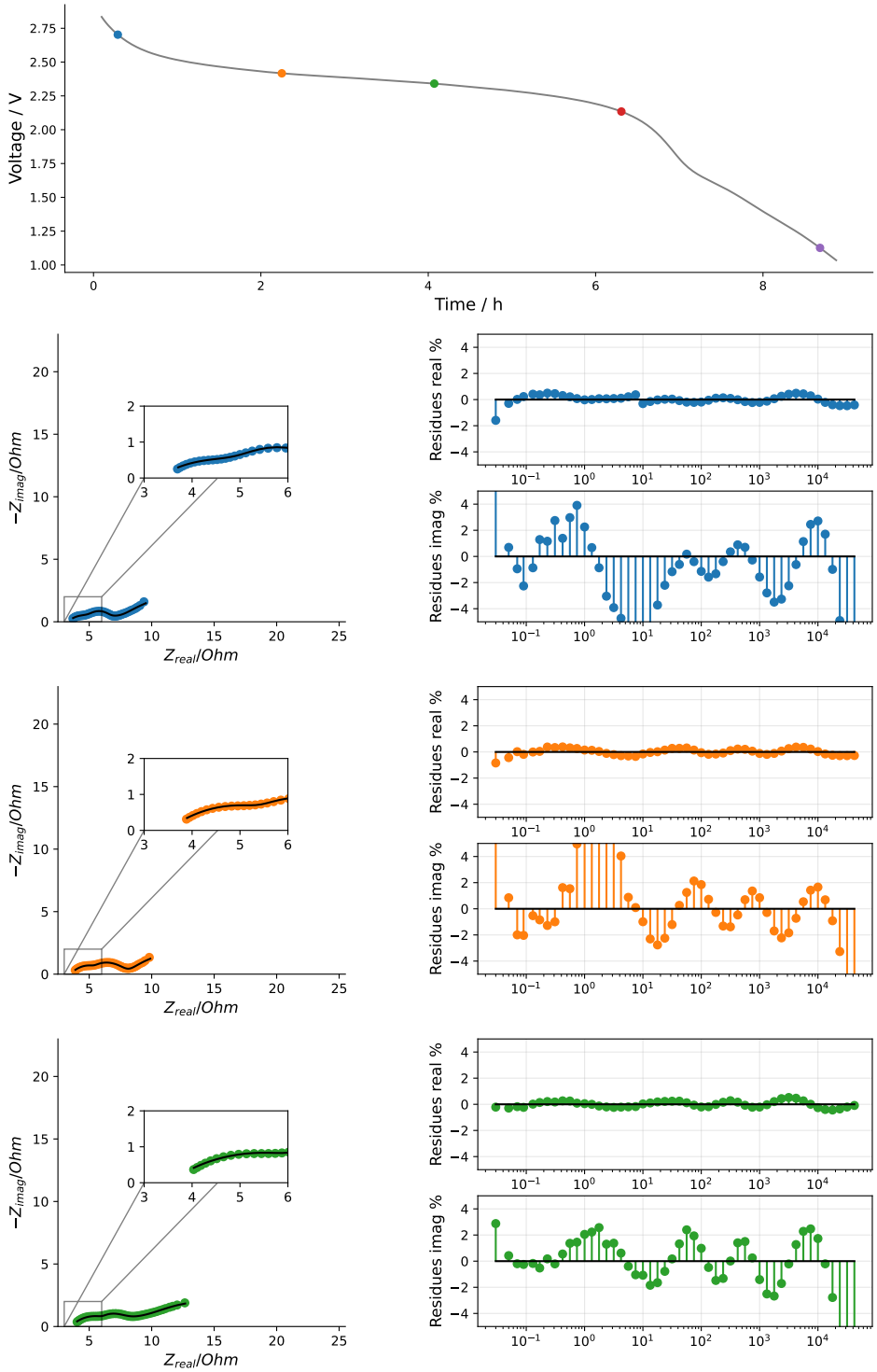


Figure 9.14: Rational polynomial fitting of dynamic impedance spectra at different states of charge (marked with colors dots) during discharge of cycle 6. The figures in the panel after the voltage time series report the impedance data (markers) and fit result (solid black line). Residuals are reported next to each spectrum. The colors of the markers represents the states of charge.

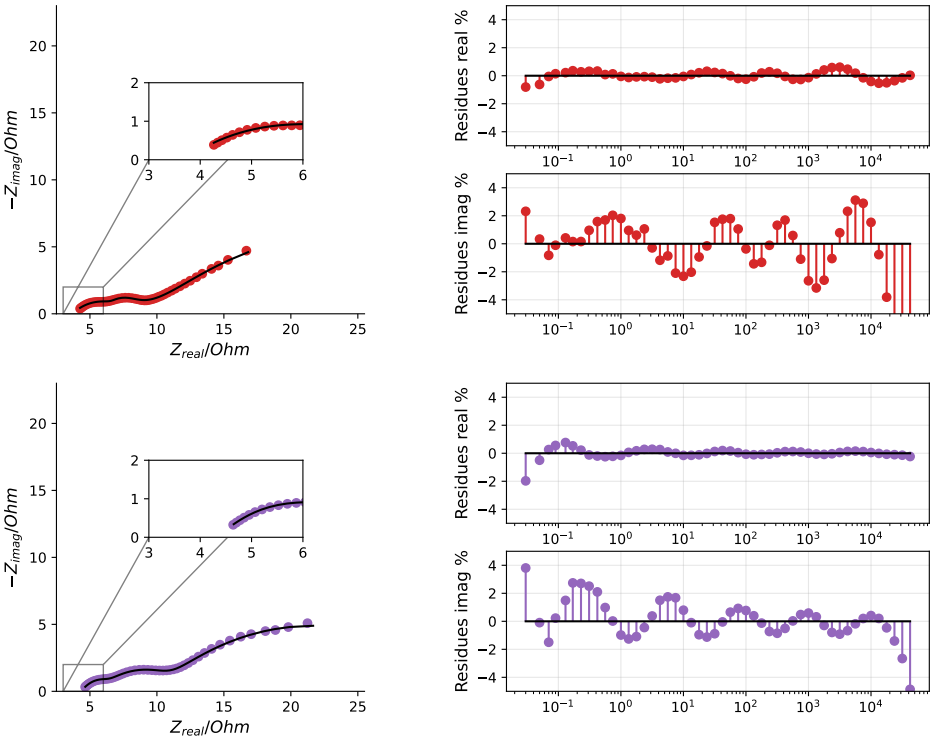


Figure 9.14: Continued

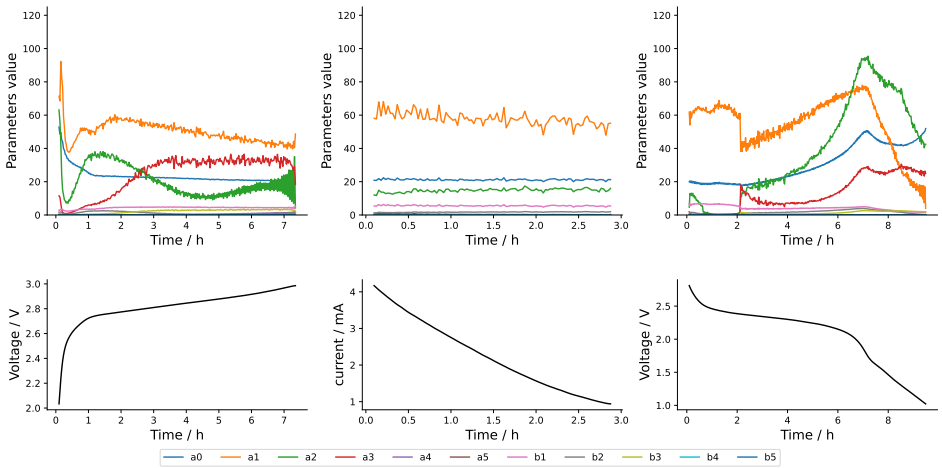


Figure 9.15: Upper row: fitted rational polynomial parameters for all techniques during cycle 2. Lower row: corresponding cell voltage and current reconstructed via DMFA. Abrupt parameter changes correspond to the emergence of new impedance features.

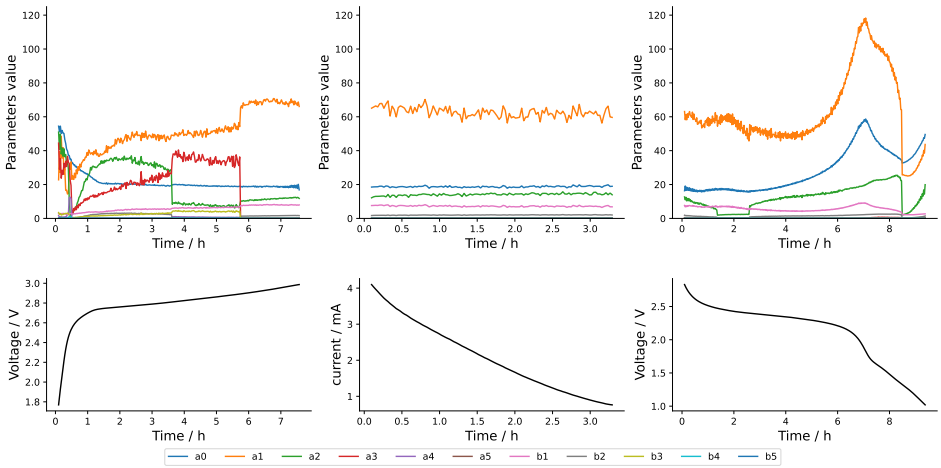


Figure 9.16: Upper row: fitted rational polynomial parameters for all techniques during cycle 5. Lower row: corresponding cell voltage and current reconstructed via DMFA.

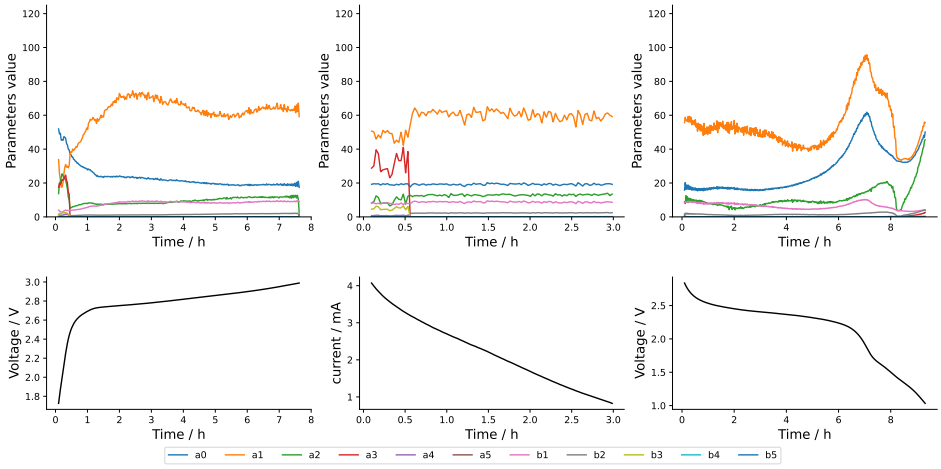


Figure 9.17: Upper row: fitted rational polynomial parameters for all techniques during cycle 10. Lower row: corresponding cell voltage and current reconstructed via DMFA.

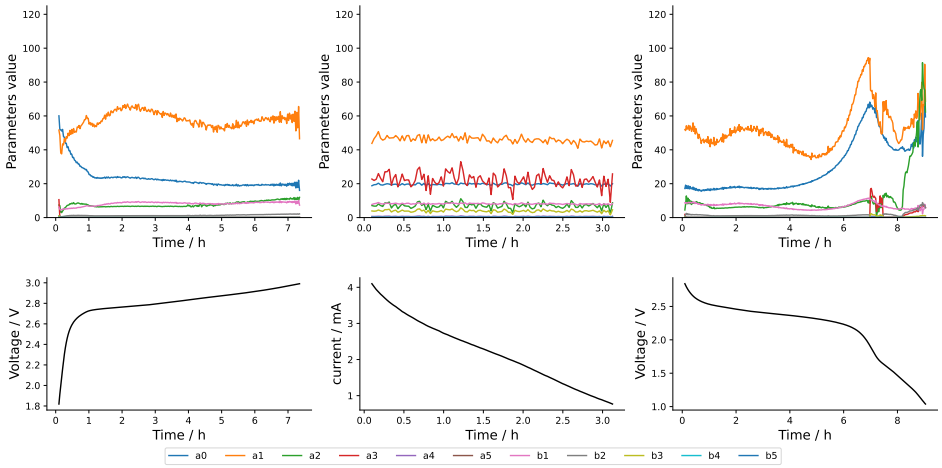


Figure 9.18: Upper row: fitted rational polynomial parameters for all techniques during cycle 15. Lower row: corresponding cell voltage and current reconstructed via DMFA.

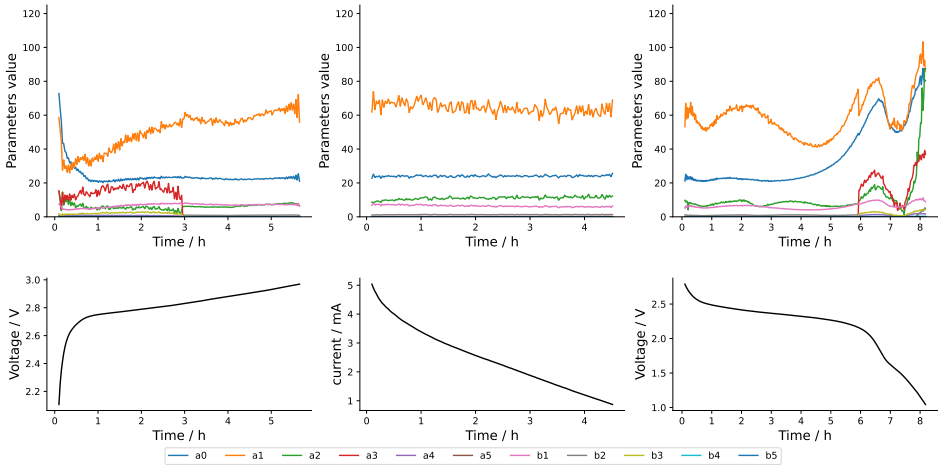


Figure 9.19: Upper row: fitted rational polynomial parameters for all techniques during cycle 20. Lower row: corresponding cell voltage and current reconstructed via DMFA.

However, the persistence of oscillatory residuals with increasing polynomial order, together with the systematic disparity between numerator and denominator coefficient magnitudes, reveals a structural limitation of the adopted identification formulation. In the least-squares estimation of rational polynomial transfer functions, the problem is *ill-posed* in the sense of inverse problem theory: small perturbations in the measured impedance (e.g., noise or minor spectral features) can produce disproportionately large variations in the estimated parameters, and multiple distinct parameter sets may generate nearly indistinguishable impedance responses [94, 95].

This behavior is reflected in the observation that denominator coefficients often assume very small magnitudes while numerator coefficients span a significantly larger dynamic range. Such scaling imbalance indicates that the optimizer compensates local spectral mismatches primarily through numerator adjustments while maintaining a weakly constrained denominator structure. Mathematically, this corresponds to a poorly conditioned Jacobian matrix and strong correlation among parameters. Under these conditions, increasing the number of coefficients does not necessar-

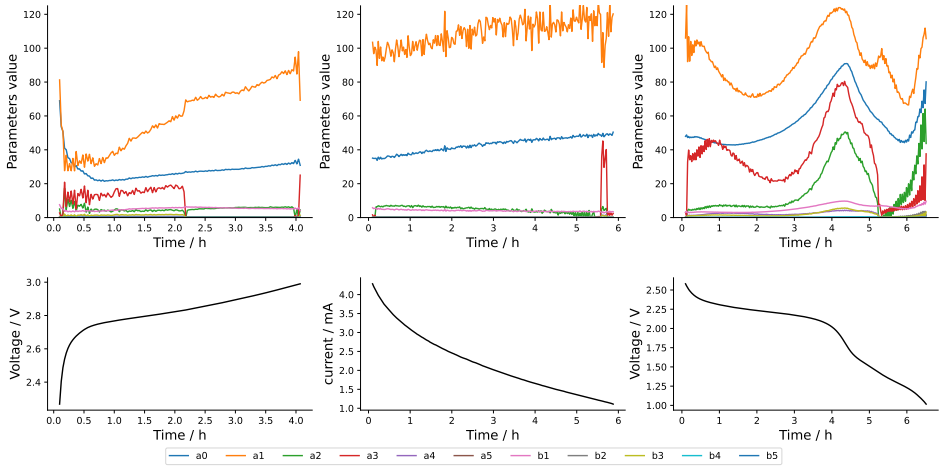


Figure 9.20: Upper row: fitted rational polynomial parameters for all techniques during cycle 22. Lower row: corresponding cell voltage and current reconstructed via DMFA.

ily improve robustness; instead, it can lead to near pole-zero cancellations, ill-scaled parameters, and structured residuals that are not consistent with purely stochastic measurement error. The impedance spectrum can often be reproduced accurately in magnitude, but the corresponding parameter values may not be uniquely determined or numerically stable. This explains why increasing model order beyond five does not eliminate oscillatory residuals and does not significantly improve parameter consistency.

Over the past three decades, alternative identification frameworks have been developed in system identification, control engineering, and RF network modeling to address precisely these issues. Methods such as vector fitting estimate poles and residues of rational approximations in a structured and numerically stable manner, allowing automatic determination of effective model order and reducing sensitivity to noise [96, 97]. Similarly, realization-based approaches such as the Loewner framework construct minimal state-space representations directly from sampled frequency data using singular value decomposition, with the model order inferred from the spectral content rather than prescribed a priori [98, 99]. These methods do not require manual smoothing parameters and generally produce more stable dynamical representations of the impedance response, but the knowledge of industrial mathematics limits the applicability in the chemical and engineering community. Nevertheless, the rational polynomial parametrization adopted here remains useful as a compact descriptive tool and as an initial framework for transforming complex impedance spectra into real-valued feature trajectories.

9.4 Summary and conclusions

In this chapter, dynamic electrochemical impedance spectroscopy was applied to commercial lithium-based coin cells operated under non-stationary charge/discharge cycling using a typical constant current–constant voltage protocol. The objective was not to extract detailed physicochemical parameters, but to validate the proposed DEIS methodology at the full-cell level and assess its consistency, robustness, and impact on battery operation.

Continuous impedance spectra were successfully acquired throughout cycling using a broadband (7 decades) multi-sine excitation combined with online and offline signal processing. The resulting dynamic impedance measurements capture the evolution of interfacial resistance, charge-transfer processes, and diffusion-related features as functions of both state of charge and state of health. Comparison with classical stationary EIS measurements performed during interrupted cycling shows good agreement within the experimental variability of the cells, indicating that at moderate C-rates (C/10), the non-stationary operating conditions do not significantly alter the electrochemical state of the interfaces.

Importantly, statistical comparison of capacity retention demonstrates that continuous superimposition of the multi-sine perturbation does not affect cell aging under the investigated conditions. This result supports the practical applicability of DEIS for long-term experiments without compromising cell integrity, at least for small-format cells and low to moderate current densities.

Dynamic impedance spectra were fitted using rational polynomial models through complex nonlinear least-squares with regularization. Although the chosen black-box model does not provide direct physical interpretation of individual electrochemical processes, it enables real-valued parametrization of impedance evolution across cycles. The smooth variation of fitted parameters with state of charge and state of health suggests that such representations may serve as effective features for data-driven aging diagnostics.

At the same time, the limitations of full-cell impedance analysis are evident. The superposition of positive and negative electrode responses leads to complex models that may not be fully represented by a rational polynomial, as evidenced by oscillatory residuals. Furthermore, a single model has limited ability to represent impedance across all operating conditions.

Nevertheless, the flexibility of rational polynomials is promising. With proper characterization of the actual measurement error structure and systematic exploration of regularization penalty factors, the robustness of the fitting can be improved. I suggest conducting a study with simulated electrochemical impedance data where the degrees of freedom and noise levels are known a priori.

Chapter 10

Studying intercalation dynamics

The previous chapter demonstrated the applicability of dynamic impedance spectroscopy for characterizing the aging of batteries as black-box systems, but the method is not limited to full cells. The rest of this thesis will show how the methodology unlocks new ways of characterizing active materials and single electrodes as white-box systems. The project discussed in this section deals with the characterization of intercalation processes in lithium sodium titanium phosphate as a negative electrode for aqueous sodium-ion batteries. The non-stationary impedance of the electrode was estimated during constant current and voltage-sweep drifts. The differences between these two approaches are analyzed along with physical interpretation. The results of this project led to publication in a peer-reviewed article [74].

10.1 Experimental section

10.1.1 Materials and cell assembly

Active material

Carbon-coated $LiTi_2(PO_4)_3$ (LTP) was synthesized by Dr. Nicolò Pianta within a scientific collaboration with the University of Milano-Bicocca, via a one-pot route [74]. A gel was prepared by dissolving ammonium dihydrogen phosphate and titanium butoxide in ethanol together with lithium acetate and glucose as lithium and carbon sources. After evaporation of the solvent, the powder was calcined at 800°C in inert atmosphere. The carbon coating is needed to enhance the electrode's electronic conductivity. LTP has a NASICON-type framework, providing fast ion diffusion and excellent stability in aqueous electrolytes. Its potential window is suitable for negative electrode operation in water-in-salt systems because it remains safely above the hydrogen evolution reaction.

Electrolyte

We used a so-called water-in-salt electrolyte based on sodium acetate ($7\ \mathrm{mol\,kg^{-1}}$) mixed with potassium acetate ($20\ \mathrm{mol\,kg^{-1}}$) in water. This high concentration extends water's electrochemical stability window by reducing solvent activity. Such super-concentrated electrolytes show relatively high ionic conductivity (around $20\ \mathrm{mS\,cm^{-1}}$ at room temperature)

and suppress side reactions typical of more dilute aqueous solutions.

Electrode preparation

A slurry of carbon-coated LTP (80 wt%), carbon black (10 wt%) and PVDF binder (10 wt%) was prepared in N-methyl-2-pyrrolidone. This slurry was dip-coated onto a carbon cloth substrate to form the working electrode with a loading of about $6\,\mathrm{mg\,cm^{-2}}$. After drying, the electrode was cut into 1×4 cm strips with 1×1 cm of exposed active material.

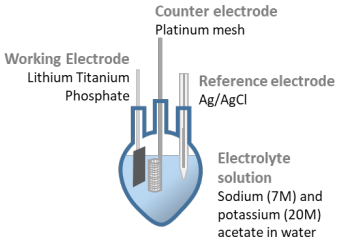


Figure 10.1: Three-electrode flooded cell configuration. Schematic of the glass cell used for DEIS measurements, showing the working electrode ($LiTi_2(PO_4)_3$), Pt mesh counter electrode, and Ag/AgCl reference electrode positioned behind the counter to minimize perturbation of the working electrode impedance.

Three-electrode flooded cell

All measurements were performed in a three-electrode glass cell. A Pt mesh was used as the counter electrode, while a Ag/AgCl electrode served as reference and was positioned behind the counter electrode. A schematic representation of the three-electrode flooded cell, including reference electrode placement relative to the counter electrode, is shown in Figure 10.1.

10.1.2 DEIS measurement

The setup for dynamic impedance measurements consisted of a Bio-Logic SP-300 potentiostat, a Keysight TrueForm 33522B arbitrary waveform generator, and a PicoTech Picoscope 4000a for voltage and current signal recording. Since this work preceded the acquisition automation, the waveform was loaded via USB into the waveform generator as a CSV file (in *.arb* format). The signal length was limited by the 16 GB of RAM of the laboratory computer, so only two cycles with DEIS could be recorded.

Multi-sine excitation

The multi-sine waveform used for these measurements was generated with a frequency band from 0.1 Hz to 10 kHz, 8 points per decade, and using only non-intermodulating integer multiples of the fundamental frequency. The amplitudes of the sinusoidal components were estimated from a preliminary EIS measurement and the phases of individual sine components were randomized to minimize the crest factor. Multi-frequency voltage and current signals were sampled at 50 kHz.

Applied drift

The multi-sine was superimposed onto a galvanostatic and a cyclic-voltammetry drift. The galvanostatic measurement was performed at $138\,\mathrm{mA\,g^{-1}}$ (approximately 1C) so that the voltage swept across the intercalation region in approximately two hours. The CV scan rate was set to $0.8\,\mathrm{mV\,s^{-1}}$ producing a similar experiment duration as the galvanostatic technique.

Impedance estimation

The impedance was estimated offline using Dynamic Multi-Frequency Analysis with the symmetric Fermi-Dirac filter of bandwidth 0.1 Hz and order $n = 8$. Hence, the time resolution of the impedance is 10 s. The DMFA was applied to the entire signal of two consecutive cycles after subtracting the straight line connecting the first and last points.

10.1.3 Analysis of the impedance

Intercalation model

The impedance of intercalation materials is modeled with porous electrode theory including the Faradaic process, originally developed by De Levie [100] for porous metal electrodes and later adapted by Bisquert [101] for insertion battery materials. The model accounts for both the ionic transport through the electrolyte-filled pores and the charge transfer/intercalation processes at the electrode-electrolyte interface. In this work, I used a physically-based porous electrode model derived from a simplified transmission-line approach [102, 103]. It accounts for ionic transport in the electrolyte-filled pores, electron conduction in the active material, and Faradaic intercalation kinetics. The overall impedance is

$$Z_T = R_s + \frac{R_p}{kl}\coth(kl), \tag{10.1}$$

where R_s is the uncompensated solution resistance, R_p combines pore-solution and solid-particle resistances, l is the length of the pore, and k is a complex factor containing the diffusion coefficient of the reaction product D, and the transport number of the cation t_+ according to the formula

$$k^2 = \frac{j\omega}{2D(1-t_+)}. \tag{10.2}$$

Figure 10.2 shows the full equivalent circuit corresponding to the porous electrode model.

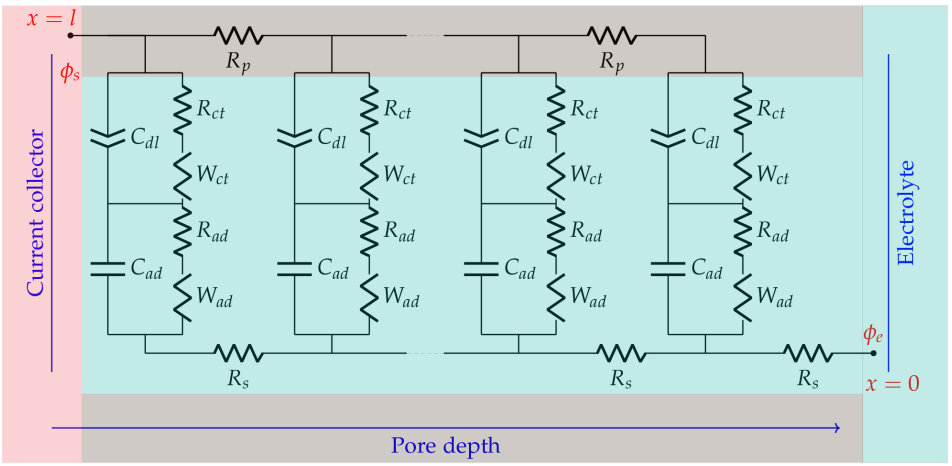


Figure 10.2: Equivalent-circuit representation of the transmission line model for porous electrodes in presence of a Faradaic reaction, developed in Reference [103]. Used under CC BY Licence [74].

For the material under study ($LiTi_2(PO_4)_3$, chapter 10), we simplified the repeating unit of the transmission line by substituting the adsorption-related Warburg element with a capacitor because it produces better fits with fewer parameters. The simplified circuit unit is reported in Figure 10.3.

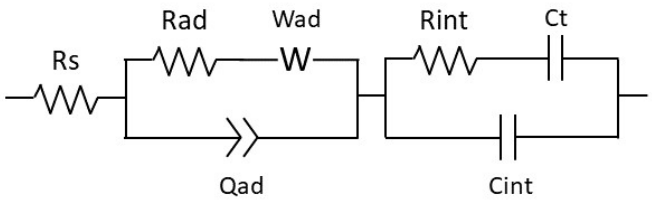


Figure 10.3: Impedance unit used for the fitting of lithium sodium titanium phosphate. It is simplified from the model developed in Reference [103] and depicted in Figure 10.2.

Fitting procedure

In this project, the objective function was written with a smoothing term, similar to Reference [67], leading to the form

$$obj = \sum_i \left(\sum_n r_{n,i}^2 + \lambda_s \sum_k \left(\frac{\partial \theta_{k,i}}{\partial i} \right)^2 \right), \tag{10.3}$$

where $r_{n,i}^2$ is the weighted complexed squared residual of the n-th point of the spectra as described in Section 4.7.1, $\theta_{k,i}$ is the k-th model parameter, i is the index of instantaneous impedances in the dataset, and λ_s is a smoothing factor that controls how much the parameter's curvature is penalized in the total objective function. The minimization was performed using the library `pymultipleis` [1]. To find proper starting parameters for the regression, it was set a smoothing factor λ_s of 1 and all the $\theta_{k,i}$ to 1. The optimized parameters were reused for a second optimization step setting this time λ_s to zero. The penalty on the curvature of the parameters across the impedances, function as an initial parameter selection leading to a good fit.

Validation strategy

To verify the alignment with the LTI conditions in the impedance datasets it is possible to use the rational polynomial functions of order m introduced in the previous chapter. The determination of the order m for a satisfactory fitting is dealt here with using a model selection criterion well known in statistics called Akaike information criterion (AIC) [104]. It is based on the statistical concept of *likelihood* defined

$$L(\theta) = \prod_{i=1}^{N} p(y_i; \theta) \tag{10.4}$$

for N independent measurements $y_1, ...y_N$ and a parametric model that predicts their joint probability density $p(y, \theta)$. The criterion is formulated as the sum of the negative maximum log-likelihood compensated by twice the number of parameters K:

$$AIC = -2 \max_{\theta} \{log(L(\theta))\} + 2K. \tag{10.5}$$

[1] https://github.com/richinex/pymultieis

In short, the AIC compensate the goodness of fit with the number of parameters.

10.2 Results and discussions

10.2.1 Impedance validation

To ensure that the estimated instantaneous impedance spectra satisfy the LTI assumption, we fitted the spectra with a general transfer function in the form of a rational polynomial of order m. To find the proper model structure different orders ranging from 3 to 8 were tested on the galvanostatic dataset, and from 3 to 9 for the sweep-voltage dataset, aiming to find the best compromise between goodness of fit and model simplicity. The AIC was calculated for each polynomial order, and its minimum value identified the optimal complexity level. As reported in Figure 10.4, orders 3 and 4 underfit several sections of the spectra, whereas orders 7 and 8 add marginal improvements but introduce unnecessary parameters. The AIC ultimately indicated order 6 to be optimal for the CC data and order 8 for the CV data.

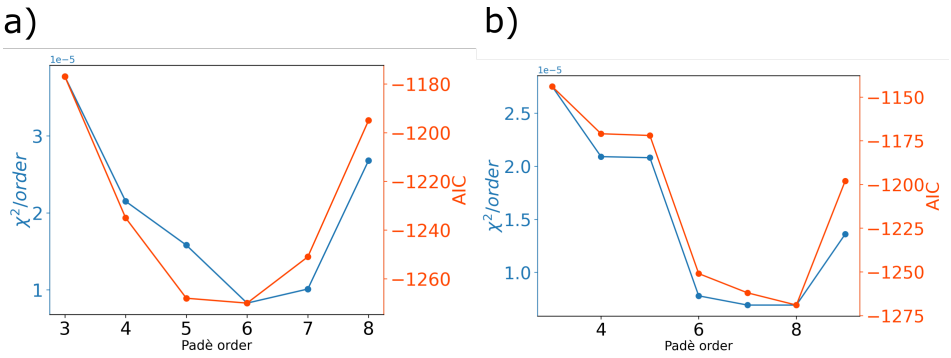


Figure 10.4: Model-order selection via Akaike Information Criterion. AIC values computed for rational polynomial fits of increasing order for galvanostatic and cyclic voltammetry datasets. The minima identify the optimal trade-off between goodness of fit and model complexity.

With the proper model structure identified, we assessed that the error was less than 0.4% of the impedance magnitude, which was a satisfactory LTI test. The results of the fitting and residuals are reported in the Appendix G.

10.2.2 DEIS with galavanostatic

During the galvanostatic experiment, the electrode crossed the entire intercalation window at a constant current.

The electrode potential evolution during galvanostatic intercalation and deintercalation was shown in Figure 10.5. The time-resolved impedance spectra obtained during the galvanostatic experiment are reported in Figure 10.6. Selected spectra illustrate the evolution of interfacial and transport-related features as the electrode crossed the intercalation plateau.

A notable observation is the contrast in impedance measured during ion insertion versus removal. The larger adsorption resistance and depressed semicircle during oxidation suggest that the energy barrier for desolvation and adsorption differs from that of desorption and re-solvation.

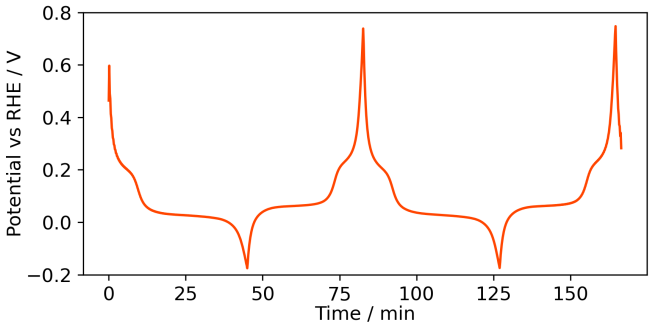


Figure 10.5: Electrode potential under galvanostatic drift. Time evolution of the working electrode potential during constant-current cycling at 138 mA/g, covering the full intercalation window of carbon-coated $LiTi_2(PO_4)_3$. Used under CC BY Licence [74].

Although both processes occurred over a similar potential region, the system appeared kinetically more hindered during oxidation. This asymmetry underscores the advantage of dynamic impedance in detecting differences that equilibrium EIS might obscure. Interestingly, an asymmetry in shape emerged when switching from intercalation (negative current) to deintercalation (positive current). The semicircle during oxidation appeared more depressed at the beginning and then recovered the same shape as reduction. The time-dependent resistance associated with ion adsorption onto the electrode surface changed more significantly during oxidation. This suggests non-symmetric free energies of ion insertion versus extraction, consistent with earlier reports of asymmetry in alkali-ion adsorption processes.

10.2.3 Fitted parameters

In order to extract quantitative information about the electrode processes, we fitted the dynamic impedance spectra with a porous-electrode-based model adapted from Reference [74]. The approach is grounded in the simplified transmission-line concept for intercalation electrodes, which includes terms for ohmic resistance, interfacial kinetics, and finite-length diffusion or pseudocapacitive behavior.

The physical model of intercalation has three main processes involved: the diffusion of the ions towards the interface, the adsorption of the ion on the inner Helmholtz plane and the ion intercalation in the lattice of the active material.

In Figure 10.7 are reported the fitted parameters of the equivalent circuit model depicted in Figure 10.3. The resistances associated with three main steps are represented in the model as R_p, R_ad and R_int. R_ad shows an asymmetry between reduction and oxidation steps (Figure 10.7d) changing from around 3 Ω to 9 Ω, while R_p (Figure 10.7g) and R_int (Figure 10.7h) remained constant. This suggests a different activation energy of the solvation and de-solvation processes while the others remained unaffected by the current direction.

Another interesting aspect observable only in the physical parameters is the effect of the change in current between reduction and oxidation. In fact, there was an open-circuit rest between the two techniques to relax the system, so the oxidation started with the interface depleted of sodium ions. As a result, all parameters reacted with a positive or negative spike before adjusting to a continuous value.

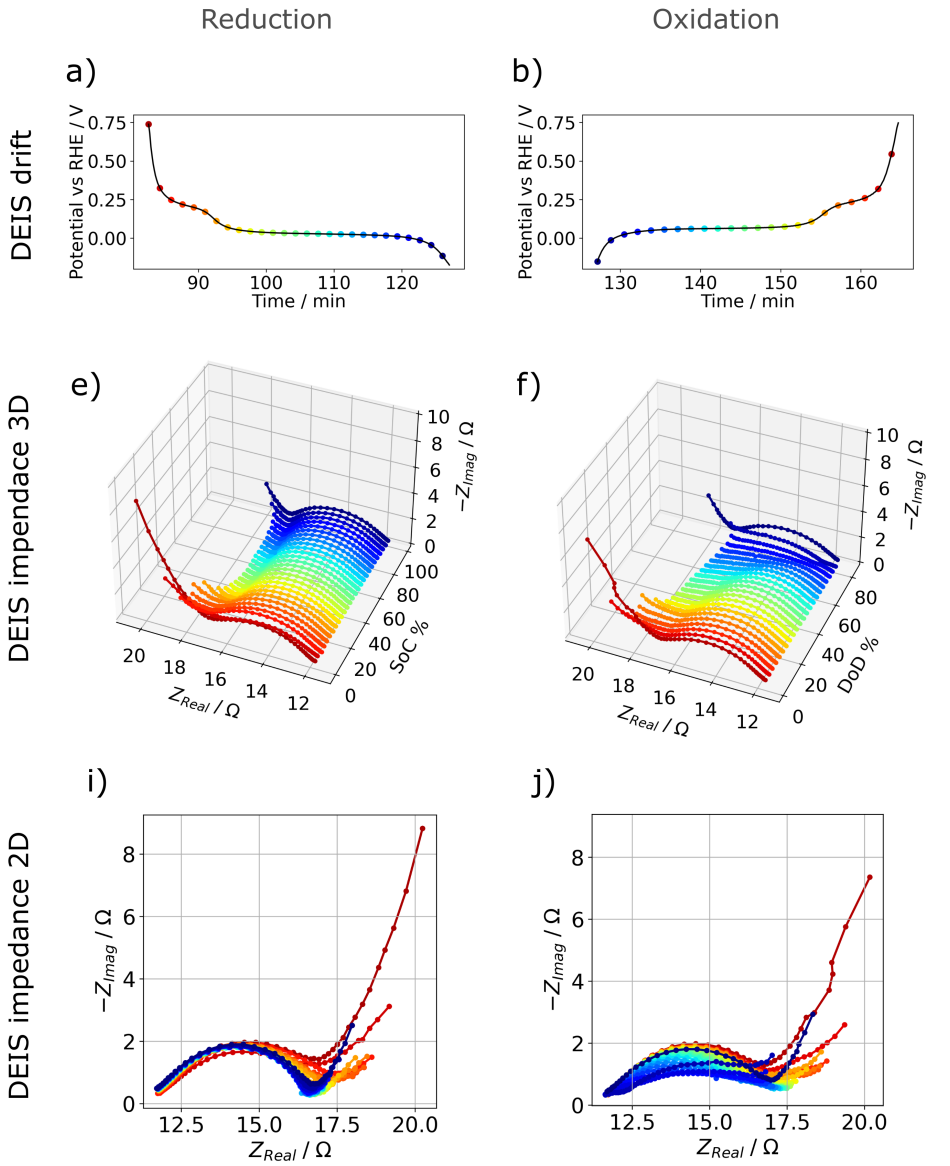


Figure 10.6: Dynamic impedance during galvanostatic intercalation. Instantaneous impedance spectra estimated via DMFA during constant-current cycling. Spectra before, during, and after the voltage plateau highlight changes in charge-transfer and diffusion-related contributions. Used under CC BY Licence [74].

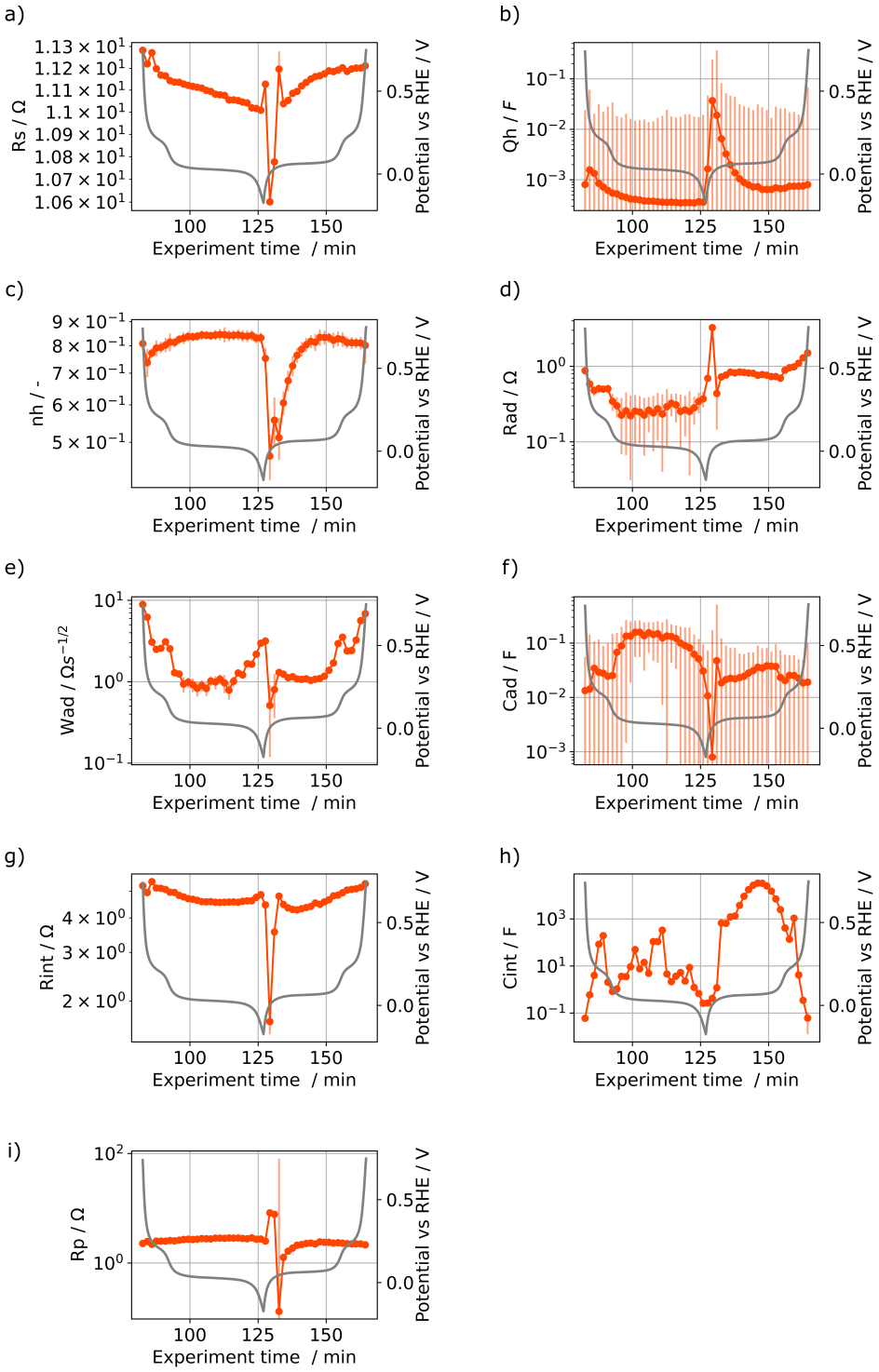


Figure 10.7: Porous-electrode model fitting. Parameters evolve with the state of charge, changing mainly when the electrode potential crosses the plateau. Abrupt jumps in some parameters are observed when the current is reversed, since the double layer has no time to relax to OCV conditions. Used under CC BY Licence [74].

10.2.4 DEIS with cyclic voltammetry

In the DEIS experiment with cyclic voltammetry, the potential scanned at $0.8\,\mathrm{mV\,s^{-1}}$, and the electrode could momentarily reach kinetic regimes strongly influenced by concentration polarization. This manifested in the DEIS data as a pronounced rise in impedance, especially around the lower plateau, where transport limitations were accentuated. The real-time spectra showed more pronounced distortions in the low-frequency region, associated with the voltage-driven, layer-by-layer intercalation behavior.

The current-potential response recorded during the cyclic voltammetry experiment is shown in Figure 10.8.

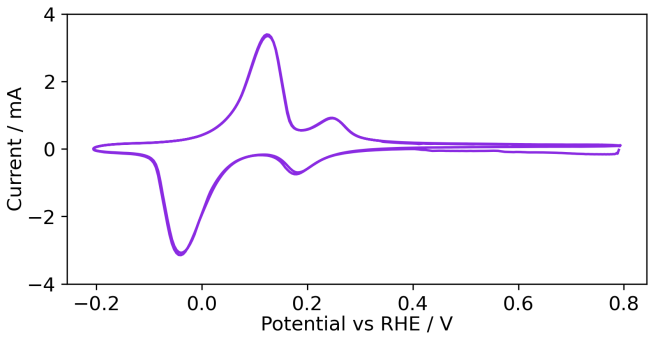


Figure 10.8: Current-potential response under voltage-sweep drift. Cyclic voltammetry of the LTP electrode at 0.8 mV/s, showing the intercalation and deintercalation plateaus probed during DEIS measurements. Used under CC BY Licence [74].

The corresponding time-resolved impedance spectra acquired during the voltage-sweep experiment were shown in Figure 10.9.

10.2.5 Comparison between the two drifts for DEIS

Comparing the galvanostatic and cyclic voltammetry results highlighted how different drifts can emphasize different kinetic contributions. Under galvanostatic conditions, the electrode had to sustain the set current, so the impedance shape largely reflected the electrode's instantaneous ability to supply or accept ions. Under sweep-voltage conditions, the potential was the control variable, and mass-transport or pseudocapacitive features could dominate as the electrode passed through distinct depletion and accumulation states.

10.3 Summary and conclusions

These experiments demonstrate that non-stationary impedance spectroscopy can separate different reaction steps for an intercalation material under realistic battery operating conditions. By comparing galvanostatic and voltage-sweep measurements, we showed that the electrode's time-resolved impedance reveals differences in diffusion, interfacial kinetics, and their directional asymmetry.

While the primary goal here was to validate DEIS rather than to fully characterize LTP, the physical model fitting revealed quantitative values for charge-transfer, adsorption, and transport processes. These results provided insight to formulate a hypothesis on asymmetry in solvation and de-

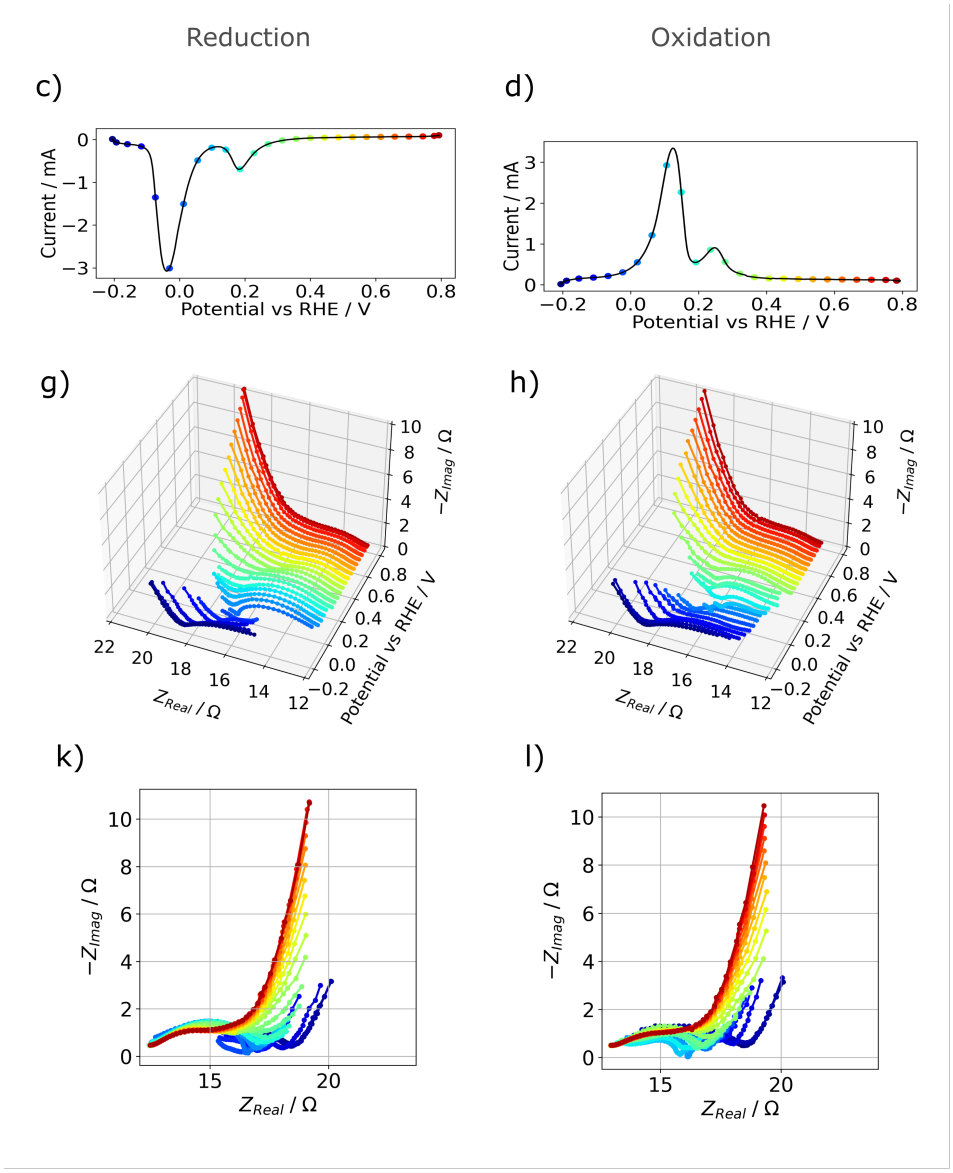


Figure 10.9: Dynamic impedance during cyclic voltammetry. Instantaneous impedance spectra estimated via DMFA during voltage-controlled operation, revealing stronger low-frequency distortions associated with concentration polarization and potential-driven interfacial dynamics. Used under CC BY Licence [74].

solvation processes that would not have been accessible via direct current methods alone.

The limitation of this study was the short measurement time, limited by the computer RAM, and also the physical model not being entirely sufficient to explain the system dynamics. Furthermore, characterization at the material level must be carried out with information from multiple techniques, not only electrochemical, to draw the full picture of the material properties.

Chapter 11

Non-Faradaic process in symmetric cell

Electrochemical double-layer capacitors are a class of energy storage devices characterized by high power and low energy density. The process of storing energy is based on adsorption of ions, a non-Faradaic process, which gives the characteristic fast kinetics at the expense of capacity; they also exhibit high self-discharge. The capacity depends on the electrode-electrolyte surface area, so materials with high porosity are particularly desirable. This section reports work on self-standing activated carbon electrodes in a three-electrode cell. The presence of the reference electrode allows separation of the instantaneous impedance of positive and negative electrodes during device operation. In a dynamic impedance spectroscopy measurement, the multi-sine was superimposed onto a cyclic voltammetry drift.

The results of this project led to a publication [**piantaEvaluationKineticParameters2023a**], and the data are available in a public repository[10].

[10] doi: 10.6084/m9.figshare.21082168

11.1 Experimental section

11.1.1 Materials and cell assembly

Electrode material

The freestanding activated carbon electrodes were prepared by Dr. Nicolò Pianta (University of Milano-Bicocca) according to reference [105] to ensure a binder-free, highly porous network with large accessible surface area. The synthesis route includes carbonization and physical activation of a precursor to achieve high microporosity. Disks of 12 mm diameter (approximately 1.54 cm^2 projected area) were punched from the freestanding membranes. Each disk weighed approximately 23.5 mg of active carbon. The high surface area allows for capacitance in the range of tens of $F\,g^{-1}$.

Electrolyte

A 10 $mol\,kg^{-1}$ KF aqueous solution was selected as the electrolyte. Compared to neutral dilute solutions, this composition (pH 10.7) offers a wider potential window and a relatively stable interface. The device stores energy mainly by the adsorption of K^+ and F^- species at the electrode double layer [**piantaEvaluationKineticParameters2023a**].

11.1.2 Three-electrode cell configuration

The electrodes were introduced into a T-shaped PEEK cell body, with two stainless-steel pistons acting as current collectors and applying mild compression to ensure low interfacial resistance. Two glass-fiber disks served as separators, thoroughly soaked in the KF solution. A key challenge was the choice and placement of the reference electrode within this compact geometry. Standard Ag/AgCl probes prove bulky and often introduce junction potential errors at high pH. Thus, we adopted a Pb/PbF_2 electrode [**piantaEvaluationKineticParameters2023a**, 106] by inserting a thin Pb wire inside the stack, where a thin layer of PbF_2 formed *in situ*, yielding a stable potential around 191 mV vs. RHE.

Reference electrode placement

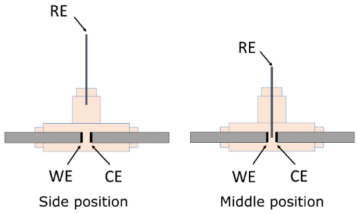


*Figure 11.1: Reference electrode placement in a compact three-electrode "Swagelock type" cell. Schematic comparison between side-positioned and centrally inserted reference electrodes, highlighting the additional electrolyte path and parasitic impedances introduced when the reference is not in the electrode stack.Reproduced from [**piantaEvaluationKineticParameters2023a**].*

Two possible positions of the reference electrode were evaluated: the "side position", where the reference is placed above or to the side of the active stack in a flooded region of the cell; and the "central position", where the Pb wire is inserted in the middle, directly between the two carbon electrodes. They are schematically illustrated in Figure 11.1. As highlighted in reference [**piantaEvaluationKineticParameters2023a**], placing the reference on the side often leads to mid- and high-frequency artifacts caused by non-uniform current paths and additional electrolyte resistances. A direct comparison of impedance spectra from the two positions showed that a loop-like artifact appears if the reference is displaced too far from the working electrodes. Consequently, for the DEIS experiments presented here, the reference was placed in the central region of the stack, minimizing parasitic inductive and capacitive pathways.

11.1.3 DEIS measurement

In this work, we aimed to measure simultaneously the impedance of both cell electrodes during operation. Since each auxiliary port of the potentiostat channel has only the working electrode voltage, we needed to use a second channel for the acquisition of the full cell. One channel was connected to the working and counter electrodes of the cell, short-circuiting the reference and counter sensing connectors. This provides the cyclic voltammetry perturbation to the cell where the voltage sweep is applied between working and counter electrode, simulating the operation of the device. A second channel was set to open-circuit mode, with its working and reference sensing connected to the working and reference electrodes of the cell for the recording of the working electrode half-cell voltage drop. The current flows equally in series connection while the voltage drop of the counter half-cell is obtained by subtraction of the total cell voltage from the working half-cell voltage.

Figure 11.2 illustrates the role of the reference electrode in separating the working electrode response from counter-electrode and electrolyte contributions, which is essential for reliable impedance measurements.

A schematic overview of the three-electrode DEIS measurement setup, including excitation and acquisition, is shown in Figure 11.3.

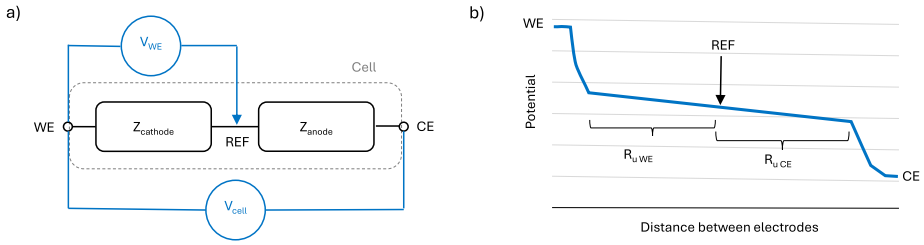


Figure 11.2: Role of the reference electrode in electrochemical impedance measurements. A reliable reference electrode design ensures that the measured impedance reflects the response of the working electrode alone, minimizing artefacts arising from counter-electrode polarization or uncompensated resistance.

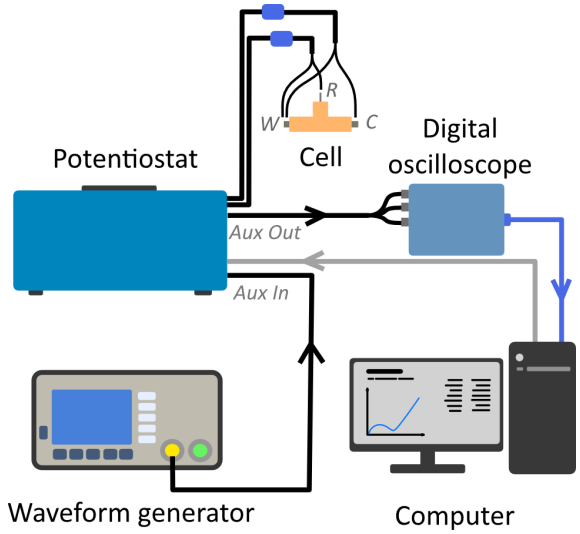


*Figure 11.3: Instrumentation for three-electrode DEIS measurements on EDLCs. Experimental configuration showing the potentiostat, arbitrary waveform generator, external data acquisition, and three-electrode cell used to resolve time-varying impedance of each carbon electrode independently. Two channels of the potentiostat are used to record the voltage and current signals from both the working half cell and the full cell simultaneously. Reproduced from [**piantaEvaluationKineticParameters2023a**].*

Multi-sine excitation

We generated a multi-sine waveform covering frequencies from about 0.1 Hz up to 100 kHz, with 8 points per decade. Frequencies were constrained to avoid intermodulation. Each sine amplitude was estimated by preliminary EIS tests to guarantee sufficiently large signals above the measurement noise, while avoiding strong nonlinearity. The phase of each sine was randomized to minimize the crest factor.

Applied drift

A quasi-linear potential sweep of $8\,\mathrm{mV\,s^{-1}}$ was superimposed onto the multi-sine. Over one full cycle (charge then discharge) the cell spans from near 0 V (completely discharged) to $\approx 1.2\,\mathrm{V}$ (fully charged) and back, requiring about 250 s. This scanning is repeated for a few consecutive cycles. Because the drift period is 250 s, we restrict ourselves to multi-sine frequencies of 0.3 Hz and higher in the final analysis, to minimize overlap with the drift fundamental frequency [67].

Impedance estimation

From the multi-frequency voltage and current, we estimated the time-resolved impedance via DMFA [**piantaEvaluationKineticParameters2023a**, 107]. Simultaneously, we reconstructed the CV-like current–voltage profiles at 1 s resolution by applying a zero-frequency filter around the drift.

11.1.4 Physical model

The non-Faradaic adsorption mechanism for EDLC electrodes can be pictured as parallel pathways of double-layer charging and pore adsorption, each limited by finite or semi-infinite transport [108]. The model is represented as the equivalent circuit model in Figure 11.4 The model is not ade-

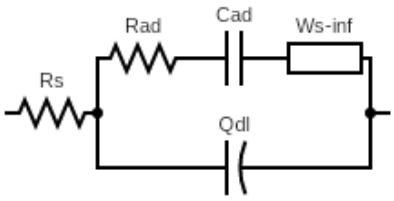


Figure 11.4: Graphic representation of the equivalent circuit model proposed in Reference [108].

quate to fit the impedance spectra of both positive and negative electrodes, so for the negative electrode the adsorption capacitance and semi-infinite Warburg element were replaced with a finite-diffusion Warburg element.

11.2 Results and discussions

11.2.1 Artifact analysis

Figure 11.5 shows the effect of the reference electrode's position in the same cell. When it is outside the cell stack, the impedance become significantly distorted. The measurement presented in the rest of the chapter are performed with the reference in the middle position.

11.2.2 Qualitative evaluation of the impedance

The DEIS data show both electrodes presenting predominantly capacitive spectra under moderate voltages (below $\approx 0.7\,\mathrm{V}$ vs. RHE). However, near the upper region ($\approx 1.2\,\mathrm{V}$ vs. RHE at the positive electrode), an additional low-frequency arc develops, signifying parasitic side reactions (likely oxygen evolution). Figure 11.6 illustrates the time evolution of the impedance modulus at representative mid-frequency (10 Hz) and low-frequency (300 mHz) components for both electrodes. The negative electrode impedance remains relatively stable, while the positive electrode experiences a marked drift in the low-frequency domain once the potential exceeds $\sim 1.0\,\mathrm{V}$ vs. RHE.

Despite the nominally identical activated carbon electrodes, the positive side consistently shows stronger frequency dispersion and larger apparent capacitance, likely due to local oxygen-containing species adsorbed upon

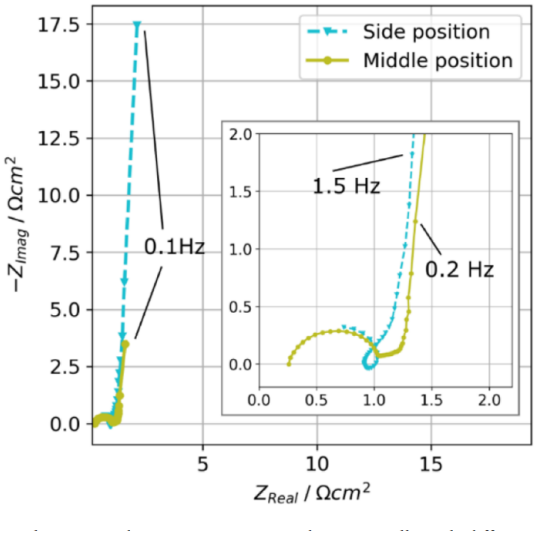


*Figure 11.5: Artifact induced from the positioning of the reference electrode in compact cells. Appearance of a loop-like feature in the mid-to-high frequency region when the reference electrode is displaced from the electrode stack, caused by non-uniform current distribution and additional electrolyte pathways. Reproduced from [***piantaEvaluationKineticParameters2023a***].*

overcharging. Conversely, the negative electrode's impedance shape more closely resembles an ideal double-layer: a single semicircle or depressed arc merges into a near-vertical line at low frequencies. Similar discrepancies appear in References [**piantaEvaluationKineticParameters2023a**, 105], confirming that potential-dependent side reactions or ionic rearrangements can occur in symmetrical carbon electrodes at different operating windows.

A direct visual comparison of the impedance spectra for the positive and negative electrodes at comparable states of charge is shown in Figure 11.7.

11.2.3 Capacitance estimation

From the reconstructed current-voltage curves, we estimated the average electrode capacitance with

$$C_{\text{avg}} = \frac{i_{\text{drift}}}{dV/dt}, \tag{11.1}$$

where i_{drift} is the net current from the potential sweep. We also extracted C_{dl}, C_{ad}, and any Warburg-like terms from equivalent circuit regressions. Both methods yield values of approximately 40–50 F g^{-1} for each electrode, consistent with free-standing activated carbons in alkaline media [105]. The positive electrode, however, shows a noticeable rise in adsorption-related capacitance above 0.9 V vs. RHE, highlighting contributions from surface-active species formed near the oxygen-evolution threshold [**piantaEvaluationKineticParameters2023a**].

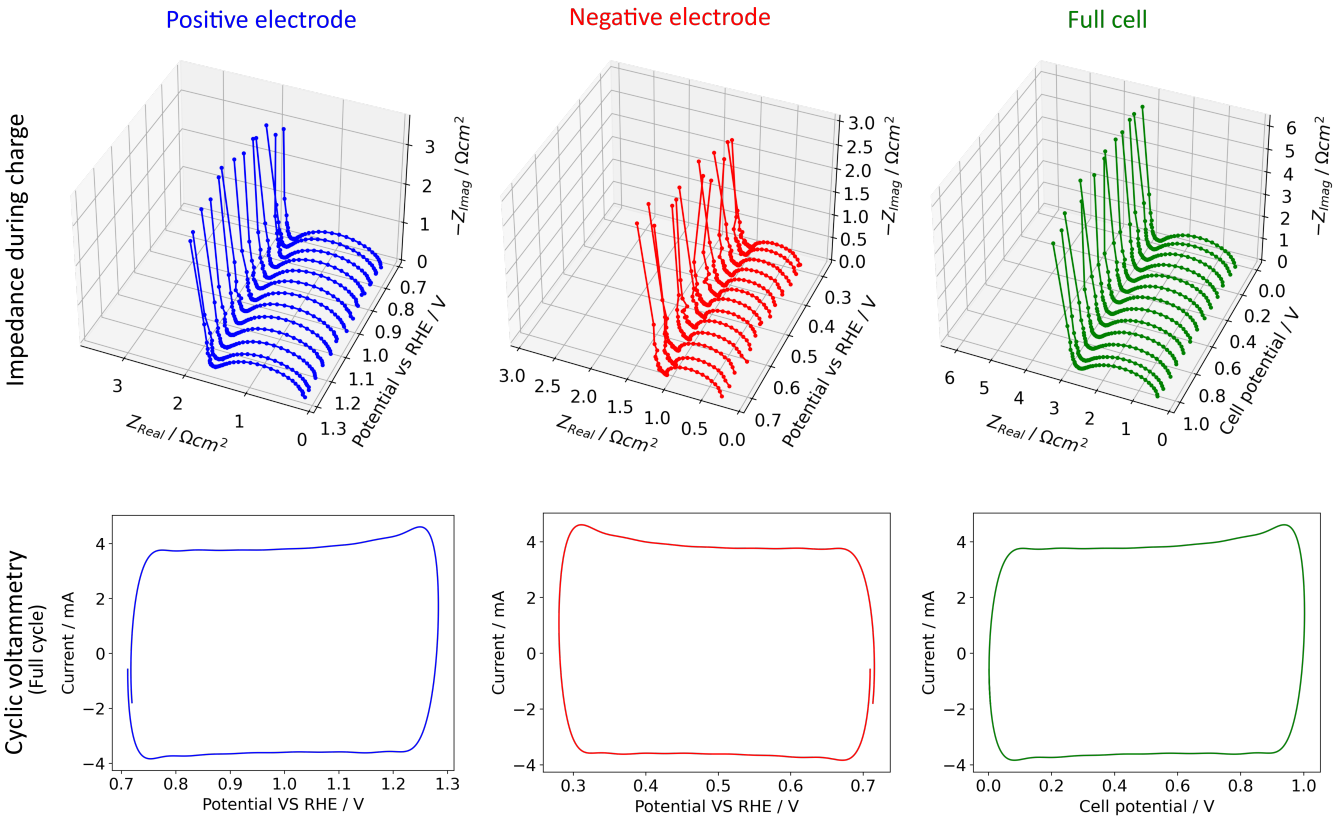


*Figure 11.6: Time evolution of selected impedance components during CV. Modulus of impedance at 10 Hz and 300 mHz for positive and negative activated carbon electrodes during cyclic voltammetry, highlighting the onset of low-frequency drift at the positive electrode near the upper potential limit. Reproduced from [***piantaEvaluationKineticParameters2023a***].*

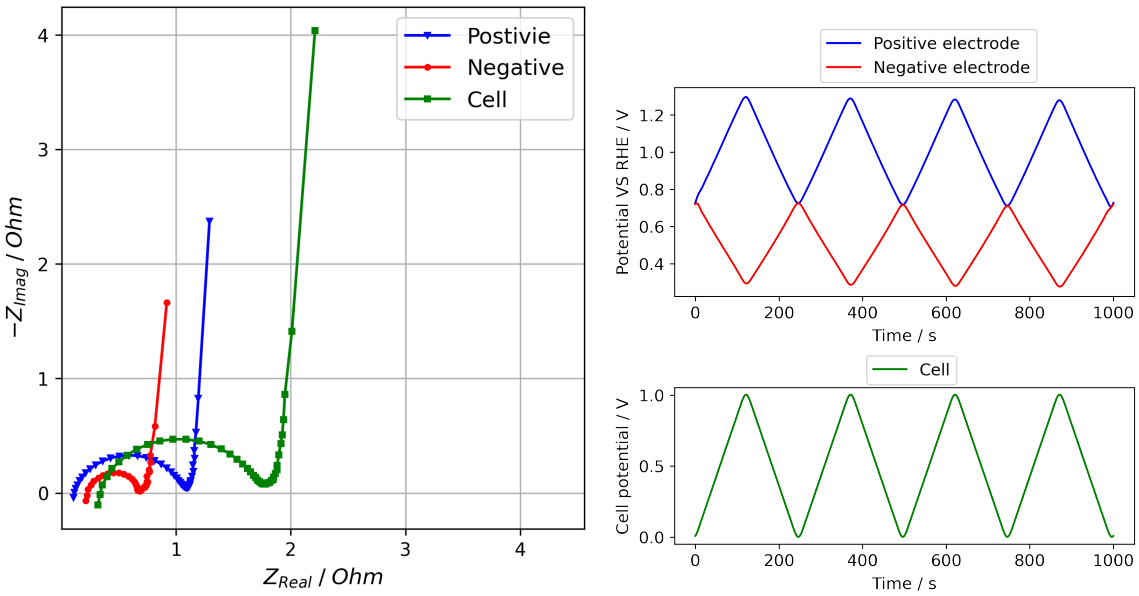


*Figure 11.7: Impedance asymmetry between nominally identical carbon electrodes at 0V cell voltage. Left: Comparison of instantaneous impedance spectra highlighting higher impedance at the positive electrode. Right: electrode potential history. Adapted from [***piantaEvaluationKineticParameters2023a***].*

11.3 Impedance fitting

Figure 11.8 reports an example of fitting for the positive and negative electrode, at the middle of an half-cycle.

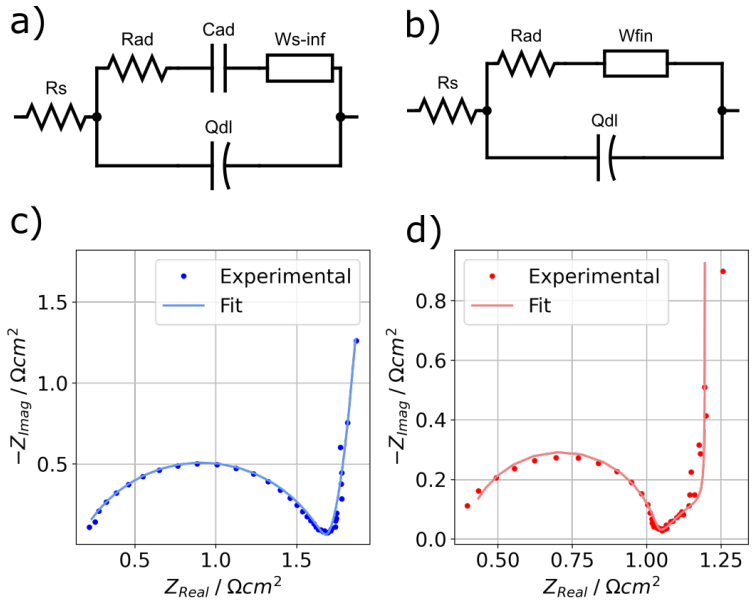


*Figure 11.8: Equivalent-circuit fitting of instantaneous EDLC impedance. Experimental impedance spectra (symbols) and fitted distributed RC models (lines) for positive and negative electrodes at selected potentials during cyclic voltammetry. Reproduced from [***piantaEvaluationKineticParameters2023a****].*

For electrochemical capacitor the capacitance is a key performance metric. For porous electrodes the impedance if fitted with the CPE elements and the effective capacitance C_{eff} can be calculated with the formula [109]

$$C_{eff} = Q_{dl}^{\frac{1}{n}} R_{ad}^{\frac{1-n}{n}}, \tag{11.2}$$

where Q_{dl} and n are the CPE parameters and R_{ad} the adsorbtion capacitance. Figure 11.9 shows the values for the CPE capacitance Q_{dl} estimated via equivalent circuit fitting and the calculated effective capacitance C_{eff} for positive and negative electrodes as a function of the potential. The effective capacitance, in contrast to the CPE capacitance, reflects properly the reversibility of the adsorption process in EDLCs.

11.4 Summary and conclusions

In this chapter, we described the simultaneous measurement of dynamic impedance spectroscopy of both half-cells for a symmetric super-capacitor based on activated nano-porous carbon electrodes. The experiment highlights that even chemically identical electrodes can have different interfacial impedance, something not apparent from direct current methods. Other than manufacturing variation, the two electrodes have different history. Analyzing such systems with the whole cell impedance doesn't inform on the specific reactions. This is important to consider for more complex systems like batteries were electrode processes might be hidden when looking at the whole cell response.

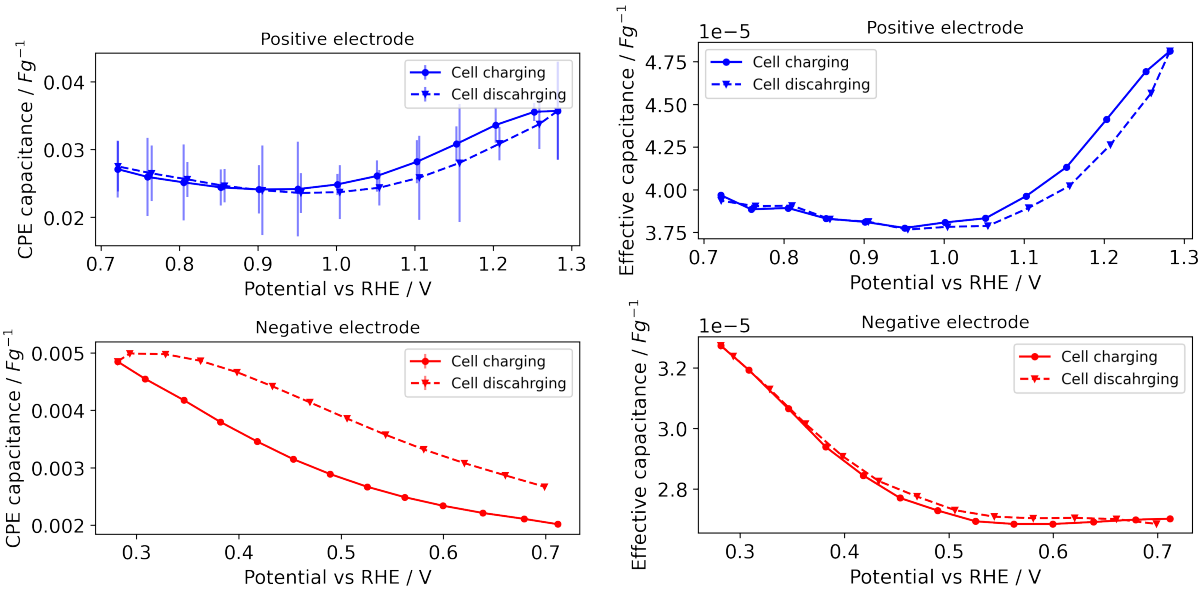


*Figure 11.9: Time variation of the CPE capacitance Q_{dl} on the left and the estimated effective capacitance on the right. Adapted from [***piantaEvaluationKineticParameters2023a****].*

Chapter 12

Exploratory anodes studies

Dynamic impedance spectroscopy measurements have demonstrated a strong correlation of the instantaneous impedance with state of charge and state of health in commercial lithium-ion batteries, as established in chapter 9. In chapters 10 and 11, three-electrode flooded and compact cells were used to isolate the impedance of individual electrodes and to relate the time-varying spectra to physically meaningful parameters. These results motivate the extension of dynamic impedance spectroscopy to batteries in three-electrode configuration with the aim of isolating and identifying degradation phenomena under operating conditions.

This chapter focuses on negative battery electrodes for lithium and sodium intercalation because, compared to positive electrodes, they are commonly associated with the onset of the most critical degradation mechanisms: metal plating and SEI growth. Both phenomena alter the local electrochemical kinetics and therefore should manifest distinct signatures in the time-resolved impedance.

To resolve the potential of the single electrodes, a micro-reference electrode can be used in compact cells [110], the typical format of battery cells. Unlike in flooded cells, the restricted volume and stack pressure limit the size and placement of reference electrodes, and improper positioning can lead to significant measurement artifacts, as demonstrated in chapter 10. For this reason, microscopic metal wires with localized exposed tips were used as reference electrodes and placed directly between the separators, enabling a closer approximation of a point-like reference potential inside the stack.

This chapter discusses DEIS measurements performed on three different half-cell configurations:

- graphite vs. lithium, using electrodeposited lithium on insulated copper wire as the reference electrode;
- SiCN(O) vs. sodium, using electrodeposited sodium on insulated copper wire as the reference electrode (two independent assemblies);
- sodium vs. sodium, using electrodeposited sodium on insulated copper wire as the reference electrode.

The overarching research question guiding this work is: Can dynamic impedance spectroscopy identify transient interfacial processes such as SEI formation, staging phenomena, or metal plating directly during electrode operation?

The following sections present the materials, cell designs, DEIS parameters, and qualitative analysis of the impedance behavior for each electrode type.

12.1 Experimental section

12.1.1 Active materials

Graphite

Graphite is the standard negative electrode material in commercial lithium-ion batteries, owing to its reversible Li^+ intercalation, good rate capability, and long-term cyclability. Its staging transitions, solid-electrolyte interphase (SEI) formation, and susceptibility to lithium plating under high-rate charging make it an ideal system for testing whether DEIS can resolve transient interfacial phenomena *in situ*.

In this work, graphite half-cells in pouch format were assembled to investigate three aspects:

1. the evolution of impedance during different intercalation stages;
2. the onset of plating-like signatures during high-current reduction;
3. the asymmetry between reduction and oxidation kinetics under non-stationary conditions.

These measurements serve as a reference point because the electrochemistry of graphite is well established and provides a benchmark against which to interpret more complex materials such as SiCN(O).

SiCN(O)

Silicon carbonitride SiCN(O) is a polymer-derived ceramic with a tunable disordered network composed of Si-N, Si-C, and C-C bonding motifs. Depending on the precursor and pyrolysis conditions, the material exhibits high microporosity, moderate electronic conductivity, and good chemical stability in non-aqueous electrolytes. These characteristics make SiCN(O) a promising negative electrode material for sodium-ion batteries, where hard carbon is currently the state of the art [111].

In contrast to graphite, SiCN(O) does not undergo intercalation into well-defined crystallographic positions. Instead, several storage mechanisms have been proposed in the literature—including surface adsorption, pore filling, and reversible metal deposition within the nano-porosity [111]. The relative contributions of these processes remain under debate.

The electrodes containing SiCN(O) as active materials were prepared by Dr.-Ing. Marco Melzi d'Eril within a collaboration with the Technical University of Darmstadt. A detailed discussion of the chemistry, synthesis routes, and structural characterization of SiCN(O) is provided in Appendix H. Here, we focus solely on the motivation for DEIS measurements.

Two properties of SiCN(O) make it particularly interesting for dynamic impedance studies:

1. its highly porous microstructure, which can host different Na storage modes that may evolve during cycling;

2. experimental evidence from galvanostatic cycling [112] suggesting that lowering the cutoff potential below 0 V vs. Na^+/Na triggers additional reversible capacity, possibly associated with sodium metal deposition within the pore network.

DEIS offers a unique opportunity to probe whether these mechanisms leave distinct, time-localized signatures in the impedance, especially during the transition between adsorption, insertion-like processes, and hypothesized metal deposition [112]. This approach complements conventional characterization methods such as ex situ NMR or Raman spectroscopy by enabling operando tracking of interfacial kinetics [113, 114].

Micro-reference electrodes

Introducing a reference electrode into a pouch-cell configuration presents significant geometric and electrochemical challenges. As discussed in chapter 10, the placement and size of the reference electrode strongly influence the measured impedance, particularly in compact cells where the electric field is highly non-uniform. Macroscopic reference electrodes, such as Ag/AgCl or metal strips, distort the potential distribution and introduce artefacts in the high- and mid-frequency regions, making them unsuitable for dynamic impedance measurements in multilayer or small-area cells.

The electrostatic distortion introduced by extended reference electrodes in compact geometries is schematically illustrated in Figure 12.1

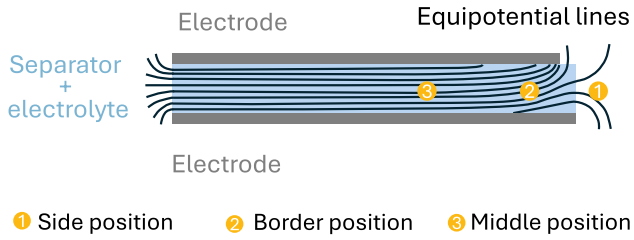


Figure 12.1: Electric field distortion at the electrodes border in compact cells. Schematic illustration showing how macroscopic reference electrodes perturb current and potential distribution near the electrode stack, motivating the use of point-like micro-reference electrodes in pouch-cell DEIS.

For this reason, I adopted microscopic reference electrodes based on insulated metal wires with only the cross-section area exposed to the electrolyte. Such electrodes approximate a point-like probe and minimize perturbation of the current distribution. When placed between two separators at the mid-plane of the stack, they provide an acceptable compromise between electrochemical stability and geometric neutrality. An illustration of a micro-reference electrode integrated into a three-electrode pouch cell is shown in Figure 12.2

In lithium systems, micro-reference electrodes are achieved by in situ alloying of gold (in the form of wire) and lithium, whereas in sodium systems few attempts were made because of sodium's inability to alloy with gold [115–118]. Sodium alloys instead with Tin but in a more complex multiphase chemistry [119].

In this exploratory work, I could not replicate the lithium-gold alloying procedure found in the literature, so I decided to proceed with electrodeposition on copper wires for both lithium and sodium. The electrochem-

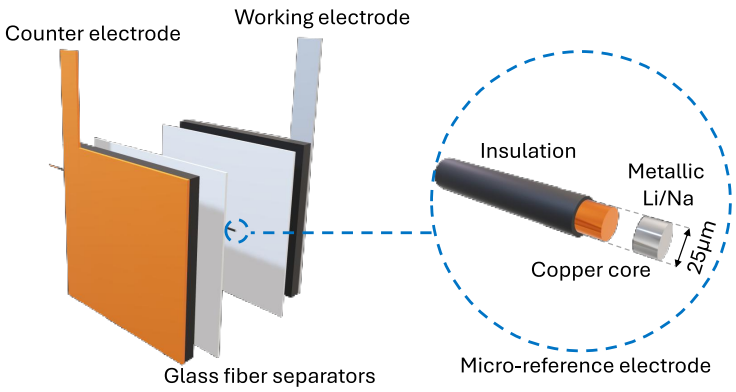


Figure 12.2: Microscopic reference electrode implemented in a pouch-cell stack. Insulated metal wire with only the cross-sectional tip exposed to the electrolyte, positioned between separators to approximate a point-like reference potential while minimizing current-path distortion.

ical stability of these microscopic electrodes is strongly dependent on the quality of the insulation and the suppression of parasitic side reactions. Although the approach enabled DEIS measurements for extended periods, reproducibility remained limited because small imperfections in insulation, wire geometry, or deposition uniformity lead to drifts of the reference potential.

A detailed account of the fabrication procedures, attempts with different substrates, stability tests, and failure modes is provided in Appendix I. In this chapter, only the essential aspects are reported, as the purpose of these measurements is not to optimize reference electrode design but to demonstrate that DEIS can be performed operando in three-electrode pouch cells.

To verify the stability of the reference electrode during the measurement, I recorded the OCV between the reference electrode and the metallic counter electrode of the cells. The voltage difference was always between 1–2 mV at the end of the experiments and even after days or weeks (cell disconnect from the instrument).

12.1.2 Electrode preparation and cell assembling

Graphite electrode

Graphite negative electrodes were prepared from an aqueous slurry containing AF-261 graphite powder, 6 wt% Super P C60 to enhance electronic conductivity, and 3 wt% carboxymethylcellulose (CMC) as binder. The slurry was cast onto copper foil using an automated doctor-blade coater and dried under vacuum at 80 °C overnight. Electrodes were then punched into 2 cm × 2 cm squares and stored in an argon atmosphere prior to cell assembly.

SiCN(O) electrode

The SiCN(O) electrodes were obtained from SiCN(O) powder (average particle size ∼40 µm) mixed with 2 wt% carbon black to increase electronic conductivity, and a binder system composed of 2 wt% CMC and 2 wt% styrene-butadiene rubber (SBR). The resulting aqueous slurry was cast

onto aluminum foil, dried under vacuum at 80 °C for 24 h, and subsequently punched into 12 mm diameter disks. The synthesis route of SiCN(O), along with bulk characterization data and electrode formulation, is described in detail in Reference [112] and is not repeated here.

Metallic sodium

Metallic sodium counter electrodes were prepared from commercial sodium-aluminum laminate disks (12 mm diameter). These disks are supplied with protective polymer films on both sides to prevent oxidation and preserve the flatness of the active surface. Prior to assembly, the protective foils were removed inside the glovebox and the disks were immediately transferred to the cell stack.

Electrolytes

Two electrolytes were used depending on the chemistry:

- 1 M $LiPF_6$ in EC:DEC (1:1 w/w) with 2 wt% vinyl carbonate (VC) for graphite/Li half-cells;
- 1 M $NaPF_6$ in EC:DEC (1:1 w/w) for SiCN(O)/Na and Na/Na cells.

All electrolytes were handled in an argon-filled glovebox (< 0.1 ppm O_2, < 0.1 ppm H_2O).

Three-electrode pouch cell

To assemble these cells, I chose the pouch cell format because it offers a high degree of geometric flexibility, larger electrode areas, and ease of assemble, characteristics that make them a standard platform for laboratory-scale battery development. The three-electrode pouch cells were assembled using two layers of glass fiber separator with the micro-reference electrode positioned between them.

The integration of the micro-reference electrode in graphite/Li pouch cells is shown in Figure 12.3 and SiCN(O)/Na cells in Figure 12.4. In both cases the current-collector tabs were put in contact to aluminum or nickel leads without welding or other adhesives, and the stack was enclosed in a laminated pouch and sealed under reduced pressure.

12.1.3 DEIS measurement

Multi-sine excitation

For all chemistries, multi-sine excitations were generated using the Keysight TrueForm 33500-series waveform generator. The signals were constructed from non-intermodulating integer multiples of the fundamental frequency, with eight frequency points per decade and randomized phases to minimize the crest factor. Current and voltage responses were recorded using a PicoTech PicoScope 4000a at sampling rates between 5 kHz and 50 kHz depending on the upper frequency of the excitation band. The multi-sine amplitude was adjusted separately for each material to maintain linearity of the response and avoid saturating the potentiostat compliance limits.

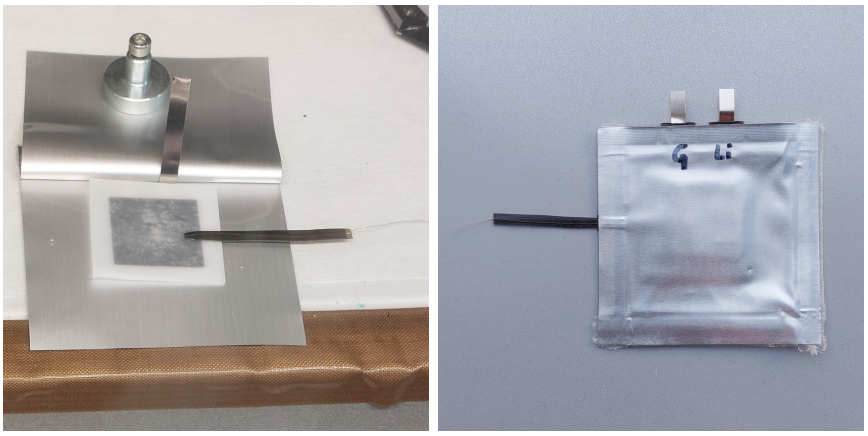


Figure 12.3: Three-electrode graphite/Li pouch cell with micro-reference electrode. Left: placement of the insulated micro-reference electrode between separators; right: sealed three-electrode pouch cell used for DEIS measurements on graphite half-cells.

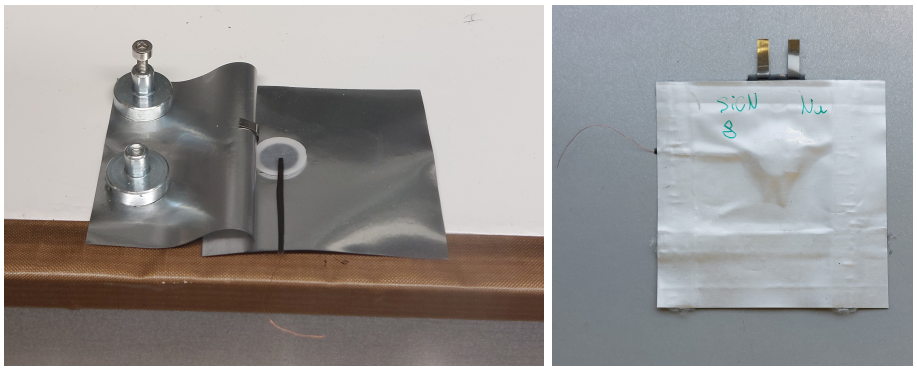


Figure 12.4: Three-electrode SiCN(O)/Na pouch cell with micro-reference electrode. Left: positioning of the microscopic reference electrode within the separator stack; right: completed three-electrode pouch cell used for operando DEIS measurements.

Applied drift and DEIS parameters for graphite

Galvanostatic cycling was performed at three C-rates (C/5, 1C, and 2C), calculated from a theoretical graphite capacity of 11 mAh. The potential limits were set to 0.8 V and 0 V vs. Li/Li^+. A multi-sine excitation containing frequency components from 0.01 Hz to 1 kHz was superimposed on the galvanostatic drift. Equal relative amplitudes were used across all sine components, and the phases were randomized to obtain the lowest crest factor. The perturbation amplitude was 40 mA_{pp} for the C/5 and 1C experiments. For the 2C experiment, the perturbation was reduced to 5 mA_{pp} to avoid non-linear distortion near the lower potential limit and to maintain the oscillation within the hardware compliance window. However, the impedance estimates at 2C were notably degraded by the reduced signal-to-noise ratio.

Applied drift and DEIS parameters for SiCN(O)

SiCN(O) electrodes were tested under galvanostatic cycling between 0 V and 2.5 V vs. Na/Na^+, using a C/10 current defined relative to the theoretical capacity reported in Reference [112]. Silicon carbonitride is an amorphous material; therefore, the maximum theoretical capacity cannot be calculated directly from its crystal structure as for typical inorganic

materials. A multi-sine excitation from 10 mHz to 1 kHz was used. A flat voltage amplitude of 80 mV was applied at the generator; the resulting current perturbation depended on the potentiostat's selected current range and was monitored during acquisition. Voltage and current were sampled at 5 kHz.

Applied drift and DEIS parameters for metallic sodium

Metallic sodium electrodes were cycled galvanostatically at 120 µA, corresponding to the C/10 rate adopted for the SiCN(O) experiments. The current was alternated between anodic and cathodic directions in 1 h intervals to induce repeated sodium plating and stripping. The multi-sine excitation ranged from 0.02 Hz to 20 kHz, with relative amplitudes selected from prior DEIS measurements on an identical cell geometry. The signals were sampled at 50 kHz to ensure adequate resolution of the highest-frequency components.

Impedance estimation

For all measurements, the instantaneous impedance was estimated offline using the DMFA method. A symmetric Fermi-Dirac function with bandwidth between 0.01 Hz and 0.02 Hz (selected according to the lowest excitation frequency) and order $n = 8$ was employed. Linear baselines were removed from both current and voltage signals prior to filtering to partially eliminate the DC offsets. The resulting time resolution of the impedance is the inverse of the bandwidth: 100 seconds (for graphite and SiCN(O)) or 50 seconds (for metallic sodium).

12.2 Results and dsiscussion

12.2.1 Graphite

General behavior and formation

Prior to the DEIS measurements, the graphite half-cells underwent a standard formation protocol followed by several galvanostatic cycles at 1C to ensure a stable SEI and reproducible electrochemical behavior. The DEIS results reported here correspond to galvanostatic cycling at different C-rates (C/5, 1C, and 2C) within the voltage window of 0–0.8 V vs. Li/Li^+. The theoretical capacity of the electrodes (11 mAh) was used to set the applied current.

Instantaneous impedance at low currents

Figure 12.5 shows the time-resolved impedance during a full C/5 cycle.

A clear evolution of the low-frequency diffusional tail is observed as the voltage crosses the characteristic graphite staging transitions. This behavior is consistent for both lithiation and de-lithiation and parallels the trends previously reported for the LTP system in Chapter 10, suggesting that changes in intercalation-induced porosity, tortuosity, or solid-phase diffusion influence the instantaneous impedance.

Towards the end of lithiation, the impedance magnitude exhibits a sudden and transient reduction, after which it rapidly returns to its typical

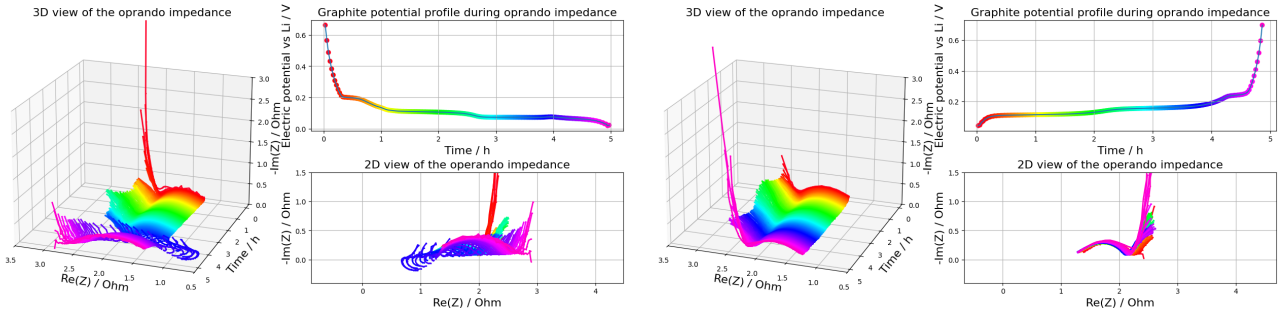


Figure 12.5: Time-resolved impedance of graphite during a C/5 galvanostatic cycle. Dynamic impedance spectra acquired during lithiation (left) and de-lithiation (right), showing systematic evolution of low-frequency features across staging transitions. The first spectrum and lowest-frequency point were excluded.

trajectory. Figure 12.6 highlights this local feature, and the instantaneous impedance spectrum at the minimum value is shown for clarity. At this

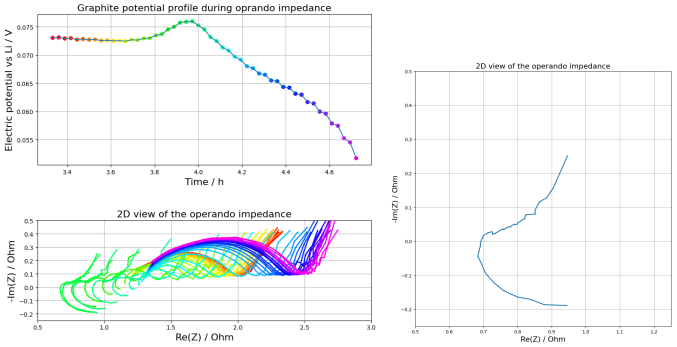


Figure 12.6: Localized transient impedance reduction during graphite lithiation. Left: zoomed view of impedance magnitude highlighting a brief drop near the end of lithiation; right: corresponding instantaneous impedance spectrum at the minimum, illustrating a temporary reduction in charge-transfer resistance.

point the voltage simultaneously increases, implying a reduction in overpotential. Importantly, the behavior is only observed under cathodic current, reinforcing the interpretation that it is not purely geometrical or diffusive in origin. It is not clear whether the origin of such response is due to a structural re-arrangement of the active material or parasitic reaction of the electrolyte or other contaminants in the cell triggered at that specific potential.

Unfortunately, the signal-to-noise ratio worsens when the impedance becomes small relative to the perturbation amplitude, and this limits the precision of the estimation in this portion of the cycle.

Instantaneous impedance at high currents

At higher rates (1C and 2C), the behavior differs markedly. As shown in Figure 12.7, neither the staging-related changes in the low-frequency region nor the transient impedance reduction are observed. At 1C, the system does not reach the third intercalation plateau due to substantial overpotential, and only approximately 70 % of the theoretical capacity is accessed. The high-frequency resistance and charge-transfer semicircle increase systematically with current rate, as is expected in regimes dominated by concentration polarization and interfacial limitations.

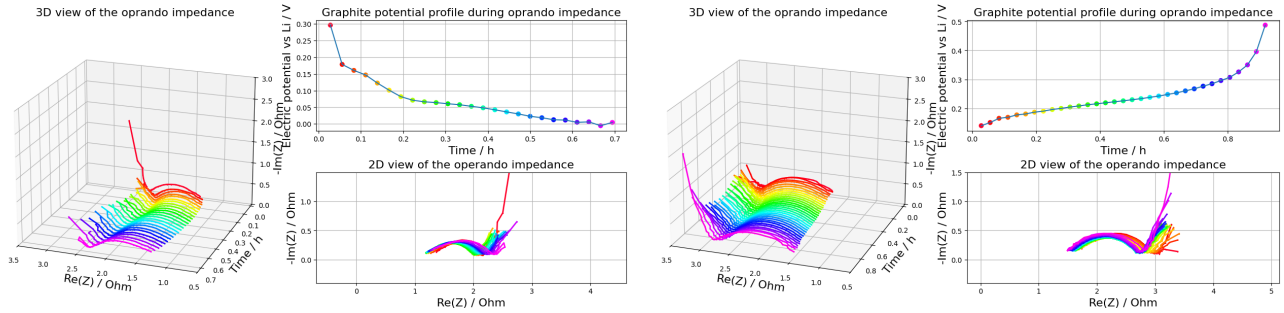


Figure 12.7: Dynamic impedance during 1C galvanostatic lithiation/de-lithiation.

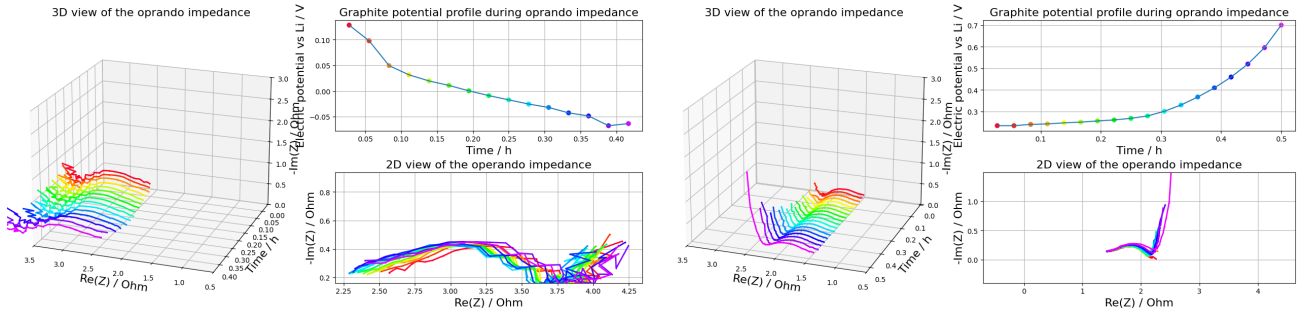


Figure 12.8: Dynamic impedance during 2C galvanostatic lithiation/de-lithiation.

To highlight the effect of current rate on the instantaneous impedance response, Figure 12.9 compares spectra at identical state of charge.

The increase in high-frequency resistance and in the semicircle diameter between 1C and 2C suggests an increase of Ohmic drop due to mass transport limitation and a variation of the charge-transfer activation energy.

At 2C the impedance becomes noisy despite reducing the excitation amplitude, indicating that the nonlinear operating point still limits the measurement quality. The sources of low precision are poor signal-to-noise ratios and strong intermodulation produced by nonlinearity in mass-transport-controlled operational regimes.

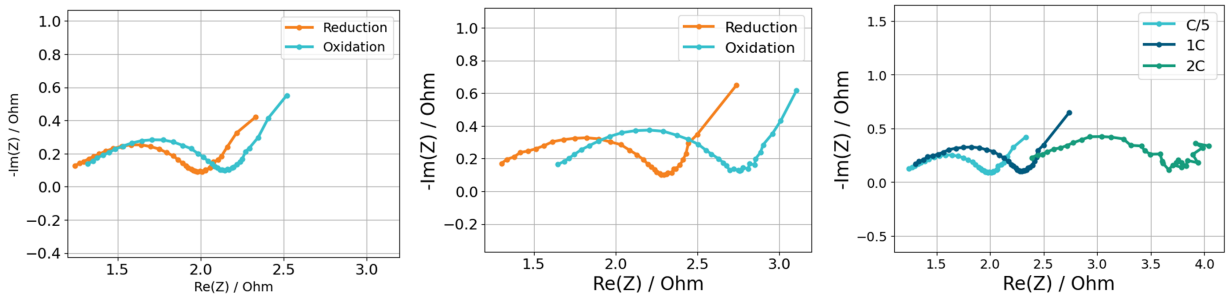


Figure 12.9: Effect of current rate on instantaneous graphite impedance. Comparison of impedance spectra at 50 % state of charge for C/5, 1C, and 2C operation, illustrating the growth of high-frequency resistance and charge-transfer arcs with increasing current.

Attempts to reproduce the experiment with different graphite electrodes were not successful due to the difficulties in fabricating micro-reference electrodes via electrodeposition when SEI is also formed on the reference electrode tip. As a consequence, confirming whether the impedance transient stems from intrinsic electrochemistry or from electrode history effects remains an open question.

12.2.2 SiCN(O)

First cycle: SEI formation

The SiCN(O) electrodes were analyzed with dynamic impedance spectroscopy from pristine state, so the first cycle includes the SEI formation, as shown in Figure 12.10. In the 1.0–0.5 V region, the impedance is large and then decreases sharply, consistent with the establishment of a passivation layer consuming Na^+. In the subsequent oxidation step, the impedance stabilizes, and in the following cycle (Figure 12.11) the electrode behavior is distinct from the formation impedance. The accessible capacity is significantly lower than the nominal value used to calculate the nominal C-rate, confirming that the effective electrochemical capacity differs from the theoretical one.

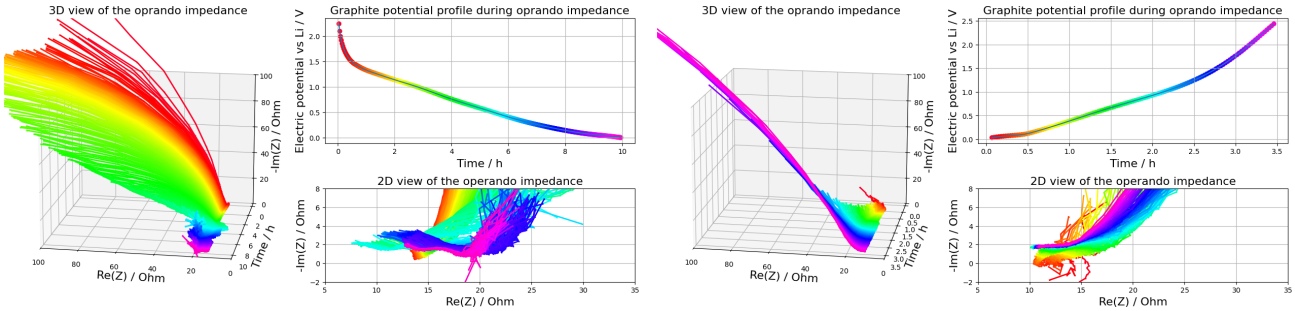


Figure 12.10: First-cycle dynamic impedance of SiCN(O) during SEI formation. Time-resolved impedance during the initial sodiation/de-sodiation cycle at nominal C/10, showing high impedance in the 1.0-0.5 V region associated with SEI formation.

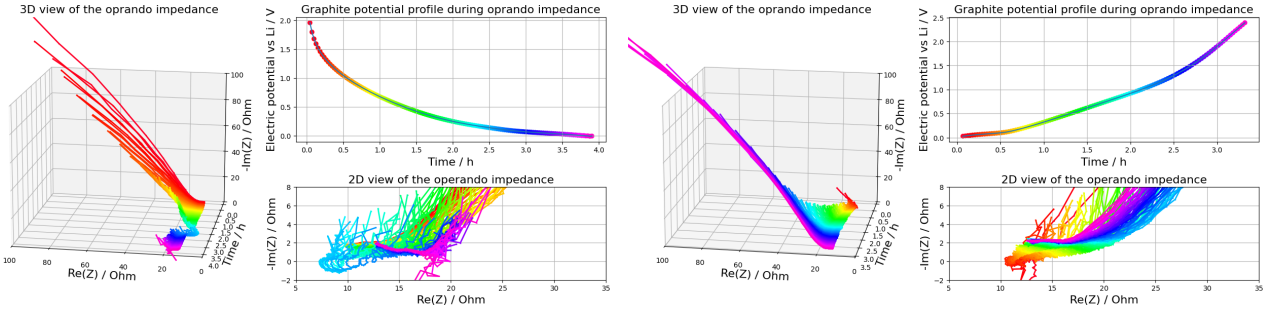


Figure 12.11: Stabilized impedance response of SiCN(O) after formation. Dynamic impedance during the second cycle, demonstrating reduced and more stable impedance compared to the formation cycle.

Transient impedance reduction

A feature analogous to that observed for graphite appears in the second SiCN(O) cycle: a sudden reduction in impedance during the cathodic sweep. Figure 12.12 illustrates this behavior and the associated instantaneous impedance spectrum. The spectral shape resembles that observed in graphite (Figure 12.6), which suggests that the underlying mechanism might be similar, although not necessarily identical. Unlike in graphite, no simultaneous increase of the voltage is seen, which implies that the mechanism may be influenced by the specific surface chemistry of SiCN(O).

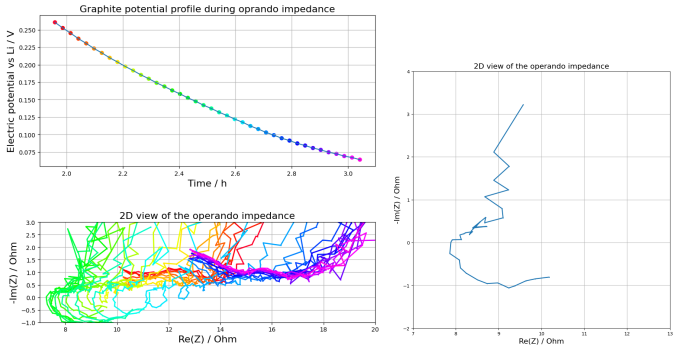


Figure 12.12: Transient impedance reduction in SiCN(O) during sodiation. Zoomed impedance magnitude (left) and corresponding instantaneous spectrum (right) highlighting a brief reduction in impedance during the second cycle, qualitatively similar to the feature observed in graphite.

A repetition of the experiment (Figures 12.13–12.15) revealed similar SEI formation behavior but did not show the transient impedance drop. This difference is plausibly attributable to variations in electrode porosity or local microstructure for the specific electrode. Indeed Dr.-Ing. Melzi D'Eril observed lack of reproducibility of the "activation peak" for reversible electrodeposition [112].

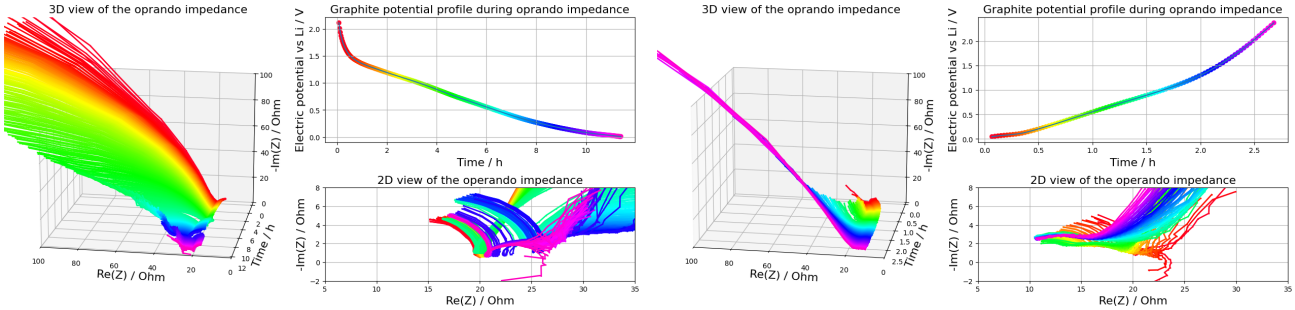


Figure 12.13: First cycle of repeated SiCN(O) experiment. Dynamic impedance during initial sodiation/de-sodiation, showing SEI formation behavior comparable to the first cell.

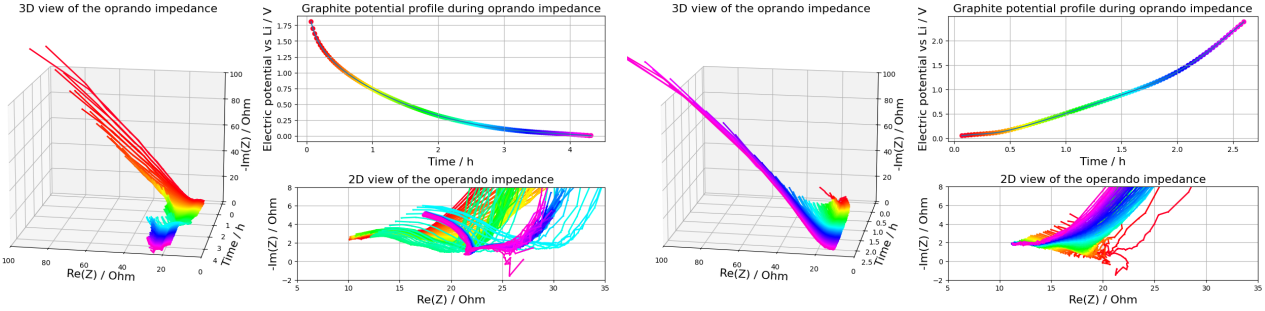


Figure 12.14: Second cycle of repeated SiCN(O) experiment. Time-resolved impedance after formation, without appearance of a transient impedance drop.

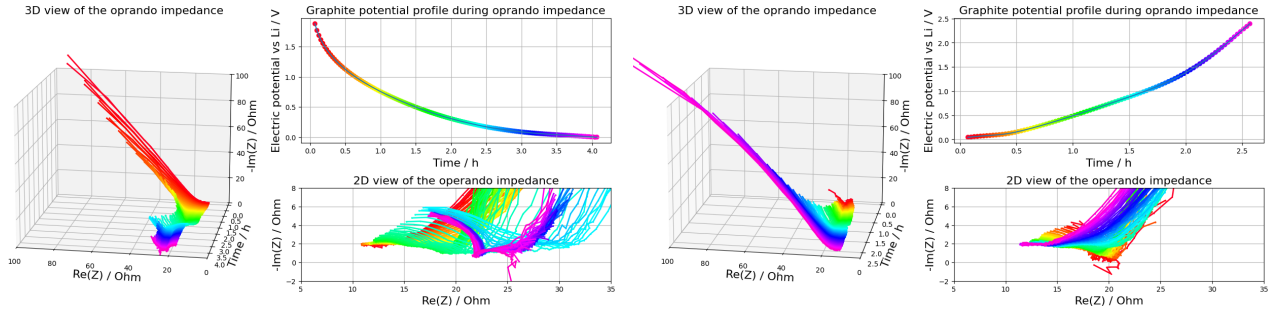


Figure 12.15: Third cycle of repeated SiCN(O) experiment. Continued cycling showing stable impedance evolution and absence of transient features.

Summary of SiCN(O) results

Both SiCN(O) electrodes exhibit SEI formation followed by a stable impedance. Only one electrode displays a transient impedance reduction, making the effect non-reproducible with the present fabrication quality. Further studies with controlled porosity and improved reference-electrode reliability are required to evaluate whether the feature is mechanistically meaningful.

12.2.3 Metallic sodium

Plating behavior

Metallic sodium plating and stripping were investigated to determine whether DEIS could detect specific mechanistic signatures associated with dendrite growth, SEI evolution, or nucleation processes. Prior to the DEIS cycle, a one-hour electrodeposition step at 120 μA was performed to generate an SEI layer. After resting for several days, plating/stripping cycles were recorded.

During plating (Figure 12.16), the impedance displays an inductive-like loop (positive imaginary part at intermediate frequencies), followed by a return toward lower resistance. The overpotential decreases from

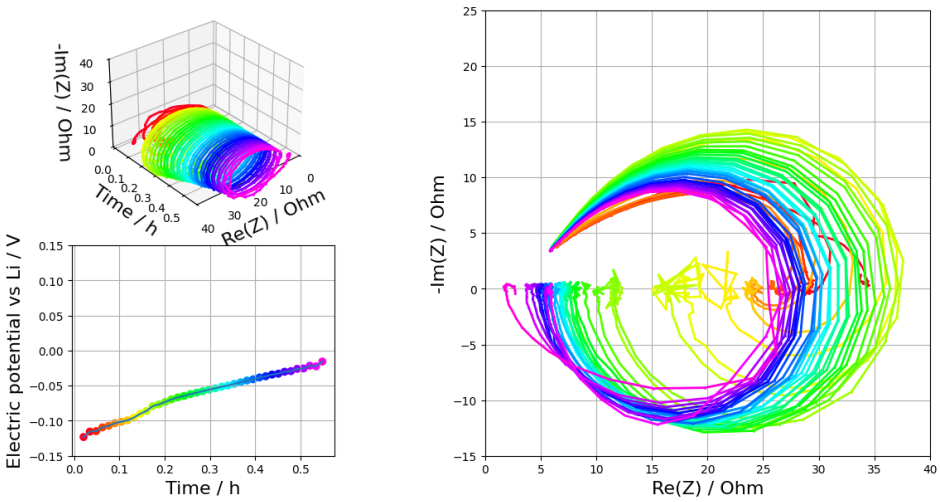


Figure 12.16: Dynamic impedance during initial sodium plating. Time-resolved impedance spectra showing an inductive-like loop at intermediate frequencies during early sodium deposition, concurrent with decreasing overpotential.

approximately −120 mV to −10 mV, consistent with an increase in electrochemically active surface area during deposition. Such positive imaginary features have been reported in systems where a slow intermediate step precedes nucleation [120–123]. The specific mechanism on metallic electrodes covered by SEI remains poorly documented, so the interpretation must remain tentative.

Stripping behavior

During stripping (Figure 12.17), the inductive loop rapidly disappears and the impedance shifts to a more conventional semicircle. The overall magnitude increases with time, indicating a reduction in active surface area as the dendritic or mossy structures formed during plating dissolve. The absence of a diffusion tail across all cycles suggests that mass transport does not limit the reaction under the chosen conditions.

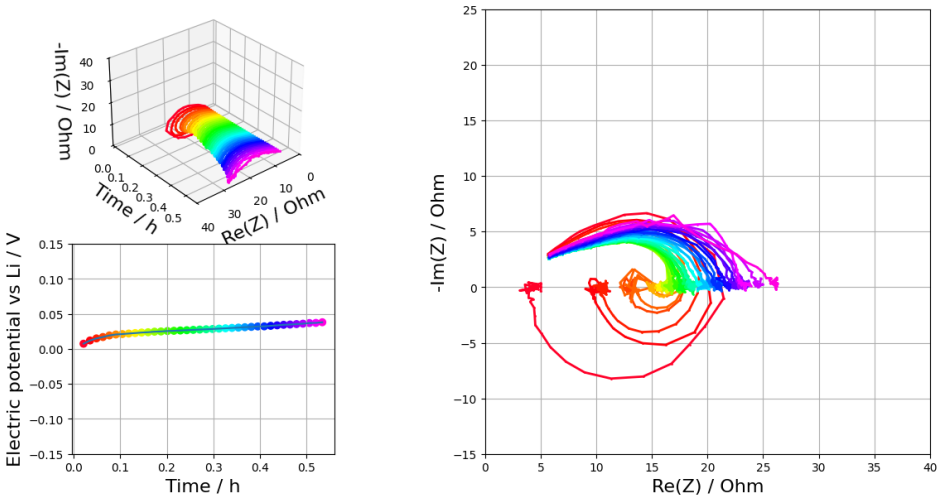


Figure 12.17: Dynamic impedance during sodium stripping. Impedance evolution during anodic dissolution of sodium, characterized by disappearance of inductive features and increasing overall resistance.

Repeated plating

A second plating step (Figure 12.18) shows no inductive loop, despite maintaining similar overpotential (≈50 mV). This suggests that the surface morphology or SEI structure no longer slows down the intermediate step that produced the positive imaginary part. Increasing impedance magnitude again correlates with increasing surface area during deposition.

Summary of sodium results

The sodium experiments highlight that DEIS is sensitive to morphology-dependent kinetic features during metal plating. The inductive-like behavior appears reproducibly in fresh sodium surfaces but disappears after the first stripping cycle, indicating that SEI and surface reconstruction strongly modify the interfacial dynamics. Access to higher frequencies would further clarify whether the origin of these features lies in nucleation kinetics, SEI viscoelasticity, or charge-transfer coupling.

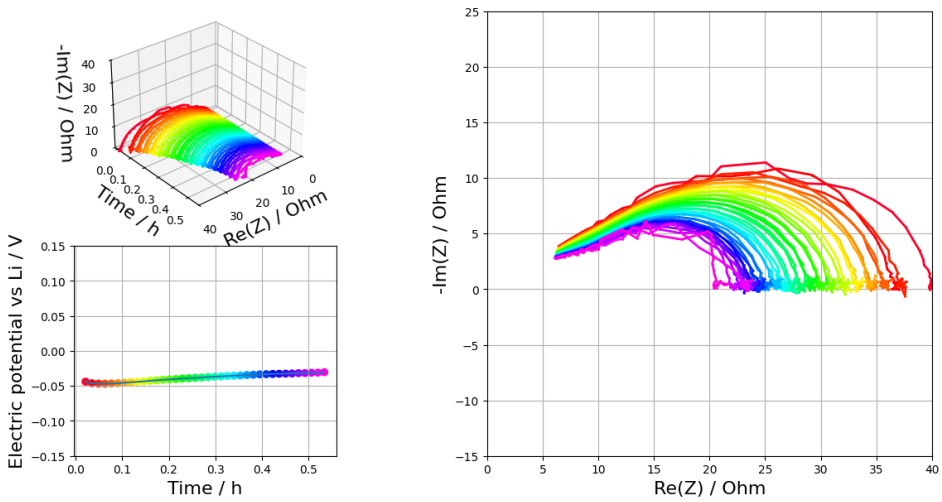


Figure 12.18: Dynamic impedance during repeated sodium plating. Second electrodeposition cycle showing absence of the inductive loop observed during initial plating, indicating modified surface or SEI properties.

12.3 Major limitations and sources of uncertainty

This work was conducted in parallel with the development of the DEIS software platform and experimental automation tools. As a consequence, several technical and methodological limitations affected data completeness, reproducibility, and interpretability. The most critical issues are summarized below.

Incomplete datasets and experimental continuity

Because the measurement software and automated workflow were still under development, several datasets are incomplete. In some cases, full cycling histories are missing, and in others the acquisition was interrupted or partially corrupted. This limits the ability to reconstruct long-term trends in dynamic impedance or to compare electrodes across multiple, controlled operating conditions.

Micro-reference electrode fabrication and reproducibility

A major source of uncertainty originates from the difficulty of fabricating reliable micro-reference electrodes. Despite extensive attempts, it was not possible to produce geometrically consistent references nor to ensure stable potentials over long experiments. As a result:

- several measurements could not be repeated under identical electrochemical conditions;
- the assignment of certain impedance features, particularly those interpreted as transient plating-related signatures, could not be verified across multiple independent cells.

Given the extreme sensitivity of impedance to interfacial heterogeneity, reproducibility requires either identical electrode surfaces or a statistically meaningful number of repeats, neither of which was achievable with the present reference-electrode designs.

Deformation of the electrodes

The micro-reference electrodes employed here had diameters comparable to those of the working electrodes (approximately $\sim 50\,\mu$m). Even when used together with glass-fiber separators, their thickness is enough to alter the planarity of the electrode stack, that might be non-negligible for electrodes with small areas. These geometric distortions may have influenced both the steady-state overpotential and the dynamic impedance response. Future implementations should consider thinner reference electrodes, meshes or grid structures to minimize localized pressure points.

Pouch-cell sealing

Certain pouch cells exhibited ineffective sealing, which was only discovered long after the experiments. In at least one case, part of the pouch was open, allowing air ingress. Oxygen and moisture contamination are well known to trigger parasitic reactions with alkali metals and to alter SEI composition [124], thereby affecting both voltage profiles and impedance.

Given the high reactivity of lithium and sodium systems, even small leaks can significantly distort the interfacial kinetics and compromise the interpretation of dynamic impedance features.

12.4 Summary and conclusions

This chapter demonstrates that dynamic electrochemical impedance spectroscopy can be successfully employed to probe electrochemical processes *in situ* under operating conditions. Specifically, I performed experiments in half-cell configurations with different negative electrodes for lithium-ion and sodium-ion batteries.

Instantaneous impedance of graphite electrodes, isolated through the use of a reference electrode, reflected the staging transitions analogous to those observed for LTP in Chapter 10 and revealed that at C/5 a reproducible transient impedance reduction appeared near the end of lithiation. Despite its unclear origin, it shows how DEIS can be effective to probe processes characteristic of operative conditions. Critically, this impedance feature appeared only during cathodic galvanostatic techniques and disappeared at higher rates (1C, 2C).

For SiCN(O) electrodes, DEIS successfully captured SEI formation during the first cycle, evidenced by high and evolving impedance in the 1.0–0.5 V region. In subsequent cycles, one electrode exhibited a transient impedance reduction qualitatively similar to graphite, while the second did not. The difference of impedance between samples indicates sensitivity to electrode microstructure. For this experiment, an electroplated micro-reference electrode of metallic sodium was successfully reproduced multiple times and demonstrated potential stability for up to dozens of hours, enabling the completion of multiple cycles of galvanostatic techniques even at low C-rates. This is remarkable for the fact that such stable electrodes have not yet been reported in the literature.

Symmetric sodium cell measurements revealed distinctive signatures of metal plating and stripping dynamics. During initial plating, a positive imaginary impedance component (inductive-like loop) appeared at intermediate frequencies while overpotential decreased from $-120\,$mV to $-10\,$mV, suggesting nucleation kinetics or SEI-mediated surface reconstruction as

rate-limiting. This feature did not reappear in subsequent plating, indicating irreversible surface morphology changes. During stripping, impedance evolved toward conventional semicircle behavior with increasing magnitude, consistent with dissolution of high-surface-area dendritic structures.

This chapter lays the foundation to fill the gap between electrode-level characterization (Chapters 10 and 11) and the monitoring of aging of commercial full cells (Chapter 9).

The inability to fully reproduce certain features highlights a fundamental challenge: electrochemical behavior is strongly influenced by preparation history, surface heterogeneity, and local microstructure. DEIS reveals this variability with high sensitivity, underscoring the need for both improved fabrication protocols and statistical approaches to impedance analysis.

Despite significant methodological challenges, particularly micro-reference electrode fabrication for lithium, the measurements reveal qualitatively consistent features that provide valuable mechanistic insights and motivate targeted future investigations. This work demonstrates the capability of DEIS to resolve electrode-specific processes in three-electrode pouch cells under different operating conditions. The characteristic impedance features observed in negative electrodes of two different chemistries suggest an underlying physical origin worthy of a future systematic investigation.

Chapter 13

Conclusions

This thesis addressed a fundamental limitation of electrochemical impedance spectroscopy when applied to realistic energy storage systems: the incompatibility between the classical assumption of stationarity and the intrinsically dynamic nature of batteries during operation. While impedance spectroscopy is widely recognized as a powerful diagnostic tool, its practical use has remained largely confined to equilibrium or quasi-equilibrium conditions, limiting its ability to probe processes that evolve continuously during cycling. The central objective of this work was therefore to establish a complete methodology for broadband impedance characterization under non-stationary conditions.

By combining frequency-domain system identification theory with electrochemical insight and digital signal processing, this thesis demonstrated that meaningful impedance information can be extracted from evolving systems. The work clarified the conceptual distinction between classical impedance and instantaneous impedance of a time-varying system, and showed how multi-frequency excitation and time-resolved estimation techniques allow this estimation in practice.

A complete experimental framework for dynamic electrochemical impedance spectroscopy was developed and validated. This framework integrates optimized multi-sine excitation, synchronized high-speed acquisition, online signal processing, and automated experiment control. Particular attention was devoted to overcoming practical limitations that have historically hindered the adoption of multi-sine-based impedance methods, including waveform memory constraints, data volume management, and long experiment duration. The resulting platform enables continuous impedance acquisition over seven decades of frequency during multi-week battery cycling experiments, while reducing raw data storage requirements by several orders of magnitude through online processing.

The methodology was validated across a range of electrochemical systems and configurations. Measurements on commercial coin cells demonstrated that continuous impedance acquisition during cycling yields results consistent with stationary measurements, while providing access to the evolution of impedance features across state of charge and state of health. Single-electrode studies further showed that the approach is applicable to both Faradaic and non-Faradaic systems, and that low-frequency processes relevant to diffusion and interfacial phenomena can be accessed under non-stationary conditions.

Beyond specific experimental results, this work delivers a set of open-source software tools that make advanced dynamic impedance measure-

ments accessible using standard laboratory equipment. By lowering the technical barrier to long-duration, broadband, non-stationary impedance experiments, this framework enables future studies to focus on physical interpretation, model development, and data-driven analysis.

Overall, this thesis establishes dynamic electrochemical impedance spectroscopy as a practical and scalable tool for operando characterization of batteries and related electrochemical systems. While the present work focused on developing and validating the measurement methodology itself, it lays the foundation for future research aimed at associating instantanous impedance features to degradation mechanisms, control strategies, and predictive models for battery health and performance.

Chapter 14

Outlook and perspectives

The work presented in this thesis establishes a complete experimental and computational framework for broadband electrochemical impedance spectroscopy under non-stationary conditions. While the methodology has been validated on laboratory-scale systems and commercial coin cells, several important directions remain open for extending its applicability, robustness, and technological relevance. This chapter outlines these perspectives, with particular emphasis on signal design, hardware integration, scalability to realistic battery systems, and implications for battery management systems (BMSs).

14.1 Further optimization of excitation signals

In this thesis, multi-sine excitation was selected as a broadband perturbation, but its realization is not unique and can be further optimized at multiple levels. First, the frequency grid itself can be refined. Alternative distributions of frequency components may reduce sensitivity to intermodulation, improve signal-to-noise ratio in targeted bands, or better align with the dominant electrochemical time constants of specific systems. In particular, adaptive or chemistry-specific grids could concentrate the power in frequency regions that are most informative for aging or transport phenomena.

Second, phase optimization strategies remain an open research direction. While stochastic phase randomization with crest-factor minimization proved sufficient for the present work, more advanced deterministic or hybrid optimization algorithms could further reduce peak-to-average power ratios, allowing higher effective excitation amplitudes while preserving linearity. This is especially relevant for systems with very low impedance, where signal amplitudes are intrinsically constrained.

Finally, the interaction between multi-sine design and impedance estimation methods (STFT, DMFA, BLTVA) deserves systematic investigation. Excitation signals optimized jointly with the estimation algorithm may enable improved low-frequency resolution or reduced sensitivity to drift, especially under strongly non-stationary conditions.

14.2 Extension to realistic lithium-ion batteries

The experimental validation presented in this thesis primarily employed small-format commercial coin cells and laboratory-scale electrodes. While

these systems are fine for methodological development and rapid iteration, extending the approach to realistic lithium-ion cells remains a crucial step.

Future work should focus on applying dynamic impedance spectroscopy to commercially relevant chemistries such as NMC/graphite and LFP/graphite cells, with capacities in the range of 1–4 Ah. These cells exhibit significantly lower internal impedance and higher nominal currents, which poses new challenges for both excitation amplitude and measurement precision.

The instrumentation used in this work is limited to a maximum current output of approximately 500 mA, which is insufficient for exciting large-format cells at representative C-rates. This limitation motivates either the integration of a current booster stage or, more fundamentally, the development of dedicated hardware platforms capable of delivering higher currents while maintaining precise small-signal perturbations. This is especially relevant when studying full-cell batteries with only two terminals. Removing the voltage follower of the reference sensing significantly reduces the complexity of the electronics. In fact, commercial battery cyclers are much cheaper than three-electrode potentiostats for analytical purposes.

14.3 Toward embedded hardware

Scaling dynamic impedance spectroscopy to higher currents and larger cells strongly suggests the need for dedicated hardware solutions. FPGA-based platforms, microcontrollers with integrated DSP blocks, or hybrid microcomputer–FPGA architectures represent promising directions.

Such platforms could:

- generate broadband excitation signals with minimal memory usage,
- perform real-time filtering directly on the hardware,
- reduce data throughput,
- enable multi-channel operation for battery modules.

Developing such integrated systems would be a necessary step toward translating the methodology from laboratory research to industrial or embedded applications.

14.4 Precision and signal-to-noise challenges

As battery size increases and impedance decreases, maintaining adequate precision becomes increasingly challenging. For large-format cells operating at high currents, the voltage response to small perturbations may approach the noise floor of the measurement system.

Future work must therefore address:

- the fundamental limits of impedance resolution under realistic operating conditions,
- the trade-off between perturbation amplitude, linearity, and signal-to-noise ratio,
- uncertainty quantification and error propagation in non-stationary impedance estimates.

These considerations are essential for assessing whether dynamic impedance spectroscopy can reliably extract informative features in practical battery systems, or whether its applicability is restricted to specific operating regimes.

14.5 From single cells to battery packs

An important open question concerns the extension of dynamic impedance characterization from single cells to battery packs composed of cells connected in series and parallel. In such configurations, impedance contributions are aggregated and potentially masked by cell-to-cell variability, contact resistances, and thermal gradients.

Future investigations should address whether aging signatures remain identifiable at the pack level, and whether dynamic impedance measurements can detect imbalance, weak cells, or localized degradation. This question is particularly relevant for BMS applications, where measurements are inherently performed at the module or pack scale. The possibility of identify a faulty or less performative battery pack inside the module could effectively extend the life-time of the battery limiting the power drained from that specific pack.

14.6 Temperature sensitivity and multi-physics correlations

Electrochemical impedance is strongly temperature-dependent, and this sensitivity represents both a challenge and an opportunity. On one hand, temperature variations complicate impedance interpretation under dynamic operating conditions. On the other hand, impedance measurements may serve as indirect temperature sensors [125].

Establishing reliable correlations between impedance features and temperature requires systematic experiments under controlled thermal conditions. Future work should therefore combine dynamic impedance spectroscopy with temperature-controlled cycling to disentangle thermal effects from aging-related changes and to assess the feasibility of impedance-based temperature estimation.

14.7 Alternative broadband excitations for BMS applications

Multi-sine excitations, while powerful, rely on relatively expensive arbitrary waveform generators and precise synchronization hardware. For large-scale deployment in BMSs, simpler broadband excitation schemes may be preferable. Pseudo-random binary sequences, maximum-length sequences, or other stochastic excitations represent promising alternatives that can be generated efficiently on low-cost embedded platforms. Evaluating their compatibility with non-stationary impedance estimation, their robustness to noise and nonlinearities, and their computational requirements remains an important topic for future research.

14.8 Compact impedance spectrometers for analytical purposes

While scaling dynamic impedance spectroscopy toward high-current battery systems presents significant engineering challenges, an equally important and complementary direction concerns low-current, three-electrode analytical setups. In recent years, numerous research groups have proposed inexpensive, microcontroller-based potentiostats capable of precision and stability comparable to commercial laboratory instruments for small-signal electrochemical measurements.

These open and modular platforms typically target currents in the microampere to milliampere range and are optimized for fundamental electrochemical studies, including redox kinetics, adsorption processes, and interfacial capacitance measurements. Their open hardware and software architectures make them particularly attractive as foundations for compact and cost-effective dynamic impedance spectrometers.

Integrating dynamic impedance capabilities into microcontroller-based potentiostats would require relatively modest extensions compared to high-current battery platforms. Broadband excitation signals can be generated digitally with minimal power requirements, and frequency-domain filtering can be implemented either on-board or through lightweight external processing. This methodology shares similar hardware requirements to audio processing and voice recognition.

Such platforms would enable compact, portable, and inexpensive dynamic impedance-enabled devices that could be deployed widely in research laboratories. They would also provide an accessible testbed for comparing excitation strategies (multi-sine, pseudo-random, chirp-based) and impedance estimation methods under well-controlled conditions before transferring validated approaches to larger-scale battery systems.

14.9 Perspective

Overall, this thesis provides a methodological and instrumental foundation for dynamic electrochemical impedance spectroscopy under realistic operating conditions. The perspectives outlined above highlight that the primary challenge ahead is no longer conceptual feasibility, but engineering scalability, robustness, and integration. Addressing these challenges will determine whether dynamic impedance spectroscopy can evolve from a powerful laboratory technique into a practical diagnostic tool for next-generation battery systems and battery management systems.

Part IV

Appendixes

Appendix A

Impedance of electrical elements and basic circuits

A.1 Basic circuit elements

The impedance of ideal lumped elements in the frequency domain is defined as follows.

Resistor

$$Z_R = R, \tag{A.1}$$

a purely real impedance, independent of frequency and without phase shift.

Capacitor

$$Z_C = \frac{1}{j\omega C}, \tag{A.2}$$

for which the current leads the voltage by 90°. Capacitors block direct current and exhibit decreasing impedance with increasing frequency.

Inductor

$$Z_L = j\omega L, \tag{A.3}$$

where the voltage leads the current by 90° and the impedance increases linearly with frequency.

A.2 Series and parallel combinations

For elements connected in series, impedances add directly:

$$Z_{\text{series}} = \sum_{k} Z_k. \tag{A.4}$$

For elements connected in parallel, admittances add:

$$\frac{1}{Z_{\text{parallel}}} = \sum_{k} \frac{1}{Z_k}. \tag{A.5}$$

A frequently used kernel for battery interfaces is the Randles circuit, consisting of a solution resistance R_s in series with a parallel combination of charge-transfer resistance R_{ct} and double-layer capacitance C_{dl}, optionally augmented by a diffusion element.

A.3 Equivalent-circuit signatures

The most common impedance features encountered in electrochemical systems can be interpreted as follows.

Ohmic resistance The high-frequency intercept on the real axis corresponds to the uncompensated ohmic resistance R_s, which includes contributions from electrolyte, separator, and electrical connections.

Charge-transfer process A charge-transfer resistance R_{ct} in parallel with a double-layer capacitance C_{dl} (or constant-phase element) produces a semicircle in the Nyquist plot, with a characteristic frequency

$$\omega \approx \frac{1}{R_{\mathrm{ct}} C_{\mathrm{dl}}}.$$

Diffusion Semi-infinite diffusion is represented by a Warburg impedance:

$$Z_W = \frac{\sigma}{\sqrt{j\omega}}, \tag{A.6}$$

which appears as a 45° line in the Nyquist plot.

For finite-length diffusion, the impedance bends toward a capacitive response at low frequency:

$$Z_{W,\mathrm{FL}} = \frac{\sigma}{\sqrt{j\omega}} \tanh\!\left(\sqrt{j\omega \tau_D}\right), \tag{A.7}$$

or equivalently with a coth formulation, depending on the boundary conditions.

A.4 Impedance representations

The impedance magnitude and phase are defined as

$$|Z(\omega)| = \sqrt{(\Re Z)^2 + (\Im Z)^2}, \qquad \phi(\omega) = \arctan\!\left(\frac{\Im Z}{\Re Z}\right). \tag{A.8}$$

In a Nyquist representation, $-\Im(Z)$ is plotted versus $\Re(Z)$, with frequency as an implicit parameter. In a Bode representation, both magnitude and phase are plotted explicitly as functions of frequency, which is particularly useful for identifying overlapping processes through their characteristic time constants.

Appendix B

Literature chronological overview

With the rapid development of digital electronics and computational capabilities in the late 1980s and early 1990s, several research groups revisited the problem of measuring electrochemical impedance under non-stationary conditions. A common experimental architecture progressively emerged, consisting of a potentiostat or galvanostat, an arbitrary waveform generator, and a digital data-acquisition system. Within this framework, broadband excitation signals and time-frequency analysis enabled the estimation of instantaneous or time-resolved impedance during electrochemical drifts.

Early multi-sine implementations (1992–1997)

The first systematic works on time-resolved impedance spectroscopy using broadband perturbations were reported by Popkirov and Schindler in the early 1990s. In 1992, they introduced a custom-built potentiostat capable of operating in two distinct frequency ranges (100 kHz–100 Hz and 0.76 Hz–0.76 mHz) and demonstrated the feasibility of impedance measurements under non-stationary conditions using multi-sine excitation [126]. Subsequent studies established that multi-sine signals composed of dozens of frequency components provide superior signal-to-noise ratio compared with alternative broadband waveforms [42].

A crucial contribution of this work was the validation of impedance estimation in the presence of nonlinearities and system non-stationarity [127, 128]. These methodological developments were applied to a wide range of electrochemical systems, including electropolymerization [129], redox processes in conducting polymers [44, 55, 130, 131], silver dissolution [132, 133], and the benzophenone/benzophenone anion radical redox couple in various solvents [52]. Both potential-sweep and galvanostatic drifts were employed, with multi-sine excitation typically spanning frequencies from 1 Hz to 31 kHz.

High-density multi-sine and experimental optimization (2000–2005)

Between 2000 and 2005, Roy, Garland, Pettit, and co-workers significantly extended the multi-sine approach by introducing high-density per-

turbations composed of up to 133 sinusoidal components. Their excitation design employed only odd harmonics, random phase distributions, and equal-amplitude components [37]. A detailed discussion of experimental constraints, parameter selection, and intrinsic limitations of the technique was provided in subsequent methodological studies [134].

This framework was applied to investigate a broad range of surface and interfacial processes on gold and other metals, including bismuth deposition [135], indium deposition with emphasis on co-adsorption phenomena [56], methanol electro-oxidation [57], redox kinetics influenced by co-adsorption [53], and the chemical reduction of nickel [136]. Additional studies focused on surface planarization and reaction pathways for thallium [137–139], copper [140–142], and silver [57]. Time-resolved impedance was also coupled with optical diagnostics to study dynamic surface processes.

Dynamic impedance via short-time Fourier analysis (2003–2020)

Starting in 2003, Darowicki, Słępski, and collaborators developed a parallel line of research focused on estimating instantaneous impedance using short-time Fourier transform (STFT) techniques. Early implementations employed exponential chirp excitation signals in the frequency range 1320–144 Hz [143, 144], later transitioning to multisine perturbations [145]. The group proposed multiple experimental configurations for dynamic impedance estimation [38].

This methodology was applied extensively to corrosion science, including the study of passive layers and pitting corrosion on stainless steels [59, 146–154], metal alloys [61, 155–157], and stress corrosion cracking [158]. Additional applications include conductive polymers [35, 159], hydrogen storage alloys, nickel-metal hydride batteries, and coating adhesion studies [160].

The frequency range in these studies typically spanned 3 Hz to 20 kHz, later extended down to approximately 30 mHz. Dynamic impedance was also combined with atomic force microscopy [161, 162] and quartz crystal microbalance techniques [163]. Further applications include fuel cells [63, 164, 165] and redox-flow batteries [64].

System identification and variance-aware impedance estimation (2004–2018)

In parallel, Schoukens, Pintelon, and co-workers introduced a rigorous system-identification framework to electrochemical impedance spectroscopy. Beginning in 2004, they applied frequency-domain identification theory to impedance measurements on corroding metals using multi-sine excitation with low-frequency components down to 10 mHz [39, 166, 167]. A detailed comparison between multi-sine and stepped-sine excitation was provided to validate measurement accuracy [168].

In 2009, the advantages of using only odd random-phase harmonics were demonstrated using multi-sine signals with up to 220 components spanning 10 mHz to 2 kHz [43]. Particular emphasis was placed on quantitative variance estimation and on separating errors arising from noise, nonlinear-

ity, and non-stationarity. Earlier foundational work on excitation signal design and stepped-sine versus broadband methods [30, 169] underpinned this approach.

Applications included corrosion studies of copper [170], aluminum and stainless steel [60, 171–173], adsorption phenomena on metals and oxides [174–176], self-assembled monolayers [177, 178], polymeric coatings [62, 179–182], ferri/ferrocyanide kinetics [183], biosensors [184], and alloy protection [185].

Dynamic impedance coupled with cyclic voltammetry

Sacci and Harrington introduced a multi-sine excitation spanning 1 Hz to 1.3 kHz, with selectively removed harmonics to suppress intermodulation effects, superimposed on cyclic voltammetry [40]. The multi-sine amplitude decreased by a factor of two every two decades, following the design principles introduced by Popkirov. This approach was applied to electrocatalytic reactions [50, 58, 186, 187] and later extended to droplet interface bilayers [188].

Analog separation of drift and perturbation

Pajkossy and Mészáros proposed an alternative architecture in which the potentiostat was custom-built to perform analog separation of the low-frequency drift and high-frequency perturbation [41]. The excitation consisted of a sweeping voltage combined with a multi-sine signal using odd prime frequencies in the range 18.3 Hz to 17.8 kHz. Extensive theoretical analysis was provided on the coupling between cyclic voltammetry and impedance for quasi-reversible redox systems [51, 54, 189].

Early embedded implementations

Earlier efforts toward compact implementations include the work of Ivaska and co-workers, who adapted Smith's technique into a smaller hardware platform [190]. Hazi, Bond, and colleagues later extended this concept using microcomputer-based instrumentation capable of admittance and impedance measurements in the range 50 Hz to 50 kHz during staircase voltammetry [36, 191].

Appendix C

Building auxiliary connectors

Each channel of BioLogic workstation (not valid for battery cyclers) has a 9-pin port for input and output for the connection of external devices. The manual provides all the details on the pins positioning such that one can build their own cable. As material we gathered a 9-pin connector, BNC connector and coaxial cable. As instrumentation was needed only a solder with tin soldering wire and sharp scissors to cut and strip the cable.
I needed to access the pins X, Y and Z, as in figure XX. I then used 3 cables, soldering the core to the proper pin and the shell together to the ground. Finally I connected the other end of the cable to the BNC connecter and close the encapsulation of the 9-pin connector after testing all the terminal points with a multimeter. Figure C.1 shows the soldering on the 9-pin connector.

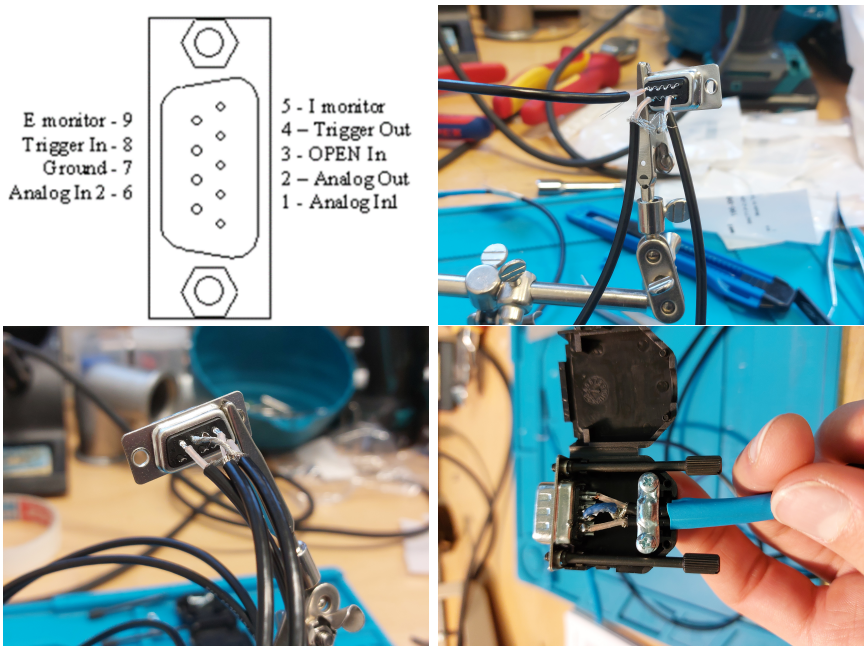


Figure C.1: In order: BioLogic 9 pin port schematics; soldering of cable for Auxiliary In 2, voltage out and current out; soldering of the common ground for all coaxial cables; final construction before closing the enclosure.

Appendix D

Multi-sine crest factor minimization

Multiple approaches have been proposed in the field to minimize the crest factor and each is a trade-off between speed and accuracy (find the absolute minimum) [192]. The right method strongly depends on the type of problem and on the following characteristics: distribution of frequency (logarithmic or linear), number of points per decades, number of frequencies in the waveform and relative amplitude (flat, linear, logarithmic, arbitrary).

The first to deal with the problem was Schroeder in 1990 whom proposed the analytical formula that carries his name:

$$\varphi_m = \varphi_0 - 2\pi \sum_{n=0}^{m-1} (m-n) \frac{|a_m|^2}{\sum_{k=0}^{M-1} |a_k|^2}.$$

Since it is fast to calculate, it is often used as both reference value and initial parameters for a following iterative optimization. Newman and Ojarand took the work forward with analytical formulas but without significant advantages [193].

As an alternative, randomization of the phases offers an equal or better crest factor but the computational time increases significantly with the number of frequencies.

Iterative algorithms achieve much better results. They belong to two families. One is based on the "clipping" method of the DFT peaks of the multi sine until the crest factor reaches a constant value [194]. The other consists in a mathematical optimization using metaheuristic approaches that perform better than derivative methods for such big parameters space (ex simulated annealing and genetic algorithm) [195].

Recently Kallel et al. proposed a mixed method, though it gives better results than the analytical formula for small multi-sine of 3-4 decades [196].

Appendix E

Creating Python script for dynamic impedance experiments

After importing the relevant libraries the single instruments are separately instantiated and configured. Then the objects are injected into another objects where the control loop is implemented and the instruments are synchronized.

E.1 Initializing the potentiostat

```
potentiostat_ip = "172.28.26.10"
eclabsdk_binary_path = "C:/EC-Lab Development Package/EC-Lab Development Package/"
potentiostat_channel = 1
config = ChannelConfig(
    live_plot=True,
    external_control=True,
    record_ece = False,
    record_charge = False,
)

# chrono-potentiometry technique
current = 0
duration = 102
vs_init = True
nb_steps = 0
record_dt = 1
record_dE = 5
repeat = 0
i_range = I_RANGE.I_RANGE_10mA
e_range = E_RANGE.E_RANGE_5V
bandwidth = BANDWIDTH.BW_9

device = BiologicDevice(potentiostat_ip, binary_path=eclabsdk_binary_path)
ca = ChronoAmperometry(
    device=device,
    voltage=voltage,
    duration=duration,
    vs_init=vs_init,
    nb_steps=nb_steps,
    record_dt=record_dt,
    record_dI=record_dI,
    repeat=repeat,
```

```
    e_range=e_range,
    i_range=i_range,
    bandwidth=bandwidth,
    xctr = generate_xctr_param(config),
)
ca.make_technique()
sequence = [
    ca,
]
writer = FileWriter(
    file_dir=Path(saving_directory),
    experiment_name=experiment_name,
)
channel1 = Channel(
    device,
    potentiostat_channel,
    writer=writer,
    config=config,
)
channel1.load_sequence(sequence)
```

E.2 Initializing the waveform generator

```
trueform_address = 'USB0::0x0957::0x4B07::MY59000581::0::INSTR'
waveforms_directory = 'E:/multisine_collection/'
waveform_name = '2505151210multisine_splitted_quasi-log_100kHz-10mHz_8ptd_flat_norm_random_phases'
multisine_high_path = waveforms_directory+waveform_name+'/high_band/'
multisine_high_name = 'ms_high'
multisine_low_path = waveforms_directory+waveform_name+'/low_band/'
multisine_low_name = 'ms_low'
amplitude_mulitsine = 0.1

awg_ch1 = TrueFormAWG(trueform_address, 1)
awg_ch1.clear_ch_mem()
multisine_high = import_awg_txt(multisine_high_path + "waveform.txt")
awg_ch1.load_awf(multisine_high_name, multisine_high)
awg_ch2 = TrueFormAWG(trueform_address, 2)
awg_ch2.clear_ch_mem()
multisine_low = import_awg_txt(multisine_low_path + "waveform.txt")
awg_ch2.load_awf(multisine_low_name, multisine_low)
awg_ch2.avalable_memory()
awg_ch1.set_offset(0)
awg_ch2.set_offset(0)
sample_rate_multisine_high = json.load(open(multisine_high_path + "waveform_metadata.json"))["Sample frequency / Hz"]
sample_rate_multisine_low = json.load(open(multisine_low_path + "waveform_metadata.json"))["Sample frequency / Hz"]
awg_ch1.set_sample_rate(sample_rate_multisine_high)
awg_ch2.set_sample_rate(sample_rate_multisine_low)
multisine_gen_high = MultisineGenerator(
    awg_ch1,
    [waveform_name]*3,
    [0, 2, 4],
    ['ms_high'] * 3,
    [sample_rate_multisine_high] * 3,
    [amplitude_galvano, amplitude_potentio, amplitude_galvano],
```

```
)
multisine_gen_low = MultisineGenerator(
    awg_ch2,
    [waveform_name]*3,
    [0, 2, 4],
    ['ms_low'] * 3,
    [sample_rate_multisine_low] * 3,
    [amplitude_galvano, amplitude_potentio, amplitude_galvano
    ],
)
multisine_gen = MultisineGeneratorCombined(
    multisine_gen_high,
    multisine_gen_low,
    [waveform_name]*3,
)
```

E.3 Initializing the scope

```
pico_samples_total = 300000000
pico_capture_size = pico_samples_total
pico_sampling_time = 4
pico_sampling_time_scale = 'PS5000A_US'
pico_bandwidth_chA = 'PS5000A_5V'
pico_bandwidth_chB = 'PS5000A_500MV'
pico_current_conversion_factor = 0.01

pico = Picoscope5000a('PS5000A_DR_14BIT')
pico.set_pico(
    pico_capture_size,
    pico_samples_total,
    pico_sampling_time,
    pico_sampling_time_scale,
    saving_path,
)
pico.set_channel(
    'PS5000A_CHANNEL_A',
    pico_bandwidth_chA,
)
pico.set_channel(
    'PS5000A_CHANNEL_B',
    pico_bandwidth_chB,
    pico_current_conversion_factor,
)
```

E.4 Dependencies injections

```
deischannel = DEISchannel(
    potentiostat = channel1,
    pico = pico,
    awg = multisine_gen,
    frequencies=frequencies,
)
deischannel.conditions.extend(cccv_software_conditions)

deischannel.start()
```

E.5 Initializing online calculator

```
frequencies = np.array(
    json.load(
        open(multisine_low_path + "waveform_metadata.json")
        )["Frequencies / Hz"]
)
```

```
frequencies = np.append(
    frequencies,
    json.load(open(multisine_high_path + "waveform_metadata.
    json")
              )["Frequencies / Hz"],
)
frequencies_fftshift = frequencies[20:]
sampling_time = 4e-6
time_window = 10

# Decimation
filter_cutoff = 9
filter_order = 25
time_experiment = 100
sampling_frequency = 2.5e5
resampling_frequency = 50
ds_factor = int(sampling_frequency / resampling_frequency)
buffer_size = int(time_experiment * resampling_frequency)

# Initialize the method for multi-frequency analysis
block_size =  int(time_window / sampling_time)
high_z_calculator = MultiFrequencyAnalysis(
    frequencies_fftshift,
    np.zeros(block_size),
    np.zeros(block_size),
    sampling_time,
)
high_z_calculator.compute_freq_axis()

# Initialize the bock calculator
fermi_dirac_low_pass = fermi_dirac_filter(
    high_z_calculator.freq_axis,
    0,
    filter_cutoff,
    filter_order
   )
block_calculator = BlockCalculator(
    input_size = block_size,
    sampling_time = sampling_time,
    high_z_calculator = high_z_calculator,
    lp_filter = fermi_dirac_low_pass,
    ds_factor = ds_factor,
    buffer_size =buffer_size,
)

# Inject block calculator into PicoCalculator

pico_calculator = PicoCalculator(
    pico = pico,
    block_calculator= block_calculator,
)
```

E.6 Dependencies injections for online calculation

```
deischannel = DEISchannel(
    potentiostat = channel1,
    pico = pico_calculator,
    awg = multisine_gen,
    frequencies=frequencies,
)
deischannel.conditions.extend(cccv_software_conditions)

deischannel.start()
```

Appendix F

Auxiliary data of coin cells

In the experiment presented in Chapter 9, nine CC-CV dynamic impedance spectroscopy experiments were performed for independent but identical cells. This chapter reports the results of all the cells, expect for the first already present in the main text, to showcase the reproducibility. The cells have the following code names:

Coin cell names
PanCoin2025_009
PanCoin2025_010
PanCoin2025_011
PanCoin2025_012
PanCoin2025_013
PanCoin2025_014
PanCoin2025_015
PanCoin2025_016
PanCoin2025_017

F.1 Cell 10

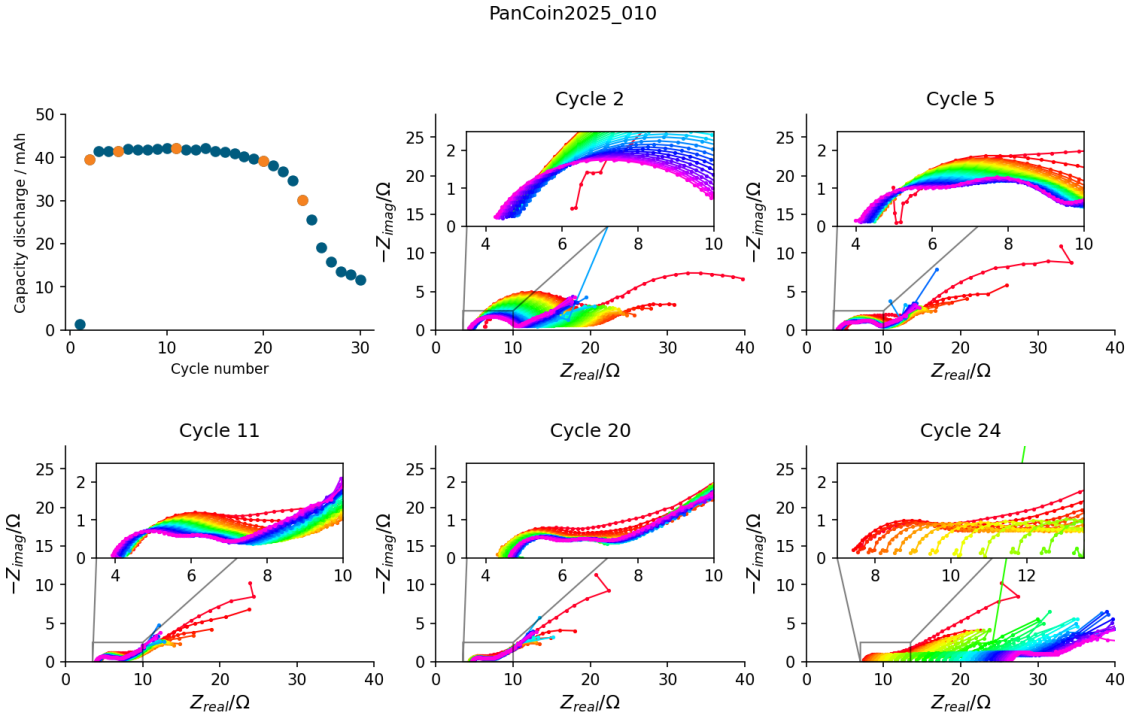


Figure F.1: Dynamic impedance spectra during charge at selected states of health for cell 10. Plotted one spectra every 10.

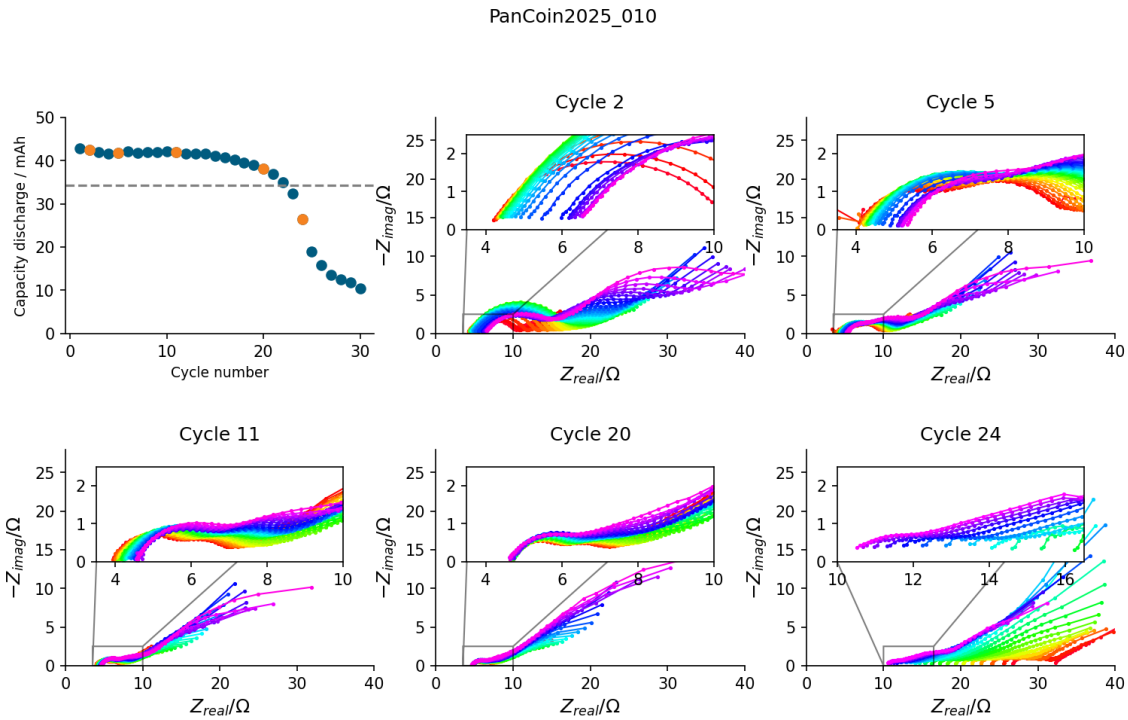


Figure F.2: Dynamic impedance spectra during discharge at selected states of health for cell 10. Plotted one spectra every 10.

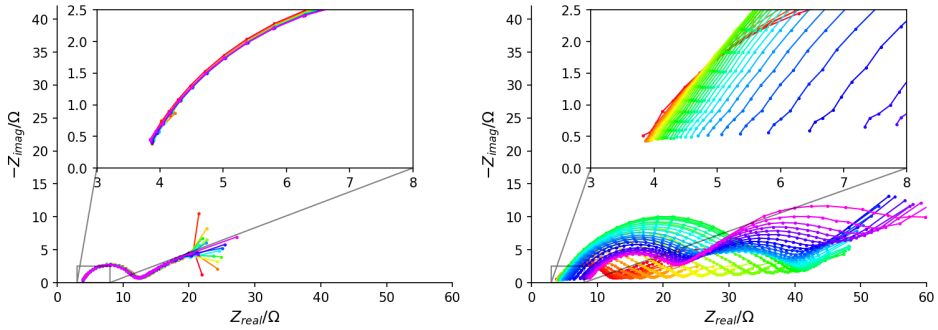


Figure F.3: Impedance spectra during the first cycle for cell 10. Left: all charge spectra; right: discharge spectra with one spectrum every ten cycles.

F.2 Cell 11

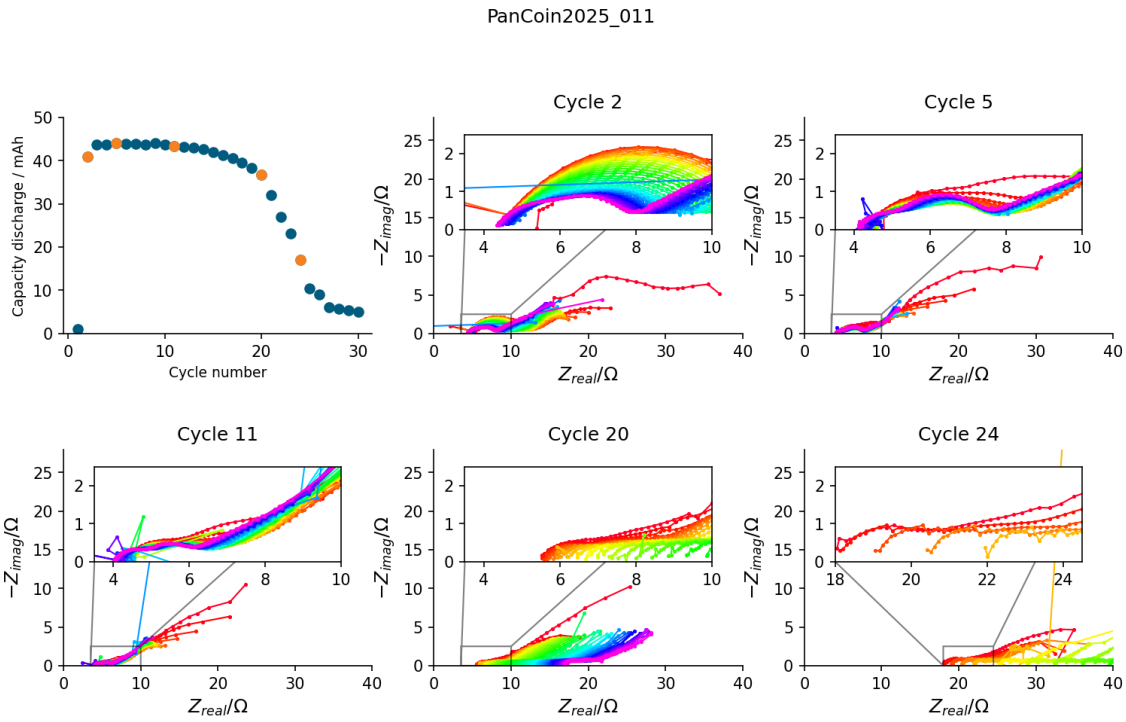


Figure F.4: Dynamic impedance spectra during charge at selected states of health for cell 11. Plotted one spectra every 10.

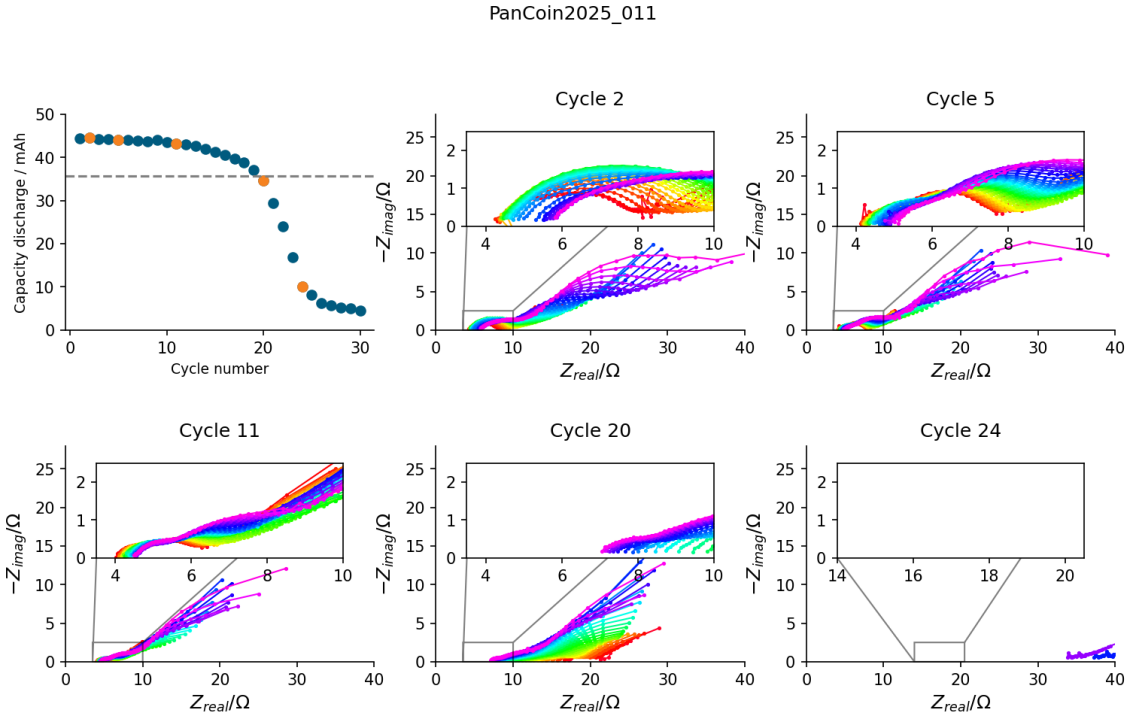


Figure F.5: Dynamic impedance spectra during discharge at selected states of health for cell 11. Plotted one spectra every 10.

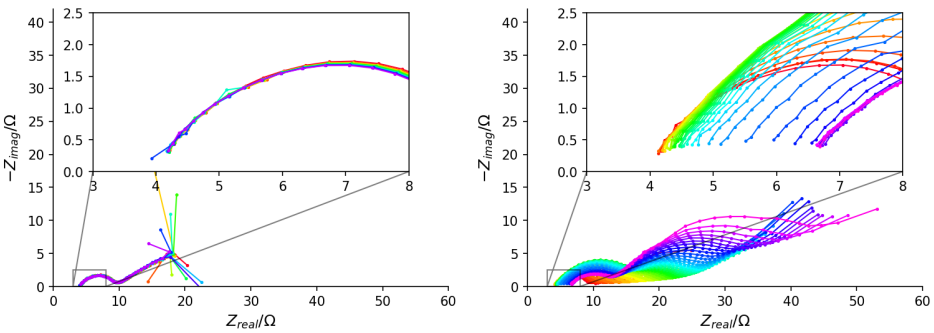


Figure F.6: Impedance spectra during the first cycle for cell 11. Left: all charge spectra; right: discharge spectra with one spectrum every ten cycles.

F.3 Cell 12

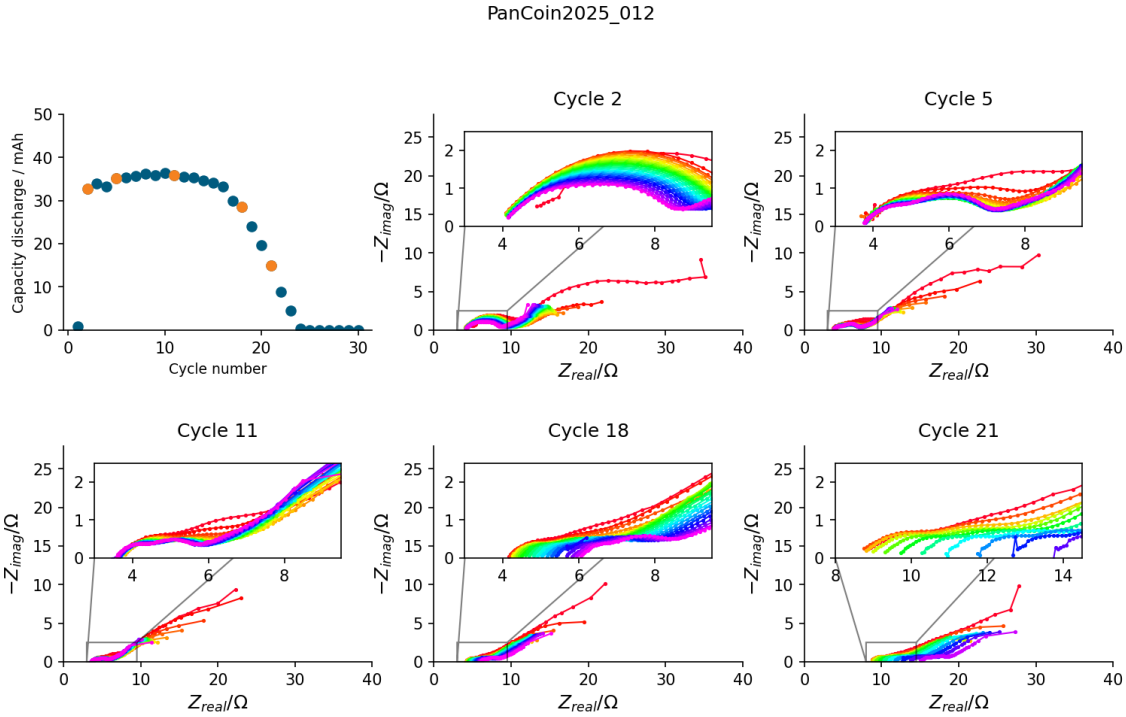


Figure F.7: Dynamic impedance spectra during charge at selected states of health for cell 12. Plotted one spectra every 10.

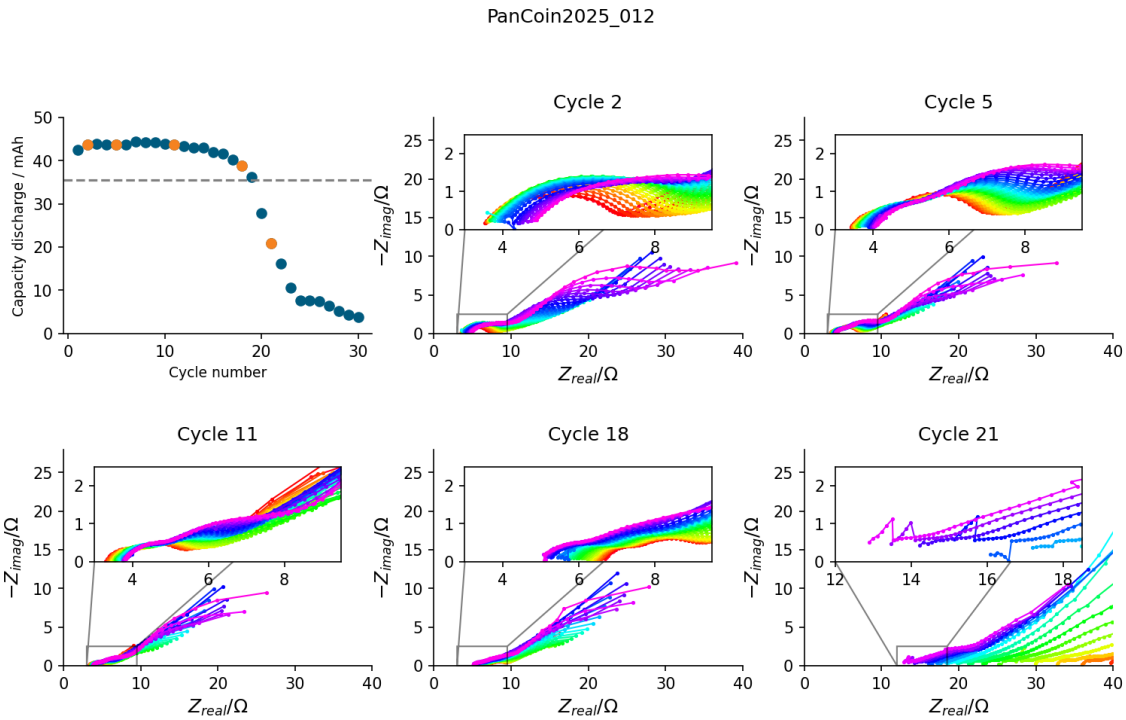


Figure F.8: Dynamic impedance spectra during discharge at selected states of health for cell 12. Plotted one spectra every 10.

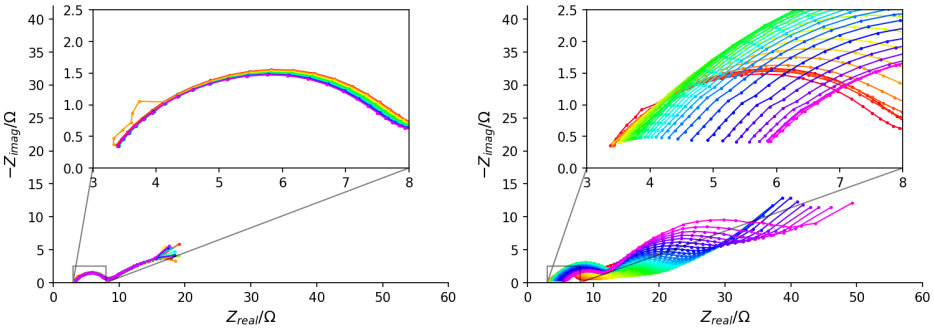


Figure F.9: Impedance spectra during the first cycle for cell 12. Left: all charge spectra; right: discharge spectra with one spectrum every ten cycles.

F.4 Cell 13

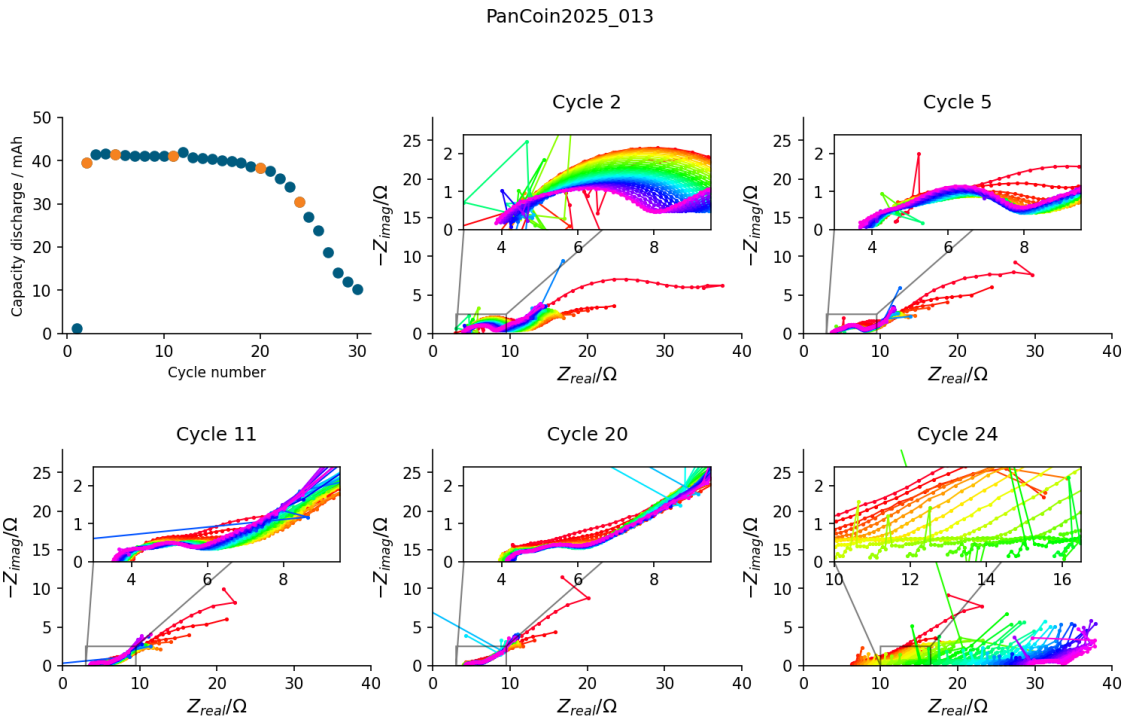


Figure F.10: Dynamic impedance spectra during charge at selected states of health for cell 13. Plotted one spectra every 10.

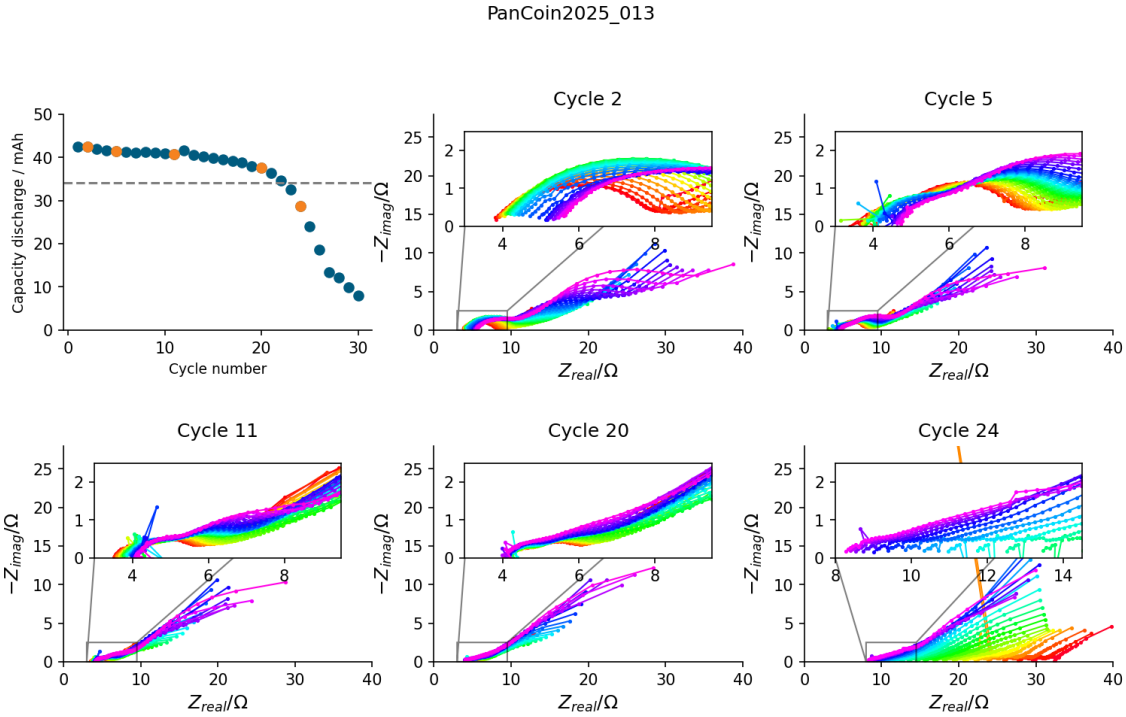


Figure F.11: Dynamic impedance spectra during discharge at selected states of health for cell 13. Plotted one spectra every 10.

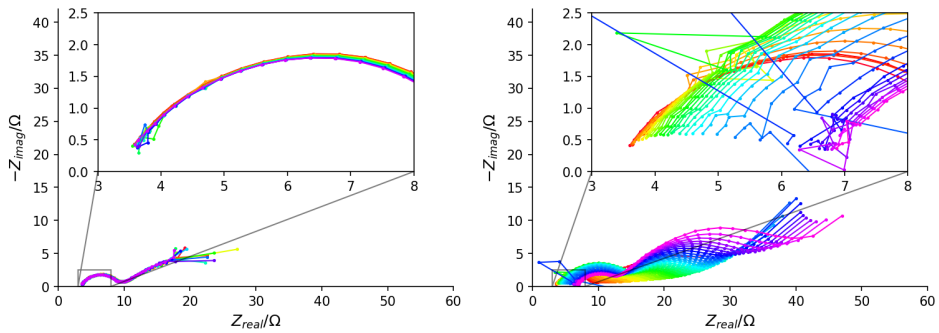


Figure F.12: Impedance spectra during the first cycle for cell 13. Left: all charge spectra; right: discharge spectra with one spectrum every ten cycles.

F.5 Cell 14

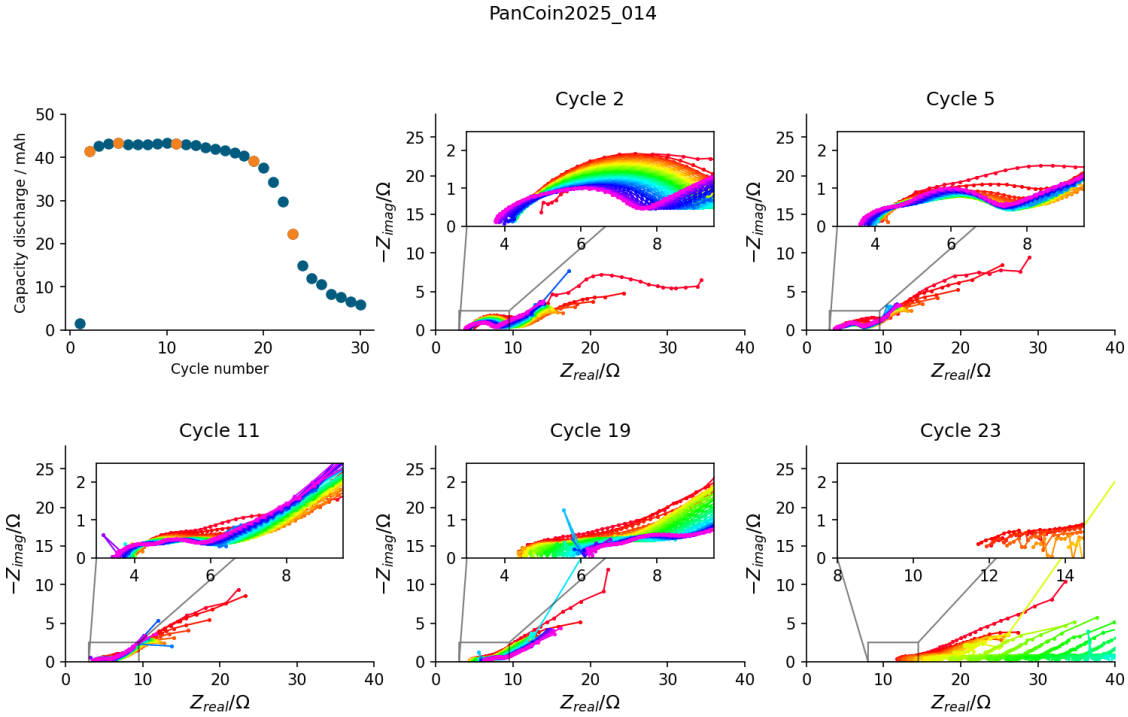


Figure F.13: Dynamic impedance spectra during charge at selected states of health for cell 14. Plotted one spectra every 10.

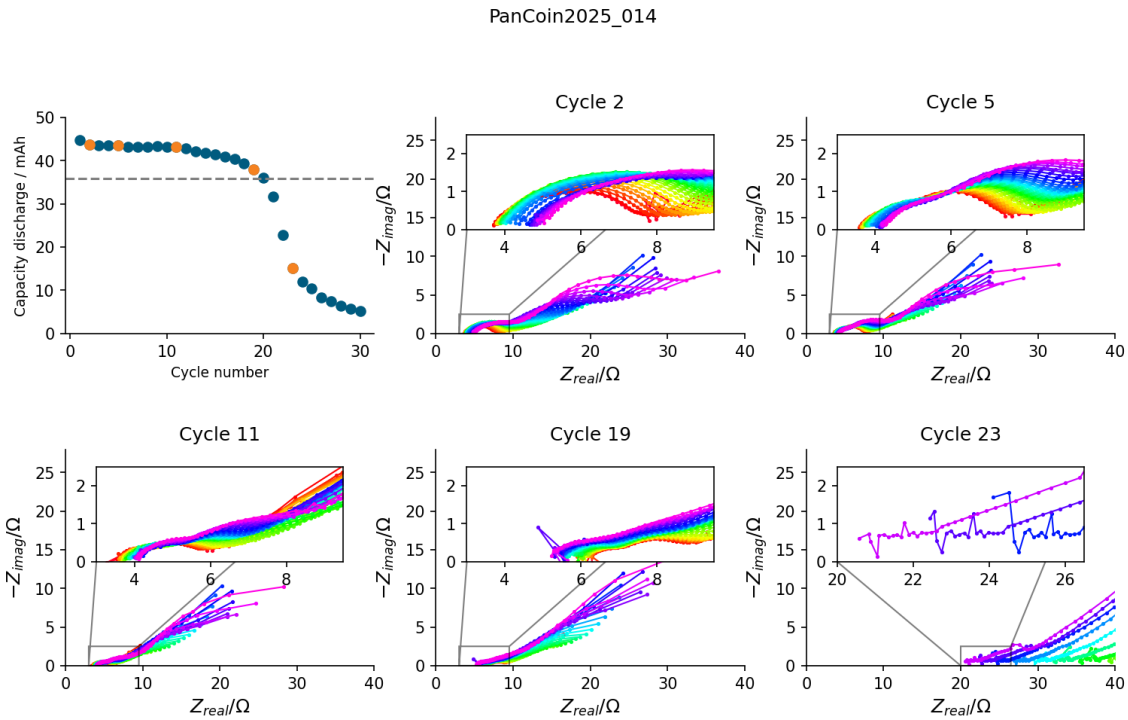


Figure F.14: Dynamic impedance spectra during discharge at selected states of health for cell 14. Plotted one spectra every 10.

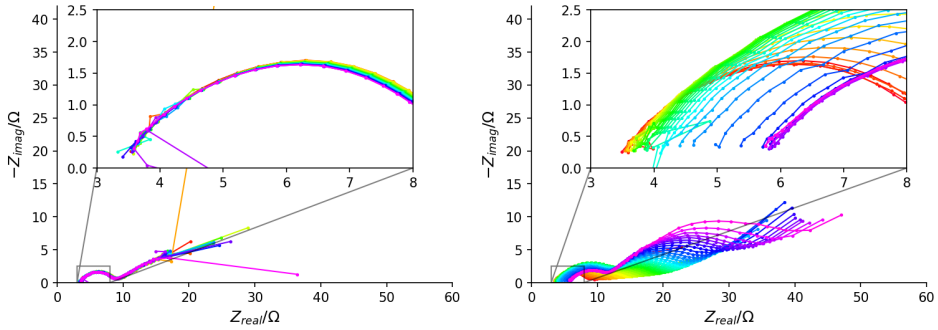


Figure F.15: Impedance spectra during the first cycle for cell 14. Left: all charge spectra; right: discharge spectra with one spectrum every ten cycles.

F.6 Cell 15

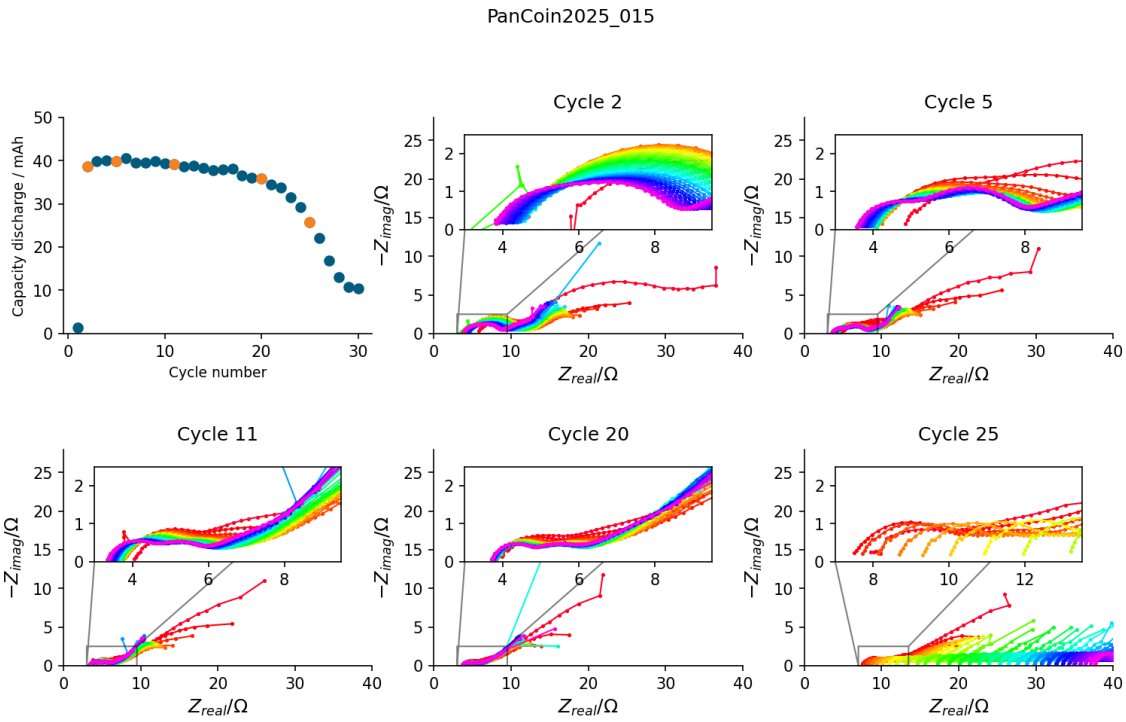


Figure F.16: Dynamic impedance spectra during charge at selected states of health for cell 15. Plotted one spectra every 10.

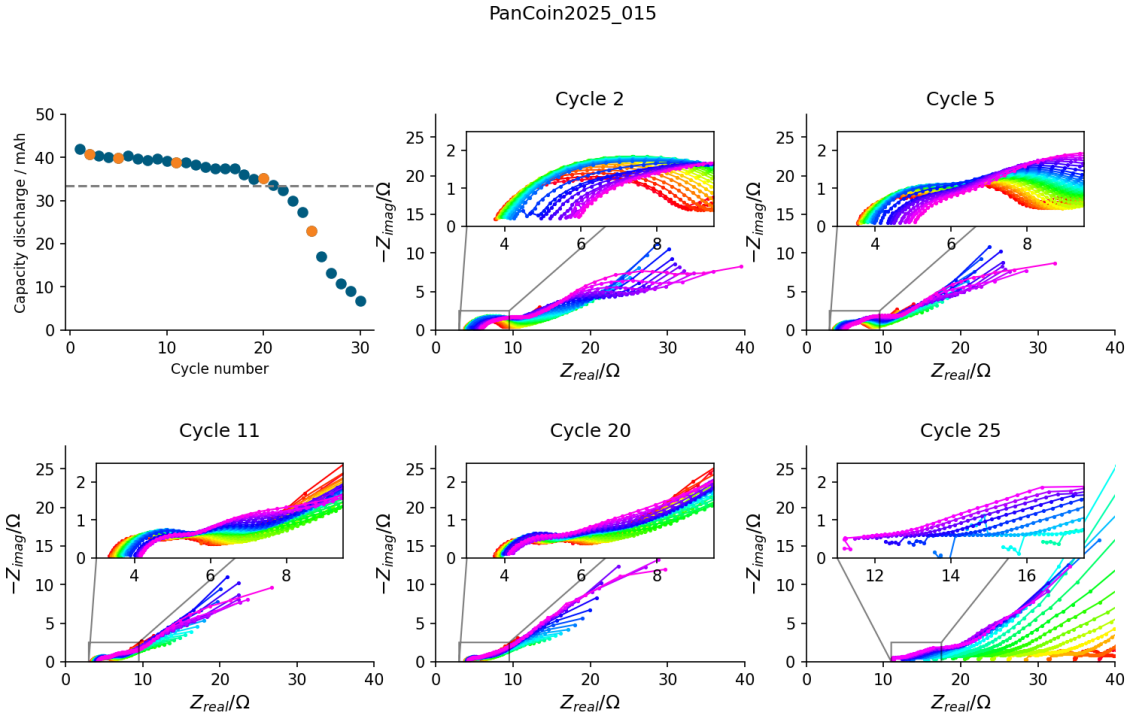


Figure F.17: Dynamic impedance spectra during discharge at selected states of health for cell 15. Plotted one spectra every 10.

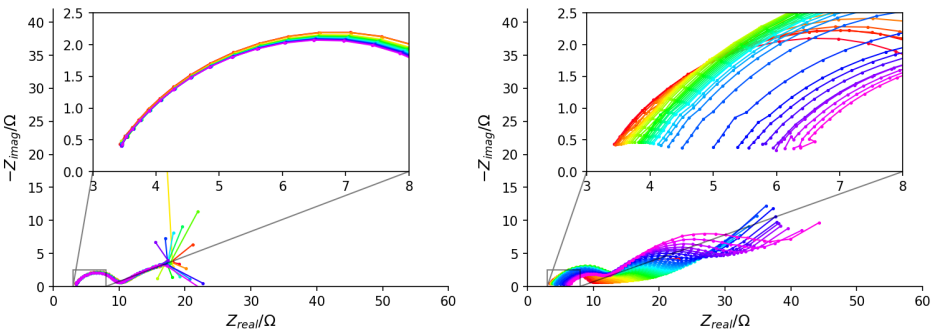


Figure F.18: Impedance spectra during the first cycle for cell 15. Left: all charge spectra; right: discharge spectra with one spectrum every ten cycles.

F.7 Cell 16

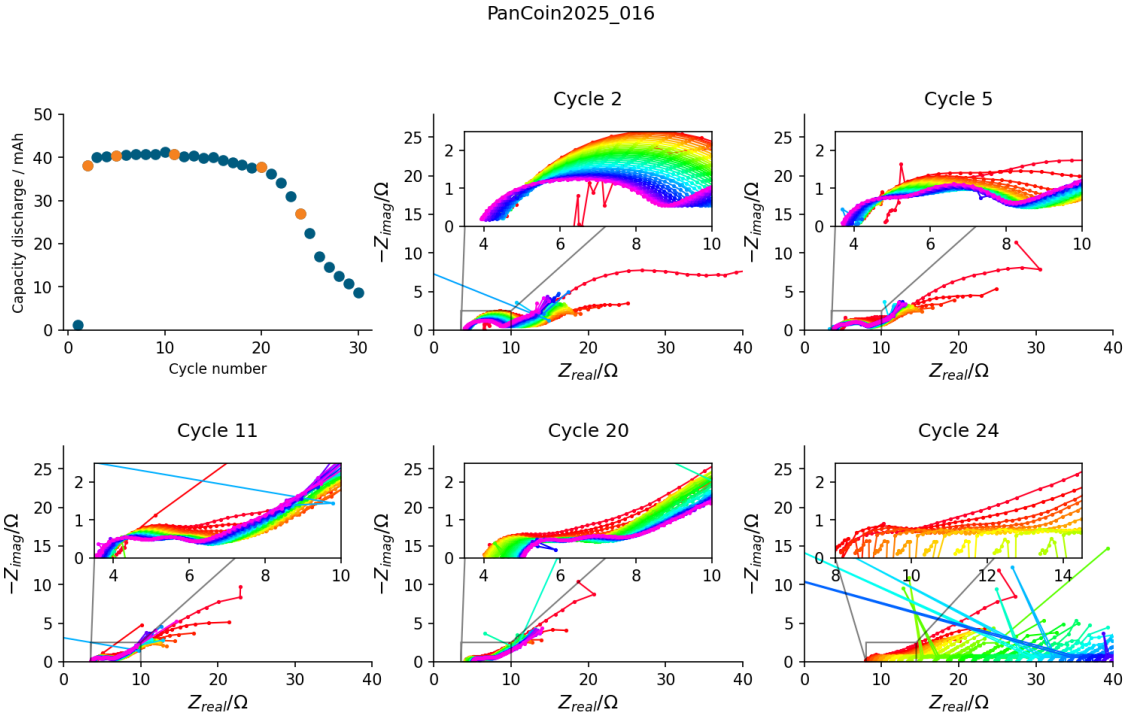


Figure F.19: Dynamic impedance spectra during charge at selected states of health for cell 16. Plotted one spectra every 10.

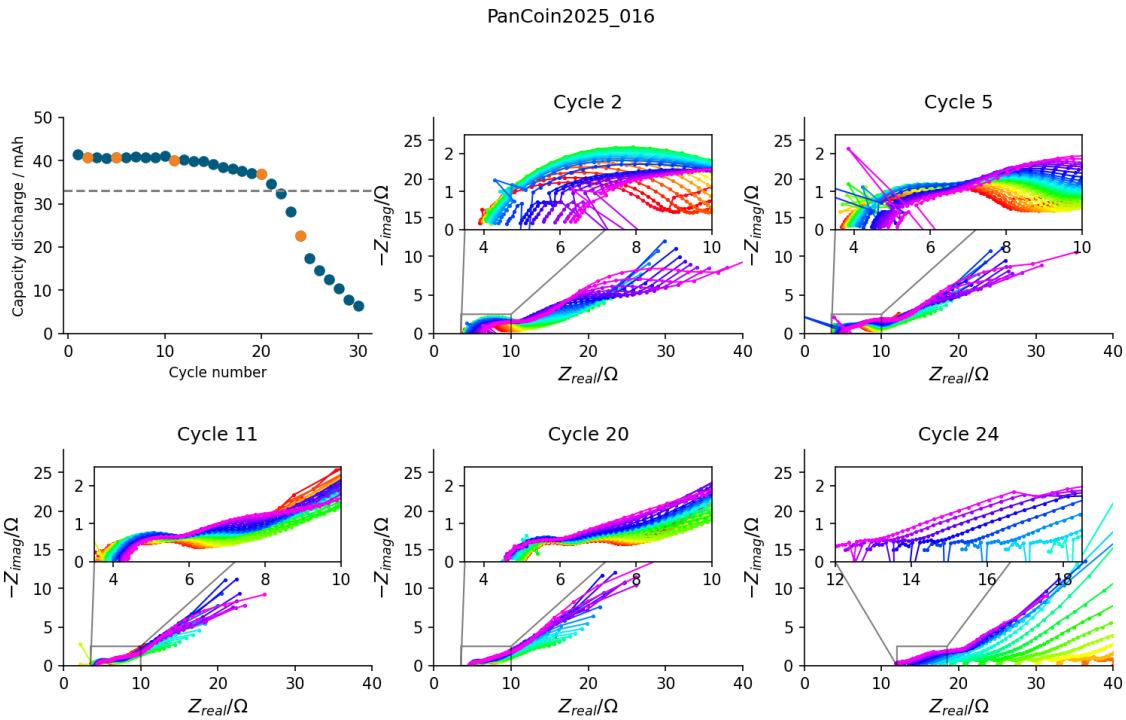


Figure F.20: Dynamic impedance spectra during discharge at selected states of health for cell 16. Plotted one spectra every 10.

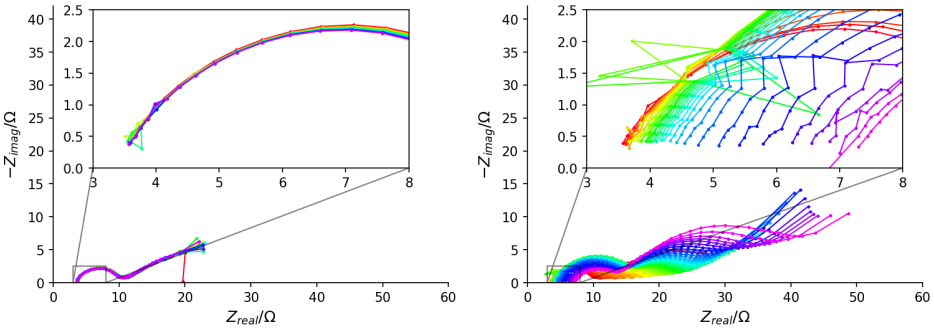


Figure F.21: Impedance spectra during the first cycle for cell 16. Left: all charge spectra; right: discharge spectra with one spectrum every ten cycles.

F.8 Cell 17

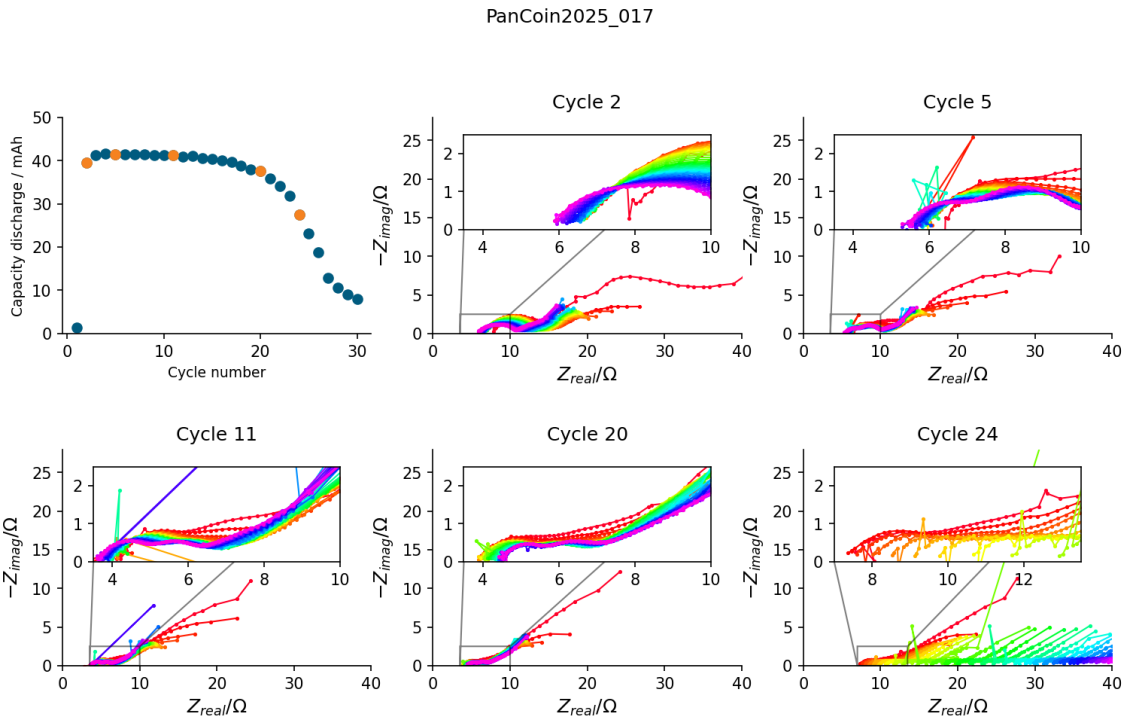


Figure F.22: Dynamic impedance spectra during charge at selected states of health for cell 17. Plotted one spectra every 10.

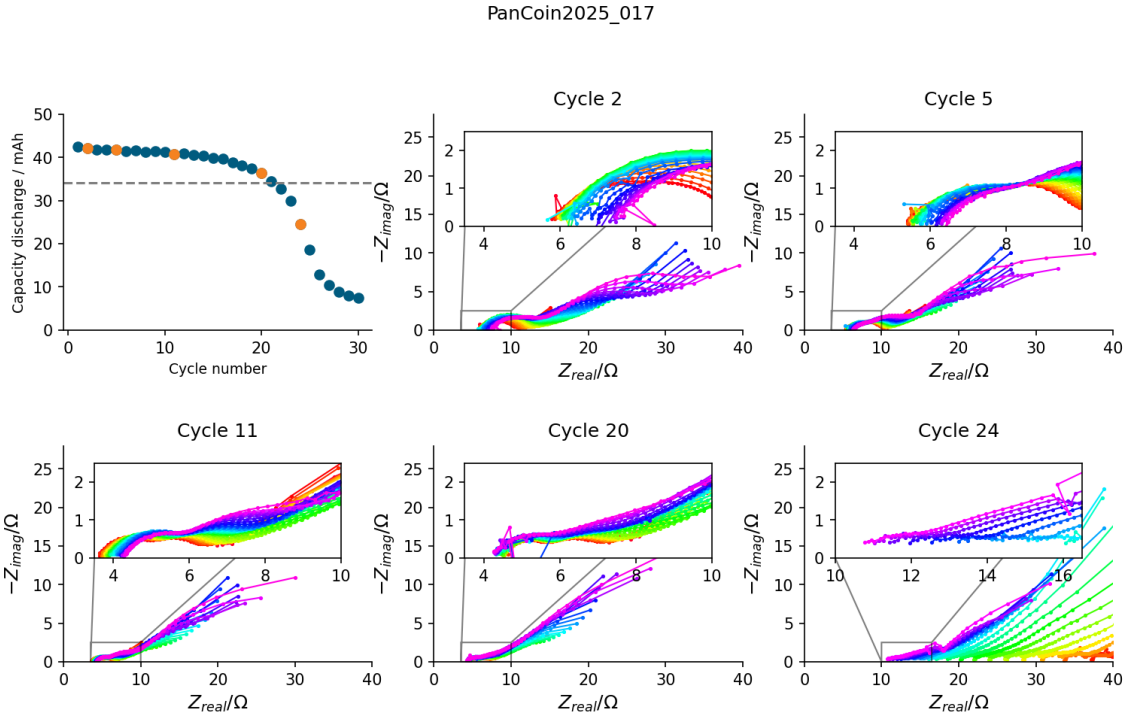


Figure F.23: Dynamic impedance spectra during discharge at selected states of health for cell 17. Plotted one spectra every 10.

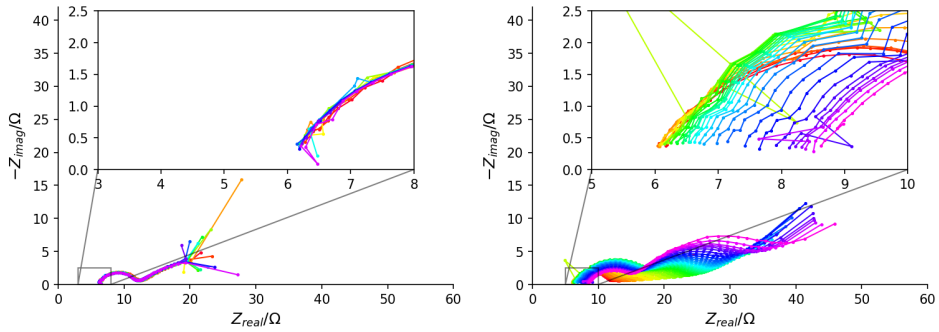


Figure F.24: Impedance spectra during the first cycle for cell 17. Left: all charge spectra; right: discharge spectra with one spectrum every ten cycles.

Appendix G

Validation of LTP data

G.1 Fit of galvanostatic data

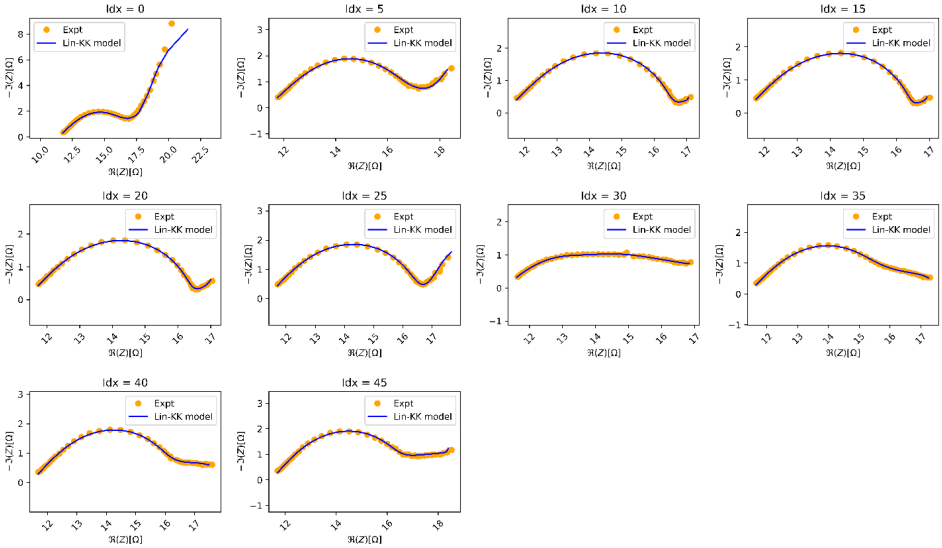


Figure G.1: Fitting of rational polynomial of order 6 on galvanostatic DEIS data.

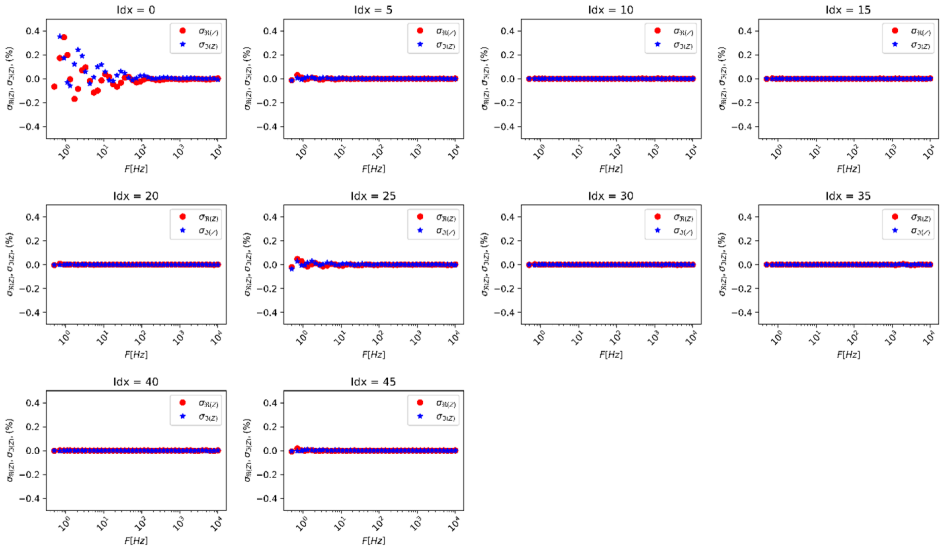


Figure G.2: Residues of the fitting of rational polynomial of order 6 on galvanostatic DEIS data.

G.2 Fit of cyclic voltammetry data

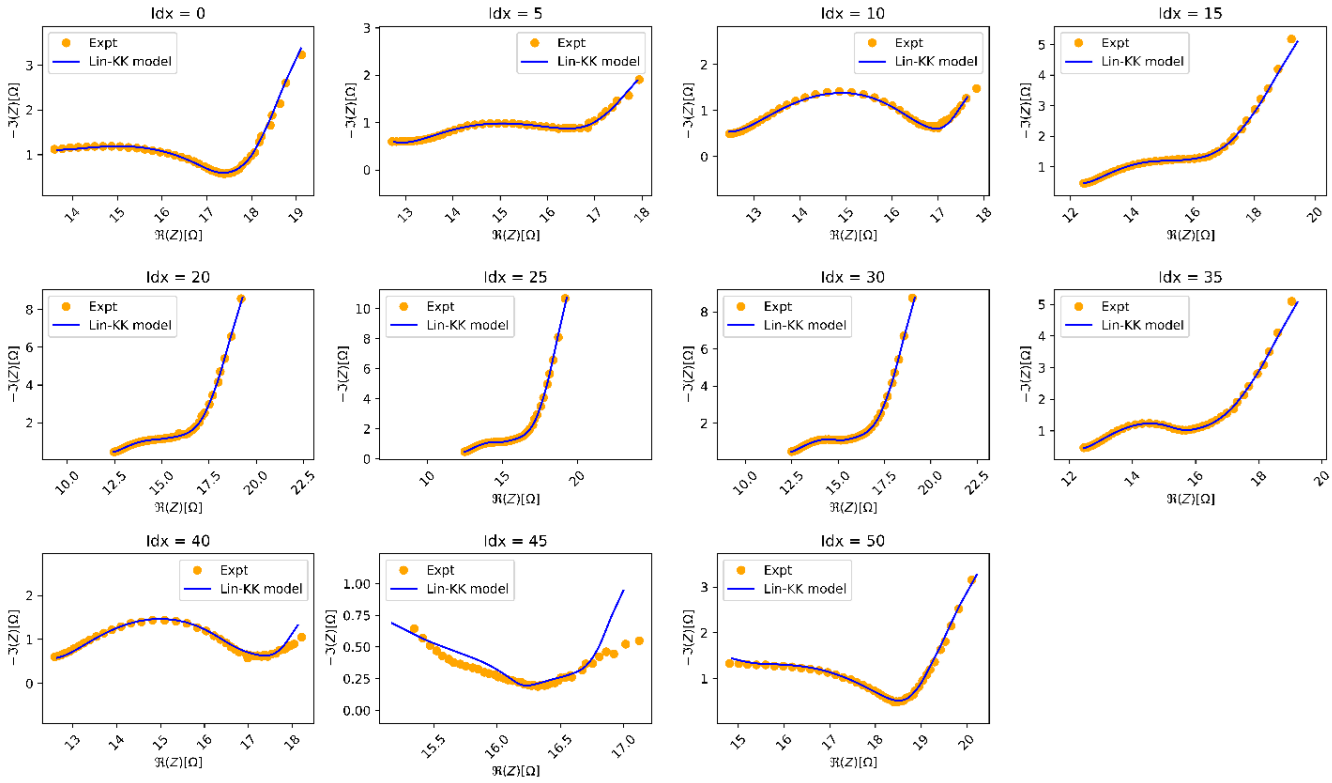


Figure G.3: Fitting of rational polynomial of order 5 on cyclic voltammetry DEIS data.

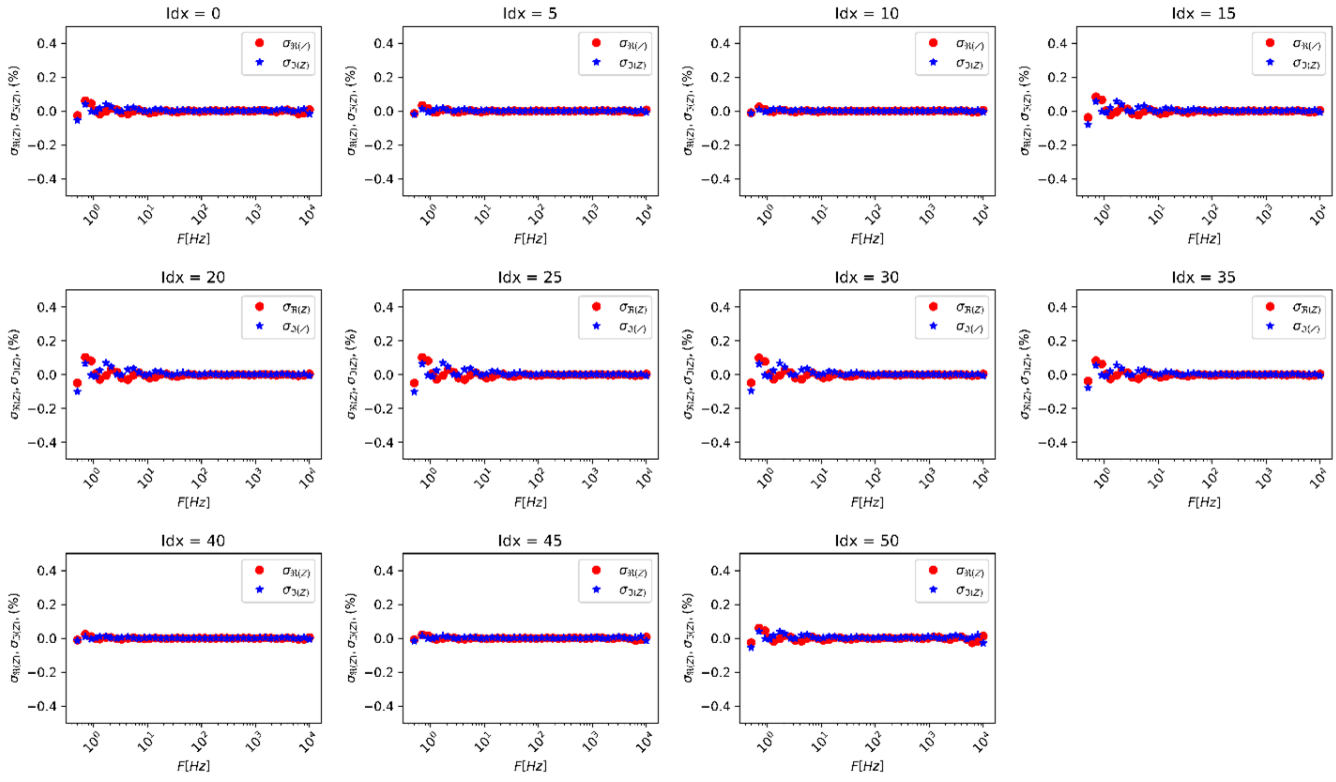


Figure G.4: Residues of the fitting of rational polynomial of order 5 on cyclic voltammetry DEIS data.

Appendix H

Silicon carbonitride as energy storage material

Silicon carbonitride (SiCN(O)) belongs to the class of polymer-derived ceramics obtained through the pyrolysis of preceramic polymers, typically polysilazanes or polysiloxanes, under inert atmosphere at temperatures in the range of 800-1200 °C. During pyrolysis, the organic precursor decomposes into an amorphous ceramic network composed of Si-N, Si-C, and, to a lesser extent, Si-O bonding environments, while simultaneously forming a carbon-rich phase with mixed sp^2/sp^3 character. The relative proportion of ceramic matrix and carbon domains, as well as the pore-size distribution, strongly depends on the precursor chemistry and on the thermal protocol used for crosslinking and pyrolysis. This synthetic route provides a higher level of structural control compared with hard carbons, which derive from the pyrolysis of organic precursors but lack the tunability offered by polymer-derived ceramics.

The resulting microstructure consists of a dual-phase material in which a turbostratic carbon phase percolates through a mechanically rigid amorphous ceramic backbone. The carbon domains facilitate electronic conduction and provide high-surface-area adsorption sites for alkali ions, whereas the ceramic network ensures chemical and mechanical stability. The incorporation of nitrogen into the ceramic framework is particularly relevant for sodium storage, because Si-N bonding increases the affinity of the material for sodium atoms and reduces their binding energy relative to pure carbon surfaces. This, together with the presence of micropores and defective carbon environments, makes SiCN(O) a promising candidate for sodium-ion battery anodes.

The storage of sodium in SiCN(O) is generally attributed to a combination of three partially overlapping mechanisms. At relatively high potentials, adsorption of Na^+ ions on defective carbon sites dominates, leading to a sloping voltage profile analogous to that of hard carbon. At intermediate potentials, ion insertion into micropores contributes significantly to the capacity, reflecting the accessible pore network created during pyrolysis. At lower potentials, and especially when cycling below 0 V vs. Na/Na^+, several studies report an additional reversible capacity that has been interpreted as the partial deposition of metallic sodium inside confined pores. This hypothesis is supported by solid-state NMR, post-mortem microscopy, and cycling experiments showing that the additional low-potential capacity disappears when the microporosity is reduced during synthesis. Nonetheless, the precise nature of the low-potential storage mode remains debated,

and may include quasi-metallic clusters, subnanometric Na aggregates, or deep insertion into defective carbon environments. Because the material is fully amorphous, these mechanisms are expected to coexist and evolve dynamically.

Characterizing SiCN(O) is challenging because the relevant electrochemical processes span a broad range of time scales and are strongly coupled to the material's heterogeneous microstructure. Adsorption, pore filling, SEI formation, and possible metal deposition all occur in partially overlapping voltage windows, and the distribution of pore sizes affects both ion transport and interfacial kinetics. The SEI itself forms on chemically distinct surfaces (carbonaceous domains and ceramic phases), leading to spatially non-uniform interphase compositions and potentially time-dependent changes during cycling. As a result, traditional techniques such as galvanostatic cycling or ex situ surface analysis often provide incomplete or ambiguous information.

In this context, dynamic electrochemical impedance spectroscopy (DEIS) is particularly well suited for studying SiCN(O). Because DEIS resolves the instantaneous impedance along a non-stationary potential trajectory, it can reveal transient kinetic events that are difficult to detect with steady-state techniques. Sudden changes in characteristic time constants, transitions between adsorption- and insertion-dominated regimes, or the onset of fast transport pathways associated with quasi-metallic phases become visible as local impedance features that coincide with specific points of the voltage profile. Moreover, DEIS can track the evolution of SEI resistance during the first cycles, distinguish formation-related impedance from intrinsic electrode impedance, and allow a direct comparison between different synthesis batches or porosity levels without relying exclusively on capacity values. In the context of this thesis, DEIS was used to investigate whether such characteristic signatures could be associated with the hypothesized sodium storage mechanisms in SiCN(O).

The purpose of this appendix is not to provide a complete review of SiCN(O) electrochemistry, but rather to supply the structural and mechanistic background necessary to interpret the DEIS measurements presented in Chapter 12. Readers interested in a comprehensive treatment of the synthesis, structure, and electrochemical behavior of polymer-derived ceramics are referred to the foundational literature [197–199].

Appendix I

Fabrication and Characterization of Microscopic Reference Electrodes

I.1 Design considerations

The goal of the microscopic reference electrode (micro-RE) is to provide a stable potential probe inside a compact cell without perturbing the electric field or contributing significant impedance artefacts. This requires minimizing the exposed electrode surface, ensuring robust insulation along the wire length, and achieving uniform metal electrodeposition on the active tip. Because the reference electrode must be inserted within the separator stack of a pouch cell, flexibility and resistance to mechanical deformation are also essential.

For lithium systems, the most straightforward approach is to electrodeposit Li onto an insulated copper wire. For sodium systems, Na can be electrodeposited on Cu, although Na does not alloy with noble metals such as Au or Al; this limits the use of stable alloy-type references commonly used in lithium-ion research.

I.2 Micro-reference electrodes

Microscopic reference electrodes introduced in pouch cells are made from an insulated metallic wire with the section exposed. This concept is the closest realization of a punctual electrode.

The goal is to probe the local potential at a single point where the field lines are equipotential. In this work I used three types of insulated wires:

- circa 130μm copper manually insulated with thermosetting polymer strips.
- 50μm diameter polymer insulated Copper from Goodfellow;
- 50μm diameter polymer insulated Gold from Goodfellow;

I.2.1 Manually insulated wire

I took a copper wire from a woven de-soldering cloth and used the polymeric thermosetting strips used for sealing pouch cells to cover the wire. Figure I.1 shows the components used to assemble the wire and the comparison with the commercial one.

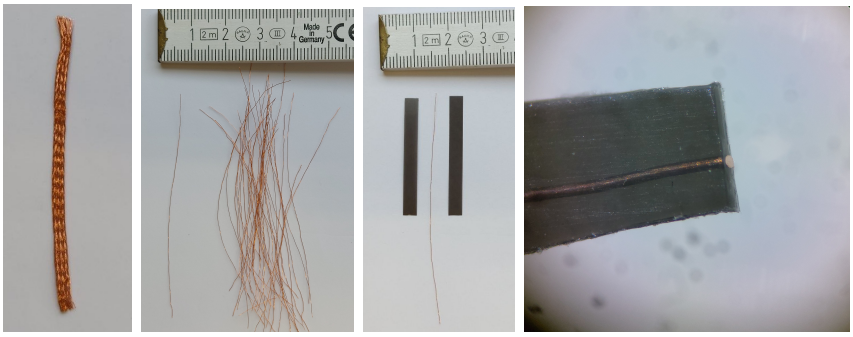


Figure I.1: From left to right: step for the preparation of insulated copper wire.

I.2.2 Commercial insulated wires

The commercial 50μm wire is too thin to clamp to potentiostat crocodile clamps and also prone to break when sealed through the pouch cell foils.

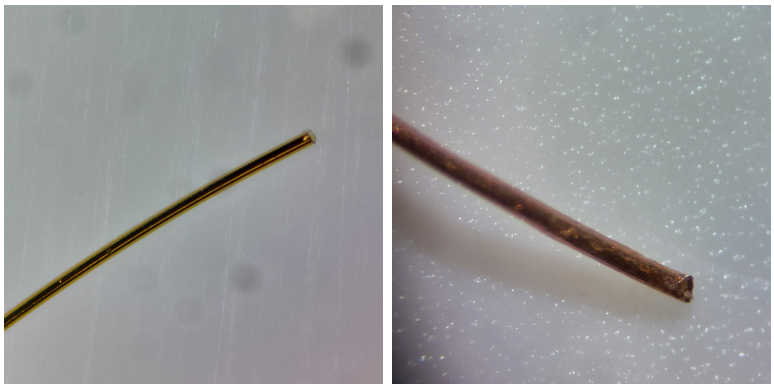


Figure I.2: Photos of 50μm wires. Left in gold and right in copper.

The solution is to remove partial of the insulation of the wire at one end and solder it to a piece of metal to act as current collector as reported in Reference [115].

Figure I.3: Gold wire with removed insulation.

In the paper they use the machine to solder the contact in micro-electronics which I didn't have at disposal.

Attempt with ultrasonic welding

As an alternative, I tried the ultrasonic welder used to solder battery electrode tabs. The device is not meant for such precision jobs but I could tune the parameters to fix the wire to the tabs as in Figure I.4

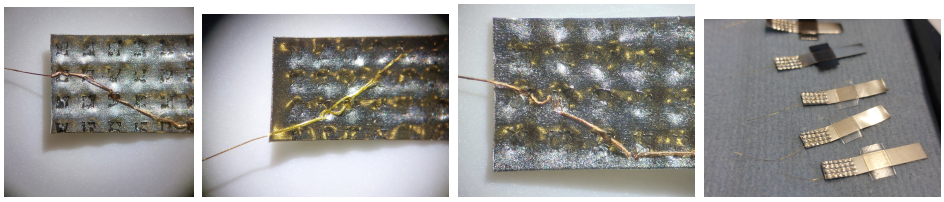


Figure I.4

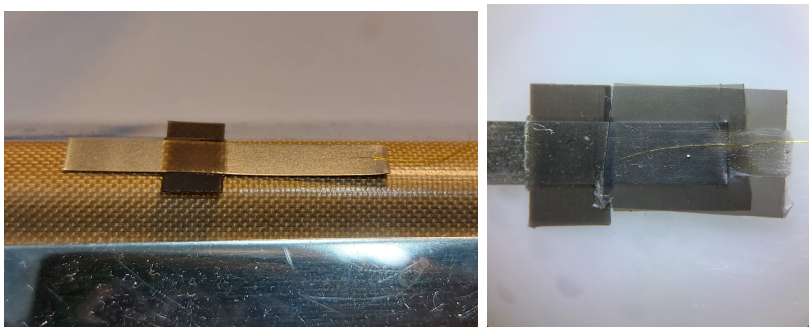
The parameters for the welding are reported in the following table:

Parameter names	value
Nachimpuls	0.3 s
Haltzeit	0.3 s
Höhe	0.56 mm
Ws	90 J
T.P.	4 PSI
W.P.	2 PSI
Oscillation	15 μm

This method proved to be ineffective due to poor mechanical stability and lack of insulation.

Attempt with thermosetting sealing

Finally I tried to put current collector and exposed wire in contact and sealed them together with the thermosetting polymer strips.



Although this provided electrical contact and I could perform electrodeposition of lithium with multiple techniques (CP, CA, CV), the deposited lithium was dissolving back into solution. I observed this because the voltage drifts from the lithium potential to the original value of copper within hours (30-90 minutes). My hypothesis is that some current-collector area remained exposed and electrodeposition occurred on its surface with a very low current density, not sufficient to form stable deposits. Furthermore,

electrodeposition in organic electrolytes is concurrent with the degradation of the electrolyte.

The insulation of the wire was accomplished by positioning the wire between two thermoplastic strips. These were sealed using an ultrasonic sealer, with the excess polymer removed with a scalpel to expose a circular wire section, as shown in Figure XX. Achieving a perpendicular cut to obtain an approximately circular section requires practice. Without microscopic verification, one might end up with an elliptical section from a non-perpendicular cut, resulting in higher surface area.

I.3 Electrodeposition procedures

For Li micro-REs, electrodeposition was performed in a two-electrode configuration using a 12 mm Li metal disk as counter electrode. A two-step protocol was adopted: 10 min of potentiostatic deposition at −100 mV, followed by 3 h of galvanostatic deposition at −50 µA. Afterward, the electrode was left at open-circuit potential. Electrodes fabricated by this method remained stable for several months, as verified by OCV tests.

Sodium deposition followed analogous protocols. However, Na deposition in organic electrolytes competes with electrolyte reduction, and the formation of a heterogeneous SEI layer frequently destabilized the deposited Na. In several cases, the Na film dissolved partially over time, causing the reference potential to drift back toward that of bare copper.

Gold produces an alloy with lithium. Alloys have better defined potential compared to metallic phase and there are less problem with the SEI formed on the surface since no new phase is grown. Stable reference electrodes were reported in the literature produced by alloying at cathodic anodic current of 50pA for 1 hour [115]. I could not replicate the same the processes with my electrodes, probably because of the ineffective current contact between wires and nickel tab.

For Au-wire substrates, alloy formation with Na was not achieved, despite attempts to reproduce literature procedures involving extremely low cathodic currents (tens of picoamperes). This difficulty is attributed to the presence of exposed regions on the metallic current collector tab.

I.4 Electrochemical stability and failure modes

Several failure modes were observed during extended cycling:

- drift of the reference potential due to partial dissolution of deposited metal;
- micro-shorting or leakage currents caused by incomplete insulation;
- mechanical deformation of the wire during pouch-cell sealing, especially for 50 μm substrates;
- artefacts in impedance spectra caused by protrusion of the wire tip into the separator or deformation of the electrode stack;
- sensitivity of the Na micro-RE to electrolyte purity and water contamination.

These issues limit reproducibility and currently prevent quantitative comparison across cells. Nonetheless, in many cases the micro-REs remained sufficiently stable over the duration of a DEIS measurement to enable meaningful qualitative analysis.

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